



Reference Manual
Generic 45nm
Salicide 1.0V/1.8V 1P 11M
Process Design Kit
and Rule Decks (PRD)
Revision 3.5

18/JULY/2011

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Appendix A A1

Appendix B B1

**Cadence Design Systems****GPDK 45 nm Mixed Signal GPDK Spec****DISCLAIMER**

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Revision History

DRC Revision History

RELEASE NOTES FOR THE 45nm GPDK

VERSION 3.5

- gpdk045 IC615 library built natively with IC6.1.5 ISR4 release code
- Added Fluid Guardring support to technology file (CCR884149)
- Modified libInit files to set various tool defaults (CCR910688)
- Added techDerivedLayers to techfile to support substrate extraction and poly cutting diffusion. Added new layers to virtuosoDefaultExtractor Setup Constraint Group (CCR875571/911171))
- Modified MOS pcell to add additional gate pin when GateConnection != None (CCR911205)
- Added ASCII techfile named tech_gpdk045.tf in the gpdk045 library (CCR869154)
- Modified PDK to have/use hspiceD views instead of hspiced (CCR887688) (only mimcap)
- Added 'soce' directory and data to support digital place and route (CCR869153)
- Corrected Assura DRC rule deck for end-of-line space (CCR868717)
- Corrected PVS LVS rules for the symmetric inductor (CCR890569)
- Moved rcx data from assura to qrc directory. It can be used for PVS/assura.

VERSION 3.0

- gpdk045 IC615 library built natively with IC6.1.5 release code
- Added WPE parameter in assura, pvs LVS decks.
- Added new antenna rules in tech file (815543)
- Added taperHalo constraint in virtuosoDefaultTaper in techfile (820934)
- Added minWidth and minSpacing rule for all 4 VT layers.
Modified techfile and DRC deck to adopt this addition.
- Modified the assura_tech.lib file and also added pvtech.lib for PVS. (847441)
- Modified sheet resistance of Metals to match the spectre model files
Change will affect the structure of Metal resistors.
- Modified sheet resistance of resnsnpoly to match the spectre model files
Change will affect the structure of resnsnpoly resistors.
- Model name of all the cells has been made unique to gpdk045 (CCR652663)
and short names. Model name change will reflect in
PCells, CDF, assura LVS/QRC, pvl LVS.
- Fixed mimcap CDF computation parameter (CCR773058)
- Modified mask numbers for layer in tech file (CCR783353)
- Fixed DRC assura decks to remove stamp errors (CCR795365)
- Description of rules ('ref arg) has been added in techfile (CCR778281)
- Added minEndOfLineSpacing rules in DRC (both assura & PVS) (CCR698679)
- Fixed the error on axlregisterCustomDeviceFilter in skill code
gpdk045_customFilter.il file (CCR 705077).
- Techfile modifications:
 - o Made prBoundary layer invalid (CCR 744942)
 - o Added layer purposed required to run abstract generator without error
and without tech.db modification (CCR 705019,797004)
 - o Added minPRBoundaryInteriorHalo constraint with half spacing values...

Generator Info

Generator Information

Sample runset for 45 nm
technology

Default Grid: 0.005

Valid Angle: 45

Flag Acute: true

Flag Self-intersecting: true

Global Parameters

Global Parameters

libName	gpdk045	Primitive Library Name	
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Introduction

This document defines the Design Rules and Electrical Parameters for a generic, foundary independent 45nm CMOS Mixed-Signal process.

This document is divided into three sections:

- * CMOS Digital Core Design Rules

describes the widths, spacings, enclosures, overlaps, etc. needed to create the physical layout of the core section of a digital CMOS design.

- * CMOS I/O Design Rules

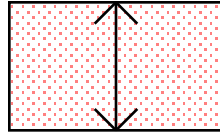
describes the widths, spacings, enclosures, overlaps, etc. needed to create the physical layout of the I/O section of a CMOS design.

- * CMOS Digital Electrical Parameters

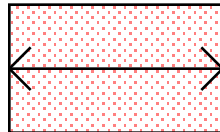
describes the electrical parameters of a digital CMOS design.

Terminology Definitions

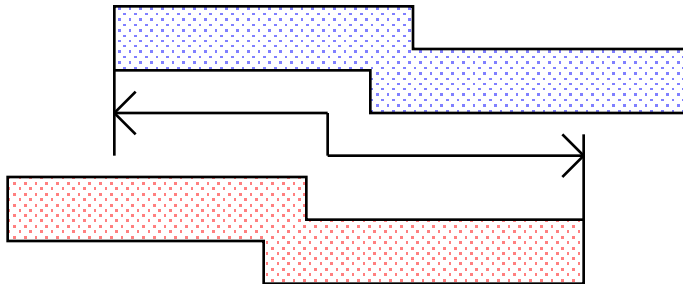
Width - shortest distance from the inside of the edge of a shape to the inside of the edge of the same shape.



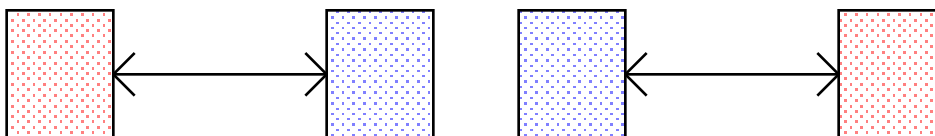
Length - opposite of Width - the measurement of the longest edge of a shape.



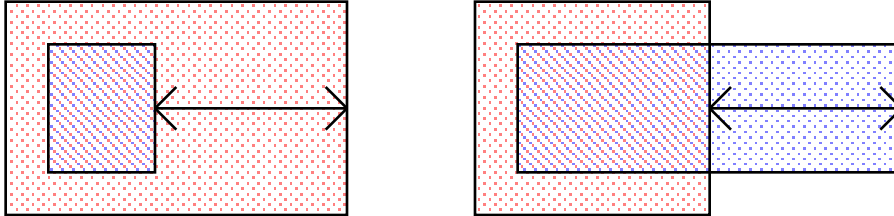
Parallel Run Length - the distance two shapes maintain a spacing less than the check value.



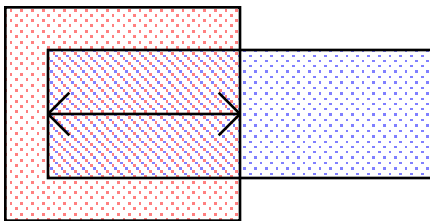
Spacing - distance from the outside of the edge of a shape to the outside of the edge of another shape.



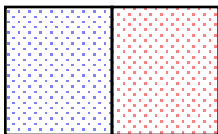
Enclosure - distance from the inside of the edge of a shape to the outside of the edge of another shape.



Overlap - distance from the inside of the edge of a shape to the inside of the edge of another shape.



Butting - outside of the edge of a shape touching the outside of the edge of another shape.



Layer Descriptions

This table describes the layers used to create devices.

Comment Table

Layer Name	GDSII Stream Number	GDSII Data Type	DFII LSW Name	DFII Layer Name	DFII Layer Purpose	DFII Layer Number	DFII Purpose Number	Description
Bondpad	36	0	Bondpad	Bondpad	drawing	95	252	Bonding Pad
CapMetal	14	0	CapMetal	CapMetal	drawing	97	252	MiM capacitor metal
Nburied	19	0	Nburied	Nburied	drawing	18	252	N+ Buried Layer
Nhvt	18	0	Nhvt	Nhvt	drawing	11	252	NMOS High Vt
Nlvt	26	0	Nlvt	Nlvt	drawing	26	252	NMOS Low Vt
Nimp	4	0	Nimp	Nimp	drawing	12	252	N+ Implant
Nwell	2	0	Nwell	Nwell	drawing	6	252	Nwell
Nzvt	52	0	Nzvt	Nzvt	drawing	15	252	NMOS Zero Vt
Oxide	1	0	Oxide	Oxide	drawing	2	252	Active Area
Oxide_thk	24	0	Oxide_thk	Oxide_thk	drawing	4	252	1.8V Active Area
Phvt	23	0	Phvt	Phvt	drawing	13	252	PMOS High Vt
Plvt	27	0	Plvt	Plvt	drawing	27	252	PMOS Low Vt
Pimp	5	0	Pimp	Pimp	drawing	14	252	P+ Implant
Poly	3	0	Poly	Poly	drawing	10	252	Poly
SiProt	72	0	SiProt	SiProt	drawing	16	252	Salicide Block

Table 1: Device Layers

This table describes the layers used to interconnect devices.

Comment Table

Layer Name	GDSII Stream Number	GDSII Data Type	DFII LSW Name	DFII Layer Name	DFII Layer Purpose	DFII Layer Number	DFII Purpose Number	Description
Cont	6	0	Cont	Cont	drawing	20	252	Metal Contact to Oxide/Poly
Metal1	7	0	Metal1	Metal1	drawing	30	252	1st Metal for interconnect
Metal2	9	0	Metal2	Metal2	drawing	34	252	2nd Metal for interconnect
Metal3	11	0	Metal3	Metal3	drawing	38	252	3rd Metal for interconnect
Metal4	31	0	Metal4	Metal4	drawing	42	252	4th Metal for interconnect
Metal5	33	0	Metal5	Metal5d	drawing	46	252	5th Metal for interconnect
Metal6	35	0	Metal6	Metal6	drawing	50	252	6th Metal for interconnect
Metal7	38	0	Metal7	Metal7	drawing	54	252	7th Metal for interconnect
Metal8	40	0	Metal8	Metal8d	drawing	58	252	8th Metal for interconnect
Metal9	42	0	Metal9	Metal9	drawing	62	252	9th Metal for interconnect
Metal10	152	0	Metal10	Metal10	drawing	66	252	10th Metal for interconnect
Metal11	162	0	Metal11	Metal11	drawing	70	252	11th metal for interconnect
Via1	8	0	Via1	Via1	drawing	32	252	Via between 1st and 2nd Metal
Via2	10	0	Via2	Via2	drawing	36	252	Via between 2nd and 3rd Metal
Via3	30	0	Via3	Via3	drawing	38	252	Via between 3rd and 4th Metal
Via4	32	0	Via4	Via4	drawing	44	252	Via between 4th and 5th Metal
Via5	34	0	Via5	Via5	drawing	48	252	Via between 5th and 6th Metal
Via6	37	0	Via6	Via6	drawing	52	252	Via between 6th and 7th Metal
Via7	39	0	Via7	Via7	drawing	54	252	Via between 7th and 8th Metal
Via8	41	0	Via8	Via8	drawing	60	252	Via between 8th and 9th Metal
Via9	151	0	Via9	Via9	drawing	64	252	Via between 9th and 10th Metal
Via10	161	0	Via10	Via10	drawing	68	252	Via between 10th and 11th Metal

Table 2: Interconnect Layers

This table describes the layers used to mark/label shapes for DRC and/or LVS..
 Comment Table

Layer Name	GDSII Stream Number	GDSII Data Type	DFII LSW Name	DFII Layer Name	DFII Layer Purpose	DFII Layer Number	DFII Purpose Number	Description
BJTdum	15	0	BJTdum	BJTdum	drawing	92	252	Marks BJT emitters
Capdum	12	0	Capdum	Capdum	drawing	96	252	Marks capacitors
Cap3dum	84	0	Cap3dum	Cap3dum	drawing	93	252	Marks capacitors 3 term
DIOdummy	22	0	DIOdum	DIOdummy	drawing	82	252	Marks diodes
INDdummy	90	0	INDdum	INDdummy	drawing	90	252	Marks inductor terminal
IND2dummy	88	0	IND2dum	IND2dummy	drawing	88	252	Marks inductor terminal
IND3dummy	114	0	IND3dum	IND3dummy	drawing	114	252	Marks inductor terminal
ESDdummy	74	0	ESDdum	ESDdummy	drawing	115	252	Marks ESD and I/O devices
Metal1_text	7	3	Metal1	Metal1	drawing	30	252	Labels Metal1 nodes
Metal2_text	9	3	Metal2	Metal2	drawing	34	252	Labels Metal2 nodes
Metal3_text	11	3	Metal3	Metal3	drawing	38	252	Labels Metal3 nodes
Metal4_text	31	3	Metal4	Metal4	drawing	42	252	Labels Metal4 nodes
Metal5_text	33	3	Metal5	Metal5	drawing	46	252	Labels Metal5 nodes
Metal6_text	35	3	Metal6	Metal6	drawing	50	252	Labels Metal6 nodes
Metal7_text	38	3	Metal7	Metal7	drawing	54	252	Labels Metal7 nodes
Metal8_text	40	3	Metal8	Metal8	drawing	58	252	Labels Metal8 nodes
Metal9_text	42	3	Metal9	Metal9	drawing	62	252	Labels Metal9 nodes
Metal10_text	52	3	Metal10	Metal10	drawing	72	252	Labels Metal10 nodes
Metal11_text	62	3	Metal11	Metal11	drawing	82	252	Labels Metal11 nodes
NPNdummy	20	0	NPNdum	NPNdummy	drawing	86	252	Marks NPN devices
PNPdummy	21	0	PNPdum	PNPdummy	drawing	84	252	Marks PNP devices
Psub	25	0	Psub	Psub	drawing	80	252	Marks separate substrate areas
Resdum	13	0	Resdum	Resdum	drawing	94	252	Marks Poly/Oxide resistor area
ResWdum	71	0	ResWdum	ResWdum	drawing	98	252	Marks Nwell resistor area
text	63	0	text	text	drawing	230	252	Text for information
SRAM	64	0	SRAM	SRAM	drawing	71	252	Memory marker layer
PO_text	65	0	PO_text	PO_text	drawing	72	252	Poly text marker layer
SEALRING	66	0	SEALRING	SEALRING	drawing	73	252	Die Sealing marker layer
LOGO	67	0	LOGO	LOGO	drawing	74	252	Chip Logo marker layer
ANALOG	68	0	ANALOG	ANALOG	drawing	75	252	Special Analog marker layer
FUSE	69	0	FUSE	FUSE	drawing	76	252	Fuse marker layer
FILLER	86	0	FILLER	FILLER	drawing	77	252	Fill cell marker layer
VIAEXCL	87	0	VIAEXCL	VIAEXCL	drawing	78	252	Via exclude marker layer

Table 3: DRC/LVS Marker/Label Layers

Device Layer Table

This table describes the layers used in each device.

- 0: the layer must not touch the device structure
- 1: the layer must enclose or straddle the device structure
- : the layer may either enclose or avoid the device structure

Comment Table

	NMOS (1.2V)	PMOS (1.2V)	LP NMOS (1.2V)	LP PMOS (1.2V)	NMOS (1.8V)	PMOS (1.8V)	Native NMOS (1.2V)	Native NMOS (1.8V)
Nburied	0	0	0	0	0	0	0	0
Nwell	0	1	0	1	0	1	0	0
Oxide	1	1	1	1	1	1	1	1
Oxide_thk	0	0	0	0	1	1	0	1
Poly	1	1	1	1	1	1	1	1
Nimp	1	0	1	0	1	0	1	1
Pimp	0	1	0	1	0	1	0	0
Nzvt	0	0	0	0	0	0	1	1
Nhvt	0	0	1	0	0	0	0	0
Phvt	0	0	0	1	0	0	0	0
SiProt	0	0	0	0	0	0	0	0

Table 4: MOS Device Layers

Comment Table

	N+/PW Diode	P+/NW Diode
Nburied	0	0
Nwell	0	1
Oxide	1	1
Oxide_thk	0	0
Poly	0	0
Nimp	1	0
Pimp	0	1
Nzvt	0	0
Nhvt	0	0
Phvt	0	0
SiProt	0	0

Table 5: Diode Device Layers

Comment Table

	Salicided N+ Poly Resistor	Salicided P+ Poly Resistor	Salicided N+ Oxide Resistor	Salicided P+ Oxide Resistor	Non- Salicided N+ Poly Resistor	Non- Salicided P+ Poly Resistor	Non- Salicided N+ Oxide Resistor	Non- Salicided P+ Oxide Resistor
Nburied	0	0	0	0	0	0	0	0
Nwell	-	-	0	1	-	-	0	1
Oxide	0	0	1	1	0	0	1	1
Oxide_thk	0	0	0	0	0	0	0	0
Poly	1	1	0	0	1	1	0	0
Nimp	1	0	1	0	1	0	1	0
Pimp	0	1	0	1	0	1	0	1
Nzvt	0	0	0	0	0	0	0	0
Nhvt	0	0	0	0	0	0	0	0
Phvt	0	0	0	0	0	0	0	0
SiProt	0	0	0	0	1	1	1	1

Comment Table

	Nwell in Oxide Resistor	Nwell in STI Resistor
Nburied	0	0
Nwell	1	1
Oxide	1	1
Oxide_thk	0	0
Poly	0	0
Nimp	1	1
Pimp	0	0
Nzvt	0	0
Nhvt	0	0
Phvt	0	0
SiProt	1	0

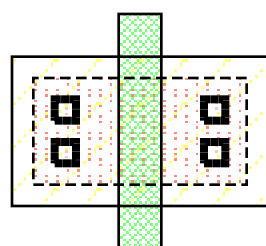
Table 6: Resistor Device Layers

Comment Table

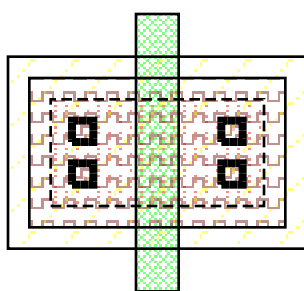
	SPNP	VNPN	Varactor (NMOSCAP)
Nburied	0	1	0
Nwell	1	1	1
Oxide	1	1	1
Oxide_thk	0	0	0
Poly	0	0	1
Nimp	1	1	1
Pimp	1	1	0
Nzvt	0	0	0
Nhvt	0	0	0
Phvt	0	0	0
SiProt	0	0	0

Table 7: Bipolar and Varactor Device Layers

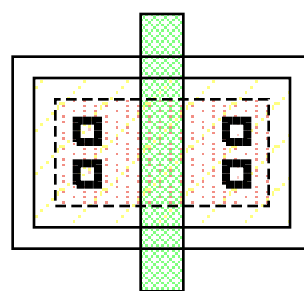
Device Layout Examples



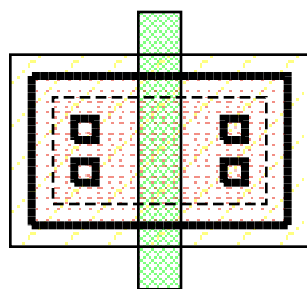
1.2V NMOS



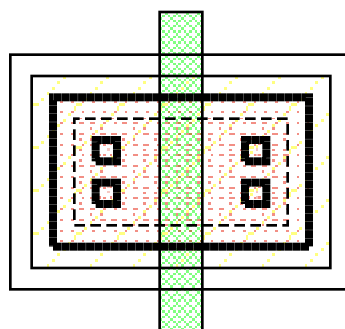
1.2V LP NMOS



1.2V Native NMOS

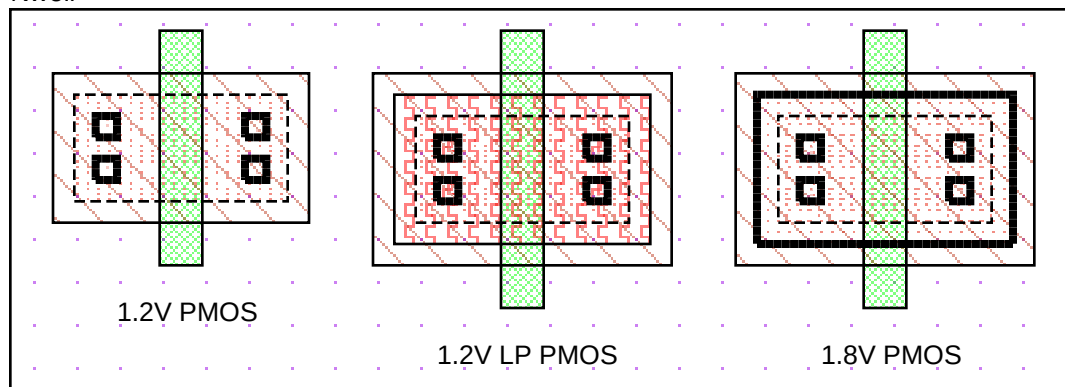


1.8V NMOS

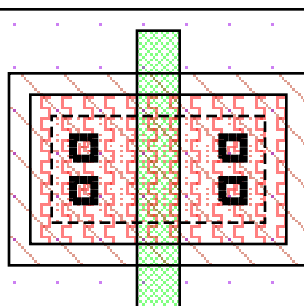


1.8V Native NMOS

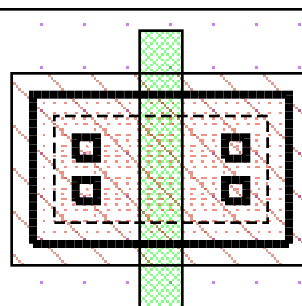
Nwell



1.2V PMOS



1.2V LP PMOS



1.8V PMOS

Nburied



Nwell



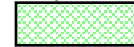
Oxide



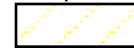
Oxide thk



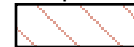
Poly



Nimp



Pimp



Nzvt



Nhvt

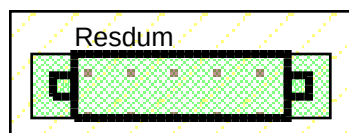


Phvt

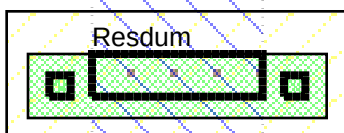


Cont

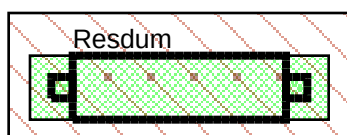




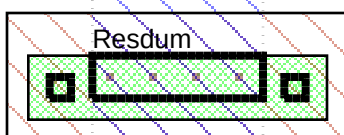
Salicided N+ Poly Resistor



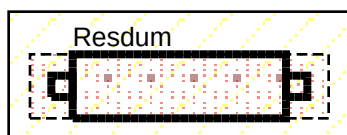
Non-Salicided N+ Poly Resistor



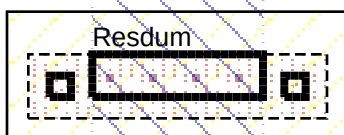
Salicided P+ Poly Resistor



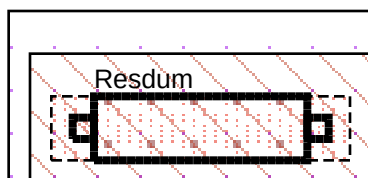
Non-Salicided P+ Poly Resistor



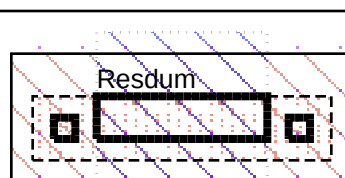
Salicided N+ Oxide Resistor



Non-Salicided N+ Oxide Resistor



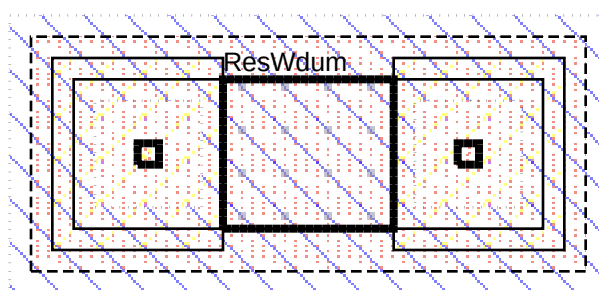
Salicided P+ Oxide Resistor



Non-Salicided P+ Oxide Resistor

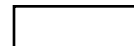


Nwell in STI Resistor



Nwell in OD Resistor

Nburied



Nwell



Oxide



Poly



Nimp



Pimp



Nzvt



Nhvt



Phvt



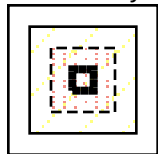
Cont



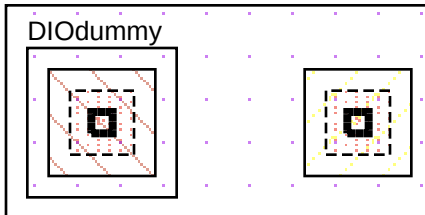
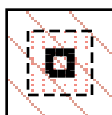
SiProt



DIOdummy

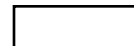


N+/PW Diode



P+/NW Diode

Nburied



Nwell



Oxide



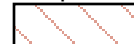
Poly



Nimp



Pimp



Nzvt



Nhvt



Phvt



Cont



SiProt

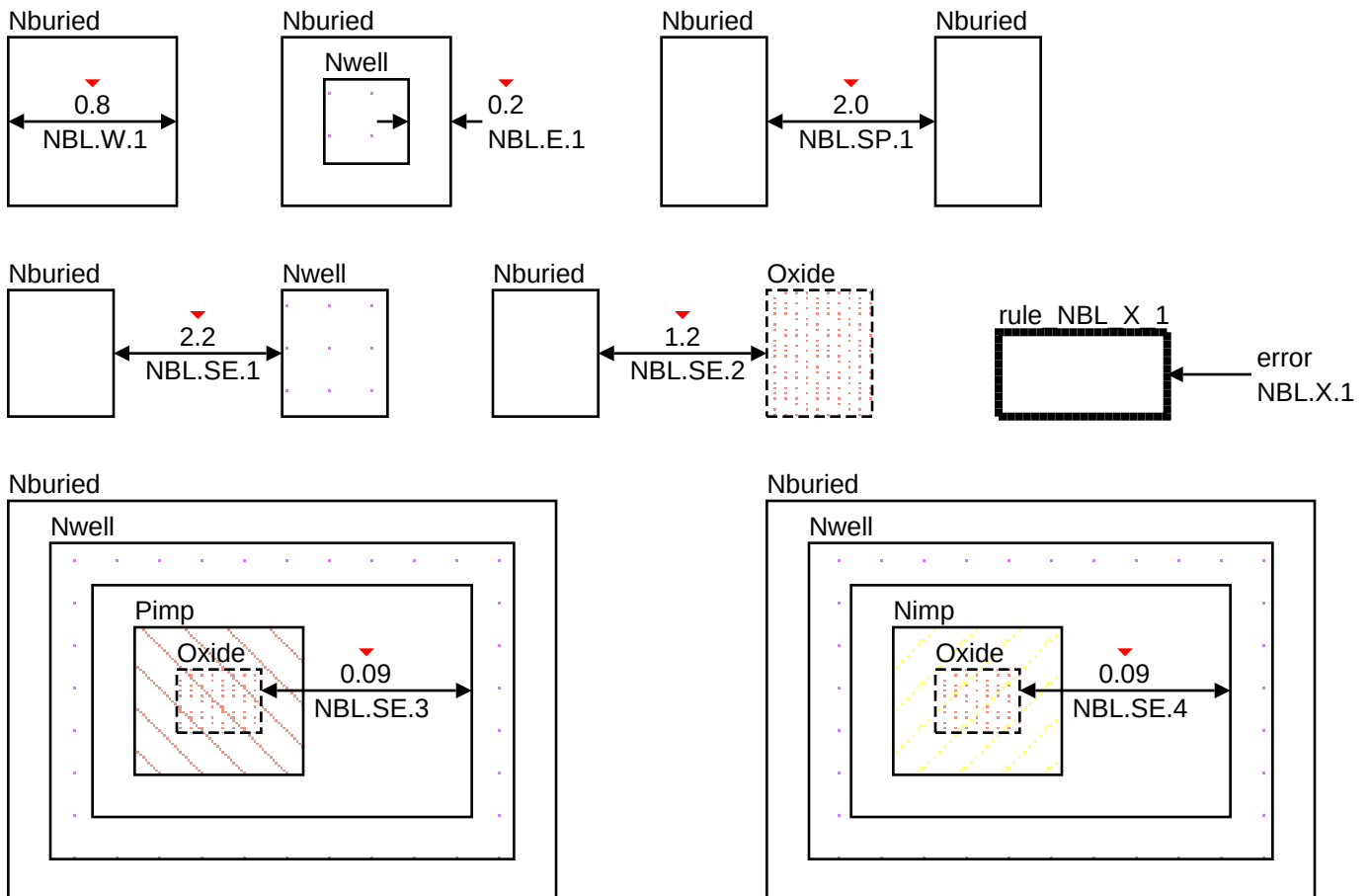


CMOS Digital Core Design Rules

N BURIED LAYER RULES

Data Table: NBL_DRC

RuleName	Description	Value
NBL.W.1	Minimum Nburied Width	0.8
NBL.E.1	Minimum Nburied to Nwell enclosure	0.2
NBL.SP.1	Minimum Nburied to Nburied spacing	2.0
NBL.SE.1	Minimum Nburied to non-related Nwell spacing	2.2
NBL.SE.2	Minimum Nburied to non-related Oxide spacing	1.2
NBL.SE.3	Minimum Nwell ring (on Nburied) to P+ Active spacing	0.09
NBL.SE.4	Minimum Nwell ring (on Nburied) to N+ Active spacing	0.09
NBL.X.1	Nwell must form isolation rings on Nburied	---



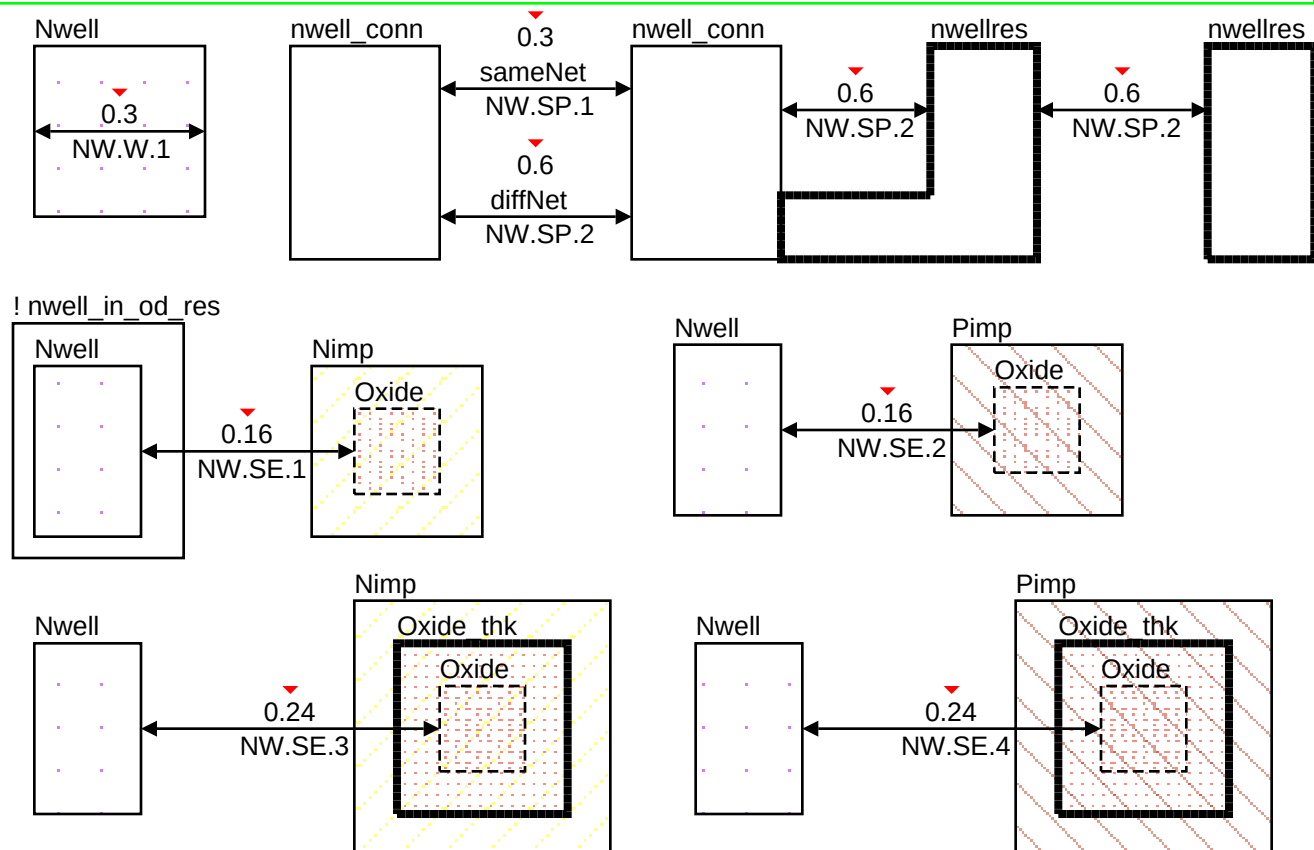
NWELL AND NWELL RESISTOR (under STI) RULES

Data Table: NWELL_DRC

RuleName	Description	Value
NW.W.1	Minimum Nwell Width	0.3
NW.SP.1	Minimum Nwell spacing to Nwell (same potential)	0.3
NW.SP.2	Minimum Nwell spacing to Nwell (different potential)	0.6
NW.SE.1	Minimum Nwell spacing to N+ Active Area	0.16
NW.SE.2	Minimum Nwell spacing to P+ Active Area	0.16
NW.SE.3	Minimum Nwell spacing to N+ 1.8V Active Area	0.24
NW.SE.4	Minimum Nwell spacing to P+ 1.8V Active Area	0.24
NW.E.1	Minimum Nwell enclosure of N+ Active Area	0.06
NW.E.2	Minimum Nwell enclosure of P+ Active Area	0.06
NW.E.3	Minimum Nwell enclosure of N+ 1.8V Active Area	0.24
NW.E.4	Minimum Nwell enclosure of P+ 1.8V Active Area	0.24
NW.A.1	Minimum Nwell area	0.18
NW.EA.1	Minimum Nwell enclosed area	0.18

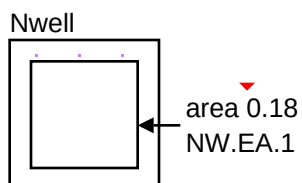
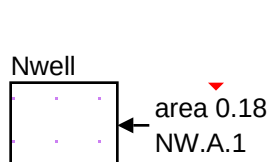
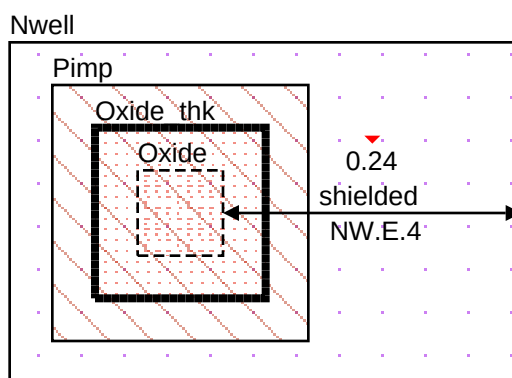
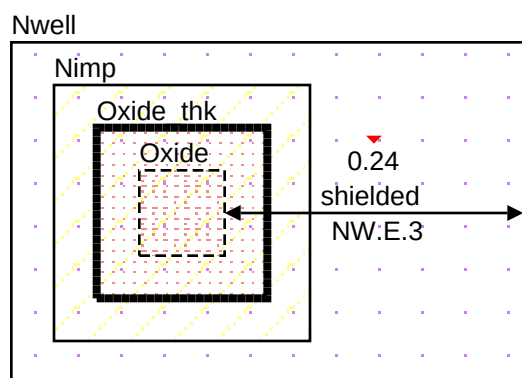
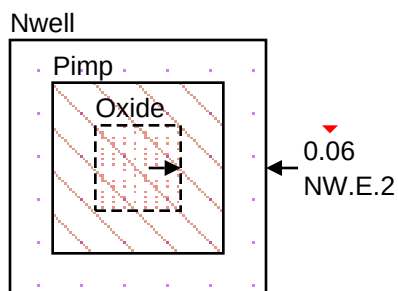
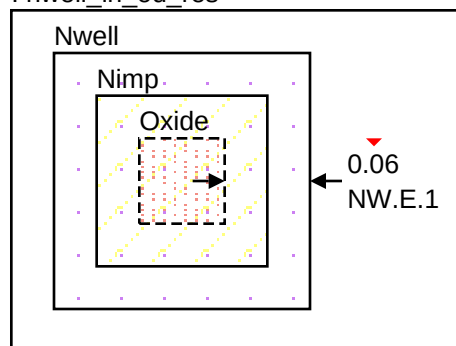
Nwell resistor is defined by the intersection of Nwell and ResWdum for DRC and LVS.

For STI Nwell resistors, the ResWdum shape must butt the N+ Oxide on both ends of Nwell the resistor and the ResWdum shape must be coincident or extend beyond the Nwell edges along the...



NWELL AND NWELL RESISTOR (under STI) RULES (continued)

! nwell_in_od_res



NWELL RESISTOR WITHIN OXIDE RULES

Data Table: NWRES_DRC

RuleName	Description	Value
NWR.E.1	Minimum Active Area to Nwell (in resistor) enclosure	0.6
NWR.E.2	Minimum salicided Nwell to Contact enclosure	0.16
NWR.SE.1	Minimum Resist Protect Oxide to Nwell spacing	0.16
NWR.E.3	Minimum Resist Protect Oxide to Oxide enclosure	0.12
NWR.O.1	Minimum N+ Implant to Resist Protect Oxide overlap	0.22
NWR.X.1	Thick Oxide is NOT allowed over Nwell Resistor	---
NWR.SP.1	Minimum Nwell resistor to other Nwell spacing	0.6

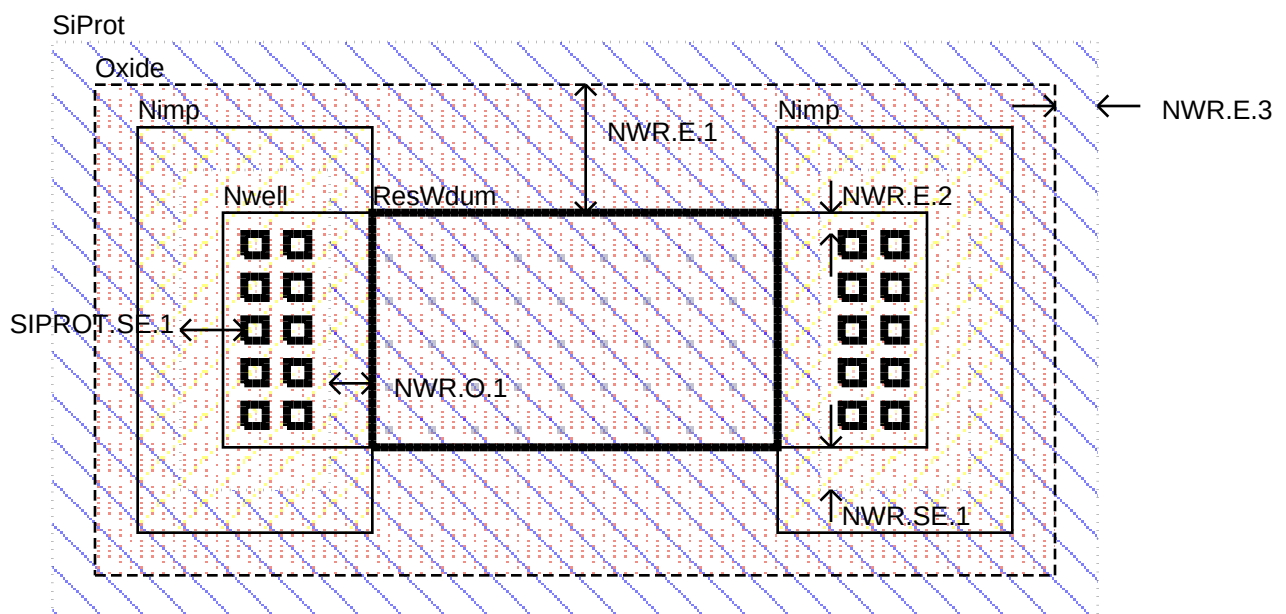
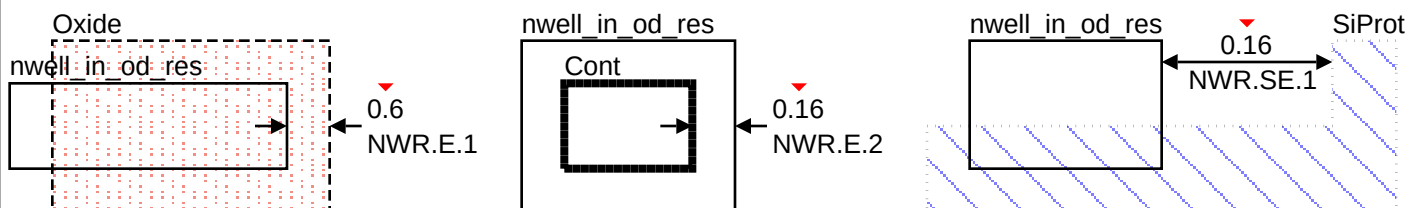


Figure 1: NWELL RESISTOR WITHIN OXIDE RULES

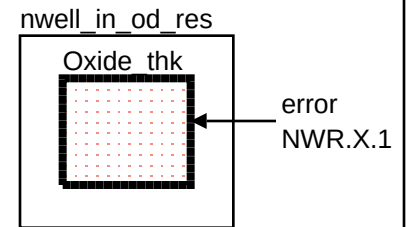
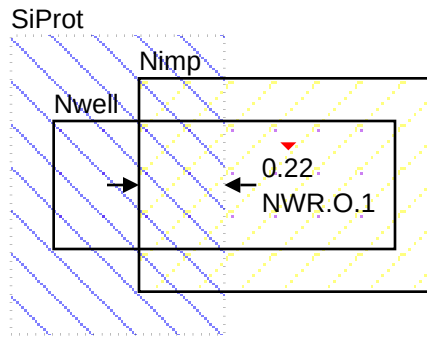
Nwell resistor in Oxide is defined by the intersection of Nwell and Resdum for DRC and LVS.

For Nwell resistor within Oxide, the ResWdum shape must butt the Nimp on both ends of the Nwell resistor and the ResWdum shape must be coincident or extend beyond the Nwell edges along the length of the Nwell resistor.



NWELL RESISTOR WITHIN OXIDE RULES (continued)

NWR.E.3 - Covered by
SIPROT.E.1.

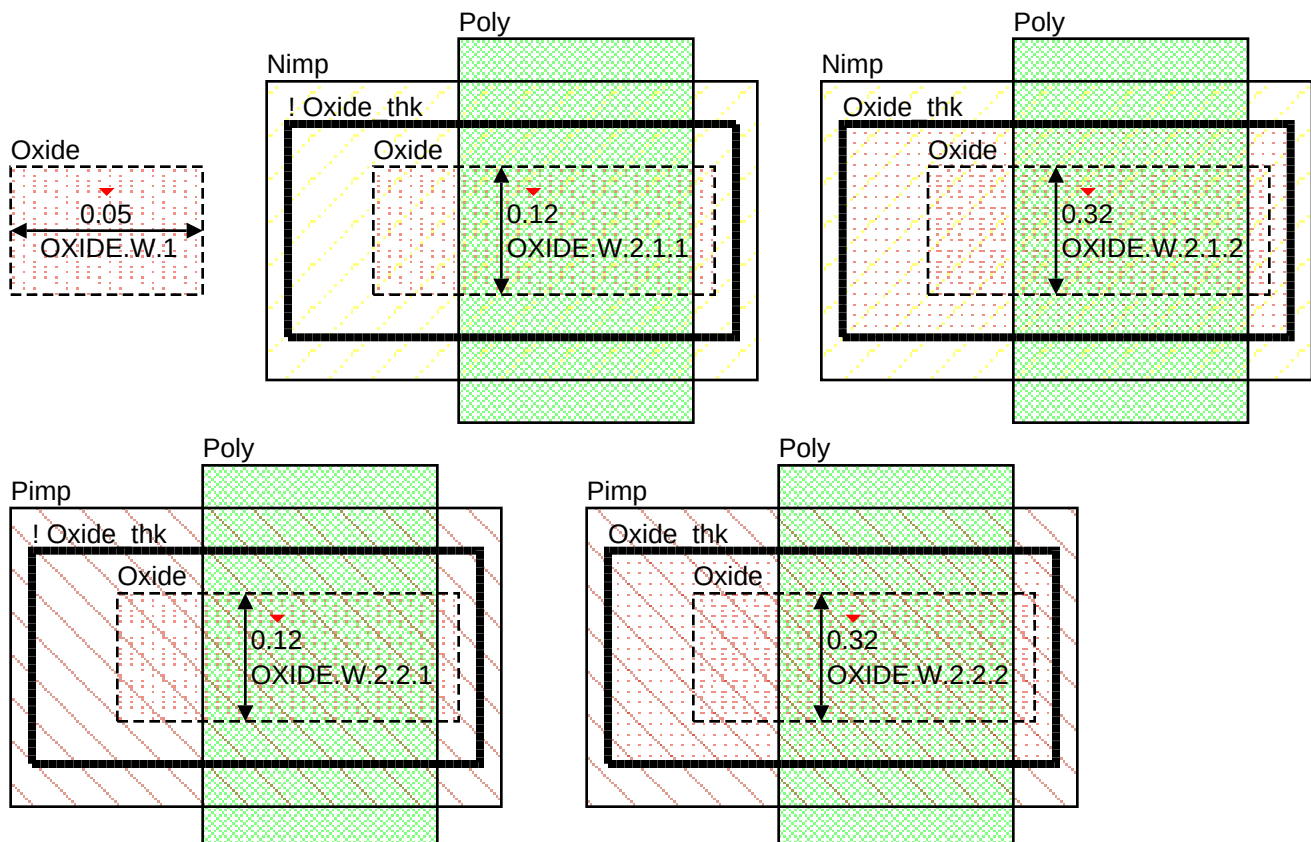


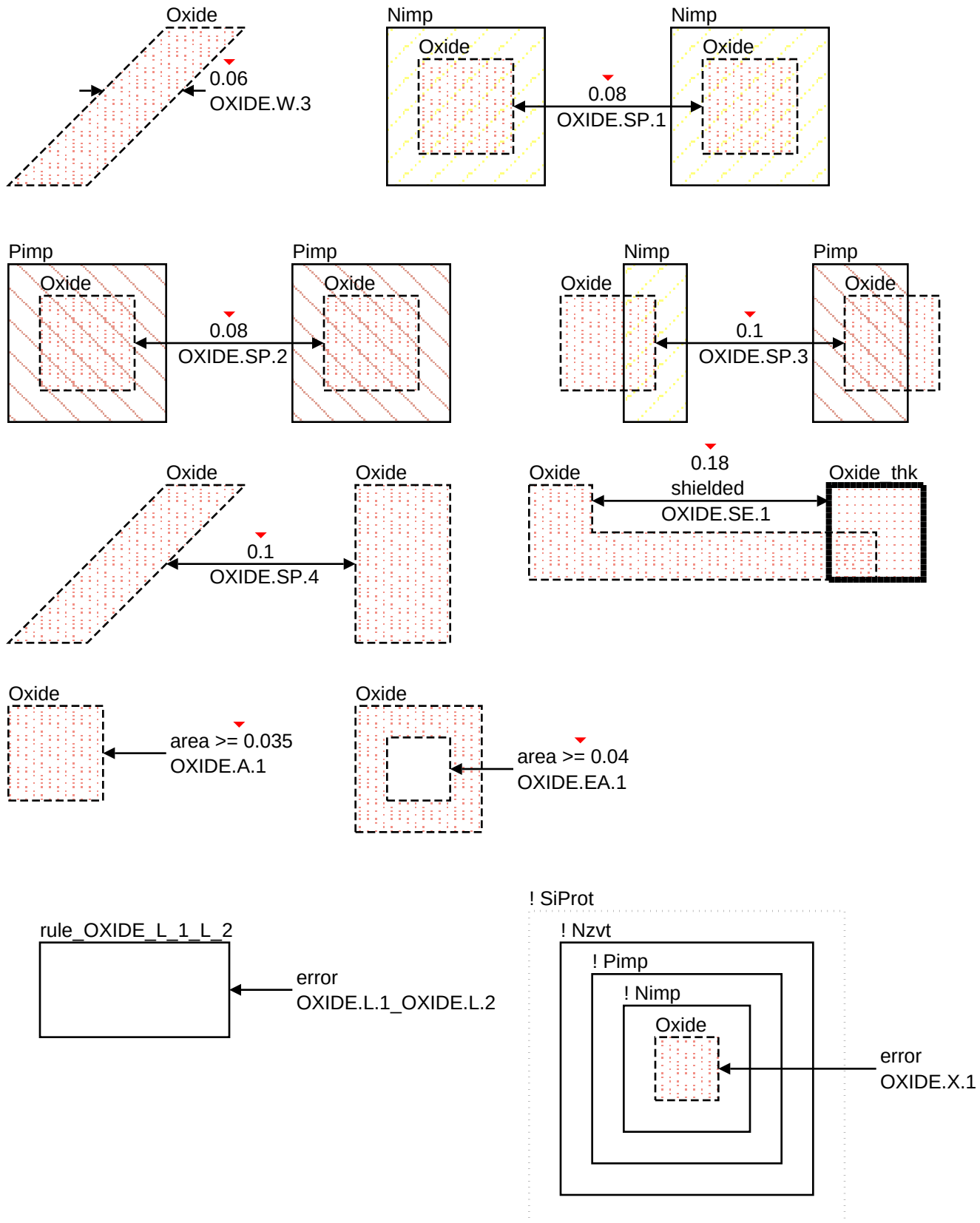
NWR.SP.1 - Covered by NW.SP.2.

ACTIVE RULES

Data Table: OXIDE_DRC

RuleName	Description	Value
OXIDE.W.1	Minimum Active Area width	0.05
OXIDE.W.2.1.1	Minimum 1.1V N-channel gate width	0.12
OXIDE.W.2.1.2	Minimum 1.8V N-channel gate width	0.32
OXIDE.W.2.2.1	Minimum 1.1V P-channel gate width	0.12
OXIDE.W.2.2.2	Minimum 1.8V P-channel gate width	0.32
OXIDE.W.3	Minimum Active Area bent 45 degree width	0.06
OXIDE.SP.1	Minimum N+ Active Area to N+ Active Area spacing	0.08
OXIDE.SP.2	Minimum P+ Active Area to P+ Active Area spacing	0.08
OXIDE.SP.3	Minimum N+ Active Area to P+ Active Area spacing	0.1
OXIDE.SP.4	Minimum Active Area bent 45 degree to Active Area spacing	0.1
OXIDE.SE.1	Minimum Active Area to Thick Active Area spacing	0.18
OXIDE.A.1	Minimum area for Active Area	0.035
OXIDE.EA.1	Minimum Active Area enclosed area	0.04
OXIDE.L.1	Maximum Oxide length between two contacts when Oxide width is $\leq 0.18\mu\text{m}$	12.0
OXIDE.L.2	Maximum Oxide length between one contact and the end of the Oxide line when Oxide width is $\leq 0.18\mu\text{m}$	6.0
OXIDE.X.1	Oxide must be covered by N+ Implant or Nzvt or Salicide Block	---
OXIDE.D.1	Full chip maximum Oxide density	> 25% < 75%
OXIDE.D.2	Local Oxide density 300x300 window stepped at 150	> 25% < 75%



ACTIVE RULES (continued)

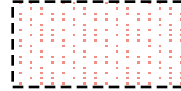
switch CHECK_DENSITY

Density

ratio > 0.30 < 0.80

id: OXIDE.D.1

Oxide



message: Oxide full chip density must be > 30% < 80%

Density

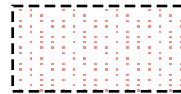
ratio > 0.25 < .75

windowSize: 300.0

stepSize: 150.0

id: OXIDE.D.2

Oxide



message: Oxide local (300x300) density must be > 25% < 75%

ACTIVE RESISTOR RULES (salicided/non-salicided)

Data Table: OXIDER_DRC

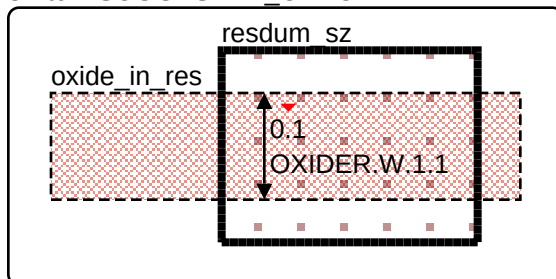
RuleName	Description	Value
OXIDER.W.1.1	Minimum Active Resistor width	0.1
OXIDER.W.1.2	Minimum suggested Active Resistor width	0.8
OXIDER.L.1	Minimum suggested Active Resistor length	4.0
OXIDER.SE.1	Minimum Salicide Block to Contact spacing	0.12
OXIDER.E.1	Minimum Salicide Block to Active Resistor enclosure	0.12
OXIDER.SE.2	Minimum Active Resistor to N+ or P+ Implant spacing	0.16
OXIDER.X.1	Active resistors must have N+ or P+ Implant	---

Active resistor is defined by the intersection of Oxide and Resdum for DRC and LVS.

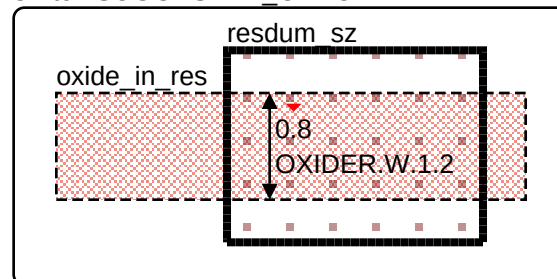
For salicided Oxide resistors, the Resdum shape must butt the contacts on both ends of Oxide the resistor and the Resdum shape must be coincident or extend beyond the Oxide edges along the length of the Oxide resistor.

For non-salicided Oxide resistors, the Resdum shape must be coincident with the edges of the Siprot that crosses the width of the Oxide resistor and the Resdum shape must be coincident or extend beyond the Oxide edges along the length of the Oxide resistor.

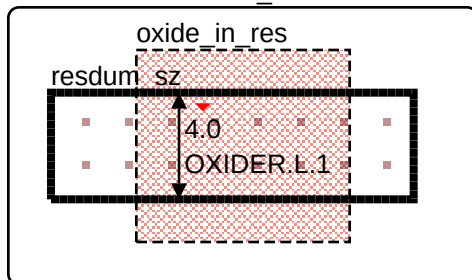
switch !SUGGESTED_CHECK



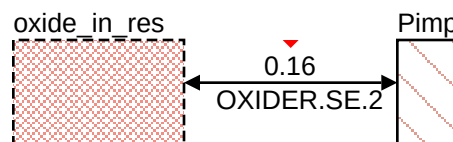
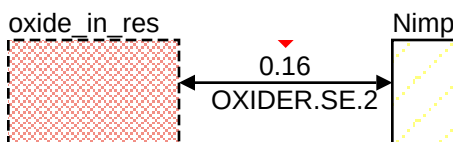
switch SUGGESTED_CHECK

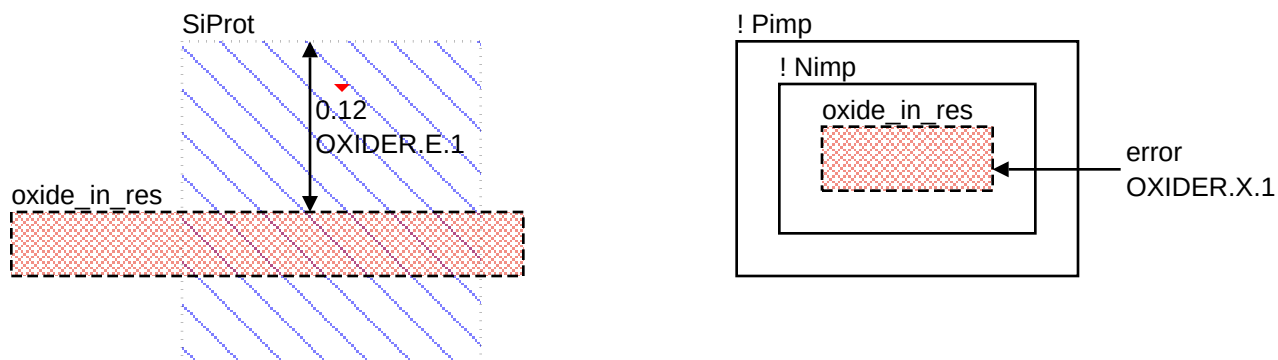


switch SUGGESTED_CHECK



**OXIDER.SE.1 is checked by
SIPROT.SE.1**



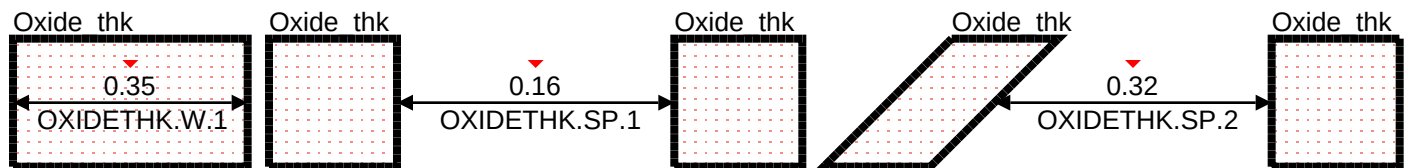
ACTIVE RESISTOR RULES (continued)

THICK ACTIVE (1.8V) RULES

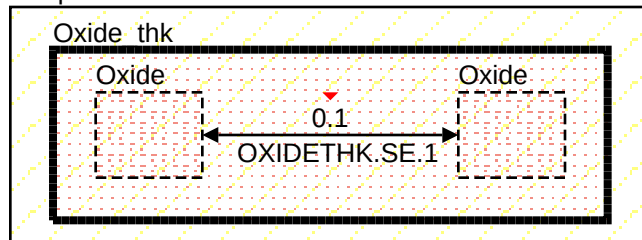
Data Table: OXIDETHK_DRC

RuleName	Description	Value
OXIDETHK.W.1	Minimum Thick Active Area width	0.35
OXIDETHK.SP.1	Minimum Thick Active Area to Thick Active Area spacing	0.16
OXIDETHK.SP.2	Minimum Thick Active Area bent 45 degrees to Thick Active Area spacing	0.32
OXIDETHK.SE.1	Minimum N+ 1.8V Active Area to 1.8V N+ Active Area spacing	0.1
OXIDETHK.SE.2	Minimum P+ 1.8V Active Area to 1.8V P+ Active Area spacing	0.1
OXIDETHK.SE.3	Minimum N+ 1.8V Active Area to 1.8V P+ Active Area spacing	0.12
OXIDETHK.SE.4	Minimum Thick Active Area to Active Area spacing	0.18
OXIDETHK.E.1	Minimum Thick Active Area to Active Area enclosure	0.16
OXIDETHK.SE.5	Minimum Thick Active Area to 1.1V Poly gate spacing	0.18
OXIDETHK.E.2	Minimum Thick Active Area to Thick Poly gate enclosure	0.18

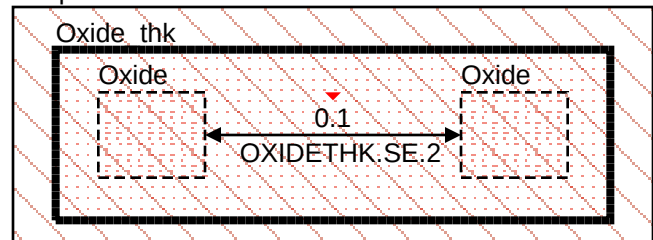
Note 1: 1.8V MOS must be defined by Active which is fully enclosed by Thick Active (with 0.0 overlap).
 Note 2: 1.2V MOS is only defined by Active without any Thick Active.



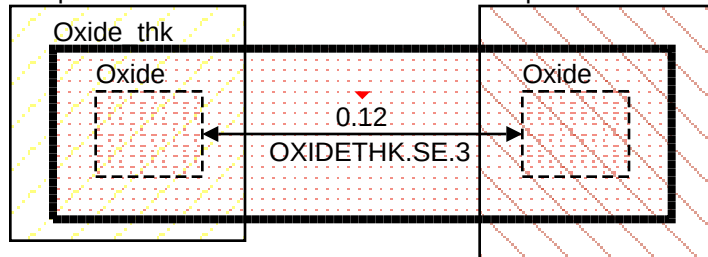
Nimp



Pimp

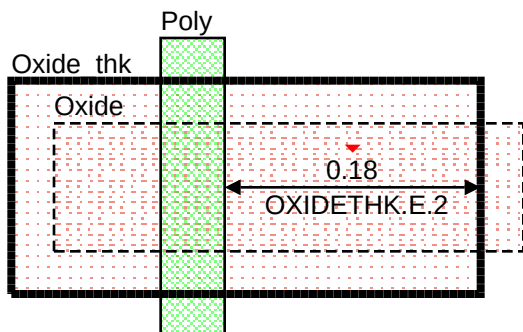
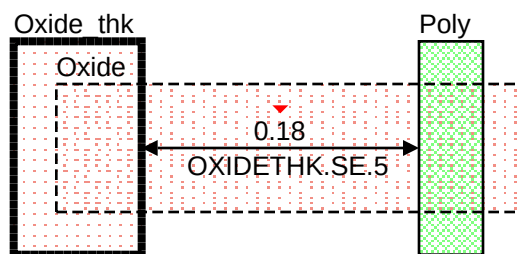
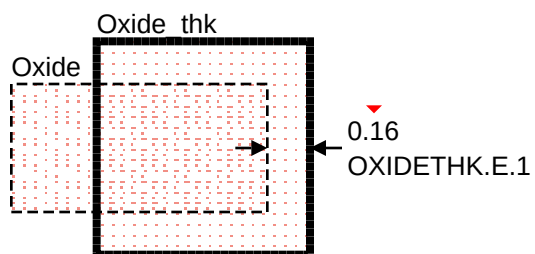


Nimp



Pimp

OXIDETHK.SE.4 - Covered by OXIDE.SE.1.

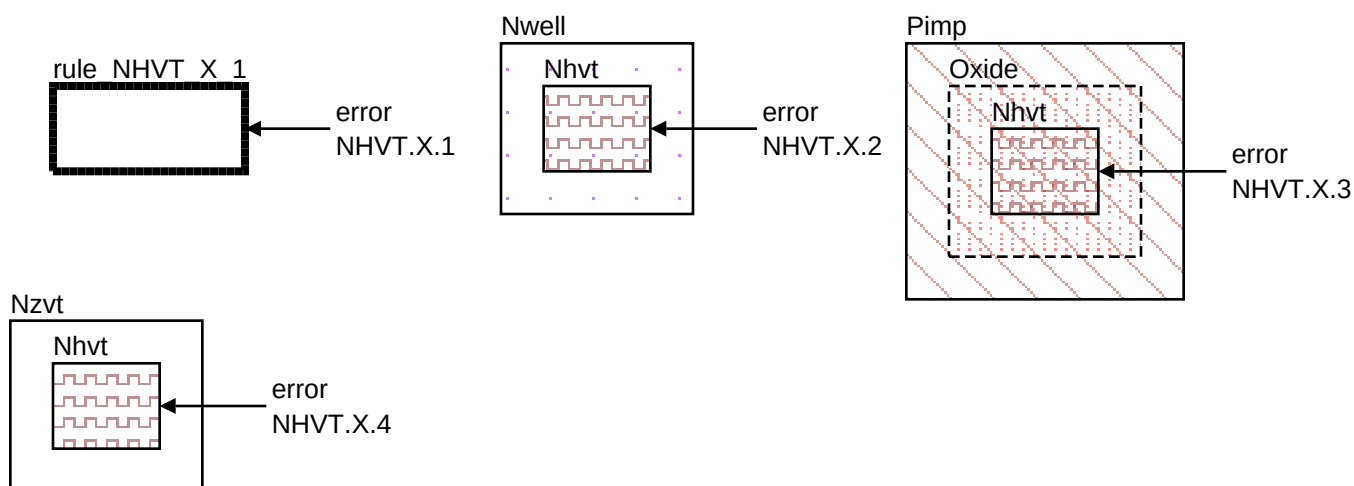
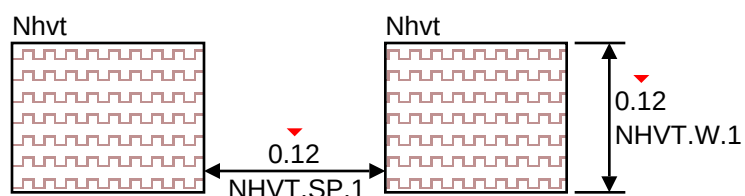
Thick ACTIVE RULES (continued)

N+ HIGH VT RULES

Data Table: NHVT

RuleName	Description	Value
NHVT.W.1	Minimum Nhvt width	0.12
NHVT.SP.1	Minimum Nhvt spacing	0.12
NHVT.X.1	Nhvt exactly matches the Oxide it is on (0.0 enclosure on all sides).	-
NHVT.X.2	Nhvt is NOT allowed on Nwell.	-
NHVT.X.3	Nhvt is NOT allowed on P+ Active.	-
NHVT.X.4	Nhvt is NOT allowed on Nzvt.	-

Note 1: Nhvt defines the 1.2V LP NMOS device.

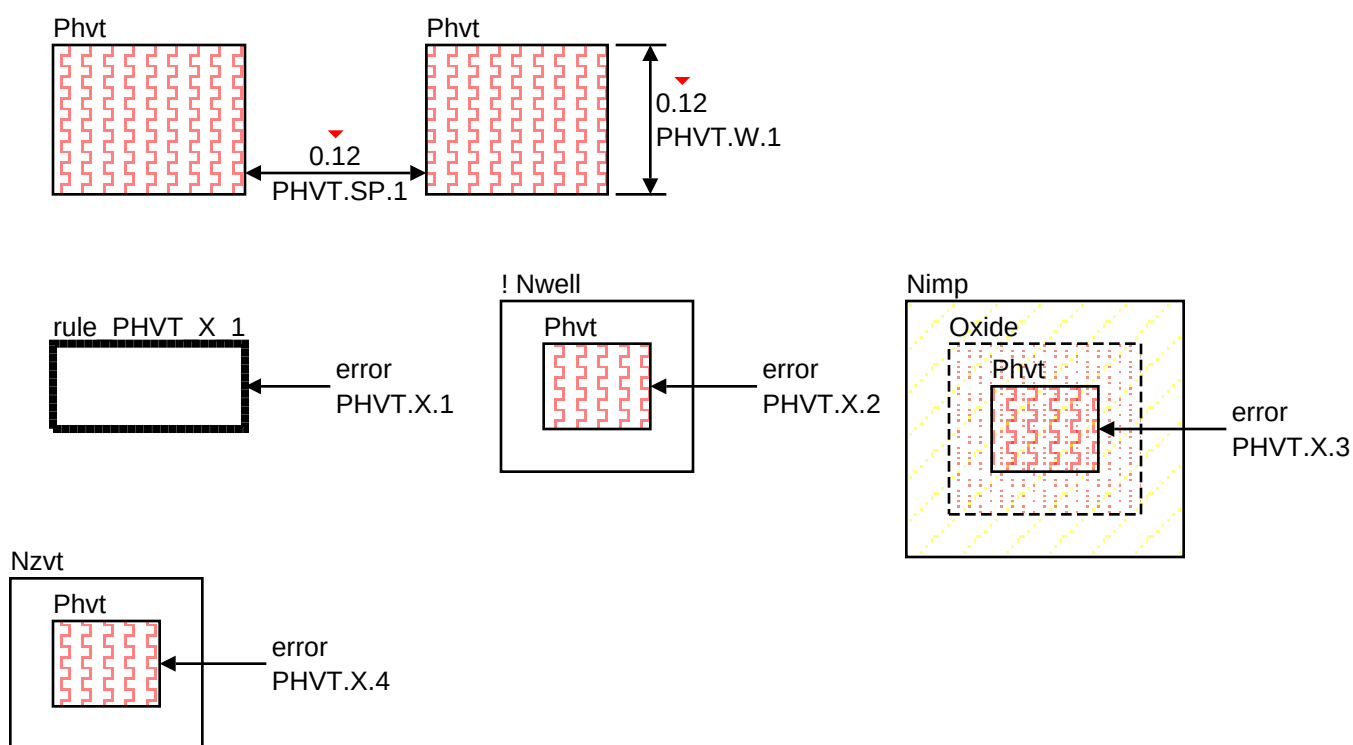


P+ HIGH VT RULES

Data Table: PHVT

RuleName	Description	Value
PHVT.W.1	Minimum Phvt width	0.12
PHVT.SP.1	Minimum Phvt spacing	0.12
PHVT.X.1	Phvt exactly matches the Oxide it is on (0.0 enclosure on all sides).	-
PHVT.X.2	Phvt is NOT allowed outside Nwell.	-
PHVT.X.3	Phvt is NOT allowed on N+ Active.	-
PHVT.X.4	Phvt is NOT allowed on Nzvt.	-

Note 1: Phvt defines the 1.2V LP PMOS device.

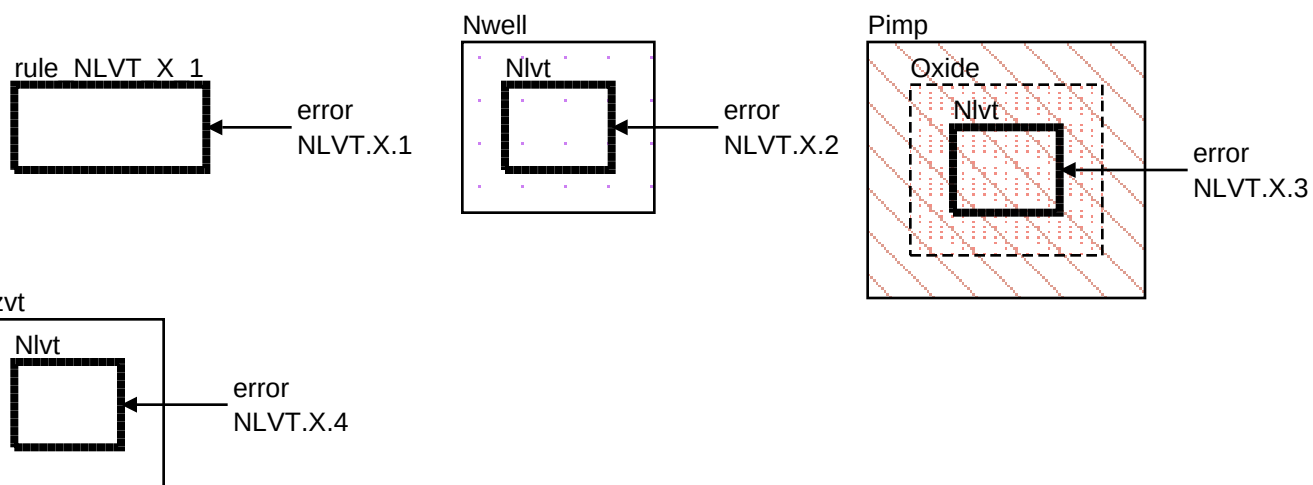
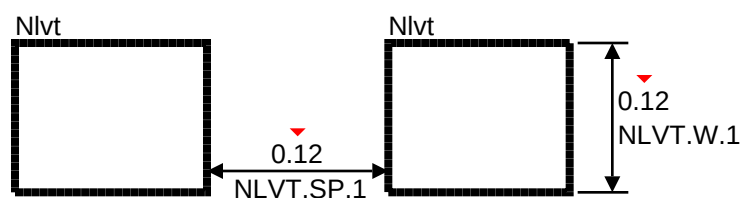


N+ LOW VT RULES

Data Table: NLVT

RuleName	Description	Value
NLVT.W.1	Minimum Nlvt width	0.12
NLVT.SP.1	Minimum Nlvt spacing	0.12
NLVT.X.1	Nlvt exactly matches the Oxide it is on (0.0 enclosure on all sides).	-
NLVT.X.2	Nlvt is NOT allowed on Nwell.	-
NLVT.X.3	Nlvt is NOT allowed on P+ Active.	-
NLVT.X.4	Nlvt is NOT allowed on Nzvt.	-

Note 1: Nlvt defines the 1.2V LP NMOS device.

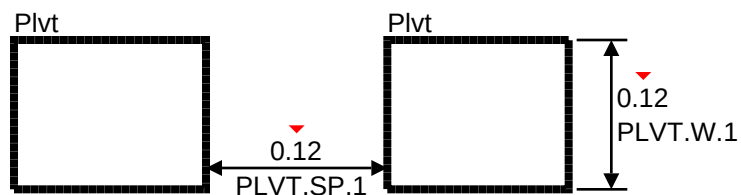


P+ LOW VT RULES

Data Table: PLVT

RuleName	Description	Value
PLVT.W.1	Minimum Plvt width	0.12
PLVT.SP.1	Minimum Plvt spacing	0.12
PLVT.X.1	Plvt exactly matches the Oxide it is on (0.0 enclosure on all sides).	-
PLVT.X.2	Plvt is NOT allowed outside Nwell.	-
PLVT.X.3	Plvt is NOT allowed on N+ Active.	-
PLVT.X.4	Plvt is NOT allowed on Nzvt.	-

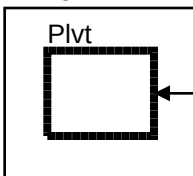
Note 1: Plvt defines the 1.2V LP PMOS device.



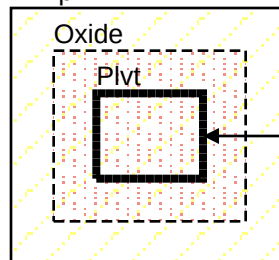
rule PLVT X 1



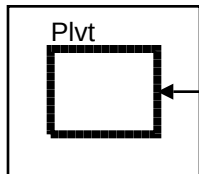
! Nwell



Nimp



Nzvt

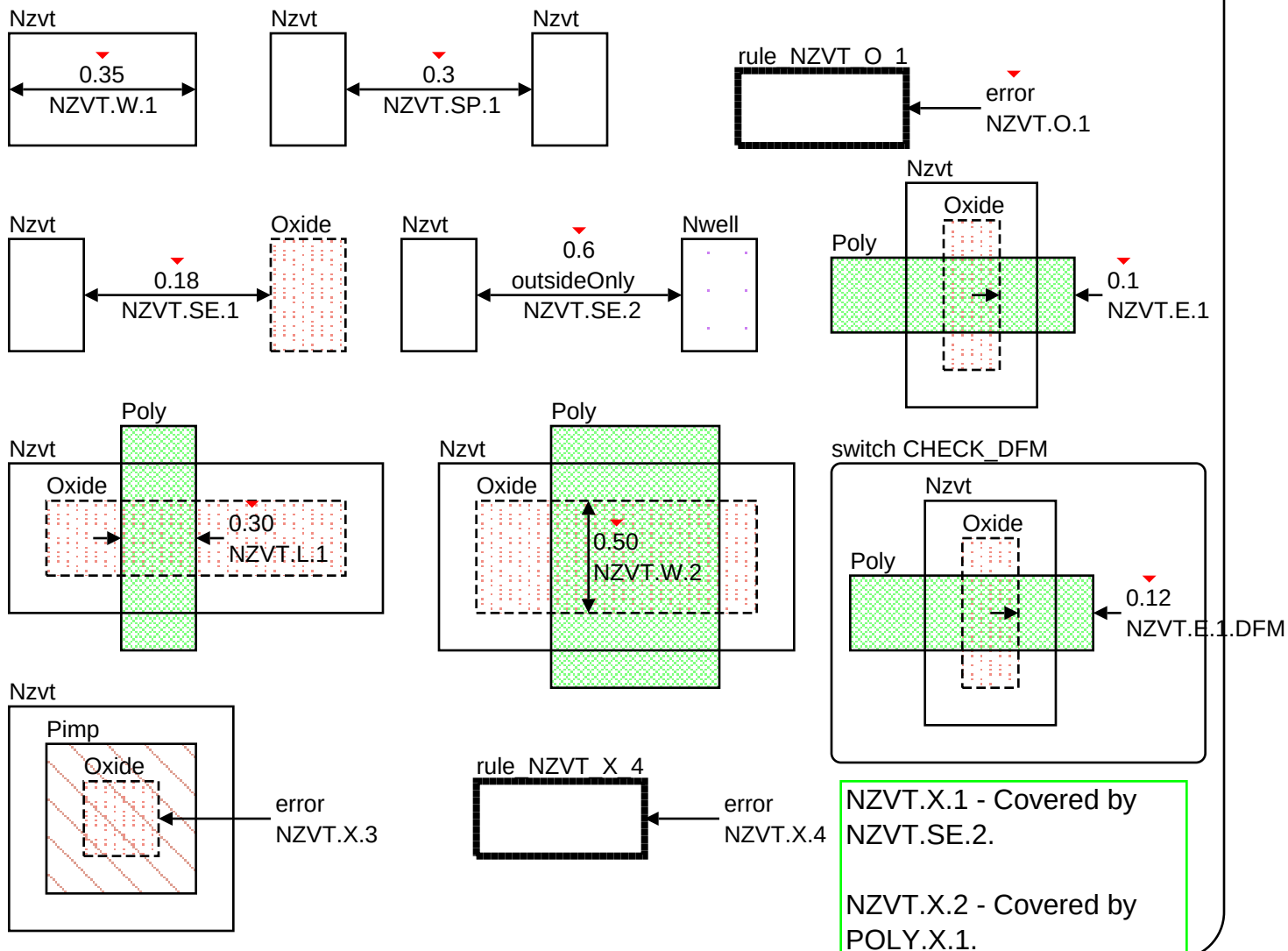


NATIVE NMOS ACTIVE RULES

Data Table: NZVT_DRC

RuleName	Description	Value
NZVT.W.1	Minimum Nzvt width	0.35
NZVT.SP.1	Minimum Nzvt to Nzvt spacing	0.3
NZVT.O.1	Minimum and maximum Nzvt to Active Area overlap	0.16
NZVT.SE.1	Minimum Nzvt to Active spacing	0.18
NZVT.SE.2	Minimum Nzvt to Nwell spacing	0.6
NZVT.E.1	Minimum N+ Poly gate end cap to Native Active Area enclosure	0.1
NZVT.E.1.DFM	Minimum N+ Poly gate end cap to Native Active Area enclosure for DFM	0.12
NZVT.L.1	Minimum Native device Poly gate length	0.30
NZVT.W.2	Minimum Native device Poly gate width	0.50
NZVT.X.1	Nzvt is NOT allowed on Nwell	---
NZVT.X.2	Bent Poly gates are NOT allowed on Nzvt	---
NZVT.X.3	P+ Active Area is NOT allowed on Nzvt	---
NZVT.X.4	Only one Active Area is allowed in an Nzvt region	---

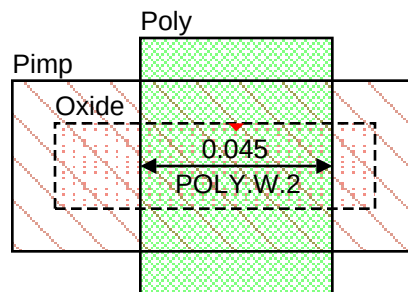
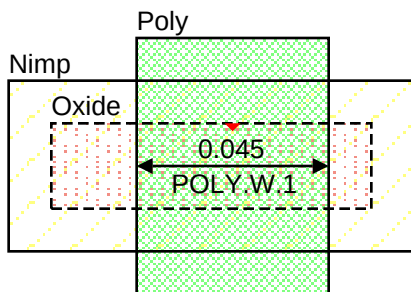
Note 1: Native NMOS is defined by Active which is full enclosed by Nzvt with 0.3um enclosure.

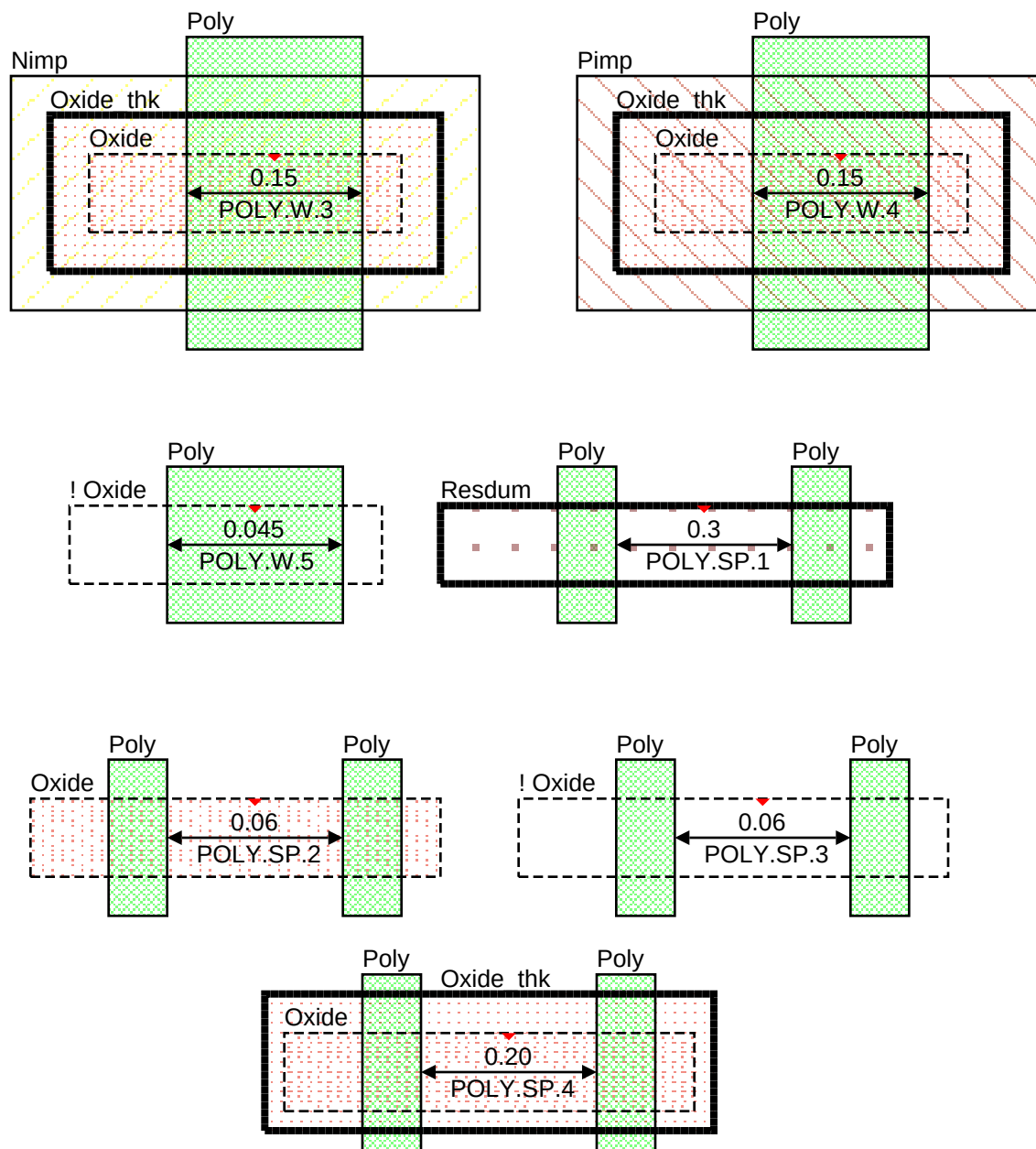


POLY RULES

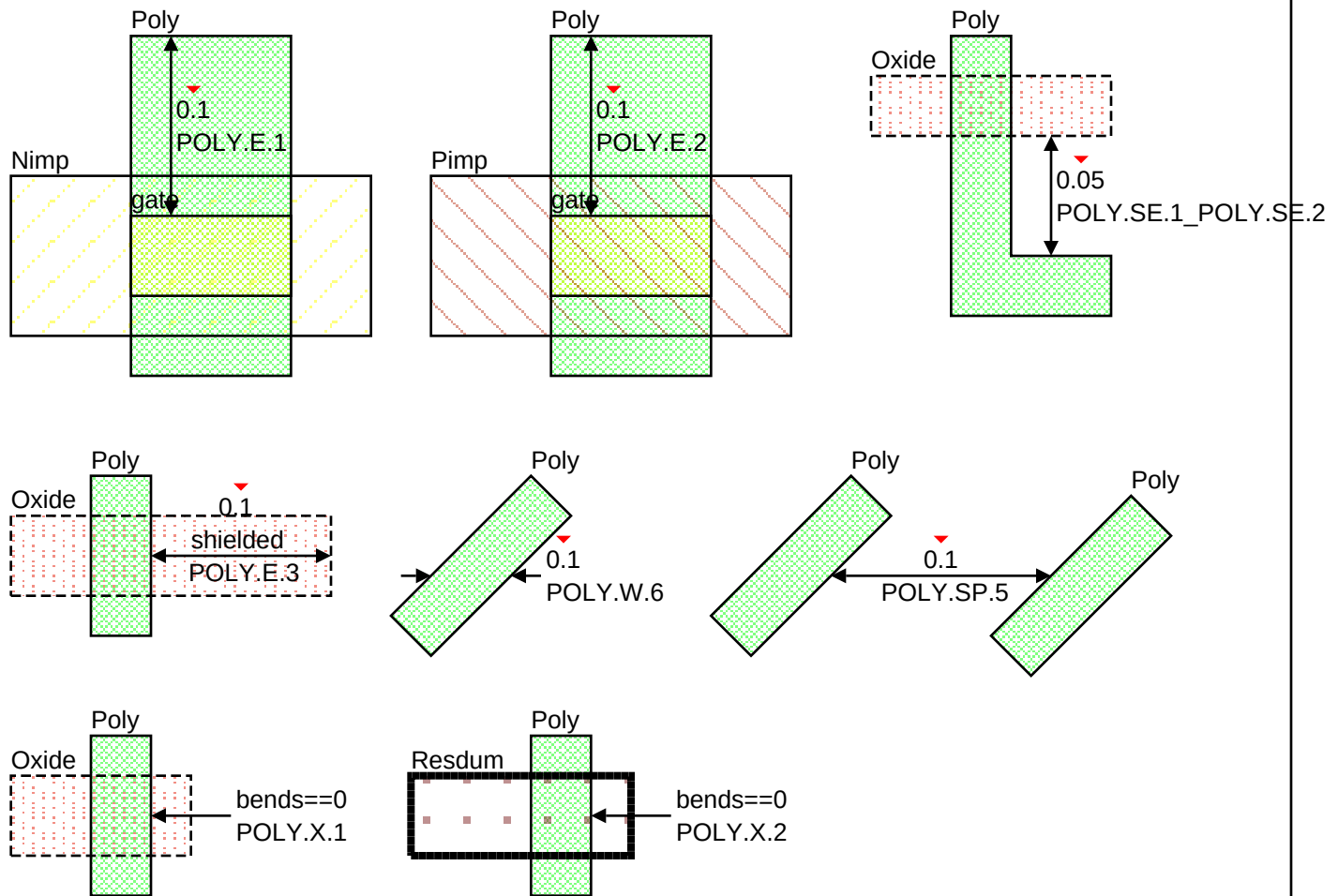
Data Table: POLY_DRC

RuleName	Description	Value
POLY.W.1	Minimum 1.1V N-channel gate length	0.045
POLY.W.2	Minimum 1.1V P-channel gate length	0.045
POLY.W.3	Minimum 1.8V N-channel gate length	0.15
POLY.W.4	Minimum 1.8V P-channel gate length	0.15
POLY.W.5	Minimum Poly interconnect width	0.045
POLY.SP.1	Minimum Poly resistor space	0.3
POLY.SP.2	Minimum Poly gate space	0.06
POLY.SP.2.DFM	Minimum Poly gate space for DFM	0.08
POLY.SP.3	Minimum Poly interconnect space	0.06
POLY.SP.4	Minimum gate space in thick active	0.20
POLY.LN.1	Maximum length of Poly (for width ≥ 0.16) between two Poly contacts or Poly line end to Poly contact	20.0
POLY.E.1	Minimum N-channel gate extension beyond Active Area	0.1
POLY.E.2	Minimum P-channel gate extension beyond Active Area	0.1
POLY.E.1.DFM	Minimum N-channel gate extension beyond Active Area for DFM	0.12
POLY.E.2.DFM	Minimum P-channel gate extension beyond Active Area for DFM	0.12
POLY.SE.1	Minimum Poly interconnect to unrelated Active Area space	0.05
POLY.SE.2	Minimum Poly interconnect to related Active Area space	0.05
POLY.E.3	Minimum Active Area (source/drain) to gate enclosure	0.1
POLY.W.6	Minimum bent Poly width	0.1
POLY.SP.5	Minimum bent Poly space	0.1
POLY.X.1	Bent gate in not allowed	---
POLY.X.2	Bent Poly resistor is not allowed	---
POLY.D.1	Maximum Poly density across full chip	50%
POLY.SE.3	Maximum Poly segment length (width < 0.14) between two contacts	12.0
POLY.A.1	Minimum area for Poly interconnect	0.02
POLY.EA.1	Minimum enclosed area for Poly interconnect	0.05





POLY RULES (continued)



assuraDRC Native Code

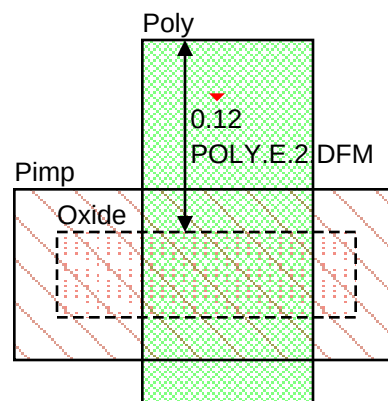
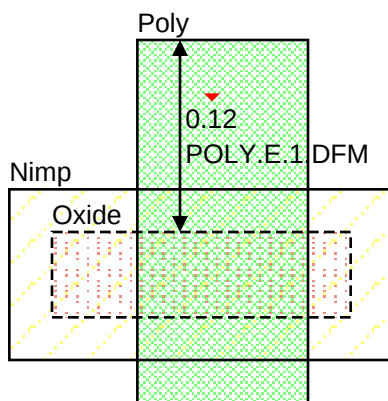
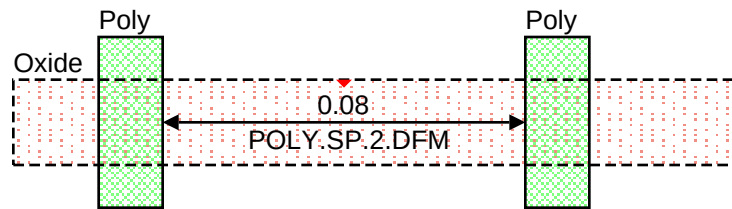
```

inpo2 = geomAndNot(Poly Oxide)
pol2_longp1 = drc( inpo2 area > 0.08* 20.0 )
pol2_longp = geomOr(pol2_longp1 gate)
copo = geomAnd(Cont Poly)
pol21_x = geomButtOrOver( pol2_longp copo )
pol22_check_copo = geomButtOrOver( copo pol21_x )
pol23_a = geomSizeAnd( pol22_check_copo pol21_x 0.11* 0.8 20.0/2 )
pol24_linen = geomButtOrOver( pol21_x pol23_a keep == 1 )
pol25_p2p = geomButtOrOver( pol21_x pol23_a keep > 1 )
pol26_b = geomButtOrOver( pol23_a pol24_linen )
pol27_c = geomSizeAnd( pol26_b pol24_linen 0.11* 0.8 20.0/2 )
linen_not1 = geomAndNot(pol24_linen pol27_c)
p2p_not1 = geomAndNot(pol25_p2p pol23_a)
pol28_bad = geomOr(linen_not1 p2p_not1)
pol29_bad_edge = geomGetEdge(pol28_bad coincident Poly)
pol210_err = drc( pol29_bad_edge width <= 0.16 )
errorLayer( geomButtOrOver( pol21_x pol210_err )
  "POLY.LN.1 Maximum Poly length [Poly width is <= 0.16 um] between two contacts as well..."

```

POLY RULES (continued)

switch CHECK_DFM



switch CHECK_DENSITY

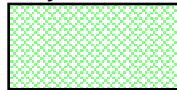
Density

ratio <= 0.5

id: POLY.D.1

message: Poly density must be <= 50%

Poly



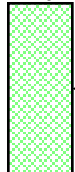
rule POLY SE 3



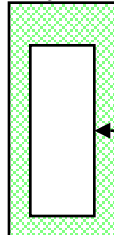
error

POLY.SE.3

Poly

area >= 0.02
POLY.A.1

Poly

area >= 0.05
POLY.EA.1

POLY RESISTOR RULES (salicided/non-salicided)

Data Table: POLYR_DRC

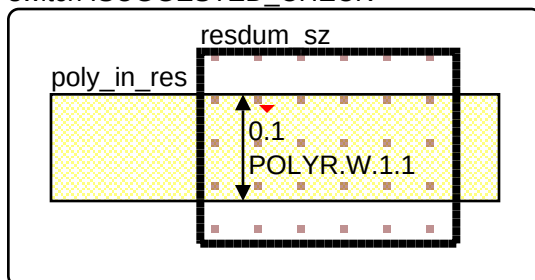
RuleName	Description	Value
POLYR.W.1.1	Minimum Poly resistor width	0.1
POLYR.W.1.2	Minimum suggested Poly resistor width	0.8
POLYR.L.1	Minimum suggested Poly resistor length	4.0
POLYR.SE.1	Minimum Salicide Block to Contact spacing	0.12
POLYR.E.1	Minimum Salicide Block to Poly resistor enclosure	0.14
POLYR.E.2	Minimum N+ Implant to Poly used in resistor enclosure	0.07
POLYR.E.3	Minimum P+ Implant to Poly used in resistor enclosure	0.07
POLYR.SE.2	Minimum Poly resistor to other Implant spacing	0.15
POLYR.X.1	Poly resistors must have N+ or P+ Implant	---

Poly resistor is defined by the intersection of Poly and Resdum for DRC and LVS.

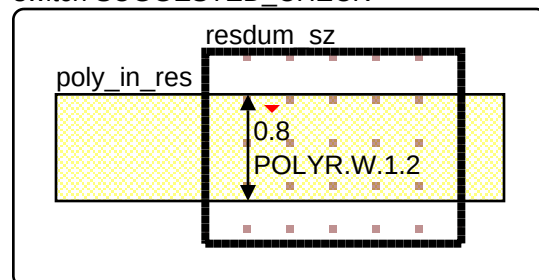
For salicided Poly resistors, the Resdum shape must butt the contacts on both ends of Poly the resistor and the Resdum shape must be coincident or extend beyond the Poly edges along the length of the Poly resistor.

For non-salicided Poly resistors, the Resdum shape must be coincident with the edges of the Siprot that crosses the width of the Poly resistor and the Resdum shape must be coincident or extend beyond the Poly edges along the length of the Poly resistor.

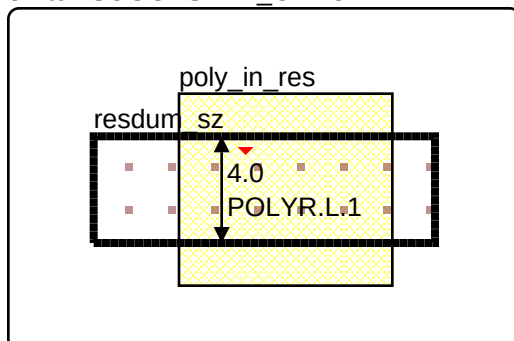
switch !SUGGESTED_CHECK



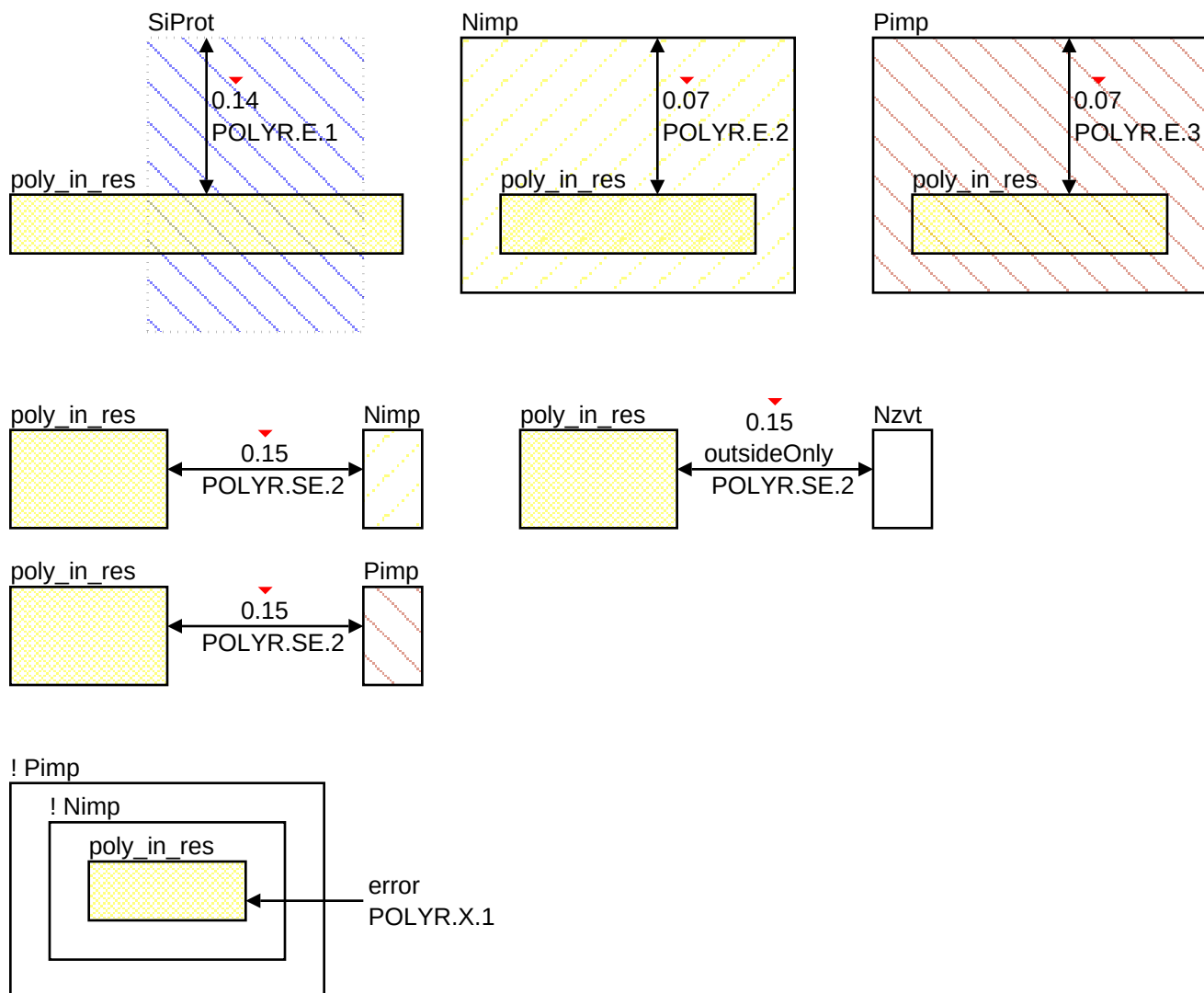
switch SUGGESTED_CHECK



switch SUGGESTED_CHECK



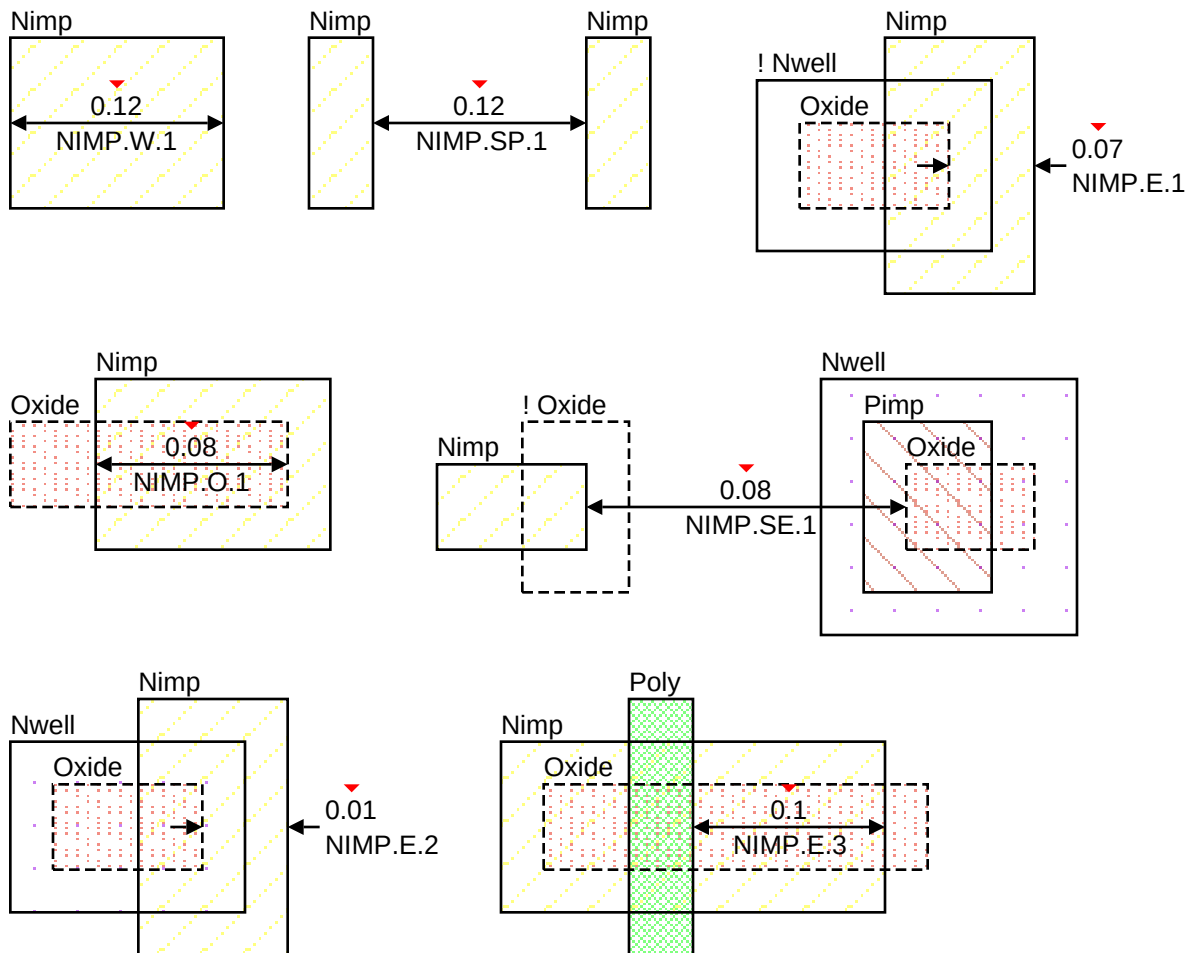
**POLYR.SE.1 is checked by
SIPROT.SE.1**

POLY RESISTOR RULES (continued)

N+ IMPLANT RULES

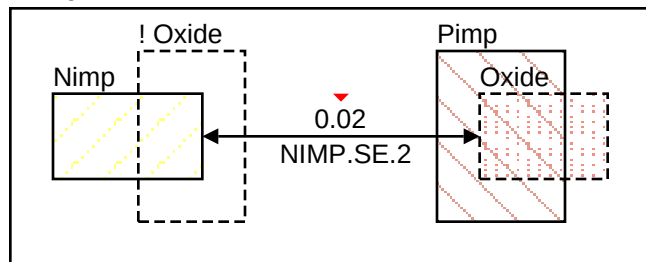
Data Table: NIMP_DRC

RuleName	Description	Value
NIMP.W.1	Minimum N+ Implant width	0.12
NIMP.SP.1	Minimum N+ Implant space	0.12
NIMP.E.1	Minimum N+ Implant to Active Area enclosure	0.07
NIMP.O.1	Minimum N+ Implant to Active Area overlap	0.08
NIMP.SE.1	Minimum N+ Implant to P+ Active Area (inside Nwell) spacing	0.08
NIMP.E.2	Minimum N+ Implant to Active Area (Nwell tie) enclosure	0.01
NIMP.E.3	Minimum N+ Implant to gate side enclosure	0.1
NIMP.SE.2	Minimum N+ Implant to P+ Active Area (substrate tie) spacing	0.02
NIMP.E.4	Minimum N+ Implant to gate (endcap) enclosure	0.1
NIMP.SE.3	Minimum N+ Implant to P+ gate side (butted Implant) spacing	0.1
NIMP.A.1	Minimum area for N+ Implant	0.018
NIMP.EA.1	Minimum N+ Implant ring enclosed area	0.04
NIMP.X.1	N+ Implant is NOT allowed over P+ Implant	---

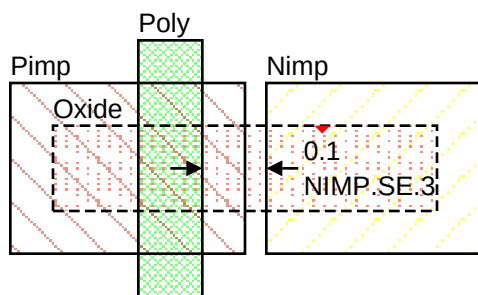
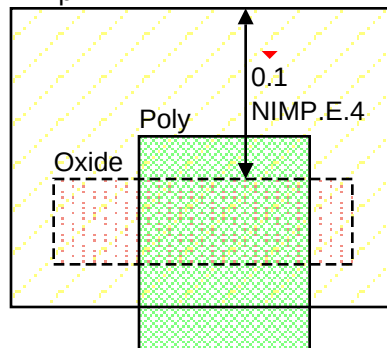


N+ IMPLANT RULES (continued)

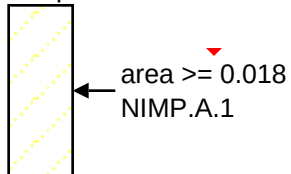
! Nwell



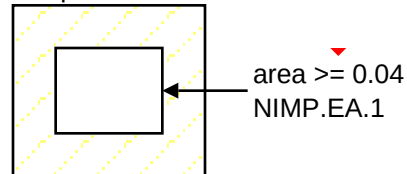
Nimp



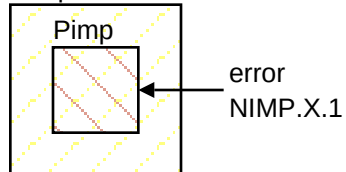
Nimp



Nimp



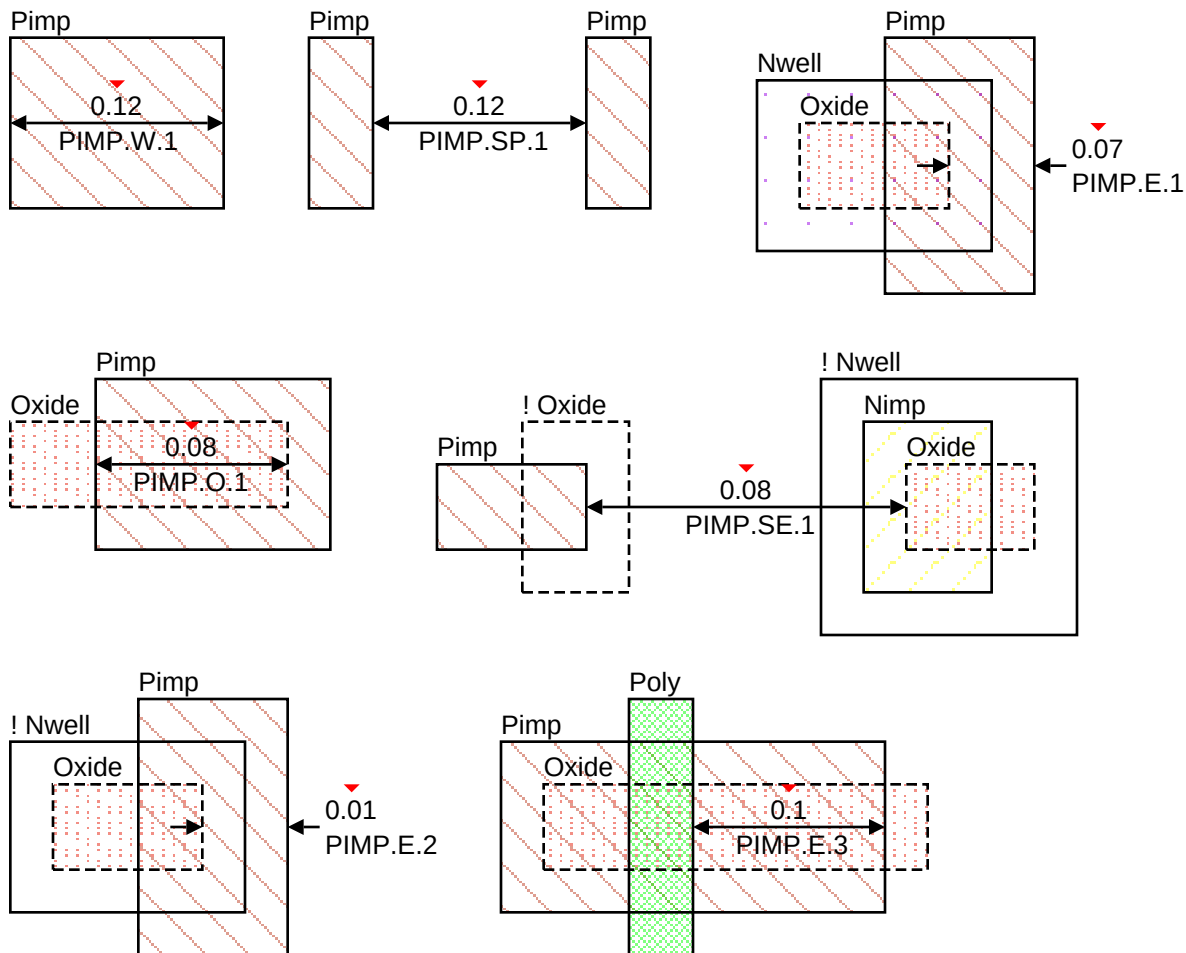
Nimp



P+ IMPLANT RULES

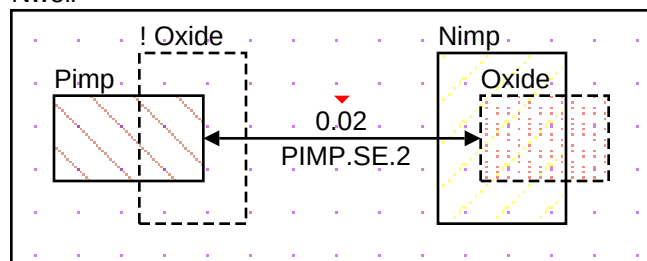
Data Table: PIMP_DRC

RuleName	Description	Value
PIMP.W.1	Minimum P+ Implant width	0.12
PIMP.SP.1	Minimum P+ Implant space	0.12
PIMP.E.1	Minimum P+ Implant to Active Area enclosure	0.07
PIMP.O.1	Minimum P+ Implant to Active Area overlap	0.08
PIMP.SE.1	Minimum P+ Implant to N+ Active Area (outside Nwell) spacing	0.08
PIMP.E.2	Minimum P+ Implant to Active Area (substrate tie) enclosure	0.01
PIMP.E.3	Minimum P+ Implant to gate side enclosure	0.1
PIMP.SE.2	Minimum P+ Implant to N+ Active Area (Nwell tie) spacing	0.02
PIMP.E.4	Minimum P+ Implant to gate (endcap) enclosure	0.1
PIMP.SE.3	Minimum P+ Implant to N+ gate side (butted Implant) spacing	0.1
PIMP.A.1	Minimum area for P+ Implant	0.018
PIMP.EA.1	Minimum P+ Implant ring enclosed area	0.04
PIMP.X.1	P+ Implant is NOT allowed over N+ Implant	---

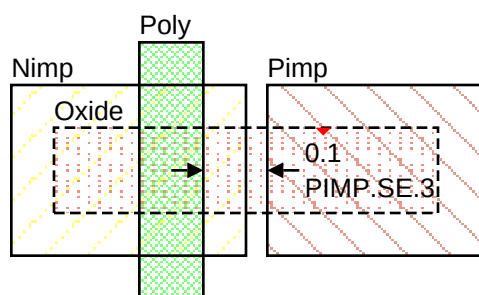
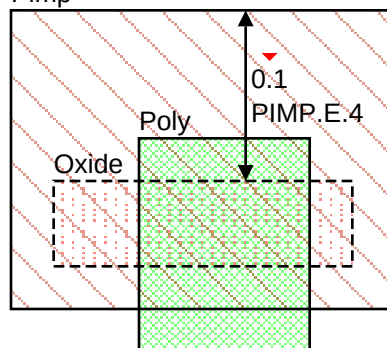


P+ IMPLANT RULES (continued)

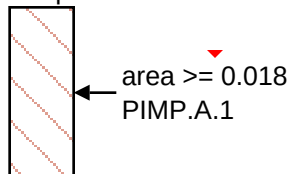
Nwell



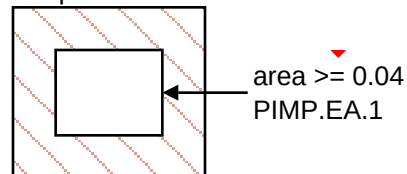
Pimp



Pimp



Pimp

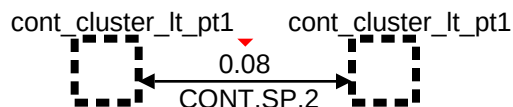
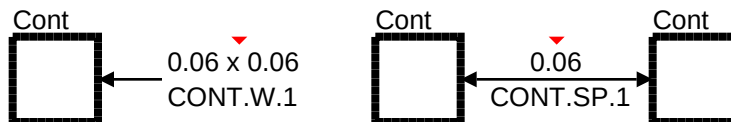


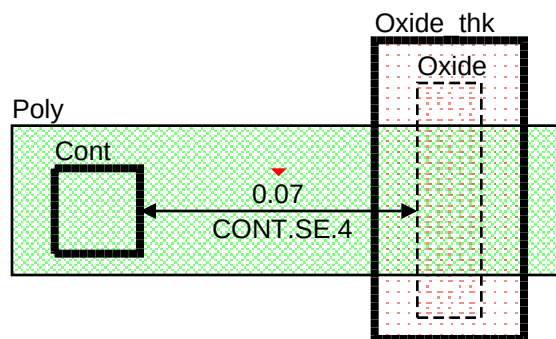
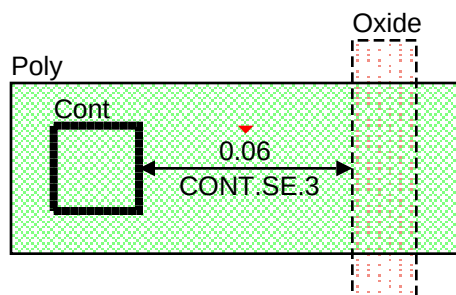
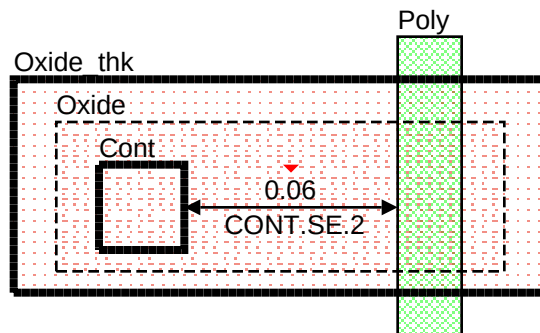
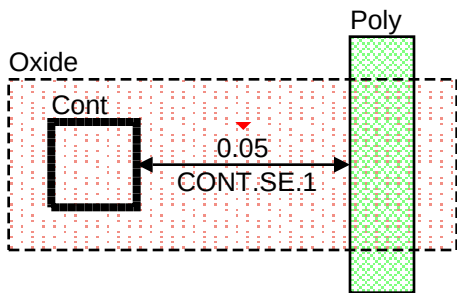
PIMP.X.1 - Covered by NIMP.X.1.

CONTACT RULES

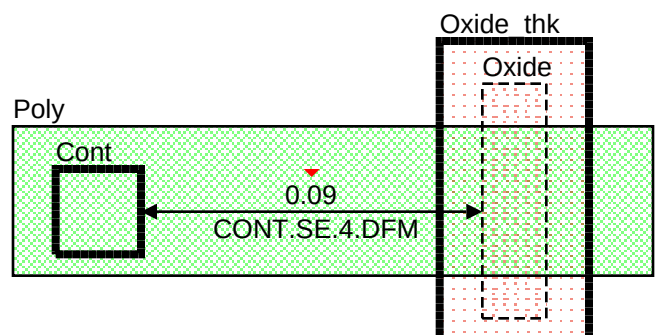
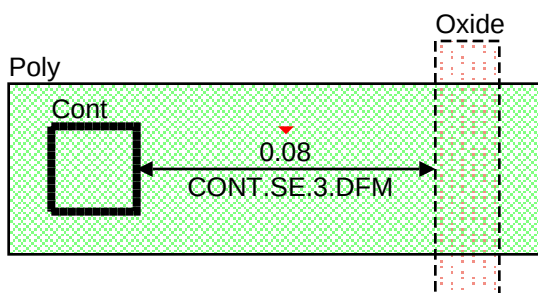
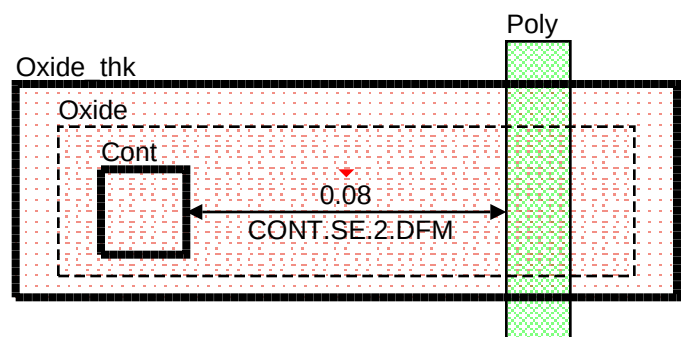
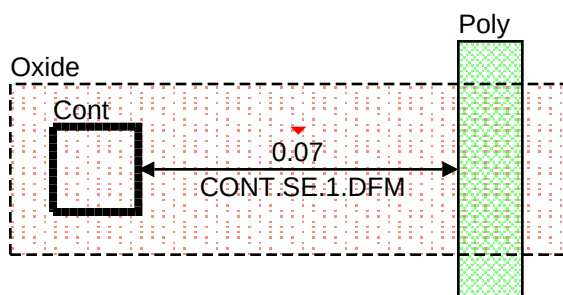
Data Table: CONT_DRC

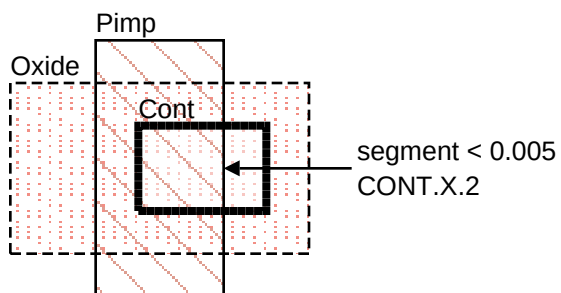
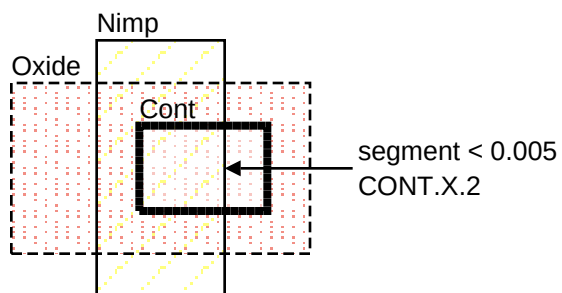
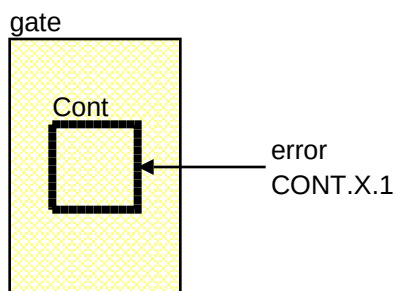
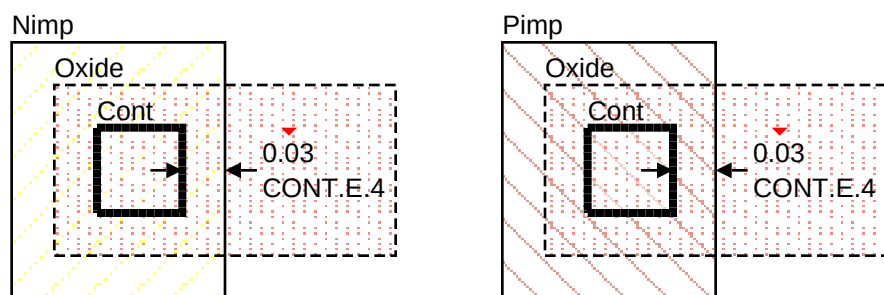
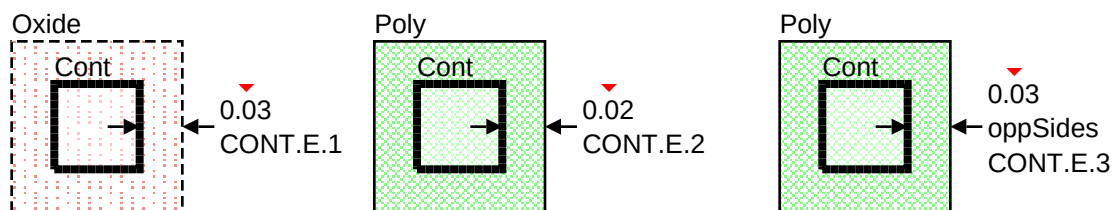
RuleName	Description	Value
CONT.W.1	Maximum and minimum Contact width/length	0.06
CONT.SP.1	Minimum Contact to Contact spacing	0.06
CONT.SP.2	Space to three adjacent Contacts (< 0.10um apart)	0.08
CONT.SE.1	Minimum Contact on Active Area to gate spacing	0.05
CONT.SE.2	Minimum Contact on 1.8V Active Area to gate spacing	0.06
CONT.SE.3	Minimum gate Contact on Active Area spacing	0.06
CONT.SE.4	Minimum 1.8V gate Contact on Active Area spacing	0.07
CONT.SE.1.DFM	Minimum Contact on Active Area to gate spacing for DFM	0.07
CONT.SE.2.DFM	Minimum Contact on 1.8V Active Area to gate spacing for DFM	0.08
CONT.SE.3.DFM	Minimum gate Contact on Active Area spacing for DFM	0.08
CONT.SE.4.DFM	Minimum 1.8V gate Contact on Active Area spacing for DFM	0.09
CONT.E.1	Minimum Active Area to Contact enclosure	0.03
CONT.E.2	Minimum Poly to Contact enclosure	0.02
CONT.E.3	Minimum Poly to Contact enclosure on at least two opposite sides	0.03
CONT.E.4	Minimum N+/P+ Implant on Active Area to Contact enclosure	0.03
CONT.X.1	Contact on gate is NOT allowed	---
CONT.X.2	Active Area Contact on N+/P+ Implant edge is NOT allowed	---
CONT.X.3	Contact must be covered by Metal1 and Active Area or Poly	---



CONTACT RULES (continued)

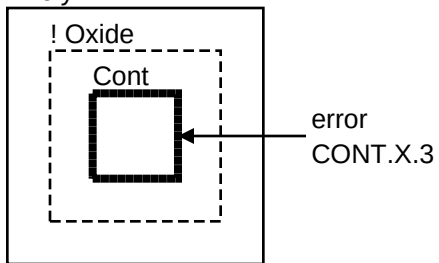
switch CHECK_DFM



CONTACT RULES (continued)

CONTACT RULES (continued)

! Poly

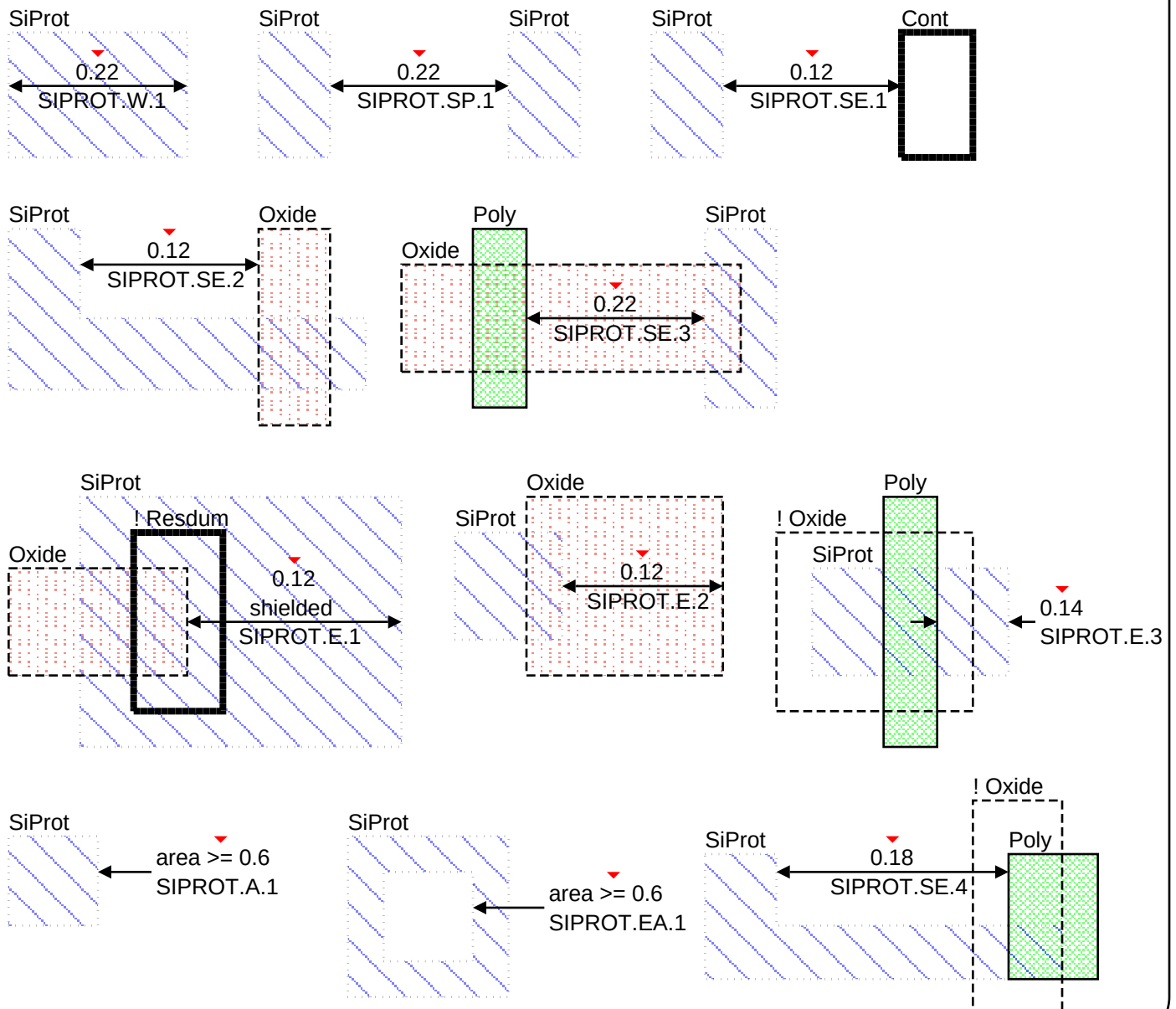


error
CONT.X.3

SALICIDE BLOCKING RULES

Data Table: SIPROT_DRC

RuleName	Description	Value
SIPROT.W.1	Minimum Salicide Block width	0.22
SIPROT.SP.1	Minimum Salicide Block space	0.22
SIPROT.SE.1	Minimum Salicide Block to Contact spacing	0.12
SIPROT.SE.2	Minimum Salicide Block to unrelated Active Area spacing	0.12
SIPROT.SE.3	Minimum Salicide Block to gate spacing	0.22
SIPROT.E.1	Minimum Salicide Block to Active Area enclosure	0.12
SIPROT.E.2	Minimum Active Area to Salicide Block enclosure	0.12
SIPROT.E.3	Minimum Salicide Block to Poly (on field) enclosure	0.14
SIPROT.A.1	Minimum Salicide Block area	0.6
SIPROT.EA.1	Minimum Salicide Block enclosed area	0.6
SIPROT.SE.4	Minimum Salicide Block to Poly (on field) spacing	0.18



METAL 1 RULES

Data Table: METAL1_DRC

RuleName	Description	Value
METAL1.W.1	Minimum Metal 1 width	0.06
METAL1.W.2	Maximum Metal 1 width	6.0
METAL1.SP.1.1	Minimum Metal 1 to Metal 1 spacing	0.06
METAL1.SP.1.2	Minimum Metal 1 to Metal 1 spacing if one metal width > 0.1 and parallel length > 0.32	0.1
METAL1.SP.1.3	Minimum Metal 1 to Metal 1 spacing if one metal width > 0.75 and parallel length > 0.75	0.25
METAL1.SP.1.4	Minimum Metal 1 to Metal 1 spacing if one metal width > 1.5 and parallel length > 1.5	0.45
METAL1.SP.1.5	Minimum Metal 1 to Metal 1 spacing if one metal width > 2.5 and parallel length > 2.5	0.75
METAL1.SP.1.6	Minimum Metal 1 to Metal 1 spacing if one metal width > 3.5 and parallel length > 3.5	1.25
METAL1.E.1	Minimum Metal 1 to Contact enclosure	0.00
METAL1.E.2	Minimum Metal 1 to Contact enclosure on two opposite sides of the Contact	0.03
METAL1.L.1	Minimum bent Metal 1 (45 degree angle) length	0.1
METAL1.SP.2	Minimum bent Metal 1 (45 degree angle) space	0.08
METAL1.SP.3	Space at Metal1 line-end (W < 0.09, Q = 0.1) (dense-line-end) If Metal1 has parallel run length with opposite Metal1 (measured with T=0.025 extension) along 2 adjacent edges of Metal1 [any one edge < Q distance from the corner of the two edges], then one of the space (S1 or S2) must be (check does not include jog with edge < 0.06u)	0.08
METAL1.W.3	Minimum bent Metal 1 (45 degree angle) width	0.07
METAL1.A.1	Minimum Metal 1 area	0.02
METAL1.EA.1	Minimum Metal 1 enclosed area	0.045
METAL1.D.1	Metal 1 Density range over any 120um x 120um area (checked by stepping in 60um increments)	> 20% < 65%
METAL1.D.2	Maximum Metal 1 Density over any 600um x 600um area (checked by stepping in 300um increments)	< 60%

METAL k (k = 2, 3, 4, 5, 6, 7, 8, 9) RULES

Metal k (k = 2, 3, 4, 5, 6, 7, 8, 9) RULES

Rule Name	Value (um)	Description
METALK.W.1	0.08	Minimum Metal k width.
METALK.W.2	6.0	Maximum Metal k width.
METALK.SP.1.1	0.07	Minimum Metal k to Metal k spacing.
		Minimum Metal k to Metal k spacing if:
METALK.SP.1.2	0.15	one Metal k width > 0.10 and parallel length > 0.32
METALK.SP.1.3	0.25	one Metal k width > 0.75 and parallel length > 0.75
METALK.SP.1.4	0.45	one Metal k width > 1.5 and parallel length > 1.5
METALK.SP.1.5	0.75	one Metal k width > 2.5 and parallel length > 2.5.
METALK.SP.1.6	1.25	one Metal k width > 3.5 and parallel length > 3.5.
METALK.E.1	0.005	Minimum Metal k enclosure of Via k-1.
METALK.E.2	0.03	Minimum Metal k enclosure of Via k-1on at least two opposite sides.
METALK.L.1	0.1	Minimum bent Metal k (45 degree angle) length.
METALK.SP.2	0.1	Minimum bent Metal k (45 degree angle) space.
METALK.SP.3	0.08	Space at Metalk line-end ($W < 0.1$, $Q = 0.1$) (dense-line-end) If Metalk has parallel run length with opposite Metalk (measured with $T=0.035$ extension) along 2 adjacent edges of Metalk [any one edge $< Q$ distance from the corner of the two edges], then one of the space (S1 or S2) must be (k=2-9) (check does not include jog with edge $< 0.07u$)
METALK.W.3	0.09	Minimum bent Metal k (45 degree angle) width.
METALK.A.1	0.02	Minimum Metal k area.
METALK.EA.1	0.055	Minimum Metal k enclosed area.
METALK.D.1	> 20% < 65%	Metal k Density range over any 120um x 120um area (checked by stepping in 60um increments).
METALK.D.2	< 60%	Maximum Metal k density over any 600um x 600um area (checked by stepping in 300um increments).

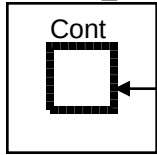
METAL k (k = 10, 11) RULES

Table 10: METAL k (k = 10, 11) RULES

Rule Name	Value (um)	Description
METALk.W.1	0.22	Minimum Metal k width.
METALk.W.2	6.0	Maximum Metal k width.
METALk.SP.1.1	0.20	Minimum Metal k to Metal k spacing.
		Minimum Metal k to Metal k spacing if:
METALk.SP.1.2	0.35	one Metal k width > 0.75 and parallel length > 0.75
METALk.SP.1.3	0.45	one Metal k width > 1.50 and parallel length > 1.50
METALk.SP.1.4	0.75	one Metal k width > 2.50 and parallel length > 2.50.
METALk.SP.1.5	1.25	one Metal k width > 3.5 and parallel length > 3.5.
METALk.E.1	0.03	Minimum Metal k overlap of Via k-1.
METALk.E.2	0.05	Minimum Metal k overlap of Via k-1 on at least two opposite sides.
METALk.A.1	0.10	Minimum Metal k area.
METALk.EA.1	0.11	Minimum Metal k enclosed area.
METALk.D.1	> 20% < 65%	Metal k Density range over any 120um x 120um area (checked by stepping in 60um increments).
METALk.D.2	< 60%	Maximum Metal k density over any 600um x 600um area (checked by stepping in 300um increments).

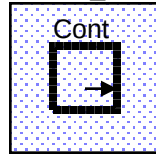
METAL RULES (continued)

! metal1_conn



error
METAL1.E.1

metal1_conn



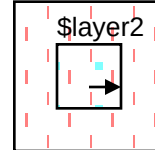
0.03
oppSides
METAL1.E.2

macro

Macro Table

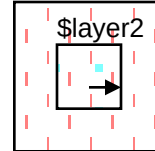
\$name1	\$layer1	\$layer2	\$id1	\$id2
Metal2	metal2_conn	Via1	METAL2.E.1	METAL2.E.2
Metal3	metal3_conn	Via2	METAL3.E.1	METAL3.E.2
Metal4	metal4_conn	Via3	METAL4.E.1	METAL4.E.2
Metal5	metal5_conn	Via4	METAL5.E.1	METAL5.E.2
Metal6	metal6_conn	Via5	METAL6.E.1	METAL6.E.2
Metal7	metal7_conn	Via6	METAL7.E.1	METAL7.E.2
Metal8	metal8_conn	Via7	METAL8.E.1	METAL8.E.2
Metal9	metal9_conn	Via8	METAL9.E.1	METAL9.E.2

\$layer1



0.005
insideOnly
\$id1

\$layer1



0.03
oppSides
\$id2

macro

Macro Table

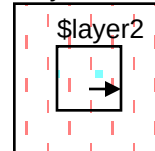
\$name1	\$layer1	\$layer2	\$id1	\$id2
Metal10	Metal10	Via9	METAL10.E.1	METAL10.E.2
Metal11	Metal11	Via10	METAL11.E.1	METAL11.E.2

\$layer1



0.03
insideOnly
\$id1

\$layer1



0.05
oppSides
\$id2

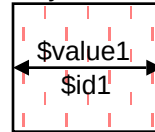
METAL RULES (continued)

macro

Macro Table

\$layer1	\$name1	\$id1	\$value1
metal1_conn	Metal1	METAL1.W.1	0.06
metal2_conn	Metal2	METAL2.W.1	0.08
metal3_conn	Metal3	METAL3.W.1	0.08
metal4_conn	Metal4	METAL4.W.1	0.08
metal5_conn	Metal5	METAL5.W.1	0.08
metal6_conn	Metal6	METAL6.W.1	0.08
metal7_conn	Metal7	METAL7.W.1	0.08
metal8_conn	Metal8	METAL8.W.1	0.08
metal9_conn	Metal9	METAL9.W.1	0.08
Metal10	Metal10	METAL10.W.1	0.22
Metal11	Metal11	METAL11.W.1	0.22

\$layer1



macro

! Bondpad

! \$layer3

! \$layer2

\$layer1

Macro Table

\$layer1	\$layer2	\$layer3	\$name1	\$id1	\$value1
metal1_conn	cont_array_zone	via1_array_zone	Metal1	METAL1.W.2	6.0
metal2_conn	via1_array_zone	via2_array_zone	Metal2	METAL2.W.2	6.0
metal3_conn	via2_array_zone	via3_array_zone	Metal3	METAL3.W.2	6.0
metal4_conn	via3_array_zone	via4_array_zone	Metal4	METAL4.W.2	6.0
metal5_conn	via4_array_zone	via5_array_zone	Metal5	METAL5.W.2	6.0
metal6_conn	via5_array_zone	via6_array_zone	Metal6	METAL6.W.2	6.0
metal7_conn	via6_array_zone	via7_array_zone	Metal7	METAL7.W.2	6.0
metal8_conn	via7_array_zone	via8_array_zone	Metal8	METAL8.W.2	6.0

! Bondpad

! via6_array_zone

Metal9

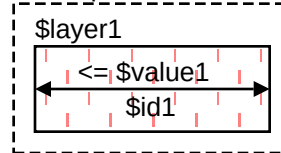
METAL RULES (continued)

macro

Macro Table

\$layer1	\$name1	\$id1	\$value1
Metal10	Metal10	METAL10.W.2	6.0
Metal11	Metal11	METAL11.W.2	6.0

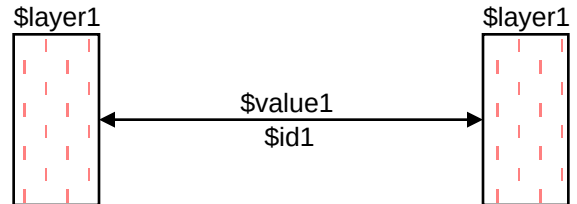
! Bondpad



macro

Macro Table

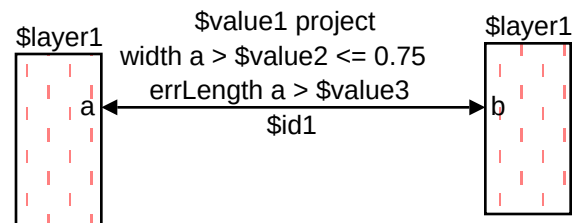
\$layer1	\$id1	\$value1
Metal1	METAL1.SP.1.1	0.06
Metal2	METAL2.SP.1.1	0.07
Metal3	METAL3.SP.1.1	0.07
Metal4	METAL4.SP.1.1	0.07
Metal5	METAL5.SP.1.1	0.07
Metal6	METAL6.SP.1.1	0.07
Metal7	METAL7.SP.1.1	0.07
Metal8	METAL8.SP.1.1	0.07
Metal9	METAL9.SP.1.1	0.07
Metal10	METAL10.SP.1.1	0.20
Metal11	METAL11.SP.1.1	0.20



macro

Macro Table

\$layer1	\$id1	\$value1	\$value2	\$value3
Metal1	METAL1.SP.1.2	0.1	0.1	0.32
Metal2	METAL2.SP.1.2	0.15	0.1	0.32
Metal3	METAL3.SP.1.2	0.15	0.1	0.32
Metal4	METAL4.SP.1.2	0.15	0.1	0.32
Metal5	METAL5.SP.1.2	0.15	0.1	0.32
Metal6	METAL6.SP.1.2	0.15	0.1	0.32
Metal7	METAL7.SP.1.2	0.15	0.1	0.32
Metal8	METAL8.SP.1.2	0.15	0.1	0.32
Metal9	METAL9.SP.1.2	0.15	0.1	0.32

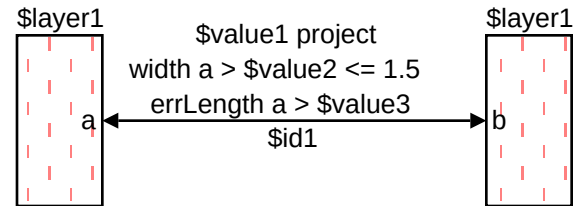


METAL RULES (continued)

macro

Macro Table

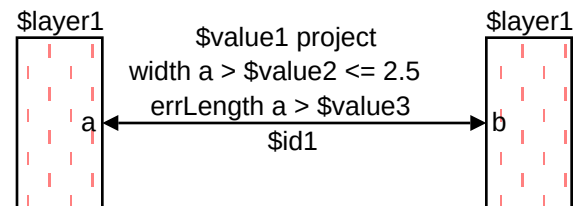
\$layer1	\$id1	\$value1	\$value2	\$value3
Metal1	METAL1.SP.1.3	0.25	0.75	0.75
Metal2	METAL2.SP.1.3	0.25	0.75	0.75
Metal3	METAL3.SP.1.3	0.25	0.75	0.75
Metal4	METAL4.SP.1.3	0.25	0.75	0.75
Metal5	METAL5.SP.1.3	0.25	0.75	0.75
Metal6	METAL6.SP.1.3	0.25	0.75	0.75
Metal7	METAL7.SP.1.3	0.25	0.75	0.75
Metal8	METAL8.SP.1.3	0.25	0.75	0.75
Metal9	METAL9.SP.1.3	0.25	0.75	0.75
Metal10	METAL10.SP.1.2	0.35	0.75	0.75
Metal11	METAL11.SP.1.2	0.35	0.75	0.75



macro

Macro Table

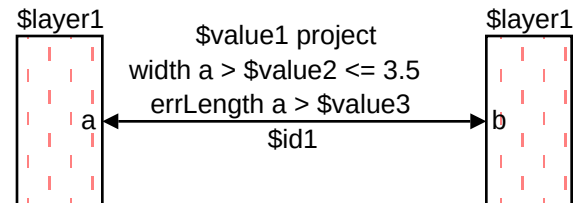
\$layer1	\$id1	\$value1	\$value2	\$value3
Metal1	METAL1.SP.1.4	0.45	1.5	1.5
Metal2	METAL2.SP.1.4	0.45	1.5	1.5
Metal3	METAL3.SP.1.4	0.45	1.5	1.5
Metal4	METAL4.SP.1.4	0.45	1.5	1.5
Metal5	METAL5.SP.1.4	0.45	1.5	1.5
Metal6	METAL6.SP.1.4	0.45	1.5	1.5
Metal7	METAL7.SP.1.4	0.45	1.5	1.5
Metal8	METAL8.SP.1.4	0.45	1.5	1.5
Metal9	METAL9.SP.1.4	0.45	1.5	1.5
Metal10	METAL10.SP.1.3	0.45	1.5	1.5
Metal11	METAL11.SP.1.3	0.45	1.5	1.5



macro

Macro Table

\$layer1	\$id1	\$value1	\$value2	\$value3
Metal1	METAL1.SP.1.5	0.75	2.5	2.5
Metal2	METAL2.SP.1.5	0.75	2.5	2.5
Metal3	METAL3.SP.1.5	0.75	2.5	2.5
Metal4	METAL4.SP.1.5	0.75	2.5	2.5
Metal5	METAL5.SP.1.5	0.75	2.5	2.5
Metal6	METAL6.SP.1.5	0.75	2.5	2.5
Metal7	METAL7.SP.1.5	0.75	2.5	2.5
Metal8	METAL8.SP.1.5	0.75	2.5	2.5
Metal9	METAL9.SP.1.5	0.75	2.5	2.5
Metal10	METAL10.SP.1.4	0.75	2.5	2.5
Metal11	METAL11.SP.1.4	0.75	2.5	2.5

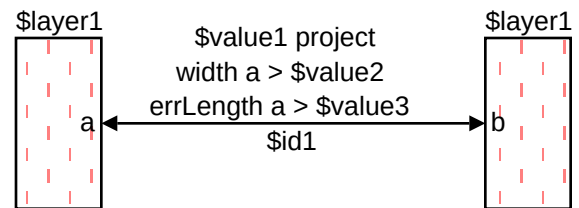


METAL RULES (continued)

macro

Macro Table

\$layer1	\$sid1	\$value1	\$value2	\$value3
Metal1	METAL1.SP.1.6	1.25	3.5	3.5
Metal2	METAL2.SP.1.6	1.25	3.5	3.5
Metal3	METAL3.SP.1.6	1.25	3.5	3.5
Metal4	METAL4.SP.1.6	1.25	3.5	3.5
Metal5	METAL5.SP.1.6	1.25	3.5	3.5
Metal6	METAL6.SP.1.6	1.25	3.5	3.5
Metal7	METAL7.SP.1.6	1.25	3.5	3.5
Metal8	METAL8.SP.1.6	1.25	3.5	3.5
Metal9	METAL9.SP.1.6	1.25	3.5	3.5
Metal10	METAL10.SP.1.5	1.25	3.5	3.5
Metal11	METAL11.SP.1.5	1.25	3.5	3.5

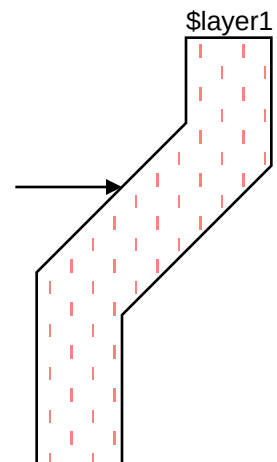


macro

Macro Table

\$layer1	\$sid1	\$value1
Metal1	METAL1.L.1	0.1
Metal2	METAL2.L.1	0.1
Metal3	METAL3.L.1	0.1
Metal4	METAL4.L.1	0.1
Metal5	METAL5.L.1	0.1
Metal6	METAL6.L.1	0.1
Metal7	METAL7.L.1	0.1
Metal8	METAL8.L.1	0.1
Metal9	METAL9.L.1	0.1

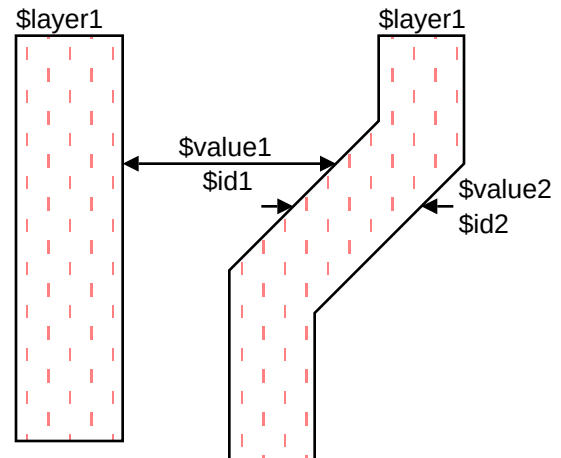
segment $\geq$ \$value1
\$sid1



macro

Macro Table

\$layer1	\$sid1	\$value1	\$sid2	\$value2
Metal1	METAL1.SP.2	0.1	METAL1.W.3	0.07
Metal2	METAL2.SP.2	0.1	METAL2.W.3	0.09
Metal3	METAL3.SP.2	0.1	METAL3.W.3	0.09
Metal4	METAL4.SP.2	0.1	METAL4.W.3	0.09
Metal5	METAL5.SP.2	0.1	METAL5.W.3	0.09
Metal6	METAL6.SP.2	0.1	METAL6.W.3	0.09
Metal7	METAL7.SP.2	0.1	METAL7.W.3	0.09
Metal8	METAL8.SP.2	0.1	METAL8.W.3	0.09
Metal9	METAL9.SP.2	0.1	METAL9.W.3	0.09



METAL1.SP.3

assuraDRC Native Code

```
metx1_a=drcDenseLineEnd(Metal1 sep < 0.08 endLength(0.065 0.09) legLength(.10) endExt(.025)
legExt(0.025))
errorLayer(metx1_a "METAL1.SP.3: Min End of Line Spacing >= 0.08")
```

METALK.SP.3 k=2-9

assuraDRC Native Code

```
metx2_a=drcDenseLineEnd(Metal2 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx2_a "METAL2.SP.3: Min End of Line Spacing >= 0.08")

metx3_a=drcDenseLineEnd(Metal3 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx3_a "METAL3.SP.3: Min End of Line Spacing >= 0.08")

metx4_a=drcDenseLineEnd(Metal4 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx4_a "METAL4.SP.3: Min End of Line Spacing >= 0.08")

metx5_a=drcDenseLineEnd(Metal5 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx5_a "METAL5.SP.3: Min End of Line Spacing >= 0.08")

metx6_a=drcDenseLineEnd(Metal6 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx6_a "METAL6.SP.3: Min End of Line Spacing >= 0.08")

metx7_a=drcDenseLineEnd(Metal7 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx7_a "METAL7.SP.3: Min End of Line Spacing >= 0.08")

metx8_a=drcDenseLineEnd(Metal8 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx8_a "METAL8.SP.3: Min End of Line Spacing >= 0.08")

metx9_a=drcDenseLineEnd(Metal9 sep < 0.08 endLength(0.075 0.1) legLength(0.1) endExt(0.035)
legExt(0.035))
errorLayer(metx9_a "METAL9.SP.3: Min End of Line Spacing >= 0.08")
```

METAL1.SP.3

pvsDRC Native Code

```
rule METAL1.SP.3 {
caption METAL1.SP.3: Min End Of Line Spacing >= 0.08;
convex_edge Metal1 -angle1 -eq 90 -angle2 -eq 90 -with_length -lt 0.09 met1_lw;
exte met1_lw Metal1 -lt 0.08 -abut -lt 90 -metric opposite_extended 0.025 met1_sp;
edge_select -inside met1_lw met1_sp met1Edge1;
inte met1Edge1 Metal1 -lt 0.1 -abut -eq 90 -intersecting only met1_q1;
edge_length met1_q1 -ge 0.06 met1_q2;
edge_expand met1_q2 -inside_by 0.001 -extend_by 0.025 met1exp1;
edge_expand met1_q2 -inside_by 0.001 met1exp2;
not met1exp1 met1exp2 met1Exp;
select -with_edge met1Exp met1_lw met1_lwEdg;
or met1_lwEdg met1exp2 met1_allEdg;
edge_select met1_allEdg met1_q2 met1_extEdg;
edge_select -not met1_extEdg met1_sp met1_last
exte met1_last Metal1 -lt 0.08 -abut -lt 90 -metric opposite -output region ;
}
```

METALK.SP.3 k=2-9

pvsDRC Native Code

```
rule METAL2.SP.3 {
caption METAL2.SP.3: Min End Of Line Spacing >= 0.08;
convex_edge Metal2 -angle1 -eq 90 -angle2 -eq 90 -with_length -lt 0.1 met2_lw;
exte met2_lw Metal2 -lt 0.08 -abut -lt 90 -metric opposite_extended 0.035 met2_sp;
edge_select -inside met2_lw met2_sp met2Edge1;
inte met2Edge1 Metal2 -lt 0.1 -abut -eq 90 -intersecting only met2_q1;
edge_length met2_q1 -ge 0.07 met2_q2;
edge_expand met2_q2 -inside_by 0.001 -extend_by 0.035 met2exp1;
edge_expand met2_q2 -inside_by 0.001 met2exp2;
not met2exp1 met2exp2 met2Exp;
select -with_edge met2Exp met2_lw met2_lwEdg;
or met2_lwEdg met2exp2 met2_allEdg;
edge_select met2_allEdg met2_q2 met2_extEdg;
edge_select -not met2_extEdg met2_sp met2_last
exte met2_last Metal2 -lt 0.08 -abut -lt 90 -metric opposite -output region ;
}

rule METAL3.SP.3 {
caption METAL3.SP.3: Min End Of Line Spacing >= 0.08;
convex_edge Metal3 -angle1 -eq 90 -angle2 -eq 90 -with_length -lt 0.1 met3_lw;
exte met3_lw Metal3 -lt 0.08 -abut -lt 90 -metric opposite_extended 0.035 met3_sp;
edge_select -inside met3_lw met3_sp met3Edge1;
inte met3Edge1 Metal3 -lt 0.1 -abut -eq 90 -intersecting only met3_q1;
edge_length met3_q1 -ge 0.07 met3_q2;
edge_expand met3_q2 -inside_by 0.001 -extend_by 0.035 met3exp1;
edge_expand met3_q2 -inside_by 0.001 met3exp2;
not met3exp1 met3exp2 met3Exp;
select -with_edge met3Exp met3_lw met3_lwEdg;
or met3_lwEdg met3exp2 met3_allEdg;
edge_select met3_allEdg met3_q2 met3_extEdg;
edge_select -not met3_extEdg met3_sp met3_last
exte met3_last Metal3 -lt 0.08 -abut -lt 90 -metric opposite -output region ;...
```

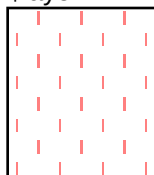
METAL RULES (continued)

macro

Macro Table

\$layer1	\$id1	\$value1
Metal1	METAL1.A.1	0.02
Metal2	METAL2.A.1	0.02
Metal3	METAL3.A.1	0.02
Metal4	METAL4.A.1	0.02
Metal5	METAL5.A.1	0.02
Metal6	METAL6.A.1	0.02
Metal7	METAL7.A.1	0.02
Metal8	METAL8.A.1	0.02
Metal9	METAL9.A.1	0.02
Metal10	METAL10.A.1	0.10
Metal11	METAL11.A.1	0.10

\$layer1



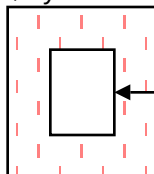
area >= \$value1
\$id1

macro

Macro Table

\$layer1	\$id1	\$value1
Metal1	METAL1.EA.1	0.045
Metal2	METAL2.EA.1	0.055
Metal3	METAL3.EA.1	0.055
Metal4	METAL4.EA.1	0.055
Metal5	METAL5.EA.1	0.055
Metal6	METAL6.EA.1	0.055
Metal7	METAL7.EA.1	0.055
Metal8	METAL8.EA.1	0.055
Metal9	METAL9.EA.1	0.055
Metal10	METAL10.EA.1	0.11
Metal11	METAL11.EA.1	0.11

\$layer1



area >= \$value1
\$id1

METAL RULES (continued)

switch CHECK_DENSITY

macro

Macro Table

\$name1	\$layer1	\$id1
Metal1	metal1_conn	METAL1.D.1
Metal2	metal2_conn	METAL2.D.1
Metal3	metal3_conn	METAL3.D.1
Metal4	metal4_conn	METAL4.D.1
Metal5	metal5_conn	METAL5.D.1
Metal6	metal6_conn	METAL6.D.1
Metal7	metal7_conn	METAL7.D.1
Metal8	metal8_conn	METAL8.D.1
Metal9	metal9_conn	METAL9.D.1
Metal10	metal10_conn	METAL10.D.1
Metal11	metal11_conn	METAL11.D.1

Density

ratio $\geq 0.20 \leq 0.65$

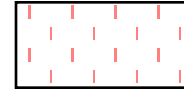
windowSize: 120.0

stepSize: 60.0

id: \$id1

message: \$name1 density must be $\geq 20\% \leq 65\%$

\$layer1



macro

Macro Table

\$name1	\$layer1	\$id1
Metal1	metal1_conn	METAL1.D.2
Metal2	metal2_conn	METAL2.D.2
Metal3	metal3_conn	METAL3.D.2
Metal4	metal4_conn	METAL4.D.2
Metal5	metal5_conn	METAL5.D.2
Metal6	metal6_conn	METAL6.D.2
Metal7	metal7_conn	METAL7.D.2
Metal8	metal8_conn	METAL8.D.2
Metal9	metal9_conn	METAL9.D.2
Metal10	metal10_conn	METAL10.D.2
Metal11	metal11_conn	METAL11.D.2

Density

ratio ≤ 0.60

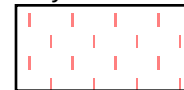
windowSize: 600.0

stepSize: 300.0

id: \$id1

message: \$name1 density must be $\leq 60\%$

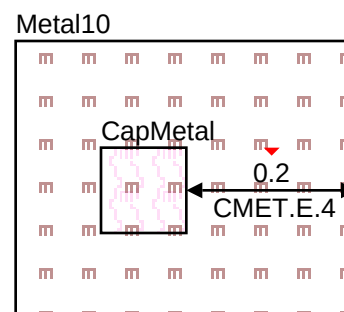
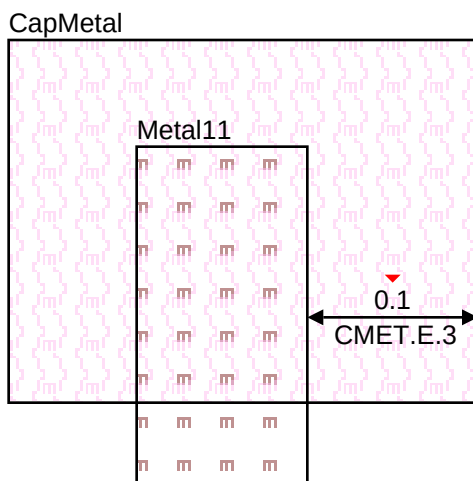
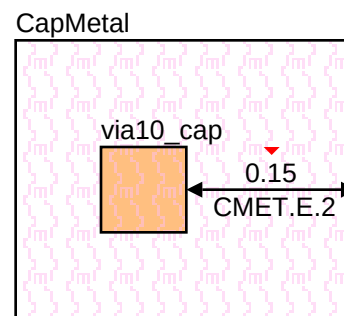
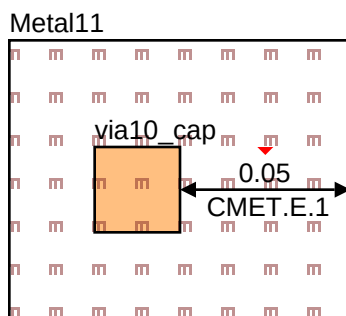
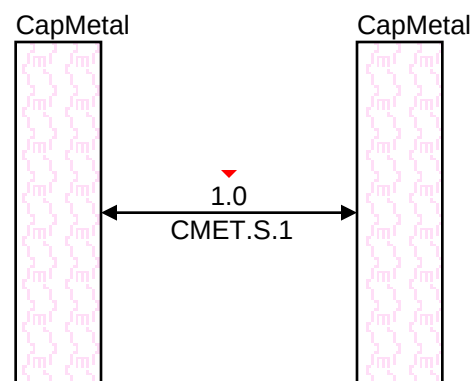
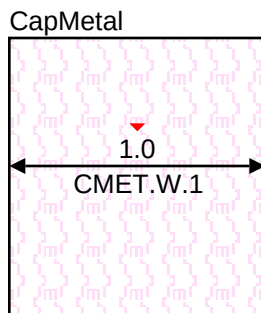
\$layer1



Capacitor Metal

Data Table: CMET_DRC

RuleName	Description	Value
CMET.W.1	Minimum width of CapMetal	1.0
CMET.S.1	Minimum space of CapMetal	1.0
CMET.E.1	Minimum Metal 1 overlap of Via 10 on CapMetal	0.05
CMET.E.2	Minimum CapMetal overlap of Via 10	0.15
CMET.E.3	Minimum CapMetal extension over Metal 11	0.1
CMET.E.4	Minimum Metal 10 overlap of CapMetal	0.2



VIA k (k = 1, 2, 3, 4, 5, 6, 7, 8) RULES

Table 8, 4, 5, 6, 7, 8) RULES

Rule Name	Value (um)	Description
VIAk.W.1	0.07	Minimum and maximum Via k width.
VIAk.SP.1	0.07	Minimum Via k to Via k spacing.
VIAk.SP.2	0.10	Space to three adjacent Via k (< 0.11 um apart)
VIAk.E.1	0.005	Minimum Metal k to Via k enclosure.
VIAk.E.2	0.03	Minimum Metal k to Via k enclosure on at least two opposite sides of Via k.
VIAk.E.3		At least 2 Via k must be used to join two Metal k when they are within 3.0um of a metal plate (Metal k or Metal k+1) when the metal plate size is has width > 1.5 and length > 1.5
VIAk.X.1	---	Minimum of two Via k with spacing <= 0.30um or four Via k with spacing <= 0.60um are required when connecting Metal k and Metal k+1 when one of the Metals has a width > 0.40um at the connection point.
VIAk.X.2	---	Minimum of four Via k with spacing <= 0.30um or nine Via k with spacing <= 0.60um are required when connecting Metal k and Metal k+1 when one of the Metals has a width > 1.0um at the connection point.
VIAk.X.3	---	Vias 1 through 8 may be consecutively stacked up to four high when only one Via is connecting two Metal layers for any level of the stack.
VIAk.X.4	---	Vias 1 through 8 may be consecutively stacked up more than four high when at least two Vias are connecting two Metal layers for all levels of the stack.

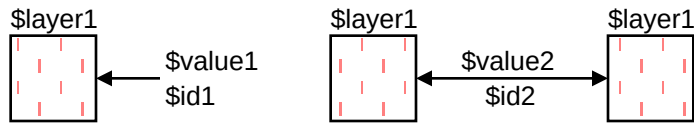
VIA 9, 10 RULES

Table 9, 10) RULES

Rule Name	Value (um)	Description
VIAk.W.1	0.18	Minimum and maximum Via k width.
VIAk.SP.1	0.18	Minimum Via k space.
VIAk.SP.2	0.20	Space to three adjacent Via k (< 0.25 um apart)
VIAk.E.1	0.015	Minimum Metal k to of Via k enclosure.
VIAk.E.2	0.04	Minimum Metal k to Via k enclosure on at least two opposite sides of Via k.

VIA RULES (continued)

macro



Macro Table

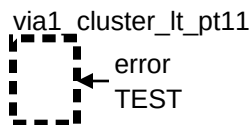
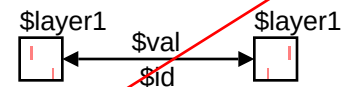
\$layer1	\$id1	\$value1	\$id2	\$value2
Via1	VIA1.W.1	0.07x0.07	VIA1.SP.1	0.07
Via2	VIA2.W.1	0.07x0.07	VIA2.SP.1	0.07
Via3	VIA3.W.1	0.07x0.07	VIA3.SP.1	0.07
Via4	VIA4.W.1	0.07x0.07	VIA4.SP.1	0.07
Via5	VIA5.W.1	0.07x0.07	VIA5.SP.1	0.07
Via6	VIA6.W.1	0.07x0.07	VIA6.SP.1	0.07
Via7	VIA7.W.1	0.07x0.07	VIA7.SP.1	0.07
Via8	VIA8.W.1	0.07x0.07	VIA8.SP.1	0.07
Via9	VIA9.W.1	0.18x0.18	VIA9.SP.1	0.18
Via10	VIA10.W.1	0.18x0.18	VIA10.SP.1	0.18

switch FALSE

macro

Macro Table

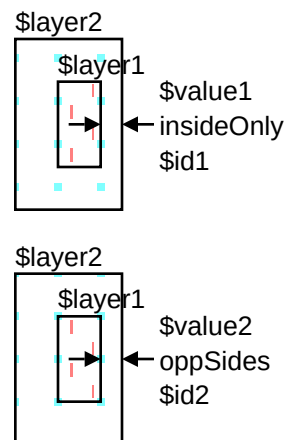
\$layer1	\$name	\$val	\$val2	\$id
via1_cluster_lt_pt11	Via1	.10	0.11	VIA1.SP.2
via2_cluster_lt_pt11	Via2	.10	0.11	VIA2.SP.2
via3_cluster_lt_pt11	Via3	.10	0.11	VIA3.SP.2
via4_cluster_lt_pt11	Via4	.10	0.11	VIA4.SP.2
via5_cluster_lt_pt11	Via5	.10	0.11	VIA5.SP.2
via6_cluster_lt_pt11	Via6	.10	0.11	VIA6.SP.2
via7_cluster_lt_pt11	Via7	.10	0.11	VIA7.SP.2
via8_cluster_lt_pt11	Via8	.10	0.11	VIA8.SP.2
via9_cluster_lt_pt25	Via9	.20	0.25	VIA9.SP.2
via10_cluster_lt_pt25	Via10	.20	0.25	VIA10.SP.2



macro

Macro Table

\$name1	\$layer1	\$layer2	\$id1	\$value1	\$id2	\$value2
Metal1	Via1	metal1_conn	VIA1.E.1	0.005	VIA1.E.2	0.03
Metal2	Via2	metal2_conn	VIA2.E.1	0.005	VIA2.E.2	0.03
Metal3	Via3	metal3_conn	VIA3.E.1	0.005	VIA3.E.2	0.03
Metal4	Via4	metal4_conn	VIA4.E.1	0.005	VIA4.E.2	0.03
Metal5	Via5	metal5_conn	VIA5.E.1	0.005	VIA5.E.2	0.03
Metal6	Via6	metal6_conn	VIA6.E.1	0.005	VIA6.E.2	0.03
Metal7	Via7	metal7_conn	VIA7.E.1	0.005	VIA7.E.2	0.03
Metal8	Via8	metal8_conn	VIA8.E.1	0.005	VIA8.E.2	0.03
Metal9	Via9	metal9_conn	VIA9.E.1	0.015	VIA9.E.2	0.04
Metal10	Via10	Metal10	VIA10.E.1	0.015	VIA10.E.2	0.04

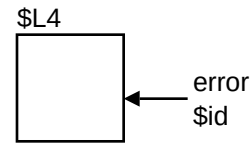


VIA RULES (continued)

macro

Macro Table

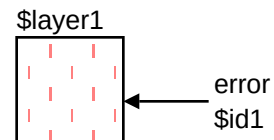
\$L1	\$L2	\$L3	\$L4	\$id
Via1	Metal1	Metal2	rule_VIA1_E_3	VIA1.E.3
Via2	Metal2	Metal3	rule_VIA2_E_3	VIA2.E.3
Via3	Metal3	Metal4	rule_VIA3_E_3	VIA3.E.3
Via4	Metal4	Metal5	rule_VIA4_E_3	VIA4.E.3
Via5	Metal5	Metal6	rule_VIA5_E_3	VIA5.E.3
Via6	Metal6	Metal7	rule_VIA6_E_3	VIA6.E.3
Via7	Metal7	Metal8	rule_VIA7_E_3	VIA7.E.3
Via8	Metal8	Metal9	rule_VIA8_E_3	VIA8.E.3



macro

Macro Table

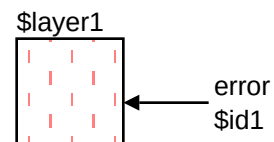
\$name1	\$name2	\$name3	\$layer1	\$id1
Via1	Metal1	Metal2	rule_VIA1_X_1	VIA1.X.1
Via2	Metal2	Metal3	rule_VIA2_X_1	VIA2.X.1
Via3	Metal3	Metal4	rule_VIA3_X_1	VIA3.X.1
Via4	Metal4	Metal5	rule_VIA4_X_1	VIA4.X.1
Via5	Metal5	Metal6	rule_VIA5_X_1	VIA5.X.1
Via6	Metal6	Metal7	rule_VIA6_X_1	VIA6.X.1
Via7	Metal7	Metal8	rule_VIA7_X_1	VIA7.X.1
Via8	Metal8	Metal9	rule_VIA8_X_1	VIA8.X.1



macro

Macro Table

\$name1	\$name2	\$name3	\$layer1	\$id1
Via1	Metal1	Metal2	rule_VIA1_X_2	VIA1.X.2
Via2	Metal2	Metal3	rule_VIA2_X_2	VIA2.X.2
Via3	Metal3	Metal4	rule_VIA3_X_2	VIA3.X.2
Via4	Metal4	Metal5	rule_VIA4_X_2	VIA4.X.2
Via5	Metal5	Metal6	rule_VIA5_X_2	VIA5.X.2
Via6	Metal6	Metal7	rule_VIA6_X_2	VIA6.X.2
Via7	Metal7	Metal8	rule_VIA7_X_2	VIA7.X.2
Via8	Metal8	Metal9	rule_VIA8_X_2	VIA8.X.2



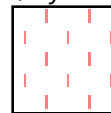
VIA RULES (continued)

macro

Macro Table

\$name1	\$name2	\$layer1	\$id1
Metal1	Metal6	rule_VIAk_X_3_X_4a	VIAk.X.3_VIAk.X.4
Metal2	Metal7	rule_VIAk_X_3_X_4b	VIAk.X.3_VIAk.X.4
Metal3	Metal8	rule_VIAk_X_3_X_4c	VIAk.X.3_VIAk.X.4
Metal4	Metal9	rule_VIAk_X_3_X_4d	VIAk.X.3_VIAk.X.4

\$layer1

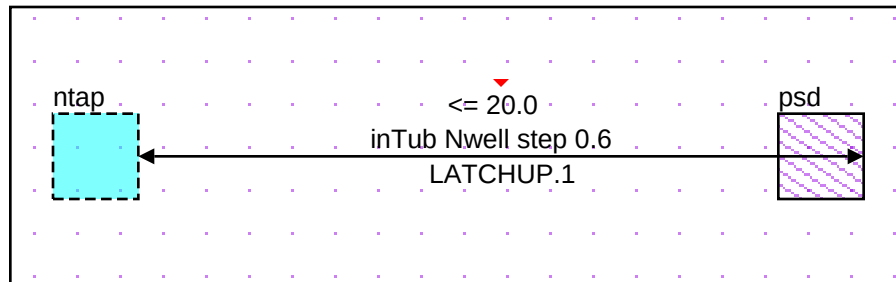
error
\$id1

LATCH-UP RULES

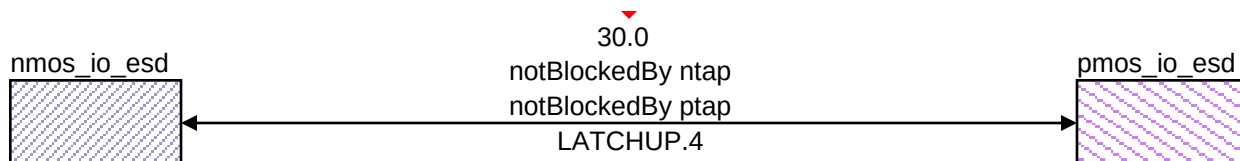
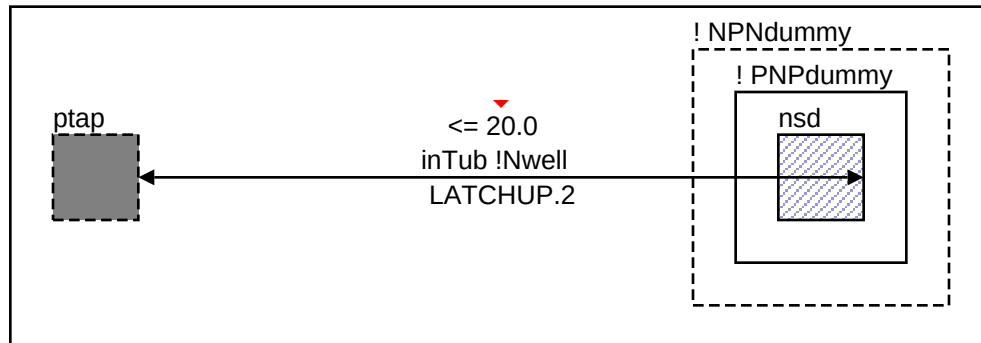
Data Table: LATCHUP_DRC

RuleName	Description	Value
LATCHUP.1	The maximum distance from any point in a P+ source/drain Active Area to the nearest Nwell pick-up in the same Nwell	20.0
LATCHUP.2	The maximum distance from any point in a N+ source/drain Active Area to the nearest Psub pick-up in the same Psub	20.0
LATCHUP.3	Minimum I/O or ESD NMOS to PMOS spacing	10.0
LATCHUP.4	Minimum I/O or ESD NMOS to PMOS spacing when not blocked by a double guarding	30.0

Nwell



! Nwell



ANTENNA RULES

ANTENNA TABLES

Rule Name	Value (um)	Description
ANT.1	275.0	Maximum ratio of Poly area to the gate area the Poly is connected to.
ANT.2	550.0	Maximum ratio of Poly sidewall area to the gate area the Poly is connected to.
ANT.3	15.0	Maximum ratio of Poly Contact area to the gate area the Contact is connected with.
ANT.4.Mx	475.0	Maximum ratio of Metal x (x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) area to the gate area the Metal x is connected to (without diode protection).
ANT.5.Vx	25.0	Maximum ratio of Via x (x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10) area to the gate area the Via x is connected with (without diode protection).
ANT.6.M1_x (x = 3, 4, 5, 6, 7, 8, 9)	1200.0	Maximum ratio of cumulative Metal areas to the gate area the Metals are connected to (without diode protection).
DIODE PROTECTION		
ANT.7.M1_x (x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)		When a protection diode with an area greater than or equal to $0.1\mu\text{m}^2$ is connected to the gate area, the maximum ratio of cumulative Metal areas (Metal 1 - Metal 8) is calculated as follows: - Ratio = (diode area X 500) + 45000 - For Metal 11 only, the ratio is calculated as follows: - Ratio = (diode area X 7500) + 55000
ANT.8.V1_x (x = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)		When a protection diode with an area greater than or equal to $0.1\mu\text{m}^2$ is connected to the gate area, the maximum ratio of cumulative Via areas (Via 1 - Via 10) is calculated as follows: Ratio = (diode area X 250) + 1000

Note 1: Source/drain diffusion areas of MOS devices are counted as part of the diode area.

Note 2: It is recommended to use one large diode with multiple Contacts rather than several smaller diodes.

Note 3: When the sum of the areas of all diodes on a net equals or exceeds $0.1\mu\text{m}^2$, then those diodes can be treated as a protection diode for ANT.7 and ANT.8.

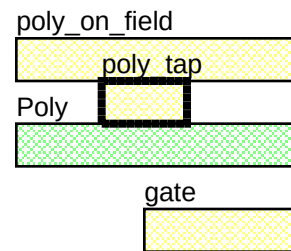
ANTENNA RULES (continued)

switch CHECK_POLY_ANT_1

Antenna

 $\text{ratio}(\text{poly_on_field.area} / \text{gate.area}) \leq 275.0$

id: ANT.1

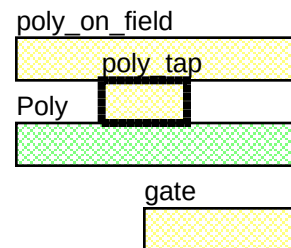
message: Field Poly area to gate area ratio must be ≤ 275.0 

switch CHECK_POLY_ANT_2

Antenna

 $\text{ratio}(\text{poly_on_field.perimeter} / \text{gate.area}) \leq 550.0$

id: ANT.2

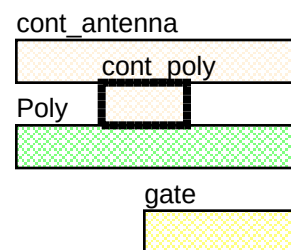
message: Field Poly perimeter to gate area ratio must be ≤ 550.0 

switch CHECK_CONT_ANT_3

Antenna

 $\text{ratio}(\text{cont_antenna.area} / \text{gate.area}) \leq 15.0$

id: ANT.3

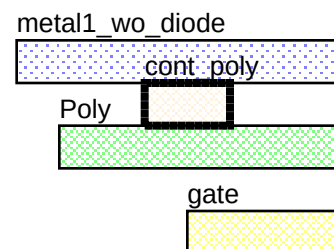
message: Poly Contact area to gate area ratio must be ≤ 15.0 

switch CHECK_METAL1_ANT_4

Antenna

 $\text{ratio}(\text{metal1_wo_diode.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M1

message: Metal1 area to gate area ratio must be ≤ 475.0 

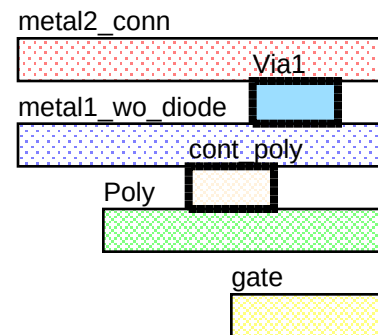
ANTENNA RULES (continued)

switch CHECK_METAL2_ANT_4

Antenna

 $\text{ratio}(\text{metal2_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M2

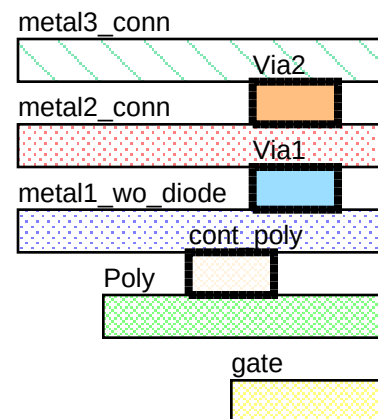
message: Metal2 area to gate area ratio must be ≤ 475.0 

switch CHECK_METAL3_ANT_4

Antenna

 $\text{ratio}(\text{metal3_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M3

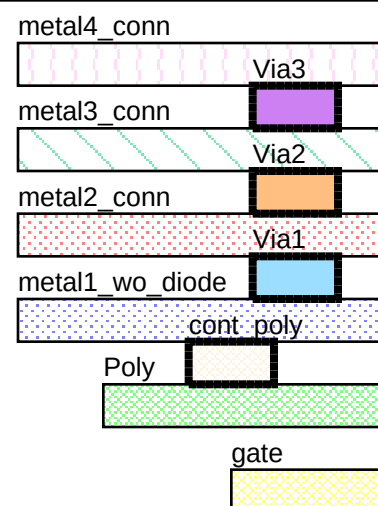
message: Metal3 area to gate area ratio must be ≤ 475.0 

switch CHECK_METAL4_ANT_4

Antenna

 $\text{ratio}(\text{metal4_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M4

message: Metal4 area to gate area ratio must be ≤ 475.0 

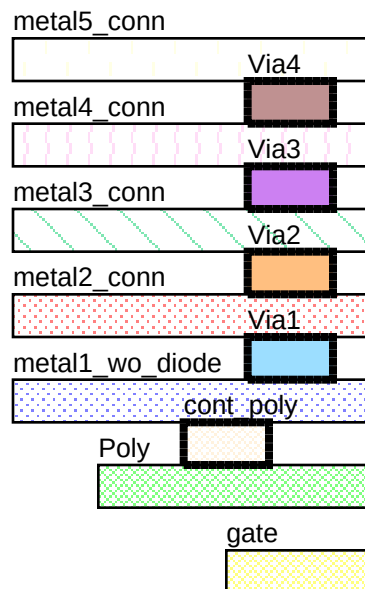
ANTENNA RULES (continued)

switch CHECK_METAL5_ANT_4

Antenna

 $\text{ratio} (\text{metal5_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M5

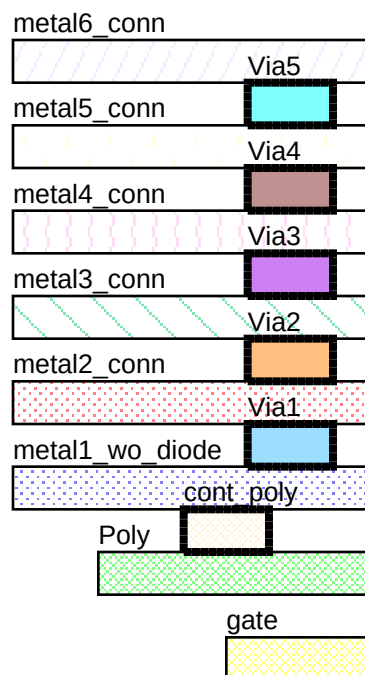
message: Metal5 area to gate area ratio must be ≤ 475.0 

switch CHECK_METAL6_ANT_4

Antenna

 $\text{ratio} (\text{metal6_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M6

message: Metal6 area to gate area ratio must be ≤ 475.0 

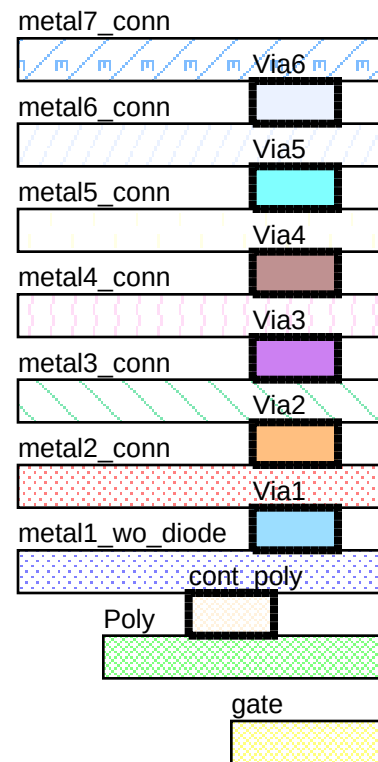
ANTENNA RULES (continued)

switch CHECK_METAL7_ANT_4

Antenna

 $\text{ratio} (\text{metal7_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M7

message: Metal7 area to gate area ratio must be ≤ 475.0 

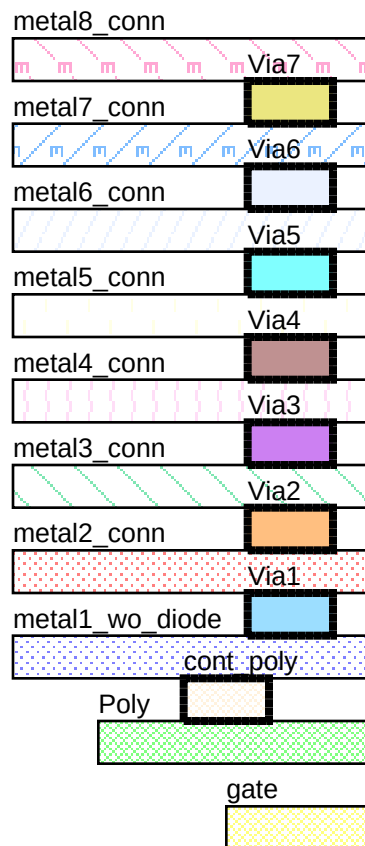
ANTENNA RULES (continued)

switch CHECK_METAL8_ANT_4

Antenna

 $\text{ratio (metal8_conn.area / gate.area)} \leq 475.0$

id: ANT.4.M8

message: Metal8 area to gate area ratio must be ≤ 475.0 

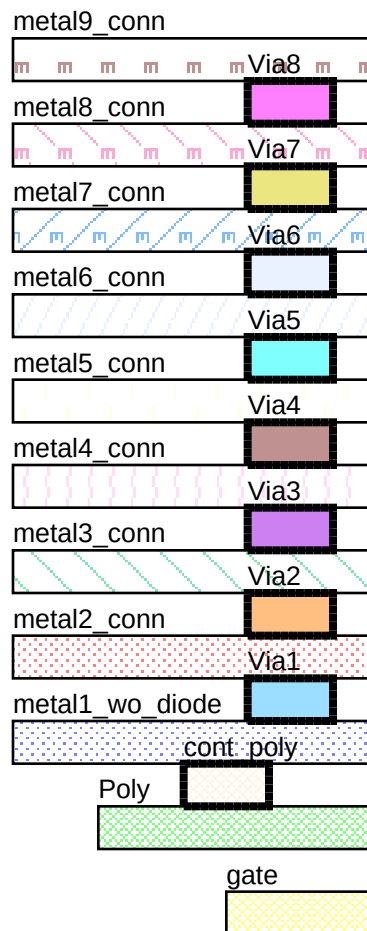
ANTENNA RULES (continued)

switch CHECK_METAL9_ANT_4

Antenna

 $\text{ratio}(\text{metal9_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M9

message: Metal9 area to gate area ratio must be ≤ 475.0 

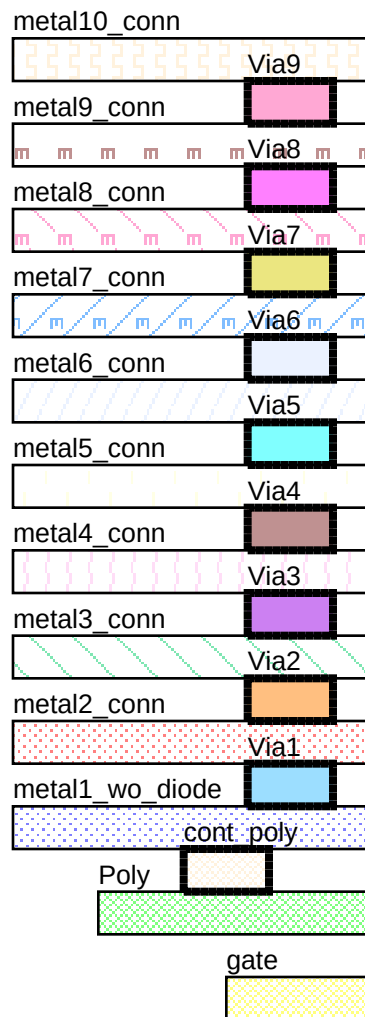
ANTENNA RULES (continued)

switch CHECK_METAL10_ANT_4

Antenna

 $\text{ratio} (\text{metal10_conn.area} / \text{gate.area}) \leq 475.0$

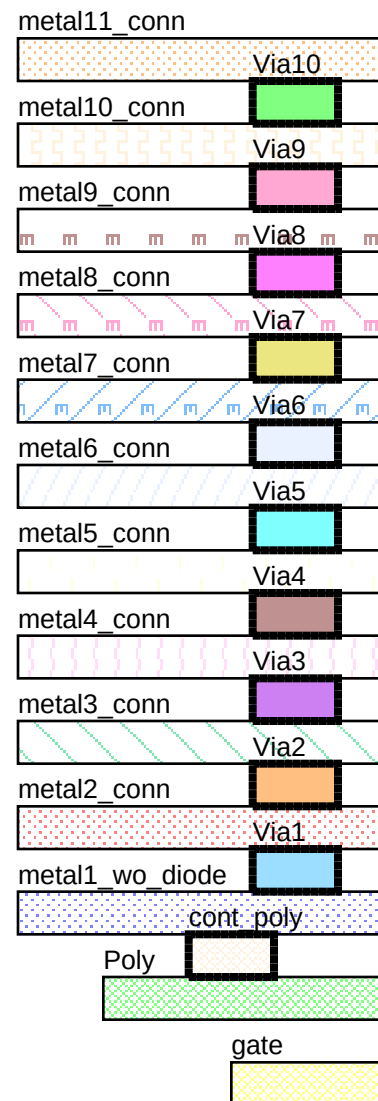
id: ANT.4.M10

message: Metal10 area to gate area ratio must be ≤ 475.0 

ANTENNA RULES (continued)

switch CHECK_METAL11_ANT_4

Antenna

 $\text{ratio}(\text{metal11_conn.area} / \text{gate.area}) \leq 475.0$

id: ANT.4.M11

message: Metal11 area to gate area ratio must be ≤ 475.0

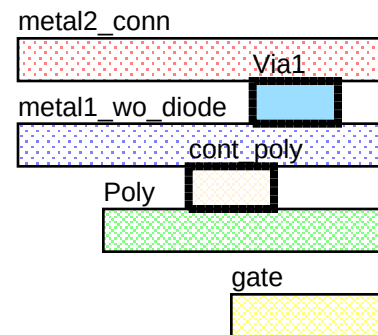
ANTENNA RULES (continued)

switch CHECK_VIA1_ANT_5

Antenna

 $\text{ratio (Via1.area / gate.area)} \leq 25.0$

id: ANT.5.V1

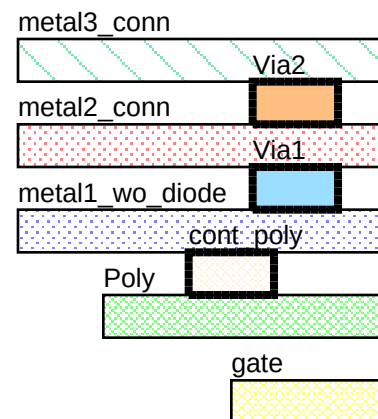
message: Via1 area to gate area ratio must be ≤ 25.0 

switch CHECK_VIA2_ANT_5

Antenna

 $\text{ratio (Via2.area / gate.area)} \leq 25.0$

id: ANT.5.V2

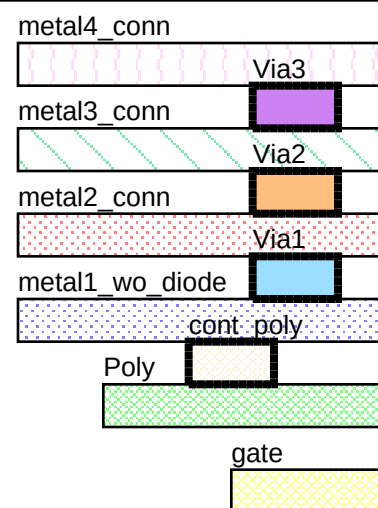
message: Via2 area to gate area ratio must be ≤ 25.0 

switch CHECK_VIA3_ANT_5

Antenna

 $\text{ratio (Via3.area / gate.area)} \leq 25.0$

id: ANT.5.V3

message: Via3 area to gate area ratio must be ≤ 25.0 

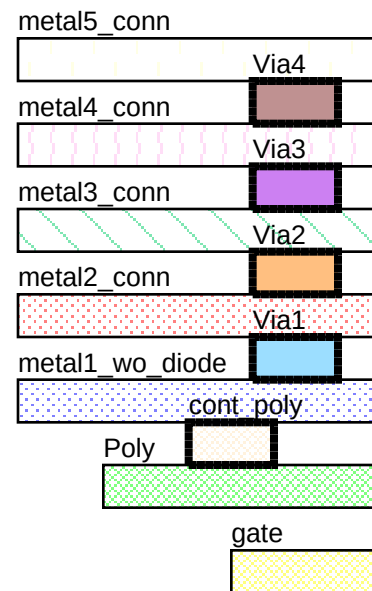
ANTENNA RULES (continued)

switch CHECK_VIA4_ANT_5

Antenna

 $\text{ratio (Via4.area / gate.area)} \leq 25.0$

id: ANT.5.V4

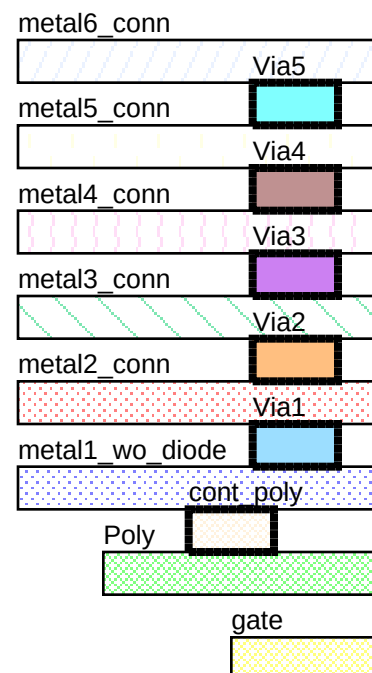
message: Via4 area to gate area ratio must be ≤ 25.0 

switch CHECK_VIA5_ANT_5

Antenna

 $\text{ratio (Via5.area / gate.area)} \leq 25.0$

id: ANT.5.V5

message: Via5 area to gate area ratio must be ≤ 25.0 

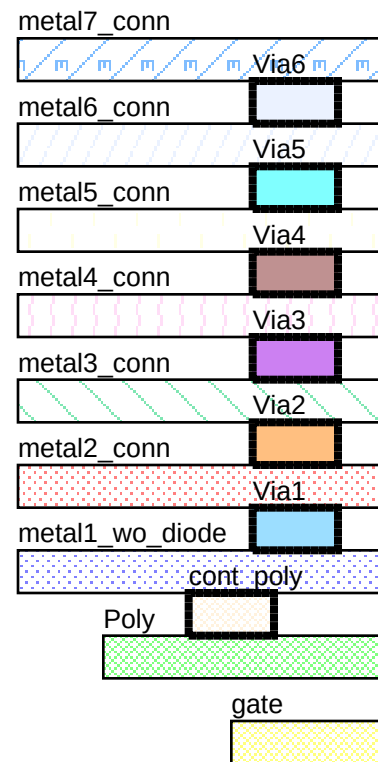
ANTENNA RULES (continued)

switch CHECK_VIA6_ANT_5

Antenna

 $\text{ratio (Via6.area / gate.area)} \leq 25.0$

id: ANT.5.V6

message: Via6 area to gate area ratio must be ≤ 25.0 

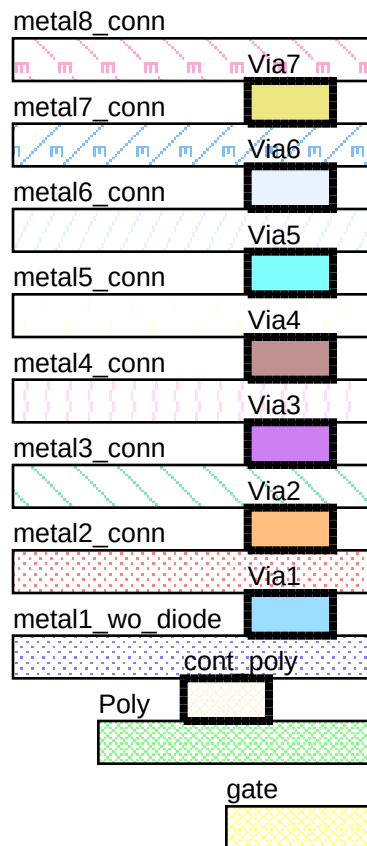
ANTENNA RULES (continued)

switch CHECK_VIA7_ANT_5

Antenna

 $\text{ratio (Via7.area / gate.area)} \leq 25.0$

id: ANT.5.V7

message: Via7 area to gate area ratio must be ≤ 25.0 

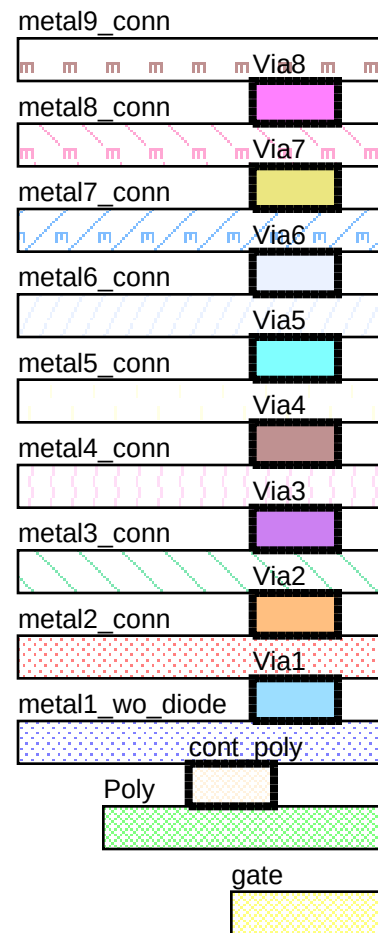
ANTENNA RULES (continued)

switch CHECK_VIA8_ANT_5

Antenna

 $\text{ratio (Via8.area / gate.area)} \leq 25.0$

id: ANT.5.V8

message: Via8 area to gate area ratio must be ≤ 25.0 

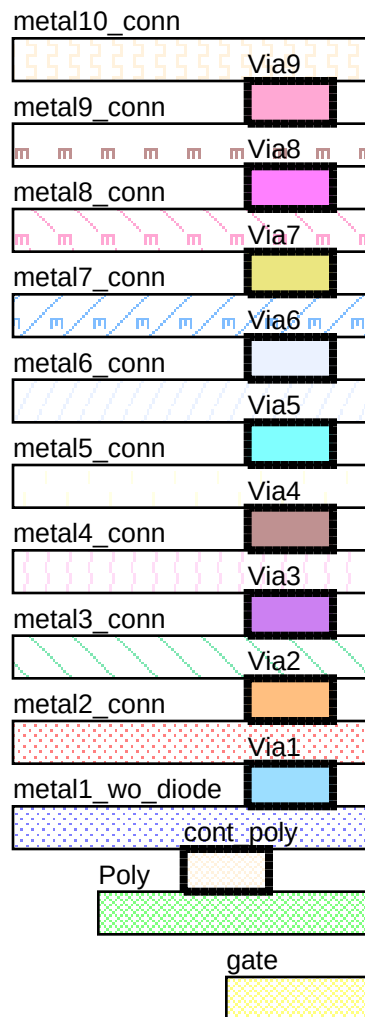
ANTENNA RULES (continued)

switch CHECK_VIA9_ANT_5

Antenna

 $\text{ratio (Via9.area / gate.area)} \leq 25.0$

id: ANT.5.V9

message: Via9 area to gate area ratio must be ≤ 25.0 

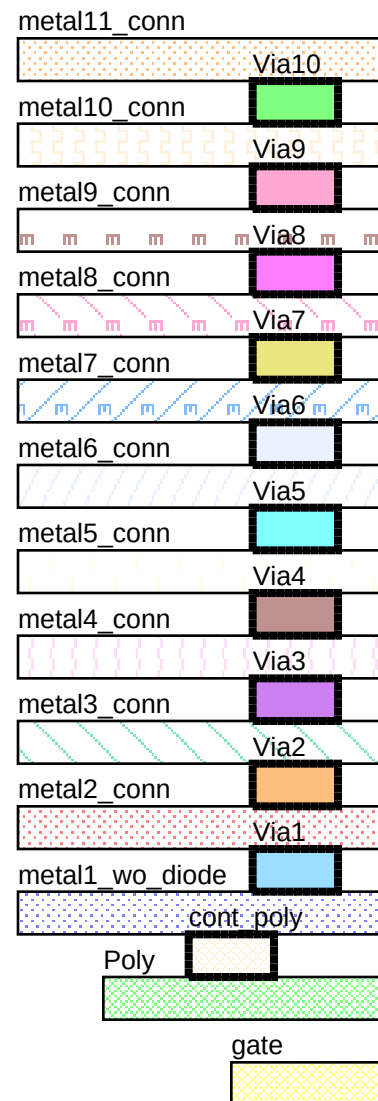
ANTENNA RULES (continued)

switch CHECK_VIA10_ANT_5

Antenna

 $\text{ratio (Via10.area / gate.area)} \leq 25.0$

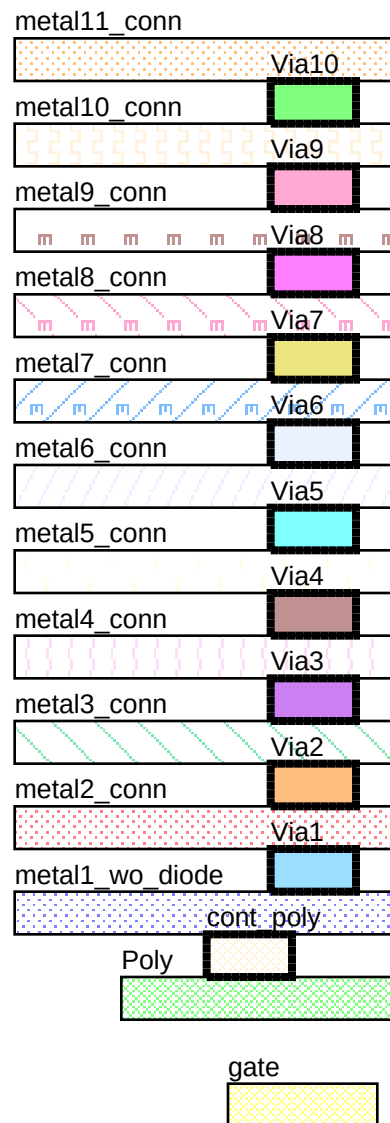
id: ANT.5.V10

message: Via10 area to gate area ratio must be ≤ 25.0 

ANTENNA RULES (continued)

switch CHECK_METAL11_ANT_6

Antenna



ratio ((metal11_conn.area + metal10_conn.area + metal9_conn.area +
metal8_conn.area + metal7_conn.area + metal6_conn.area +
metal5_conn.area + metal4_conn.area +
metal3_conn.area + metal2_conn.area + metal1_wo_diode.area) /
gate.area) <= 1200.0

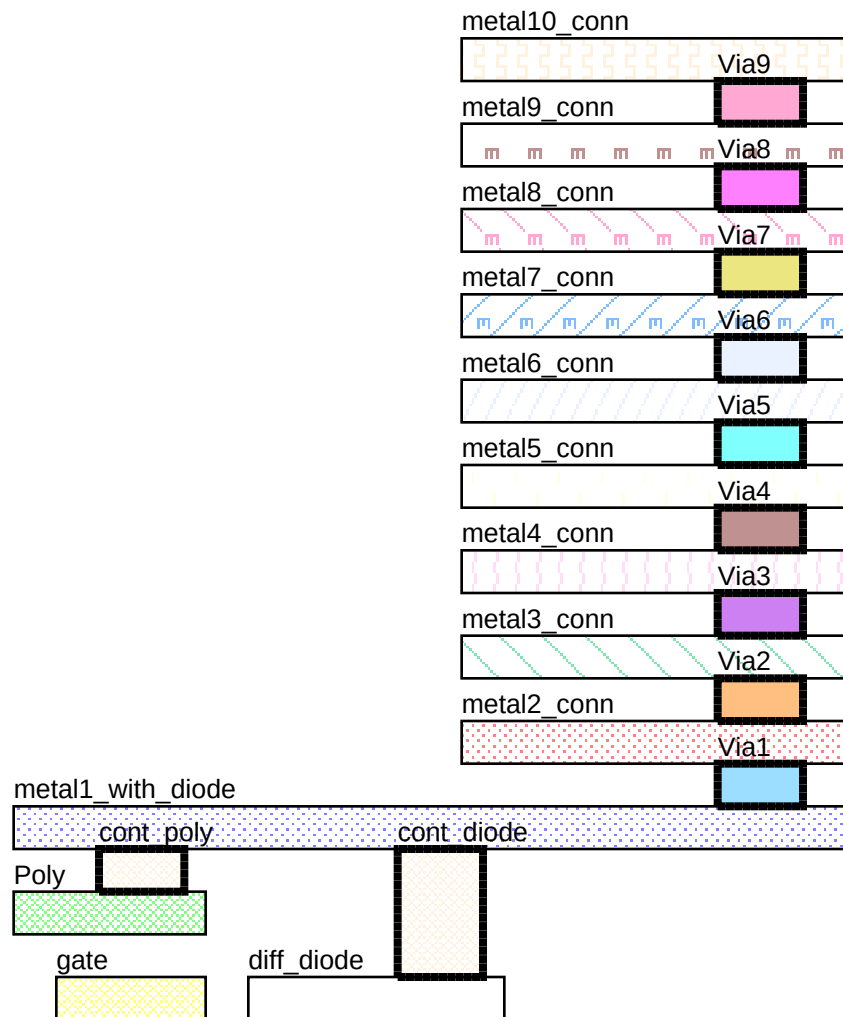
id: ANT.6.M1_11

message: Cumulative Metal1 through Metal11 area to gate area ratio must be <= 1200.0

ANTENNA RULES (continued)

switch CHECK_METAL10_ANT_7

Antenna



ratio (((metal1_with_diode.area + metal2_conn.area + metal3_conn.area +
metal4_conn.area + metal5_conn.area + metal6_conn.area +
metal7_conn.area + metal8_conn.area + metal9_conn.area +
metal10_conn.area) / gate.area) - ((diff_diode.area * 500.0) + 45000.0)) <= 0.0

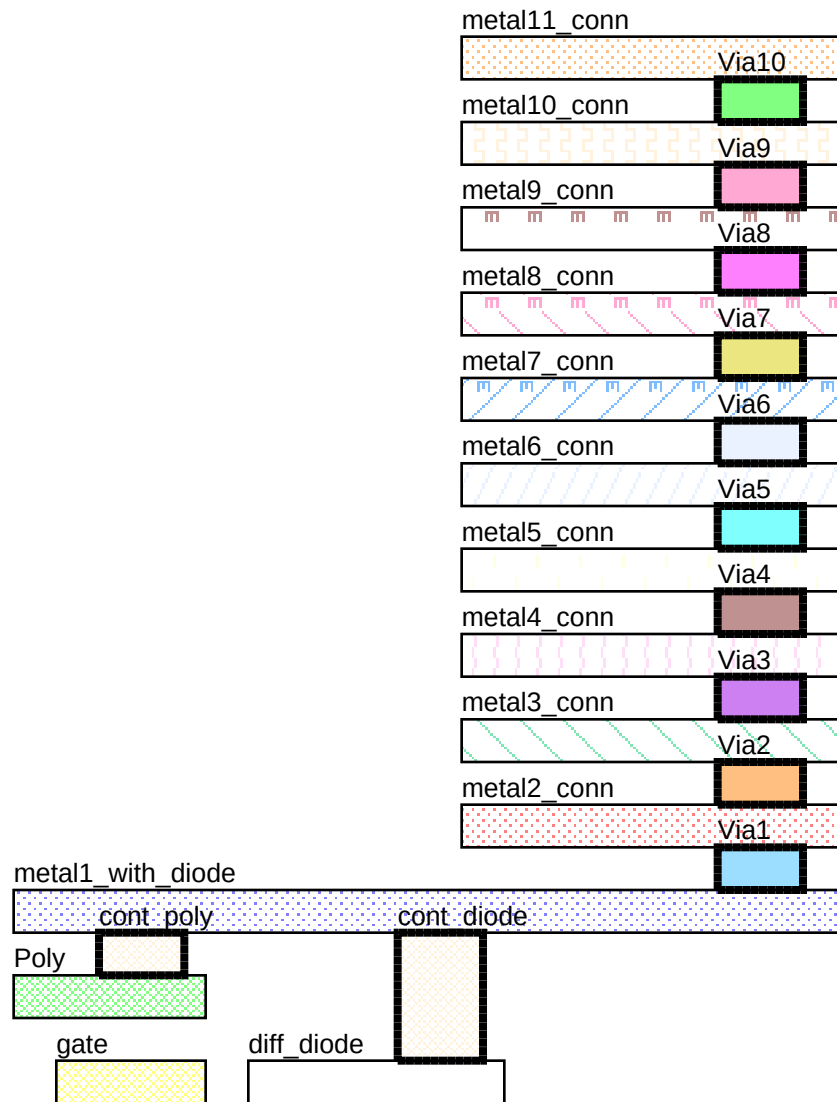
id: ANT.7.M1_10

message: Cumulative Metal1 through Metal10 area to gate area ratio must be
<= 45000 + (diode area * 500)

ANTENNA RULES (continued)

switch CHECK_METAL11_ANT_7

Antenna



id: ANT.7.M1_11

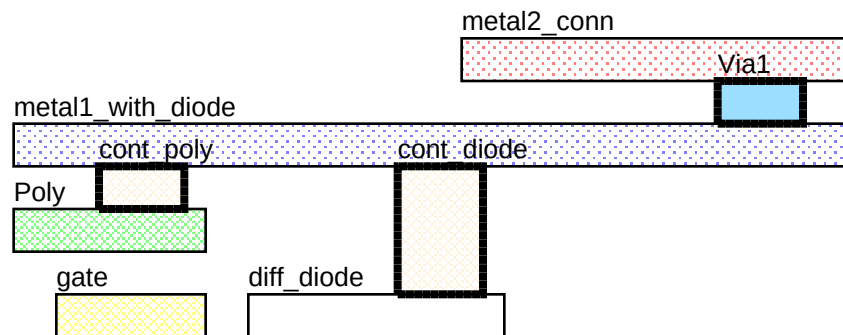
ratio ((metal1_with_diode.area + metal2_conn.area + metal3_conn.area +
metal4_conn.area + metal5_conn.area + metal6_conn.area +
metal7_conn.area + metal8_conn.area + metal9_conn.area +
metal10_conn.area + metal11_conn.area / gate.area) -
((diff_diode.area * 7500.0) + 55000.0)) <= 0.0

message: Cumulative Metal1 through Metal11 area to gate area ratio must be
<= 55000 + (diode area * 7500)

ANTENNA RULES (continued)

switch CHECK_VIA1_ANT_8

Antenna



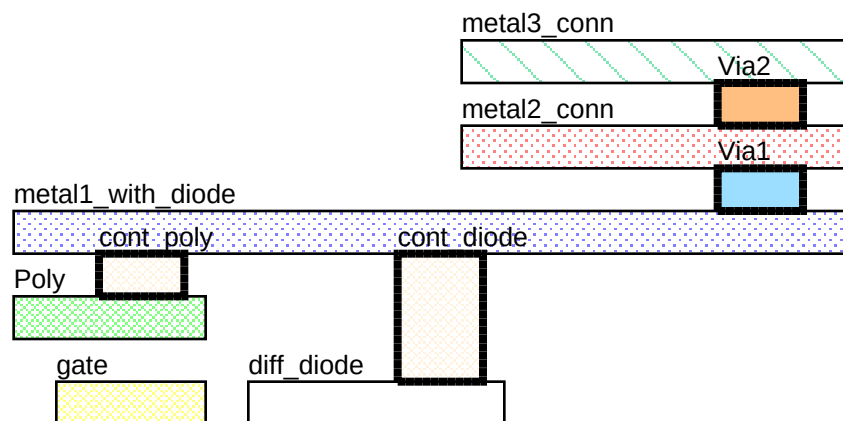
$$\text{ratio} ((\text{Via1.area} / \text{gate.area}) - ((\text{diff_diode.area} * 250.0) + 1000.0)) \leq 0.0$$

id: ANT.8.V1_1

message: Via1 area to gate area ratio must be $\leq 1000 + (\text{diode area} * 250)$

switch CHECK_VIA2_ANT_8

Antenna



$$\text{ratio} (((\text{Via1.area} + \text{Via2.area}) / \text{gate.area}) - ((\text{diff_diode.area} * 250.0) + 1000.0)) \leq 0.0$$

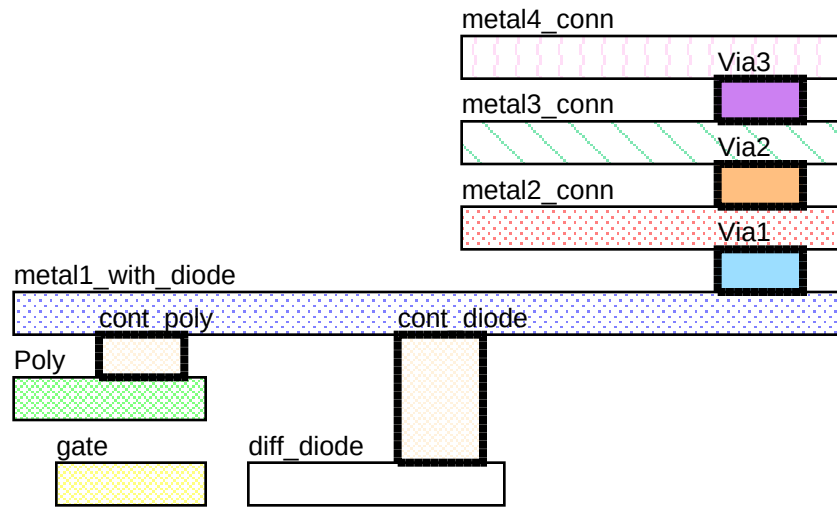
id: ANT.8.V1_2

message: Cumulative Via1 through Via2 area to gate area
ratio must be $\leq 1000 + (\text{diode area} * 250)$

ANTENNA RULES (continued)

switch CHECK_VIA3_ANT_8

Antenna



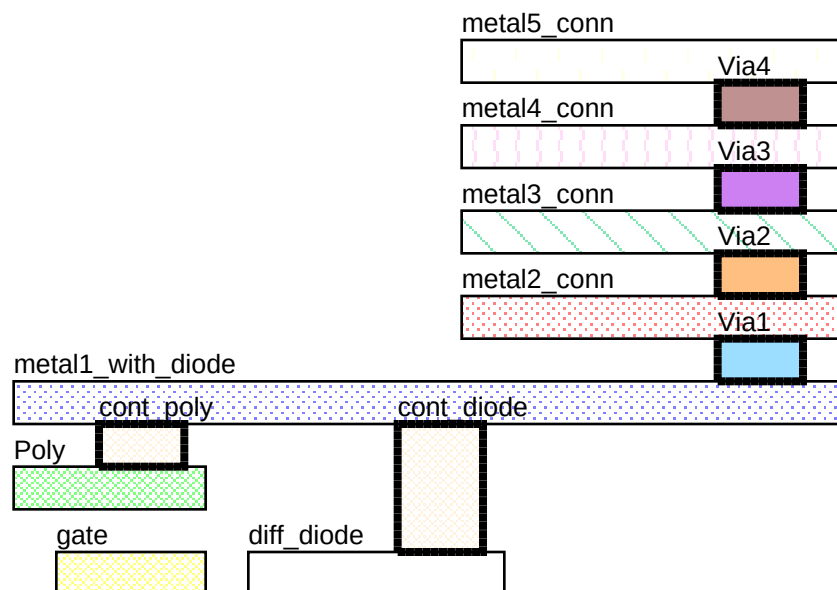
ratio (((Via1.area + Via2.area + Via3.area) / gate.area) -
((diff_diode.area * 250.0) + 1000.0)) <= 0.0

id: ANT.8.V1_3

message: Cumulative Via1 through Via3 area to gate area
ratio must be <= 1000 + (diode area * 250)

switch CHECK_VIA4_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area) /
gate.area) - ((diff_diode.area * 250.0) + 1000.0)) <= 0.0

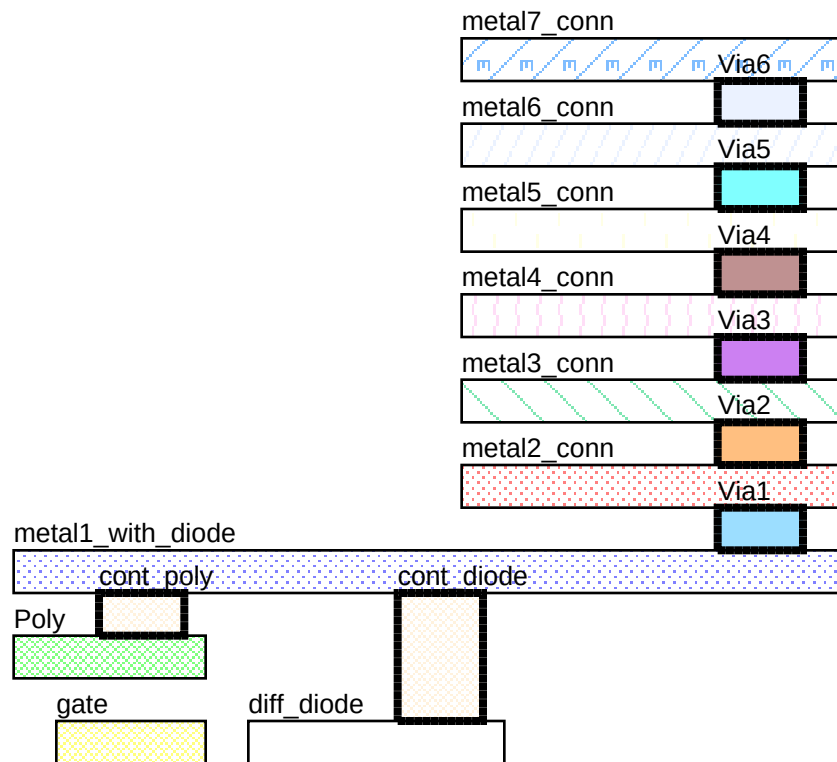
id: ANT.8.V1_4

message: Cumulative Via1 through Via4 area to gate area
ratio must be <= 1000 + (diode area * 250)

ANTENNA RULES (continued)

switch CHECK_VIA6_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area + Via5.area + Via6.area) / gate.area) - ((diff_diode.area * 250.0) + 1000.0)) <= 0.0

id: ANT.8.V1_6

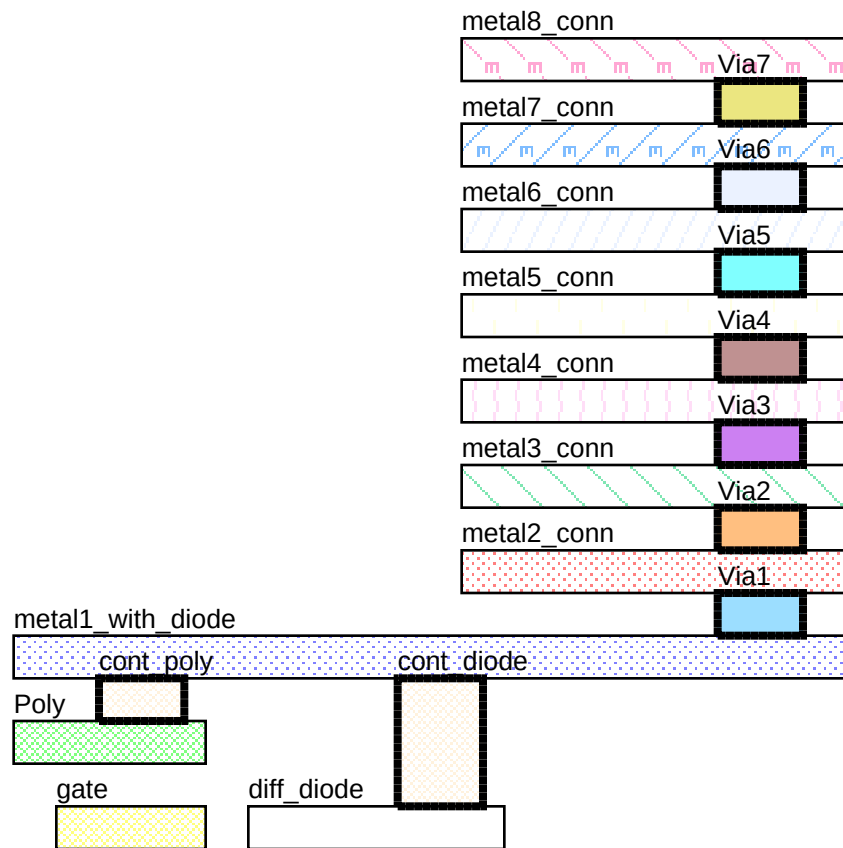
message: Cumulative Via1 through Via6 area to gate area

ratio must be <= 1000 + (diode area * 250)

ANTENNA RULES (continued)

switch CHECK_VIA7_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area +
Via5.area + Via6.area + Via7.area) / gate.area) - ((diff_diode.area * 250.0) + 1000.0))
≤ 0.0

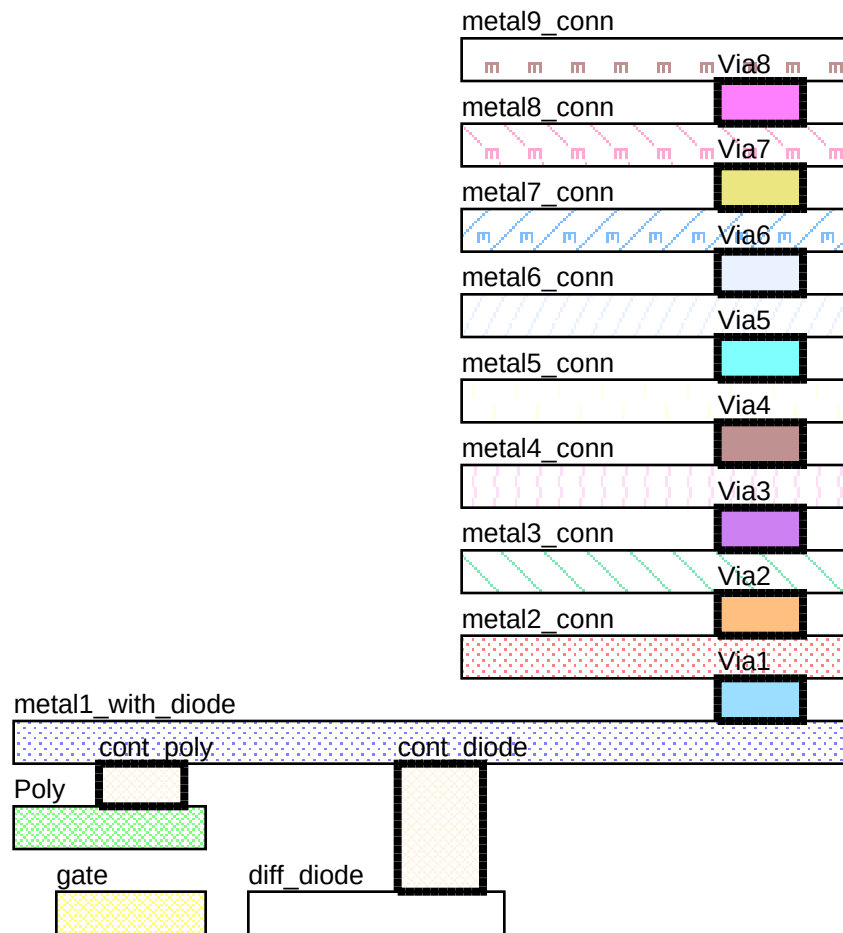
id: ANT.8.V1_7

message: Cumulative Via1 through Via7 area to gate area
ratio must be ≤ 1000 + (diode area * 250)

ANTENNA RULES (continued)

switch CHECK_VIA8_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area + Via5.area +
Via6.area + Via7.area + Via8.area) / gate.area) - ((diff_diode.area
* 250.0) + 1000.0)) <= 0.0

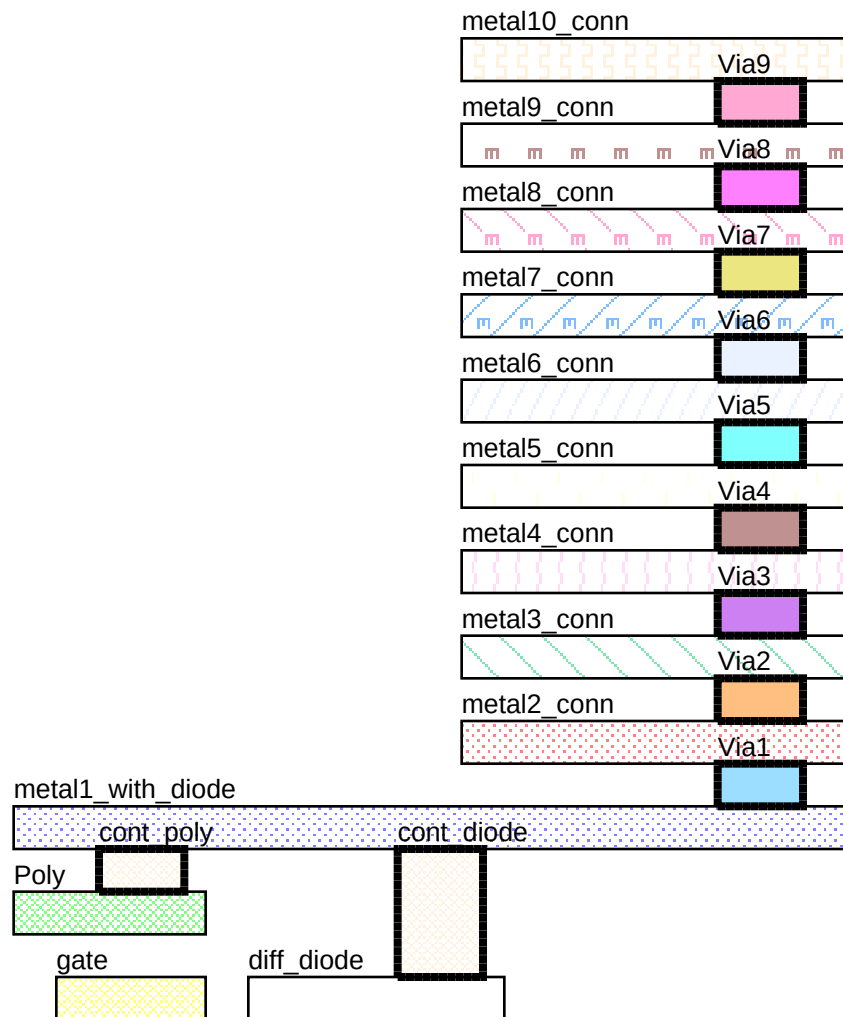
id: ANT.8.V1_8

message: Cumulative Via1 through Via8 area to gate area
ratio must be <= 1000 + (diode area * 250)

ANTENNA RULES (continued)

switch CHECK_VIA9_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area + Via5.area +
Via6.area + Via7.area + Via8.area + Via9.area) / gate.area) - ((diff_diode.area
* 250.0) + 1000.0)) <= 0.0

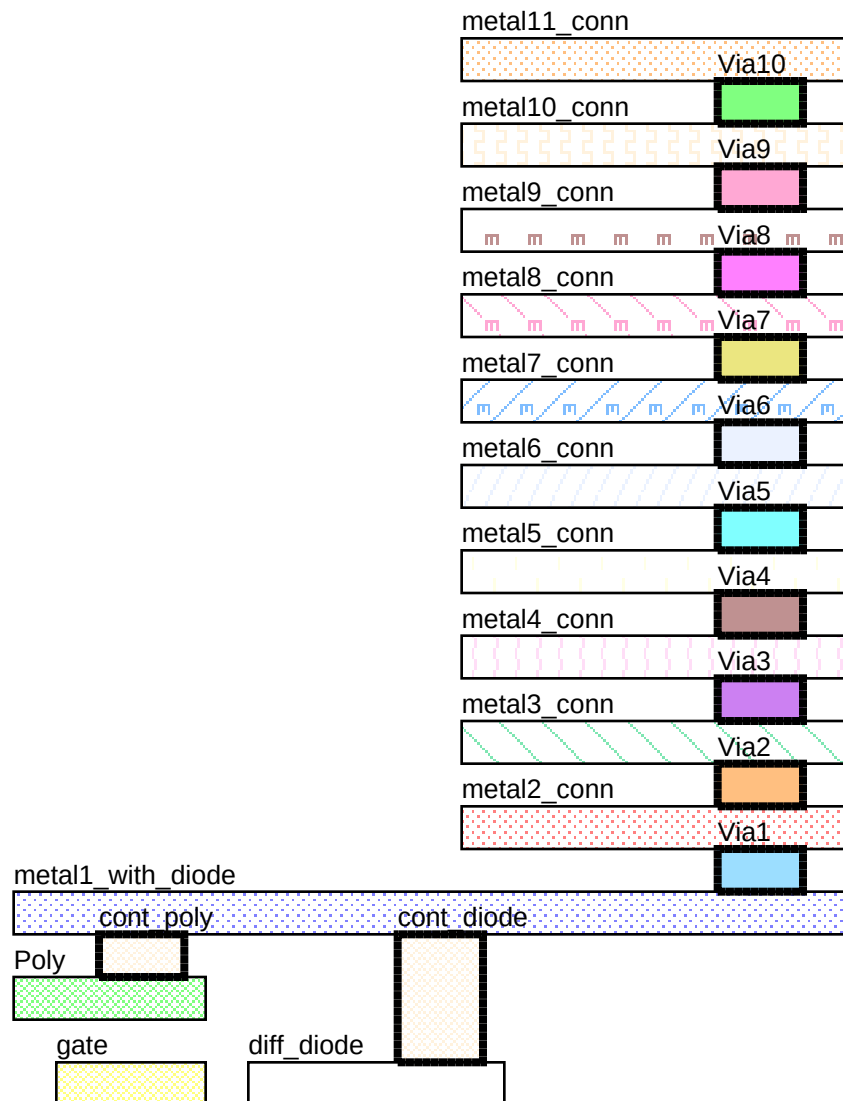
id: ANT.8.V1_9

message: Cumulative Via1 through Via9 area to gate area
ratio must be <= 1000 + (diode area * 250)

ANTENNA RULES (continued)

switch CHECK_VIA10_ANT_8

Antenna



ratio (((Via1.area + Via2.area + Via3.area + Via4.area + Via5.area +
Via6.area + Via7.area + Via8.area + Via9.area + Via10.area)
/ gate.area) - ((diff_diode.area * 250.0) + 1000.0)) <= 0.0

id: ANT.8.V1_10

message: Cumulative Via1 through Via10 area to gate area
ratio must be <= 1000 + (diode area * 250)

CMOS I/O Design Rules

ESD Design Rules

The "ESDdummy" marker layer must be used to mark I/O ESD circuitry. If the "ESDdummy" layer is not used, the correct DRC checks of I/O ESD circuitry will not take place.

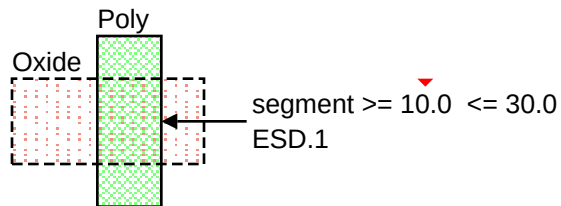
NMOS and PMOS devices used for ESD protection follow a strict finger structure using specific finger dimensions and layout.

ESD Design Rules

Rule Name	Value (um)	Description
ESD.1	10 - 30	Width of each finger of NMOS and PMOS in I/O buffers and in Vdd to Vss ESD protection.
ESD.2	210	Minimum NMOS combined finger width for I/O buffers and for Vdd to Vss ESD protection.
ESD.3	210	Minimum PMOS combined finger width for I/O buffers.
ESD.4		Outer Oxide area of NMOS and PMOS in I/O buffers and in Vdd to Vss ESD protection must be Source or connected to Bulk to prevent parasitic bipolars and unwanted discharge paths during ESD zapping.
ESD.5		NMOS ESD protection devices must be surrounded by a P+ Guard Ring.
ESD.6		PMOS ESD protection devices must be surrounded by an N+ Guard Ring.
ESD.7		NMOS and PMOS in ESD protection can NOT have butted taps.
ESD.8		NMOS and PMOS in an I/O buffer must have non-salicided Drains. The Contacts still must be salicided.
ESD.9		A P+ Oxide strap should be placed between N+ Oxides of different I/O and ESD devices when both connect to different pads.
ESD.10		An N+ Oxide strap should be placed between P+ Oxides of different I/O and ESD devices when both connect to different pads.
ESD.11	0.05	Minimum SiProt to Poly gate overlap in NMOS and PMOS drains.
ESD.12	0.9	Minimum enclosure of SiProt edge to Poly gate edge in NMOS and PMOS I/O drains.
ESD.13	0.9	Minimum SiProt to Oxide overlap in NMOS and PMOS I/O drains.
ESD.14	0.2	Exact gate length of NMOS and PMOS in I/O buffers and in Vdd to Vss ESD protection.
ESD.15	0.12	Minimum Poly gate to Contact spacing in NMOS and PMOS in I/O buffers and in Vdd to Vss ESD protection.

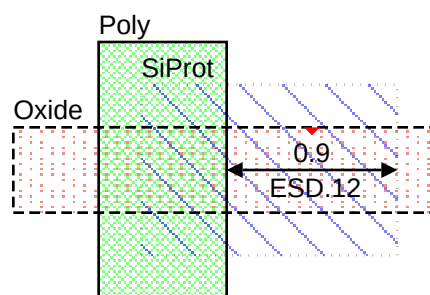
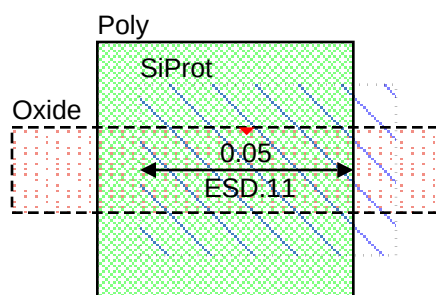
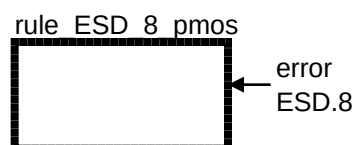
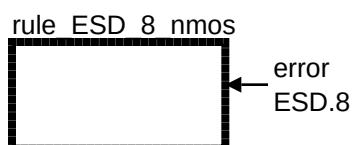
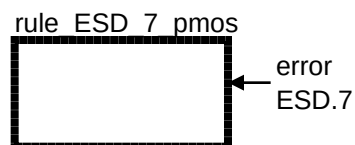
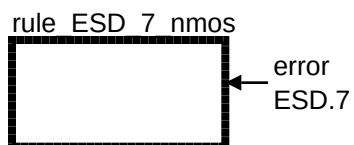
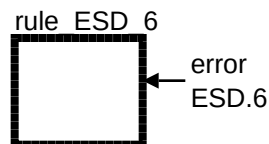
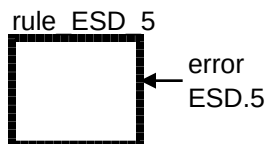
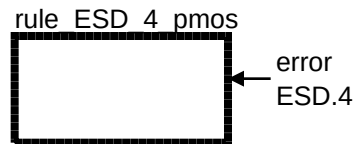
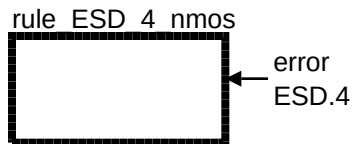
ESD Design Rules (continued)

ESDdummy



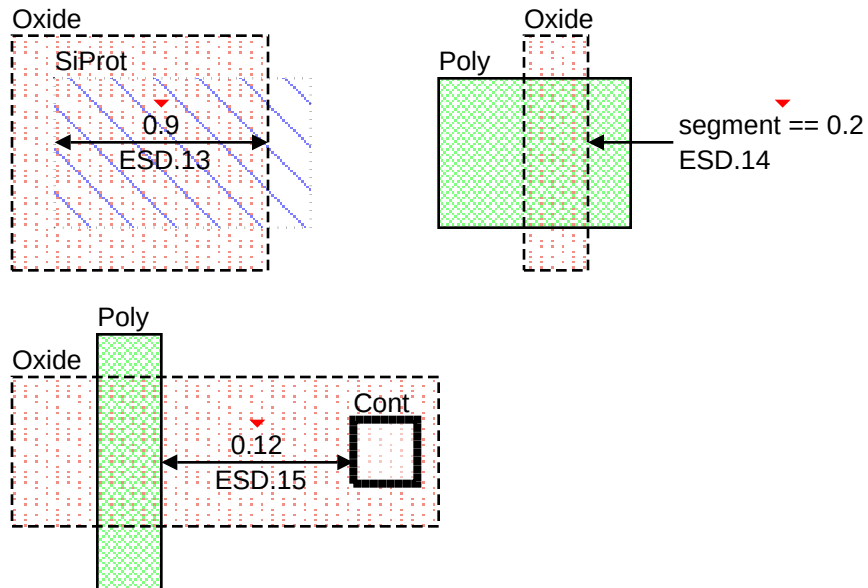
ESD.2 - Checked during LVS.

ESD.3 - Checked during LVS.



ESD Design Rules (continued)

ESDdummy



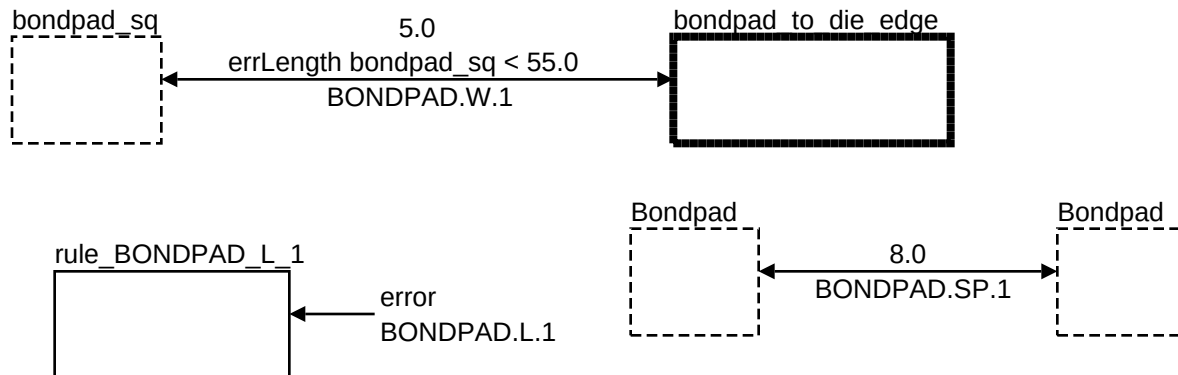
Bond Pad Design Rules

- 1) The bond pad structure must contain all Metal levels and all Via levels.
- 2) Metals over the Bonpad area must NOT have stress relief slots.
- 3) Vias on odd levels should be on top of each other. Vias on even levels should be on top of each other.

Table 1: Bond Pad Design Rules

Rule Name	Value (um)	Description
BONDPAD.W.1	55.0	Minimum Bondpad width of edges parallel to the die edge.
BONDPAD.L.1	68.0	Minimum Bondpad length of edges perpendicular to the die edge.
BONDPAD.SP.1	8.0	Minimum Bondpad to Bondpad spacing.
BONDPAD.E.1	2.0	Minimum Metal (all levels) enclosure of Bondpad.
BONDPAD.SP.2	3.0	Minimum Bondpad Metal to Metal (including Bondpad Metal) spacing.
BONDPAD.B.1	1.8~3.2	Minimum length of Bonpad Metal beveled corner. All Bondpad Metal corners must be beveled at 45 degrees.
BONDPAD.W.2	0.14	Minimum and maximum Bondpad Via k width (k = 1, 2, 3, 4, 5, 6).
BONDPAD.W.3	0.36	Minimum and maximum Bondpad Via k width (k = 7, 8).
BONDPAD.SP.3	0.22	Minimum Bondpad Viak to Bondpad Viak spacing (k = 1, 2, 3, 4, 5, 6).
BONDPAD.SP.4	0.54	Minimum Bondpad Viak to Bondpad Viak spacing (k = 7, 8).
BONDPAD.E.2	0.05	Minimum Bondpad Metalk to Bondpad Viak enclosure (k = 1, 2, 3, 4, 5, 6, 7, 8). Minimum Bondpad Metalk+1 to Bondpad Viak enclosure (k = 1, 2, 3, 4, 5, 6, 7, 8).
BONDPAD.E.3	0.09	Maximum Bondpad Metalk to Bondpad Viak enclosure (k = 7, 8). Maximum Bondpad Metalk+1 to Bondpad Viak enclosure (k = 7, 8).
BONDPAD.R.1	16.0	Minimum Bondpad Viak inside Metalk to Metalk+1 crossing (k = 1, 2, 3, 4, 5, 6).
BONDPAD.R.2	4.0	Minimum Bondpad Viak inside Metalk to Metalk+1 crossing (k = 7, 8).
BONDPAD.SP.5	1.5	Minimum and Maximum Pad Metal slot to Pad Metal slot spacing.
BONDPAD.W.4	1.0	Minimum and Maximum Pad Metal slot width (expect first slot on each edge of Pad).
BONDPAD.W.5	5.0	Minimum and Maximum Pad Metalk width in outer ring of Pad Metalk (expect for the bevelled corners) (k = 1, 2, 3, 4, 5, 6, 7, 8).
BONDPAD.SP.6	1.0~3.5	Minimum and Maximum Pad Metalk ring to nearest Pad Metalk across first slot (k = 1, 2, 3, 4, 5, 6, 7, 8).
BONDPAD.SP.7	1.1	Minimum Pad Viak array to Pad Viak array spacing (k = 1, 2, 3, 4, 5, 6, 7, 8).

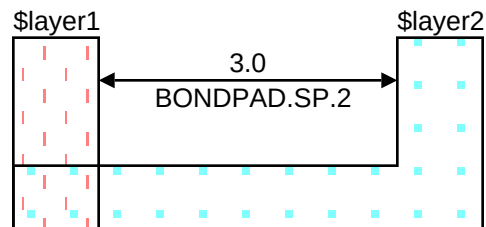
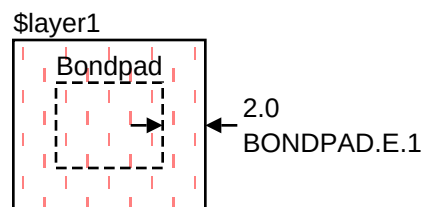
Bond Pad Design Rules (continued)



macro

Macro Table

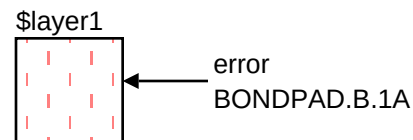
\$layer1	\$layer2
bondpad_metal1_filled	Metal1
bondpad_metal2_filled	Metal2
bondpad_metal3_filled	Metal3
bondpad_metal4_filled	Metal4
bondpad_metal5_filled	Metal5
bondpad_metal6_filled	Metal6
bondpad_metal7_filled	Metal7
bondpad_metal8_filled	Metal8
bondpad_metal9_filled	Metal9
bondpad_metal10_filled	Metal10
bondpad_metal11_filled	Metal11



macro

Macro Table

\$layer1	\$name1
rule_BONDPAD_B_1_m1	Metal1
rule_BONDPAD_B_1_m2	Metal2
rule_BONDPAD_B_1_m3	Metal3
rule_BONDPAD_B_1_m4	Metal4
rule_BONDPAD_B_1_m5	Metal5
rule_BONDPAD_B_1_m6	Metal6
rule_BONDPAD_B_1_m7	Metal7
rule_BONDPAD_B_1_m8	Metal8
rule_BONDPAD_B_1_m9	Metal9
rule_BONDPAD_B_1_m10	Metal10
rule_BONDPAD_B_1_m11	Metal11

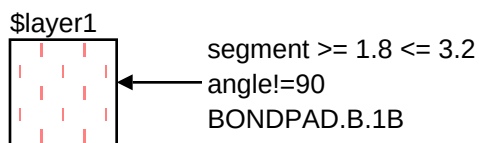


Bond Pad Design Rules (continued)

macro

Macro Table

\$layer1	\$name1
bondpad_metal1	Metal1
bondpad_metal2	Metal2
bondpad_metal3	Metal3
bondpad_metal4	Metal4
bondpad_metal5	Metal5
bondpad_metal6	Metal6
bondpad_metal7	Metal7
bondpad_metal8	Metal8
bondpad_metal9	Metal9
bondpad_metal10	Metal10
bondpad_metal11	Metal11



Bond Pad Design Rules (continued)

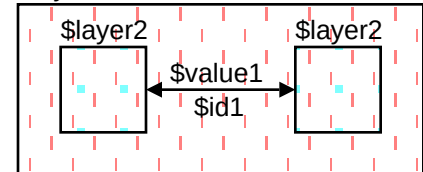
BONDPAD.W.2 and BONDPAD.W.3 - covered by VIAk.W.1.

macro

Macro Table

\$layer1	\$layer2	\$value1	\$id1
bondpad_metal1	Via1	0.22	BONDPAD.SP.3
bondpad_metal2	Via2	0.22	BONDPAD.SP.3
bondpad_metal3	Via3	0.22	BONDPAD.SP.3
bondpad_metal4	Via4	0.22	BONDPAD.SP.3
bondpad_metal5	Via5	0.22	BONDPAD.SP.3
bondpad_metal6	Via6	0.22	BONDPAD.SP.3
bondpad_metal7	Via7	0.22	BONDPAD.SP.4
bondpad_metal8	Via8	0.22	BONDPAD.SP.4
bondpad_metal9	Via9	0.54	BONDPAD.SP.4
bondpad_metal10	Via10	0.54	BONDPAD.SP.4

\$layer1

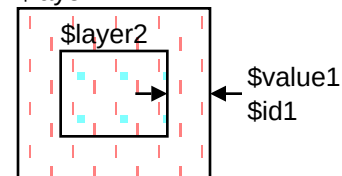


macro

Macro Table

\$layer1	\$layer2	\$name1	\$value1	\$id1
bondpad_metal1	Via1	Metal1	0.05	BONDPAD.E.2
bondpad_metal2	Via2	Metal2	0.05	BONDPAD.E.2
bondpad_metal3	Via3	Metal3	0.05	BONDPAD.E.2
bondpad_metal4	Via4	Metal4	0.05	BONDPAD.E.2
bondpad_metal5	Via5	Metal5	0.05	BONDPAD.E.2
bondpad_metal6	Via6	Metal6	0.05	BONDPAD.E.2
bondpad_metal7	Via7	Metal7	0.05	BONDPAD.E.3
bondpad_metal8	Via8	Metal8	0.05	BONDPAD.E.3
bondpad_metal9	Via9	Metal9	0.09	BONDPAD.E.3
bondpad_metal10	Via10	Metal9	0.09	BONDPAD.E.3

\$layer1

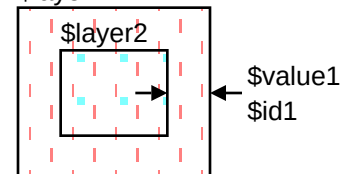


macro

Macro Table

\$layer1	\$layer2	\$name1	\$value1	\$id1
bondpad_metal2	Via1	Metal2	0.05	BONDPAD.E.2
bondpad_metal3	Via2	Metal3	0.05	BONDPAD.E.2
bondpad_metal4	Via3	Metal4	0.05	BONDPAD.E.2
bondpad_metal5	Via4	Metal5	0.05	BONDPAD.E.2
bondpad_metal6	Via5	Metal6	0.05	BONDPAD.E.2
bondpad_metal7	Via6	Metal7	0.05	BONDPAD.E.2
bondpad_metal8	Via7	Metal8	0.05	BONDPAD.E.3
bondpad_metal9	Via8	Metal9	0.05	BONDPAD.E.3
bondpad_metal10	Via9	Metal10	0.09	BONDPAD.E.3
bondpad_metal11	Via10	Metal11	0.09	BONDPAD.E.3

\$layer1



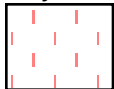
Bond Pad Design Rules (continued)

macro

Macro Table

\$layer1	\$name1	\$name2	\$name3	\$value	\$id1
rule_BONDPAD_R_1_via1	Via1	Metal1	Metal2	16.0	BONDPAD.R.1
rule_BONDPAD_R_1_via2	Via2	Metal2	Metal3	16.0	BONDPAD.R.1
rule_BONDPAD_R_1_via3	Via3	Metal3	Metal4	16.0	BONDPAD.R.1
rule_BONDPAD_R_1_via4	Via4	Metal4	Metal5	16.0	BONDPAD.R.1
rule_BONDPAD_R_1_via5	Via5	Metal5	Metal6	16.0	BONDPAD.R.1
rule_BONDPAD_R_1_via6	Via6	Metal6	Metal7	16.0	BONDPAD.R.1
rule_BONDPAD_R_2_via7	Via7	Metal7	Metal8	16.0	BONDPAD.R.1
rule_BONDPAD_R_2_via8	Via8	Metal8	Metal9	16.0	BONDPAD.R.1
rule_BONDPAD_R_2_via9	Via9	Metal9	Metal10	4.0	BONDPAD.R.2
rule_BONDPAD_R_2_via10	Via10	Metal10	Metal11	4.0	BONDPAD.R.2

\$layer1

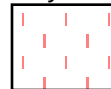
error
\$id1

macro

Macro Table

\$layer1	\$name1
rule_BONDPAD_SP_5_metal1	Metal1
rule_BONDPAD_SP_5_metal2	Metal2
rule_BONDPAD_SP_5_metal3	Metal3
rule_BONDPAD_SP_5_metal4	Metal4
rule_BONDPAD_SP_5_metal5	Metal5
rule_BONDPAD_SP_5_metal6	Metal6
rule_BONDPAD_SP_5_metal7	Metal7
rule_BONDPAD_SP_5_metal8	Metal8
rule_BONDPAD_SP_5_metal9	Metal9
rule_BONDPAD_SP_5_metal10	Metal10

\$layer1

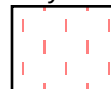
error
BONDPAD.SP.5

macro

Macro Table

\$layer1	\$name1
rule_BONDPAD_W_4_metal1	Metal1
rule_BONDPAD_W_4_metal2	Metal2
rule_BONDPAD_W_4_metal3	Metal3
rule_BONDPAD_W_4_metal4	Metal4
rule_BONDPAD_W_4_metal5	Metal5
rule_BONDPAD_W_4_metal6	Metal6
rule_BONDPAD_W_4_metal7	Metal7
rule_BONDPAD_W_4_metal8	Metal8
rule_BONDPAD_W_4_metal9	Metal9
rule_BONDPAD_W_4_metal10	Metal10

\$layer1

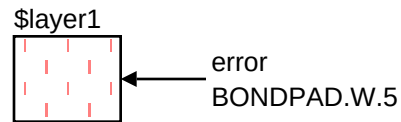
error
BONDPAD.W.4

Bond Pad Design Rules (continued)

macro

Macro Table

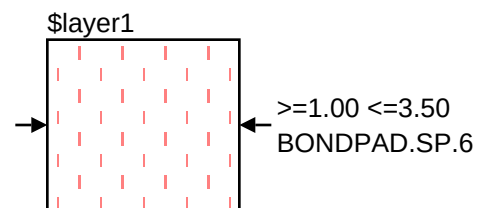
\$layer1	\$name1
rule_BONDPAD_W_5_metal1	Metal1
rule_BONDPAD_W_5_metal2	Metal2
rule_BONDPAD_W_5_metal3	Metal3
rule_BONDPAD_W_5_metal4	Metal4
rule_BONDPAD_W_5_metal5	Metal5
rule_BONDPAD_W_5_metal6	Metal6
rule_BONDPAD_W_5_metal7	Metal7
rule_BONDPAD_W_5_metal8	Metal8
rule_BONDPAD_W_5_metal9	Metal9
rule_BONDPAD_W_5_metal10	Metal10



macro

Macro Table

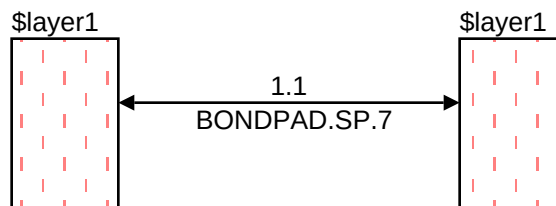
\$layer1	\$name1
bondpad_metal1_slot_on_edge	Metal1
bondpad_metal2_slot_on_edge	Metal2
bondpad_metal3_slot_on_edge	Metal3
bondpad_metal4_slot_on_edge	Metal4
bondpad_metal5_slot_on_edge	Metal5
bondpad_metal6_slot_on_edge	Metal6
bondpad_metal7_slot_on_edge	Metal7
bondpad_metal8_slot_on_edge	Metal8
bondpad_metal9_slot_on_edge	Metal9
bondpad_metal10_slot_on_edge	Metal10



macro

Macro Table

\$layer1	\$name1
bondpad_via1_array	Via1
bondpad_via2_array	Via2
bondpad_via3_array	Via3
bondpad_via4_array	Via4
bondpad_via5_array	Via5
bondpad_via6_array	Via6
bondpad_via7_array	Via7
bondpad_via8_array	Via8
bondpad_via9_array	Via9
bondpad_via10_array	Via10



CMOS Digital Electrical Parameters

Sheet Resistances The units are ohms/square

Global Parameters

R_metal1	0.0736	Metal 1 sheet resistance	
R_metal2_7	0.0604	Metal 2,3,4,5,6,7 sheet resistance	
R_metal8_10	0.0214	Metal 8,9,10 sheet resistance	
R_metal11	0.021	Metal 11 sheet resistance	
R_snpoly	15	Salicide N+ Poly sheet resistance	
R_sppoly	15	Salicide P+ Poly sheet resistance	
R_nsnpoly	200	Non-salicide N+ Poly sheet resistance	
R_nsppoly	600	Non-salicide P+ Poly sheet resistance	
R_snactive	18	Salicide N+ Oxide sheet resistance	
R_spactive	15	Salicide P+ Oxide sheet resistance	
R_nsnactive	100	Non-salicide N+ Oxide sheet resistance	
R_nspactive	200	Non-salicide P+ Oxide sheet resistance	
R_nwell	450	Nwell sheet resistance	
R_pwell	1000	Pwell sheet resistance	

Contact/Via Resistances The units are ohms/contact or ohms/via

Global Parameters

R_via10	0.4	Via 10 resistance	
R_via8_9	0.28	Via 8,9 resistance	
R_via1_7	0.5	Via 1,2,3,4,5,6,7 resistance	
R_metal1-contact	1	Metal 1 to Contact resistance	
R_poly-contact	45	Poly to Contact resistance	
R_pplus-contact	62	P+ Oxide to Contact resistance	
R_nplus-contact	75	N+ Oxide to Contact resistance	

Current Densities The units are ma/um

Global Parameters

L_metal10_11	8	Metal 10,11 current density	
L_metal1_9	2	Metal 1,2,3,4,5,6,7, 8, 9 current density	

Contact/Via Current Densities The units are ohm/contact or ma/via

Global Parameters

I_via1_8	0.1	Via 1,2,3,4,5,6 current density	
I_Via9_10	0.8	Via 7,8 current density	
I_metal-contact-poly	0.1	Metal 1 Contact to Poly current density	
I_metal-contact-oxide	0.1	Metal 1 Contact to Oxide current density	

Layer and Dielectric Thickness

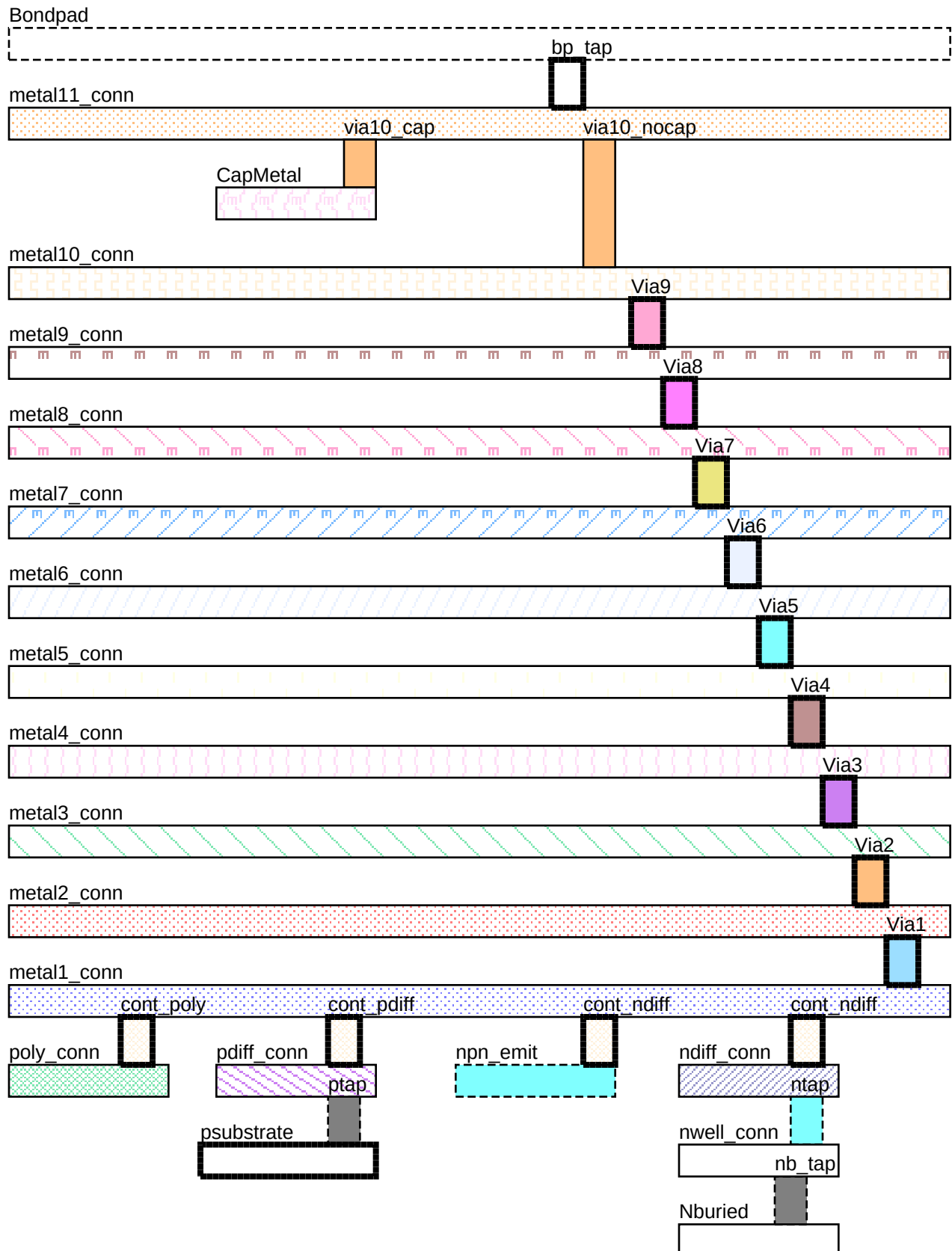
Comment Table

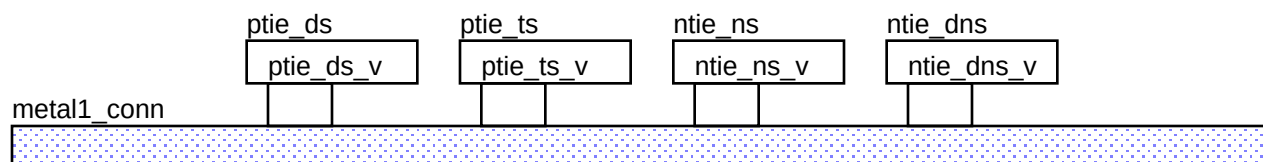
Layer	Thickness (A)	Description
Pass2	7500	k= 8.0
Pass1	6500	k = 4.2
IMD11a	14000	K = 4.2
Metal11	14000	Cu
IMD10b	6000	K = 4.2
IMD10a	10000	K = 4.2
Metal10	10000	Cu
IMD9b	6000	K = 4.2
IMD9a	10000	K = 4.2
Metal9	10000	Cu
IMD8b	3000	K = 2.6
IMD8a	1500	K = 4.2
Metal8	1500	Cu
IMD7b	3000	K = 2.6
IMD7a	1500	K = 4.2
Metal7	1500	Cu
IMD6b	3000	K = 2.6
IMD6a	1500	K = 4.2
Metal6	1500	Cu
IMD5b	3000	K = 2.6
IMD5a	1500	K = 4.2
Metal5	1500	Cu
IMD4b	3000	K = 2.6
IMD4a	1500	K = 4.2
Metal4	1500	Cu
IMD3b	3000	K = 2.6
IMD3a	1500	K = 4.2
Metal3	1500	Cu
IMD2b	3000	K = 2.6
IMD2a	1500	K = 4.2
Metal2	1500	Cu
IMD1b	3000	K = 2.6
IMD1a	1500	K = 4.2
Metal1	1500	Cu
ILD	3000	k = 4.2
LINER	1200	k = 7
Poly	1200	
STI (FOX)	3500	k=3.9

**For further information on electrical parameters and Model parameters, Please look into the following document.
gpdk045_PDK_Model_Report.pdf**

Connectivity Definition

Connectivity





 \$L1	dummy	
 \$L2	dummy	
 \$L3	dummy	
 \$L4	dummy	
 \$diff	dummy	
 \$enc	dummy	
 \$layer1	dummy	
 \$layer2	dummy	
 \$layer3	dummy	
 \$metal	dummy	
 \$recLayer	dummy	
 \$res	dummy	
 \$tap	dummy	
 \$via	dummy	
 BJTdum	input 15;0 df2order 72 (BJTdum drawing) packet zbip	
 Bondpad	input 36;0 df2order 54 (Bondpad drawing) packet pass	
 Cap3dum	input 84;0 df2order 56 (Cap3dum drawing) packet zcap	
 CapMetal	input 14;0 df2order 39 (CapMetal drawing) packet mcap	
 Capdum	input 12;0 df2order 55 (Capdum drawing) packet zcap	
 Cont	input 6;0 df2order 11 (Cont drawing) packet cw	via
 DIOfdummy	input 22;0 df2order 75 (DIOfdummy drawing) packet zdiode	
 ESDdummy	input 74;0 df2order 77 (ESDdummy drawing) packet esddum	
 FOX	bulk andnot (Oxide or Oxide_thk)	
 IND2dummy	input 88;0 packet zind2	
 IND3dummy	input 114;0 packet zind3	
 INDdummy	input 90;0 packet zind	
 M1Resdum	input 75;0 (M1Resdum drawing) df2order 63 packet zrm1	
 M2Resdum	input 76;0 (M2Resdum drawing) df2order 64 packet zrm2	
 M3Resdum	input 77;0 (M3Resdum drawing) df2order 65 packet zrm3	
 M4Resdum	input 78;0 (M4Resdum drawing) df2order 66 packet zrm4	
 M5Resdum	input 79;0 (M5Resdum drawing) df2order 67 packet zrm5	
 M6Resdum	input 80;0 (M6Resdum drawing) df2order 68 packet zrm6	
 M7Resdum	input 81;0 (M7Resdum drawing) df2order 69 packet zrm7	
 M8Resdum	input 82;0 (M8Resdum drawing) df2order 70 packet zrm8	
 M9Resdum	input 83;0 (M9Resdum drawing) df2order 71 packet zrm9	
 M10Resdum	input 93;0 (M10Resdum drawing) df2order 178 packet zrm10	
 M11Resdum	input 103;0 (M11Resdum drawing) df2order 80 packet zrm11	
 Metal1	input 7;0 df2order 12 (Metal1 drawing) packet m1	
 Metal2	input 9;0 df2order 15 (Metal2 drawing) packet m2	
 Metal3	input 11;0 df2order 18 (Metal3 drawing) packet m3	

	Metal4	input 31;0 df2order 21 (Metal4 drawing) packet m4	
	Metal5	input 33;0 df2order 24 (Metal5 drawing) packet m5	
	Metal6	input 35;0 df2order 27 (Metal6 drawing) packet m6	
	Metal7	input 38;0 df2order 30 (Metal7 drawing) packet m7	
	Metal8	input 40;0 df2order 33 (Metal8 drawing) packet m8	
	Metal9	input 42;0 df2order 36 (Metal9 drawing) packet m9	
	Metal10	input 152;0 df2order 64 (Metal10 drawing) packet m10	
	Metal11	input 162;0 df2order 66 (Metal11 drawing) packet m11	
	NOD	SNA Oxide and Nimp	
	NPN2dum	input 110;0 packet znnp2 fillStyle outline	
	NPN5dum	input 111;0 packet znnp5 fillStyle outline	
	NPN10dum	input 112;0 packet znnp10 fillStyle outline	
	NPNdummy	input 20;0 df2order 73 (NPNdummy drawing) packet znnp	
	Nburied	input 19;0 df2order 41 (Nburied drawing) packet npblk	
	Nhvt	input 18;0 df2order 6 (Nhvt drawing) packet nhvt	
	Nimp	input 4;0 df2order 7 (Nimp drawing) packet nplus	
	Nlvt	input 26;0 (Nlvt drawing) df2order 43 packet nlvt	
	Nwell	input 2;0 df2order 1 (Nwell drawing) packet nwell	
	Nzvt	input 52;0 df2order 9 (Nzvt drawing) packet Nzvt	
	Oxide	input 1;0 df2order 2 (Oxide drawing) packet tox	
	Oxide_thk	input 24;0 df2order 3 (Oxide_thk drawing) packet Oxide_thk	
	PNPdummy	input 21;0 df2order 74 (PNPdummy drawing) packet znpn	
	POD	SNA Oxide and Pimp	
	PWdummy	input 85;0 df2order 85 (PWdummy drawing) packet zpw	
	Phvt	input 23;0 df2order 8 (Phvt drawing) packet phvt	
	Pimp	input 5;0 df2order 5 (Pimp drawing) packet pplus	
	Plvt	input 27;0 (Plvt drawing) df2order 44 packet plvt	
	Poly	input 3;0 df2order 4 (Poly drawing) packet poly1	
	Psub	input 25;0 (Psub drawing) df2order 42 packet psub	
	ResWdum	input 71;0 df2order 62 (ResWdum drawing) packet zrwll	
	Resdum	input 13;0 df2order 61 (Resdum drawing) packet zrpoly	
	SiProt	input 72;0 df2order 10 (SiProt drawing) packet siprot	
	VNP2dum	input 60;0 packet zvpnp2 fillStyle outline	
	VNP5dum	input 61;0 packet zvpnp5 fillStyle outline	
	VNP10dum	input 62;0 packet zvpnp10 fillStyle outline	
	Via1	input 8;0 df2order 14 (Via1 drawing) packet v1	via
	Via2	input 10;0 df2order 17 (Via2 drawing) packet v2	via
	Via3	input 30;0 df2order 20 (Via3 drawing) packet v3	via
	Via4	input 32;0 df2order 23 (Via4 drawing) packet v4	via
	Via5	input 34;0 df2order 26 (Via5 drawing) packet v5	via
	Via6	input 37;0 df2order 29 (Via6 drawing) packet v6	via
	Via7	input 39;0 df2order 32 (Via7 drawing) packet v7	via
	Via8	input 41;0 df2order 35 (Via8 drawing) packet v8	via
	Via9	input 151;0 df2order 63 (Via9 drawing) packet v9	via
	Via10	input 161;0 df2order 65 (Via10 drawing) packet v10	via

1 Introduction

This report describes the predictive SPICE models developed by Accelicon Technologies, Inc. for a generic 45nm technology based on semiconductor research. The extracted model list is attached in the later sections followed by the predicted simulation PCM targets (I_{dsat} , V_{th}) of all corners. The delivered models are in the format of SPECTRE.

2 MOSFET Models

This section lists the details of the MOSFET models developed for this project. The target definition for V_{th} and I_{dsat} is listed below:

V_{th} is obtained using constant current method, the intercept current = $I_{con} * W/L$ ($I_{con} = 1e-8A$ for 1.1V device and $I_{con} = 1e-7A$ for 1.8V device), V_{th} unit is V.

I_{dsat} is obtained from the drain current at $V_{ds} = V_{dd}$, $V_{gs} = V_{dd}$ and $V_{bs} = 0$, I_{dsat} unit is A. I_{dsat}/W is the normalized I_{dsat} value, Unit is A/m.

2.1: MOSFET Intrinsic Models

There are total of 10 MOSFET models developed for this project. The binning model approach is adopted according to customer request. The model naming, valid geometry range, temperature range and V_{dd} are listed in table 2-1.

Device	Model Name	Vdd (V)	Bins	Temp scale (°C)	Wmin(um)	Wmax(um)	Lmin(um)	Lmax(um)
1.1V Standard-Vt	nch	1.1	30	-40 to 125	0.12	10	0.04	10
	pch	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V High-Vt	nch_hvt	1.1	30	-40 to 125	0.12	10	0.04	10
	pch_hvt	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V low-Vt	nch_lvt	1.1	30	-40 to 125	0.12	10	0.04	10
	pch_lvt	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V native	nch_na	1.1	9	-40 to 125	0.5	10	0.3	10
1.8V	nch_18	1.8	12	-40 to 125	0.32	10	0.15	10
	pch_18	1.8	12	-40 to 125	0.32	10	0.15	10
1.8V native	nch18_na	1.8	9	-40 to 125	0.5	10	0.8	10

Table2.1 List of MOSFET Models

Device	Model Name	Vdd (V)	Bin s	Temp scale (°C)	Wmin (um)	Wmax (um)	Lmin (um)	Lmax (um)
1.1V Standard-Vt	nch_mis	1.1	30	-40 to 125	0.12	10	0.04	10
	nch_mis	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V High-Vt	nch_hvt_mis	1.1	30	-40 to 125	0.12	10	0.04	10
	pch_hvt_mis	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V low-Vt	nch_lvt_mis	1.1	30	-40 to 125	0.12	10	0.04	10
	pch_lvt_mis	1.1	30	-40 to 125	0.12	10	0.04	10
1.1V native	nch_na_mis	1.1	9	-40 to 125	0.5	10	0.3	10
1.8V	nch_18_mis	1.8	12	-40 to 125	0.32	10	0.15	10
	pch_18_mis	1.8	12	-40 to 125	0.32	10	0.15	10
1.8V native	nch18_na_mis	1.8	9	-40 to 125	0.5	10	0.8	10

Table2.2 List of MOSFET Statistical Models and Mismatch Models.

2.2: MOSFET Corner and Statistical Models

Corner Models and Statistical Models (with Mismatch) are also developed, the model list is shown in Table 2-3.

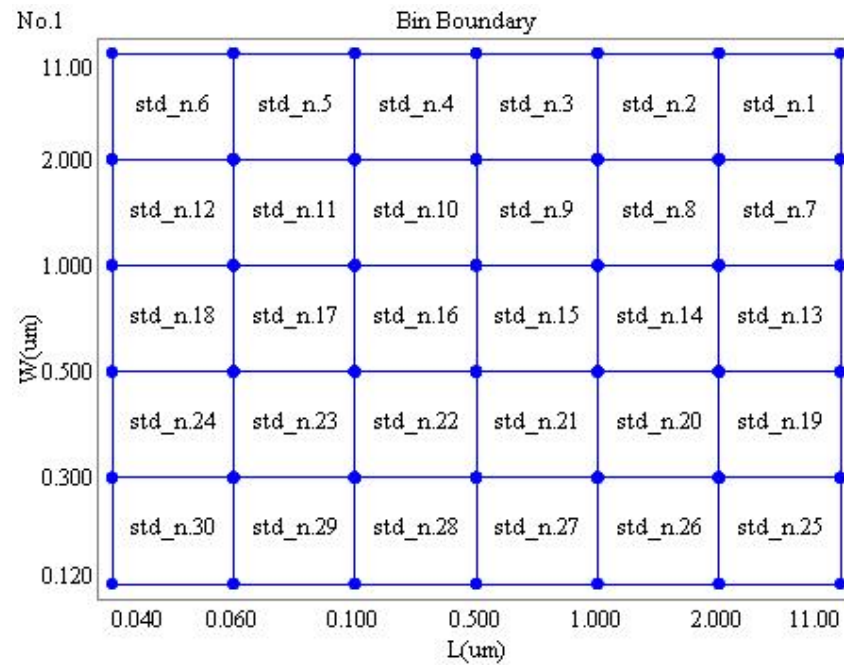
Device	Typical Nmos & PMOS	Fast Nmos & PMOS	Slow Nmos & PMOS	Fast Nmos & Slow mos	Slow mos & Fast Pmos
1.1V Standard Vt	tt	ff	ss	fs	sf
1.1V High Vt	tt_hvt	ff_hvt	ss_hvt	fs_hvt	sf_hvt
1.1V Low Vt	tt_lvt	ff_lvt	ss_lvt	fs_lvt	sf_lvt
1.1v native	tt_na	ff_na	ss_na	fs_na	sf_na
1.8 v	tt_18	ff_18	ss_18	fs_18	sf_18
1.8v Native	tt_na18	ff_na18	ss_na18	fs_na18	sf_na18

Table2.3 List of MOSFET Corner Models

2.3: MOSFET Model Characteristics

2.3.1: 1.1V Standard Vt NMOS

Binning Scheme:



Model Spec:

W(um)	L(um)	Vth	Idsat
10	10	0.36	9.01E-05
0.12	0.04	0.36	7.40E-05
10	0.04	0.38	5.96E-03
0.12	10	0.14	1.21E-06

Table 2.4 1.1v Standard Vt NMOS Model Spec

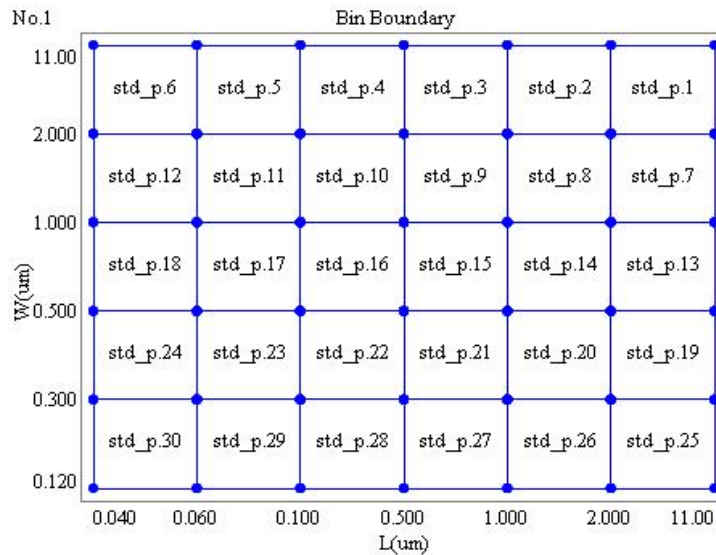
Corner Model Spec

Target	W(um)	L (um)	SS	SF	TT	FS	FF
Vth	10	10	0.39	0.38	0.36	0.34	0.32
	0.12	0.04	0.46	0.41	0.36	0.31	0.26
	10	0.04	0.43	0.41	0.38	0.35	0.32
	0.12	10	0.2	0.17	0.14	0.11	0.08
Idsat	10	10	7.83E-05	8.42E-05	9.01E-05	9.58E-05	1.02E-04
	0.12	0.04	5.54E-05	6.47E-05	7.40E-05	8.32E-05	9.24E-05
	10	0.04	5.23E-03	5.59E-03	5.96E-03	6.31E-03	6.68E-03
	0.12	10	8.73E-07	1.04E-06	1.21E-06	1.38E-06	1.56E-06

Table 2.5 1.1v Standard Vt NMOS Corner Model Spec

2.3.2: 1.1V Standard Vt PMOS

Binning Scheme:



Model Spec:

W(um)	L(um)	Vth	Idsat
10	10	-0.3	-6.48E-05
0.12	0.04	-0.4	-4.27E-05
10	0.04	-0.43	-3.06E-03
0.12	10	-0.22	-8.67E-07

Table 2.6 1.1v Standard Vt PMOS Model Spec

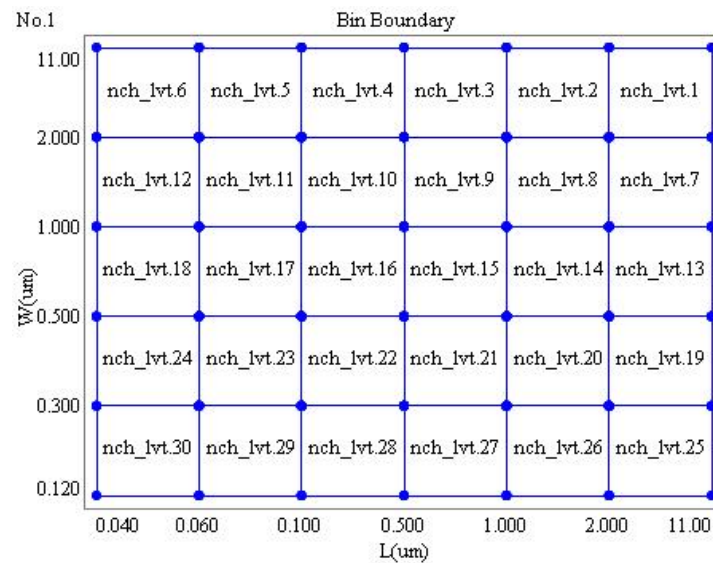
Corner Model SPEC

Target	W (um)	L (um)	SS	FS	TT	SF	FF
Vth	10	10	-0.33	-0.31	-0.3	-0.28	-0.27
	0.12	0.04	-0.49	-0.45	-0.4	-0.36	-0.31
	10	0.04	-0.48	-0.46	-0.43	-0.41	-0.38
	0.12	10	-0.28	-0.25	-0.22	-0.19	-0.16
Idsat	10	10	-5.63E-05	-6.06E-05	-6.48E-05	-6.89E-05	-7.33E-05
	0.12	0.04	-3.22E-05	-3.56E-05	-4.27E-05	-4.81E-05	-5.34E-05
	10	0.04	-2.69E-03	0.00E+00	-3.06E-03	-3.24E-03	-3.42E-03
	0.12	10	-6.34E-07	-7.49E-07	-8.67E-07	-9.84E-07	-1.10E-06

Table 2.7 1.1v Standard Vt PMOS Corner Model Spec

2.3.3: 1.1V Low Vt NMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	0.26	1.17E-04
0.12	0.04	0.26	9.78E-05
10	0.04	0.28	7.76E-03
0.12	10	0.05	1.53E-06

Table 2.8 1.1v Low Vt NMOS Model Spec

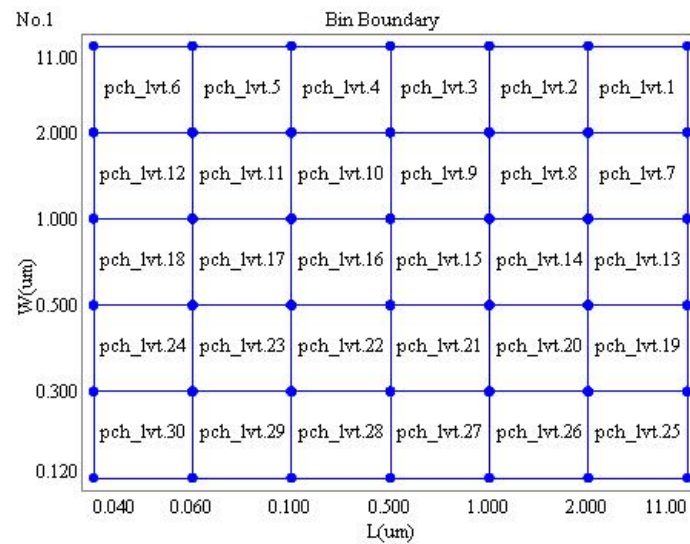
Corner Model SPEC

Target	W (um)	L (um)	TT_LVT	FF_LVT	SS_LVT	FS_LVT	SF_LVT
Vth	10	10	0.26	0.22	0.29	0.24	0.27
	0.12	0.04	0.26	0.16	0.36	0.21	0.31
	10	0.04	0.28	0.23	0.34	0.25	0.31
	0.12	10	0.05	-0.01	0.11	0.02	0.08
Idsat	10	10	1.17E-04	1.32E-04	1.02E-04	1.24E-04	1.09E-04
	0.12	0.04	9.78E-05	1.23E-04	7.34E-05	1.10E-04	8.55E-05
	10	0.04	7.76E-03	8.69E-03	6.82E-03	8.22E-03	7.29E-03
	0.12	10	1.53E-06	1.95E-06	1.12E-06	1.75E-06	1.32E-06

Table 2.9 1.1v Low Vt NMOS Corner Model Spec

2.3.4: 1.1V Low Vt PMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	-0.21	-8.67E-05
0.12	0.04	-0.31	-5.15E-05
10	0.04	-0.34	-3.69E-03
0.12	10	-0.13	-1.13E-06

Table 2.10 1.1v Low Vt PMOS Model Spec

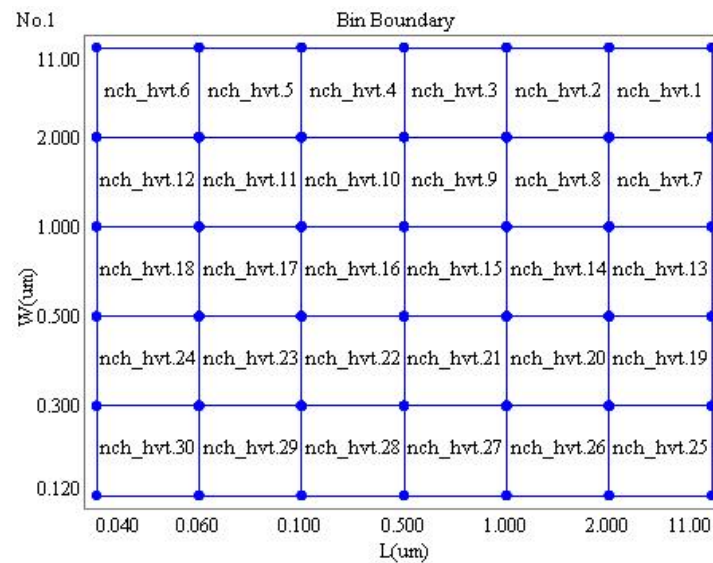
Corner Model SPEC

Target	W (um)	L (um)	TT_LVT	FF_LVT	SS_LVT	FS_LVT	SF_LVT
Vth	10	10	-0.21	-0.18	-0.24	-0.22	-0.2
	0.12	0.04	-0.31	-0.21	-0.41	-0.36	-0.26
	10	0.04	-0.34	-0.29	-0.39	-0.36	-0.31
	0.12	10	-0.13	-0.07	-0.19	-0.16	-0.1
Idsat	10	10	-8.67E-05	-9.83E-05	-7.56E-05	-8.12E-05	-9.24E-05
	0.12	0.04	-5.15E-05	-6.42E-05	-3.86E-05	-4.50E-05	-5.78E-05
	10	0.04	-3.69E-03	-4.13E-03	-3.24E-03	-3.46E-03	-3.91E-03
	0.12	10	-1.13E-06	-1.44E-06	-8.27E-07	-9.79E-07	-1.29E-06

Table 2.11 1.1v Low Vt PMOS Corner Model Spec

2.3.5: 1.1V High Vt NMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	0.43	6.81E-05
0.12	0.04	0.46	5.01E-05
10	0.04	0.48	3.97E-03
0.12	10	0.2	9.42E-07

Table 2.12 1.1v High Vt NMOS Model Spec

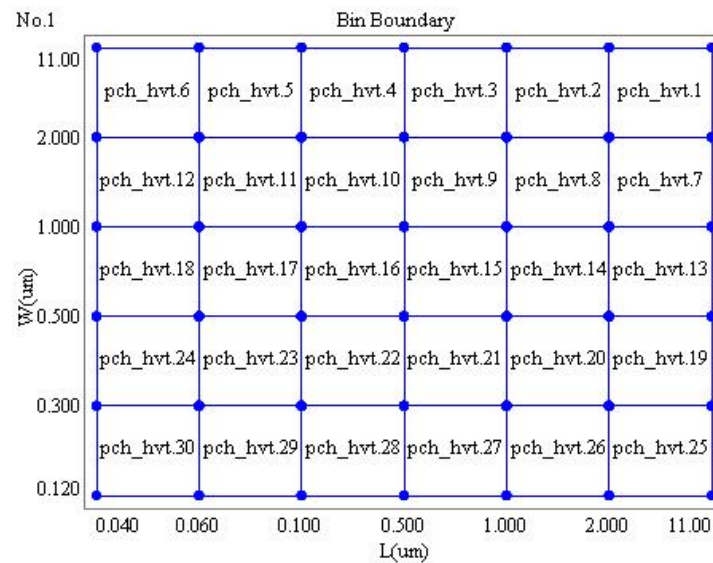
Corner Model SPEC

Target	W (um)	L (um)	TT_HVT	FF_HVT	SS_HVT	FS_HVT	SF_HVT
Vth	10	10	0.43	0.39	0.46	0.41	0.45
	0.12	0.04	0.46	0.37	0.55	0.42	0.51
	10	0.04	0.48	0.43	0.53	0.46	0.51
	0.12	10	0.2	0.14	0.26	0.17	0.23
Idsat	10	10	6.81E-05	7.73E-05	5.95E-05	7.25E-05	6.37E-05
	0.12	0.04	5.01E-05	6.26E-05	3.76E-05	5.63E-05	4.38E-05
	10	0.04	3.97E-03	4.45E-03	3.50E-03	4.21E-03	3.74E-03
	0.12	10	9.42E-07	1.20E-06	6.88E-07	1.07E-06	8.10E-07

Table 2.13 1.1v High Vt NMOS Corner Model Spec

2.3.6: 1.1V High Vt PMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	-0.38	-4.76E-05
0.12	0.04	-0.49	-3.00E-05
10	0.04	-0.52	-2.00E-03
0.12	10	-0.3	-6.52E-07

Table 2.14 1.1v High Vt PMOS Model Spec

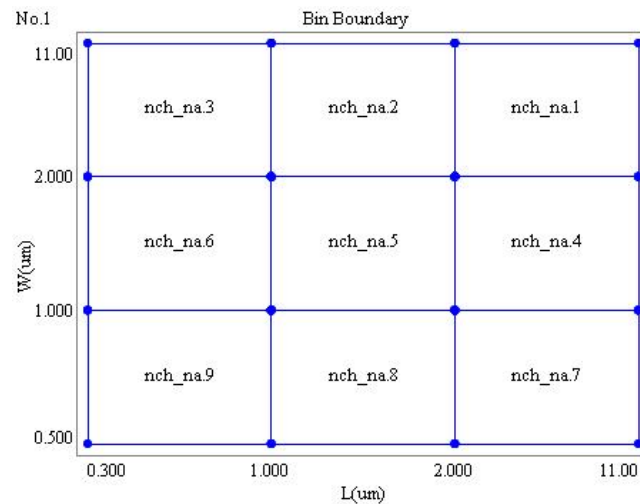
Corner Model SPEC

Target	W (um)	L (um)	TT_HVT	FF_HVT	SS_HVT	FS_HVT	SF_HVT
Target 1 Vth	10	10	-0.38	-0.35	-0.4	-0.39	-0.36
	0.12	0.04	-0.49	-0.42	-0.58	-0.53	-0.45
	10	0.04	-0.52	-0.47	-0.57	-0.54	-0.49
	0.12	10	-0.3	-0.24	-0.35	-0.33	-0.27
Target 2 Idsat	10	10	-4.76E-05	-5.38E-05	-4.14E-05	-4.45E-05	-5.06E-05
	0.12	0.04	-3.00E-05	-3.75E-05	-2.25E-05	-2.62E-05	-3.37E-05
	10	0.04	-2.00E-03	-2.24E-03	-1.76E-03	-1.87E-03	-2.12E-03
	0.12	10	-6.52E-07	-8.27E-07	-4.77E-07	-5.63E-07	-7.39E-07

1. Table 2.15 1v High Vt PMOS Corner Model Spec

2.3.7: 1.1V Native NMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	0.05	2.26E-04
0.5	0.3	-0.07	2.39E-04
10	0.3	-2.71E-03	4.59E-03
0.5	10	-0.16	1.20E-05

Table 2.16 1.1v Native NMOS Model Spec

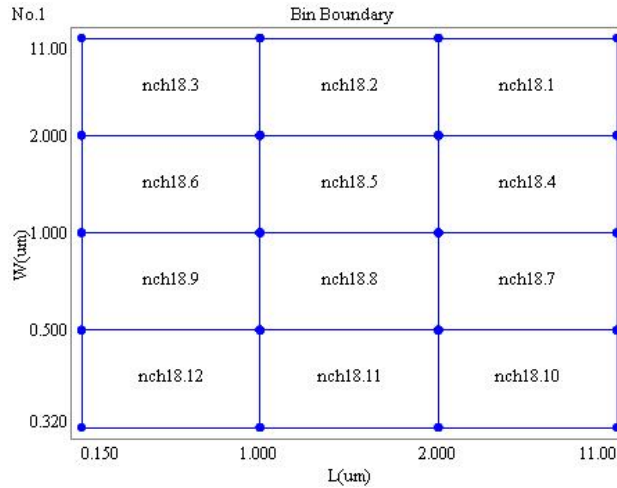
Corner Model SPEC

Target	W (um)	L (um)	TT_NA	FF_NA	SS_NA	FS_NA	SF_NA
Vth	10	10	0.05	0.01	0.08	0.03	0.07
	0.5	0.3	-0.07	-0.15	-9.87E-03	-0.11	-0.03
	10	0.3	-2.71E-03	-0.06	0.06	-0.03	0.03
	0.5	10	-0.16	-0.2	-0.12	-0.18	-0.14
Idsat	10	10	2.26E-04	2.46E-04	2.06E-04	2.36E-04	2.16E-04
	0.5	0.3	2.39E-04	2.75E-04	2.03E-04	2.57E-04	2.19E-04
	10	0.3	4.59E-03	5.14E-03	4.04E-03	4.87E-03	4.32E-03
	0.5	10	1.20E-05	1.35E-05	1.06E-05	1.27E-05	1.13E-05

Table 2.17 1.1v Native NMOS Corner Model Spec

2.3.8: 1.8V IO NMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	0.43	2.39E-04
0.32	0.15	0.45	2.20E-04
10	0.15	0.49	6.79E-03
0.32	10	0.32	7.86E-06

Table 2.18 1.8v NMOS Model Spec

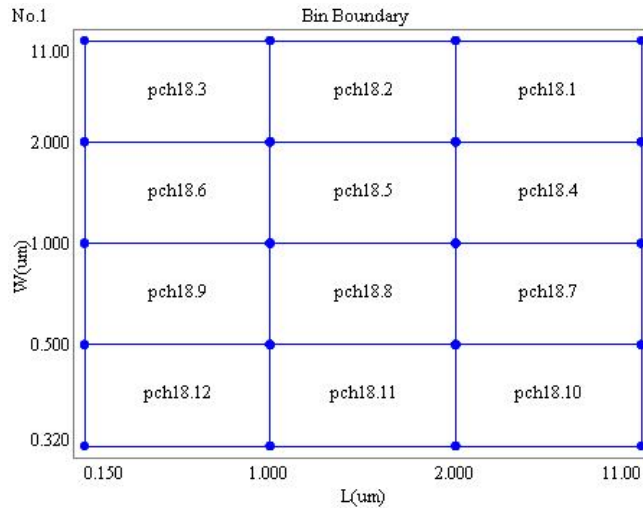
Corner Model SPEC

Target	W (um)	L (um)	TT_18	FF_18	SS_18	FS_18	SF_18
Vth	10	10	0.43	0.39	0.47	0.41	0.45
	0.32	0.15	0.45	0.36	0.54	0.4	0.49
	10	0.15	0.49	0.42	0.55	0.46	0.52
	0.32	10	0.32	0.26	0.38	0.29	0.35
Idsat	10	10	2.39E-04	2.69E-04	2.08E-04	2.54E-04	2.23E-04
	0.32	0.15	2.20E-04	2.59E-04	1.80E-04	2.39E-04	2.00E-04
	10	0.15	6.79E-03	7.59E-03	5.95E-03	7.17E-03	6.36E-03
	0.32	10	7.86E-06	9.43E-06	6.29E-06	8.64E-06	7.07E-06

Table 2.19 1.8v NMOS Corner Model Spec

2.3.9: 1.8V IO PMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	-0.37	-1.04E-04
0.12	0.04	-0.49	-1.02E-04
10	0.04	-0.51	-3.00E-03
0.12	10	-0.37	-3.76E-06

Table 2.20 1.8v PMOS Model Spec

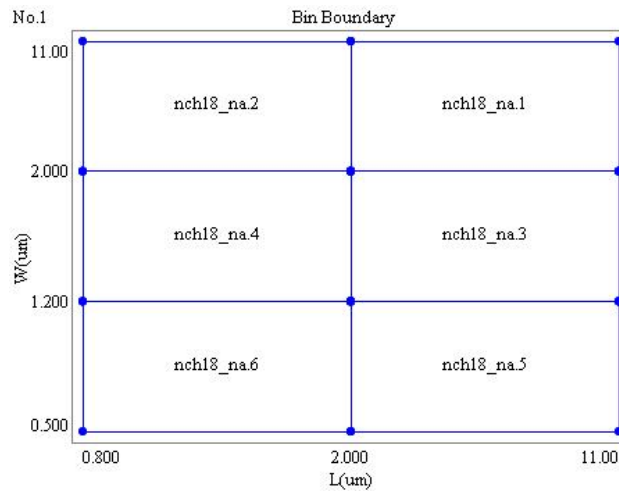
Corner Model SPEC

Target	W (um)	L (um)	TT_18	FF_18	SS_18	FS_18	SF_18
Vth	10	10	-0.37	-0.33	-0.4	-0.39	-0.35
	0.32	0.15	-0.49	-0.41	-0.57	-0.53	-0.45
	10	0.15	-0.51	-0.44	-0.58	-0.54	-0.48
	0.32	10	-0.37	-0.32	-0.42	-0.39	-0.34
Idsat	10	10	-1.04E-04	-1.19E-04	-8.81E-05	-9.59E-05	-1.11E-04
	0.32	0.15	-1.02E-04	-1.20E-04	-8.34E-05	-9.26E-05	-1.11E-04
	10	0.15	-3.00E-03	-3.40E-03	-2.56E-03	-2.77E-03	-3.19E-03
	0.32	10	-3.76E-06	-4.51E-06	-3.01E-06	-3.38E-06	-4.13E-06

Table 2.21 1.8v PMOS Corner Model Spec

2.3.10: 1.8V Native NMOS

Binning Scheme



SPEC Table

W(um)	L(um)	Vth	Idsat
10	10	6.69E-03	5.03E-04
0.5	0.8	-0.05	2.41E-04
10	0.8	-0.02	4.72E-03
0.5	10	-0.11	2.54E-05

Table 2.22 1.8v Native NMOS Model Spec

Corner Model SPEC

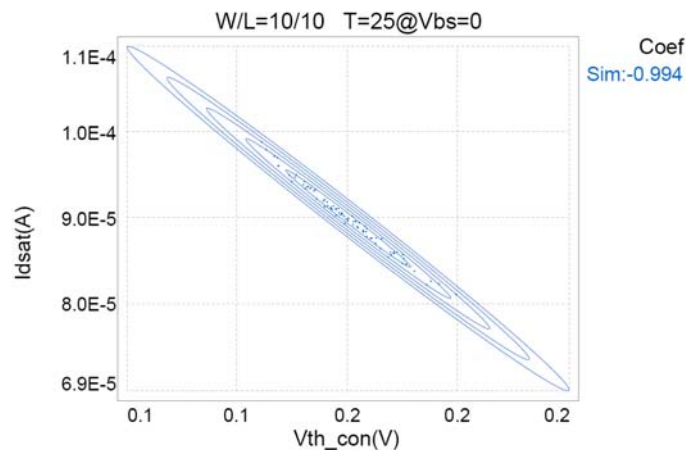
Target	W (um)	L (um)	TT_NA18	FF_NA18	SS_NA18	FS_NA18	SF_NA18
Vth	10	10	6.69E-03	-0.04	0.06	-0.02	0.03
	0.5	0.8	-0.05	-0.14	0.04	-0.1	-7.54E-03
	10	0.8	-0.02	-0.1	0.05	-0.06	0.01
	0.5	10	-0.11	-0.17	-0.05	-0.14	-0.08
Idsat	10	10	5.03E-04	5.38E-04	4.67E-04	5.20E-04	4.85E-04
	0.5	0.8	2.41E-04	2.78E-04	2.05E-04	2.59E-04	2.23E-04
	10	0.8	4.72E-03	5.29E-03	4.16E-03	5.01E-03	4.44E-03
	0.5	10	2.54E-05	2.86E-05	2.21E-05	2.70E-05	2.37E-05

Table 2.23 1.8v Native NMOS Corner Model Spec

2.4 MOSFET Statistical Model and Mismatch Model

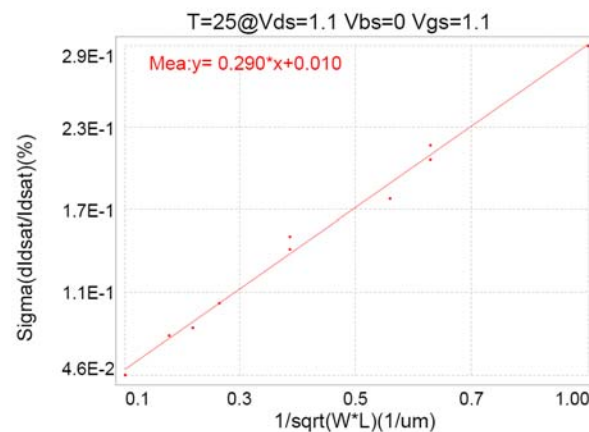
2.4.1: Statistical SPICE Model

Due to the lack of wafer data for PCA (Principle Component Analysis), the 3σ of I_{dsat} , V_{th} and the key model parameters such as T_{ox} , X_L and X_W are obtained from corner SPEC. Correlation coefficients between key model parameters are obtained by fitting the 3σ of I_{dsat} , V_{th} .



2.4.2: Mismatch Model

Mismatch model is also available, TOX , XL , XW and V_{th0} are used to account for the local random variation.



3 BJT Models

There are total of 8 bipolar models are developed with two types, Vertical NPN and Vertical PNP, each with models for 3 emitter size and a scalable model. The model nominal temperature is 25°C, while the valid temperature varied from -40 to 125°C.

BJT Device list is shown below:

	2*2	5*5	10*10	Scalable model
PNP	pnp2	pnp5	pnp10	pnp_scalable
NPN	nnp2	nnp5	nnp10	nnp_scalable

Table 3.1 Bipolar model list

3.1 BJT Intrinsic Models

3.1.1: Vertical PNP

Skew Table and Corner SPEC

Parameters	TYPICAL PNP10	TYPICAL PNP5	TYPICAL PNP2	SLOW %	FAST %
BF	1.022	1.226	1.851	-10	10
IS(A)	8.491E-18	2.440E-18	5.047E-19	-20	20
NF	1.015	1.015	1.015	0.30	-0.30
RE(Ohm)	3.48	6.52	9.56	20	-20
RB(Ohm)	88.42	92.64	140.5	20	-20
RC(Ohm)	7.12	8.58	10.01	20	-20
RBM(Ohm)	0.1	0.1	0.1	20	-20
CJE(F)	1.17e-13	2.848E-14	5.1E-15	10	-10
CJC(F)	2.55e-13	5.512E-14	3.8E-14	10	-10

Table 3.2 Skew parameters table for vertical PNP models

3.1.2: Vertical NPN

Skew Table and Corner SPEC

Parameters	TYPICAL NPN10	TYPICAL NPN5	TYPICAL NPN2	SLOW %	FAST %
BF	4.049	4.560	5.064	-10	10
IS(A)	1.995E-17	5.569E-18	1.085E-18	-20	20
NF	1.014	1.014	1.014	0.3	-0.3
RE(Ohm)	4.312	6.512	8.386	20	-20
RB(Ohm)	112.22	176.08	323.5	20	-20
RC(Ohm)	5.98	7.26	13.42	20	-20
RBM(Ohm)	0.1	0.1	0.1	20	-20
CJE(F)	1.2e-13	2.88E-14	5.38E-15	10	-10
CJC(F)	2.74e-13	1.480E-13	8.29E-14	10	-10

Table 3.3 Skew parameters table for vertical NPN models

3.2 BJT Statistical Model

Due to the lack of wafer data for PCA (Principle Component Analysis), the 3σ of skewed model parameters such as IS, BF, etc are obtained from corner model parameters.

4 Diode Models

There are total of 10 diode models developed for this project, the device list is shown below:

	1.1v standard	1.1v high vt	1.1v low vt	1.8v	1.1v native	1.8v native
Lib name	dio	dio_hvt	dio_lvt	dio_18	dio_na	dio_na18
N+/PW	ndio	ndio_hvt	ndio_lvt	ndio_18	ndio_na	ndio_na18
P+/NW	pdio	pdio_hvt	pdio_lvt	pdio_18		

Table 4.1 List of junction diode models

4.1 Intrinsic Diode Models

Skew table

Parameters	Typical N+P Diode	Typical P+N Diode	SLOW %	FAST %
IS(A/m2)	1.985E-7	1.658E-7	-20	20
ISW(A/m2)	7.240E-13	6.075E-13	-60	60
N	1.241	1.224	0.3	-0.3
RS(Ohm)	8.701E-11	1.1E-10	15	-15
CJO(F/m2)	1.177E-3	1.308E-3	8	-8
CJSW(F/m2)	6.562E-11	7.604E-11	8	-8

Table 4.2 Skew parameters table for 1.1V Standard Vt junction diodes.

Parameters	Typical N+P Diode	Typical P+N Diode	SLOW %	FAST %
IS(A/m2)	1.897E-7	1.343E-7	-20	20
ISW(A/m2)	5.987E-13	4.216E-13	-60	60
N	1.199	1.116	0.3	-0.3
RS(Ohm)	9.016E-11	1.307E-10	15	-15
CJO(F/m2)	7.978E-4	1.349E-3	8	-8
CJSW(F/m2)	4.4E-11	7.636E-11	8	-8

Table 4.3 Skew parameters table for 1.1V high Vt junction diodes

Parameters	Typical N+P Diode	Typical P+N Diode	SLOW%	FAST%
IS(A/m2)	2.392E-7	1.658E-7	-20	20
ISW(A/m2)	4.460E-13	6.075E-13	-60	60

N	1.326	1.224	0.3	-0.3
RS(Ohm)	1.424E-10	1E-10	15	-15
CJO(F/m2)	1.177E-3	1.308E-3	8	-8
CJSW(F/m2)	6.562E-11	8.253E-11	8	-8

Table 4.4 Skew parameters table for 1.1V low Vt junction diodes.

Parameters	Typical N+P Diode	Typical P+N Diode	SLOW %	FAST %
IS(A/m2)	5.386E-7	4.319E-7	-20	20
ISW(A/m2)	3.452E-12	1.835E-12	-60	60
N	1.232	1.224	0.3	-0.3
RS(Ohm)	1E-10	1.100E-10	15	-15
CJO(F/m2)	1.154E-3	1.352E-3	8	-8
CJSW(F/m2)	1.761E-10	8.582E-11	8	-8

Table 4.5 Skew parameters table for 1.8V junction diodes.

Parameters	Typical N+P Diode	SLOW%	FAST%
IS(A/m2)	7.893E-7	-20	20
ISW(A/m2)	1.068E-12	-60	60
N	1.225	0.3	-0.3
RS(Ohm)	1.200E-10	15	-15
CJO(F/m2)	1.668E-4	8	-8
CJSW(F/m2)	1.60E-10	8	-8

Table 4.6 Skew parameters table for 1.1V Native junction diodes.

Parameters	Typical N+P Diode	SLOW%	FAST%
IS(A/m2)	1.076E-6	-20	20
ISW(A/m2)	1.670E-12	-60	60
N	1.224	0.3	-0.3
RS(Ohm)	1.210E-10	15	15
CJO(F/m2)	1.668E-4	8	-8
CJSW(F/m2)	2.344E-10	8	-8

Table 4.7 Skew parameters table for 1.8V Native junction diodes.

4.2 Statistical Diode Models

Due to the lack of wafer data for PCA (Principle Component Analysis), the 3σ of skewed model parameters such as IS, N, etc are obtained from corner model parameters.

5 Resistor Models

There are total of 30 resistor models developed for this project, including 10 2-terminal resistors, 10 3-terminal resistors and 11 metal resistors. The device list is shown below:

Model name	structure	rsh value
resnwsti	N-well Resistor under STI	1000
resnwoxide	N-well Resistor under Oxide	450
ressndiff	N+ Diffused Resistor (salicide)	18
resspdiff	P+ Diffused Resistor (salicide)	15
ressnpoly	N+ Poly Resistor (salicide)	15
ressppoly	P+ Poly Resistor (salicide)	15
resnsndiff	N+ Diffused Resistor (w/o salicide)	100
resnspdiff	P+ Diffused Resistor (w/o salicide)	200
resnsnpoly	N+ Poly Resistor (w/o salicide)	200
resnsppoly	P+ Poly Resistor (w/o salicide)	600

Table 5.1 list of 2-terminal resistor model

Model name	structure	rsh value
resnwsti_m	NW diff(under STI)	1000
resnwoxide_m	NW diff(under OD)	450
ressndiff_m	N+diff w/i silicide	18
resspdiff_m	P+diff w/i silicide	15
ressnpoly_dis	N+Poly w/i silicide	15
ressppoly_dis	P+Poly w/i silicide	15
resnsndiff_m	N+diff w/o silicide	100
resnspdiff_m	P+diff w/o silicide	200
resnsnpoly_dis	N+Poly w/o silicide	200
resnsppoly_dis	P+Poly w/o silicide	600

Table 5.2 list of 3-terminal resistor model

Model name	structure	rsh value
resm1	Metal 1 Resistor	0.0736
resm2	Metal 2 Resistor	0.0604
resm3	Metal 3 Resistor	0.0604
resm4	Metal 4 Resistor	0.0604
resm5	Metal 5 Resistor	0.0604
resm6	Metal 6 Resistor	0.0604
resm7	Metal 7 Resistor	0.0604
resm8	Metal 8 Resistor	0.0214
resm9	Metal 9 Resistor	0.0214
resm10	Metal 10 Resistor	0.0214
resm11	Metal 11 Resistor	0.021

Table 5.3 list of metal resistor model

6 Capacitor Models

6.1 MOS Capacitor

Model name	structure
s1p8v_nch18_mac	Mosfet capacitance for 1.8v nmos
s1p8v_pch18_mac	Mosfet capacitance for 1.8v pmos
s1v_nch_mac	Mosfet capacitance for 1.1v nmos
s1v_nch_mac	Mosfet capacitance for 1.1v pmos

Table 6.1 list of MOS capacitor model

6.2 MIM Capacitor

Model name	structure
mimcap	MiM capacitance Model

Table 6.2 list of MIM capacitor model

7 Inductor Models

Device	Model name
inductor	ind_m

Table 7.1 list of inductor model



Reference Manual
Generic 45nm
Salicide 1.0V/1.8V 1P 11M
Process Design Kit
and Rule Decks (PRD)
Revision 3.5

18/JULY/2011

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1 Overview

The purpose of this Reference Manual is to describe the technical details of the 45nm Generic Process Design Kit (“GPDK045”) provided by Cadence Design Systems, Inc. (“Cadence”).

1.1 Software Environment

The GPDK045 has been designed for use within a Cadence software environment that consists of the following tools -

GPDK045 Cadence IC615 Database

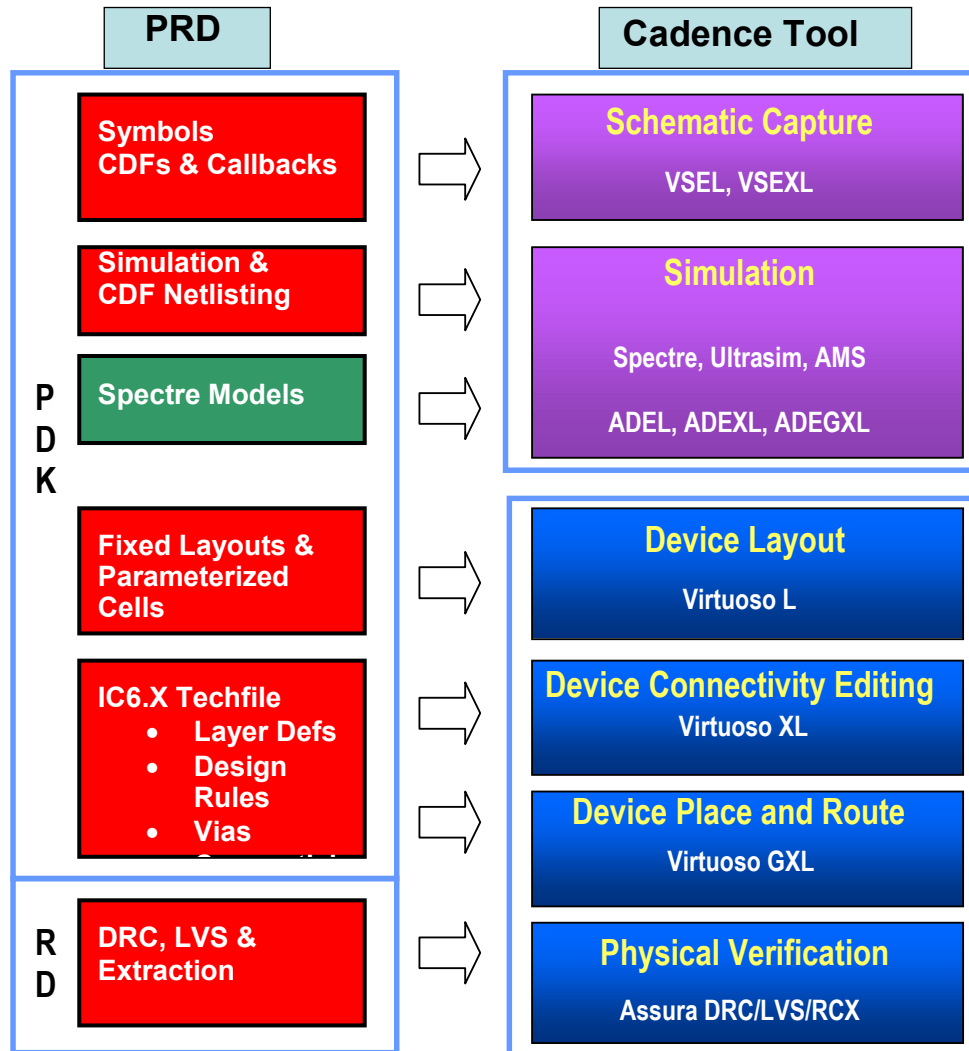
Software Release Stream	Key Products
IC615	Cadence Virtuoso Design Environment, Analog Design and Simulation, Physical Design
FINALE72	Cadence Space Based Router
IUS81	Cadence AMS Designer, AMS/Ultra
MMSIM71	Cadence Spectre, Ultrasim
ASSURA41	Cadence DRC, LVS
PVS10.1	Cadence DRC, LVS
EXT911	Cadence QRC Extraction (L, XL, GXL)
ANLS71	Cadence VoltageStorm
SOC81	Cadence SOC Encounter

2 Documents

Documents Used		Rev
Design Rule Document	gpdk045_drc.pdf	3.5
Design Rule Document	gpdk045_drc.pdf	3.0
Design Rule Document	gpdk045_drc.pdf	2.0
Design Rule Document	gpdk045_drc.pdf	1.0

3 What makes up a PRD?

A PRD contains the process technology and needed information to do chip-level design in the Cadence IC6.X environment. The diagram shows the relationship between the PRD and the Cadence IC6.X tool suite that it enables.



Process Design Kits (PDK) contains the following:

- Symbols & CDFs & Callbacks
- Simulation & CDFs
- Spectre Models
- Fixed Layouts & Parameterized Cells
- IC6.X Technology File

Rule Decks (RD) contain the following

- DRC Rule Decks
- LVS Rule Decks
- Extraction Files

4 Installation of the PRD

The user who will own and maintain the PRD should logon to the computer.

Choose a disk and directory under which the PRD will be installed. This disk should be exported to all client machines and must be mounted consistently across all client machines.

Connect to the directory where the PRD will be installed:

```
cd <PRD_install_directory>
```

Extract the PRD from the archive using the following commands:

```
gzip -dc <path_to_PRD_tar_file>/gpdk045_<version>.tar.gz | tar xf -
```

The default permissions on the PRD have already been set to allow only the owner to have write, read and execute access. Other users will have only read and execute access.

This PRD requires the following UNIX environmental variables:

“CDS_Netlisting_Mode” to be set to “Analog”

“CDSHOME” to be set to the Cadence DFII installation path

5 PRD Install Directory Structure/Contents

Within the <**PRD_install_directory**> directory there are several directories to organize the information associated with the PRD.

assura_tech.lib - File containing the Cadence Assura PV initialization path

pvtech.lib - File containing the Cadence PVS initialization path

cds.lib - File containing the Cadence library definition file.

docs - Directory containing the Cadence PRD documentation and the Process

gpdk045 - The IC615 version of the PDK library

assura - Directory containing the Physical Verification Rule Decks for Assura

lib.defs - File containing the Cadence library definition file.

models - Directory containing the device spectre models

pvs - containing the Physical Verification Rule Decks for PVS

qrc - contains rcx/qrc data for assura-QRC or PVS-QRC flow

6 Creation of a Design Project

A unique directory should be created for each circuit design project. The following command can be executed in UNIX:

```
mkdir ~/circuit_design
```

```
cd ~/circuit_design
```

All work by the user should be performed in this circuit design directory.

The user should create a "cds.lib" file. Using any text editor the following entry should be put in the cds.lib file:

```
INCLUDE <PRD_install_directory>/cds.lib
```

Where "PRD_install_directory" is the path to where the GPDK045 PRD was installed.

The following UNIX links are optional but may aid the user in entering certain forms with the Cadence environment. In UNIX the following command can be used:

```
ln -s <PRD_install_directory>/models
```

Where, again, "PRD_install_directory" is the path to where the GPDK045 PRD was installed.

7 Technology File Methodology

The GPDk045 Library techfile will be designated as the **master** techfile. This techfile will contain all required techfile information. An ASCII version of this techfile is shipped with the PRD. This ASCII version represents the techfile currently compiled into the gpdk045 library

The **attach** method should be used for any design library that is created. This allows the design database techfile to be kept in sync with the techfile in the process PRD. To create a new library that uses an attached techfile, use the command *File->New->Library* from either the CIW or library manager and select the *Attach to an existing techfile* option. Select the gpdk045 library when asked for the name of the *Attach To Technology Library*.

Note: This PRD is using 2000uu/dbu for all layout views.

8 Customizing Layer Display Properties

The display.drf file is automatically loaded by the liblnit.il file whenever the gpdk045 library is opened.

To auto-load your own display.drf file at Cadence start-up time put the display.drf file in the Cadence start-up directory. To manually load the display.drf file (or load a new version), choose *Tools->Display Resources->Merge Files...* from the CIW and enter the location of the display.drf file that you want to use. If the display.drf file is not auto-loaded and you do not manually load it, you will get error messages about missing packets when you try to open a schematic or layout view and you will not be able to see any process specific layers.

A display.drf file for the GPDK045 can be found in the PRD install directory under techFiles directory.

Listed below are the packet, color, lineStyle, and stipplePattern definitions for a metal3 drawing layer. The packet info references predefined color, lineStyle, and stipplePattern definitions. Any of these can be changed to suit an individual user's preferences in the project copy of the display.drf file.

drDefinePacket(

```
;( DisplayName      PacketName      Stipple      LineStyle
  Fill      Outline )
( display      m3      dots      solid      green green )
)
```

drDefineColor(

```
;( DisplayName      ColorName      Red      Green      Blue      Blink )
( display      green      0      204      102      nil )
)
```

drDefineLineStyle(

```
;( DisplayName      LineStyle      Size      Pattern )
( display      solid      1      (1 1 1) )
)
```

drDefineStipple(

```
;( DisplayName      StippleName      Bitmap )
( display      dots      (
```

(0 1 0 0 0 1 0 0 0 1 0 0 0 1 0 0)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 1 0 0 0 1 0 0 0 1 0 0 0 1 0 0)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 1 0 0 0 1 0 0 0 1 0 0 0 1 0 0)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 1 0 0 0 1 0 0 0 1 0 0 0 1 0 0)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)
(0 0 0 1 0 0 0 1 0 0 0 1 0 0 0 1)
(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0)

)

9 Schematic Design

The user should follow the guidelines listed below while building schematics using Composer:

Project libraries should list the primitive PRD library as a reference library in the library properties form.

Users can add instances from the PRD library to designs stored in the project libraries.

When performing hierarchical copy of schematic designs, care should be taken to preserve the references to the PRD libraries. These references should not be copied locally to the project directories and the references set to the local copy of PRD cells. This would prevent your designs from inheriting any fixes done to the PRD library from an upgrade.

Users should exercise caution when querying an instance and changing the name of the cell and replacing it with a reference to another cell. While similar parameters will inherit values, callbacks are not necessarily executed. This would cause dependent parameters to have incorrect values.

Schematics should be designed with schematic driven layout methodology in mind. Partitioning of schematics, hierarchical design, input and output ports, should be done in a clean and consistent fashion.

10 Library Device Setup

10.1 Resistors

The resistors in the library consist of three types; *diffused*, *insulated*, and *metal*. The diffused types include p+ and n+ and come in three-terminal varieties. The insulated resistors are those that are isolated from silicon by an insulator (oxide) such as poly resistors. These resistors are also three-terminal devices. The metal resistors are those resistors that are used as interconnect and feed-throughs; they are 2-terminal devices. Serpentine resistor layouts are not allowed.

Units:

The length and width are specified in meters for schematic simulation. Design variables are supported for both the length and width parameters.

Calculation:

The user has two choices in determining how the final resistor configuration is calculated. The user may request the calculation of either the resistor length or the resistor value. In both cases, the calculated values are determined based upon a combination of the length, width, resistance value, number of resistor segments (series or parallel), and contact resistance.

The width and length are snapped to grid, and the resistances are recalculated and updated on the component form based on actual dimensions.

Simulation:

Subcircuit definitions are used to model the resistors

10.2 Mosfets

All mosfets in the PRD library are 4 terminals.

Units:

Length and width are in meters, with areas and perimeters in meters squared and meters, respectively. Design variables are allowed for specifying parameter values on mosfet devices.

Calculation:

The area and perimeter parameters for the sources and drains are calculated from the width and the number of fingers used. This calculation assumes that the drain will always have the less capacitance (area) when there are an even number of fingers (odd number of diffusion areas). The finger width is calculated by dividing the total width by the number of fingers. Depending upon which value is entered into the form by the user, either the total width or the finger width will be calculated using the aforementioned calculation.

Simulation:

These mosfets are netlisted as their predefined device names for simulation purposes.

10.3 Bipolar Transistors

All BJT's in the PRD library are 3 terminal.

Units:

Only fixed size devices are allowed. Emitter sizes of 2, 5, and 10 are available.

Calculation:

The area is calculated from the emitter size of the device.

Simulation:

These BJTs are netlisted as their predefined device names for simulation purposes.

10.4 Diodes

All diodes in the PRD library are two-terminal.

Units:

Length and width are in meters. Design variables are allowed for Length and Width entries.

Calculation:

The area is calculated from the width and length entered.

Simulation:

These diodes are netlisted as their predefined device names for simulation purposes.

10.5 Capacitor

The metal capacitor is three-terminal. The MOS capacitors are captured as two-terminal devices in schematics, four-terminal MOS in layout.

Units:

Length and width are in meters. Design variables are allowed for Length and Width entries.

Calculation:

The capacitance is calculated from the width and length entered.

Simulation:

These capacitors are netlisted as their predefined device names for simulation purposes.

10.6 Inductor

The inductor is three-terminal. It has symmetric and asymmetric inductors.

Units:

Radius and width are in meters.

Calculation:

The inductance value is computed using radius, width, space (not a variable) and number of turns.

Simulation:

These inductors are netlisted as their predefined device names for simulation purposes.

11 Supported Devices

11.1 Mosfets

- nmos1v - 1.1 volt nominal Vt NMOS transistor
- nmos1v_3 - 1.1 volt nominal Vt NMOS transistor with inherited BULK
- nmos1v_hvt - 1.1 volt high Vt NMOS transistor
- nmos1v_hvt_3 - 1.1 volt high Vt NMOS transistor with inherited BULK
- nmos1v_lvt - 1.1 volt low Vt NMOS transistor
- nmos1v_lvt_3 - 1.1 volt low Vt NMOS transistor with inherited BULK
- nmos1v_nat - 1.1 volt native Vt NMOS transistor
- nmos1v_nat_3 - 1.1 volt native Vt NMOS transistor with inherited BULK
- nmos2v - 1.8 volt nominal Vt NMOS transistor
- nmos2v_3 - 1.8 volt nominal Vt NMOS transistor with inherited BULK
- nmos2v_nat - 1.8 volt native Vt NMOS transistor
- nmos2v_nat_3 - 1.8 volt native Vt NMOS transistor with inherited BULK
- pmos1v - 1.1 volt nominal Vt PMOS transistor
- pmos1v_3 - 1.1 volt nominal Vt PMOS transistor with inherited BULK
- pmos1v_hvt - 1.1 volt high Vt PMOS transistor
- pmos1v_hvt_3 - 1.1 volt high Vt PMOS transistor with inherited BULK
- pmos1v_lvt - 1.1 volt low Vt PMOS transistor
- pmos1v_lvt_3 - 1.1 volt low Vt PMOS transistor with inherited BULK
- pmos2v - 1.8 volt nominal Vt PMOS transistor
- pmos2v_3 - 1.8 volt nominal Vt PMOS transistor with inherited BULK

11.2 Resistors

- resnsndiff - N+ diffused resistor w/o salicide
- resnsndiff_2 - N+ diffused resistor w/o salicide with inherited BULK
- resnspdiff - P+ diffused resistor w/o salicide
- resnspdiff_2 - P+ diffused resistor w/o salicide with inherited BULK
- ressndiff - N+ diffused resistor w/i salicide
- ressndiff_2 - N+ diffused resistor w/i salicide with inherited BULK

- resspdiff - P+ diffused resistor w/i salicide
- resspdiff_2 - P+ diffused resistor w/i salicide with inherited BULK
- resnwsti - N-Well resistor under STI
- resnwsti_2 - N-Well resistor under STI with inherited BULK
- resnwoxide - N-Well resistor under OD
- resnwoxide_2 - N-Well resistor under OD with inherited BULK
- resnsnpoly - N+ Poly resistor w/salicide
- resnsnpoly_2 - N+ Poly resistor w/salicide with inherited BULK
- resnsppoly - P+ Poly resistor w/salicide
- resnsppoly_2 - P+ Poly resistor w/salicide with inherited BULK
- ressnpoly - N+ Poly resistor w/o salicide
- ressnpoly_2 - N+ Poly resistor w/o salicide with inherited BULK
- ressppoly - P+ Poly resistor w/o salicide
- ressppoly_2 - P+ Poly resistor w/o salicide with inherited BULK
- resm<k> - Metal <k> resistor (k=1..11)

11.3 Capacitor

- mimcap - CapMetal (Intermediate Metal 11) to Metal 10 cap
- nmoscap1v - 1.1 volt Nmos cap
- pmoscap1v - 1.1 volt Pmos cap
- nmoscap2v - 1.8 volt Nmos cap
- pmoscap2v - 1.8 volt Pmos cap

11.4 Bipolars

- vnpn2 - Vertical substrate PNP 2x2 Emitter
- vnpn5 - Vertical substrate PNP 5x5 Emitter
- vnpn10 - Vertical substrate PNP 10x10 Emitter
- npn2 - Vertical substrate NPN 2x2 Emitter
- npn5 - Vertical substrate NPN 5x5 Emitter
- npn10 - Vertical substrate NPN 10x10 Emitter

11.5 Diodes

- pdio - 1.1 volt P+/nwell diode
- pdio_hvt - 1.1 volt high Vt P+/nwell diode
- pdio_lvt - 1.1 volt low Vt P+/nwell diode
- pdio_2v - 1.8 volt P+/nwell diode
- ndio - 1.1 volt N+/psub diode
- ndio_hvt - 1.1 volt high Vt N+/psub diode
- ndio_lvt - 1.1 volt low Vt N+/psub diode
- ndio_nvt - 1.1 volt native Vt N+/psub diode
- ndio_2v - 1.8 volt N+/psub diode
- ndio_2v_nvt - 1.8 volt native Vt N+/psub diode

11.6 Inductors

- ind_a - Asymmetric Inductor
- ind_s - Symmetric Inductor

12 Views provided

The following table explains the use of the cellviews provided as part of this PRD:

symbol	Used in Composer schematics
spectre	Simulation / netlisting view for the Spectre & UltraSim simulator
hspiceD	Simulation / netlisting view for the hspiceD simulator
schematic	Simulation / netlisting view for all simulators; Mixed-mode and logic resistors use schematic to call other simulator resistor views. It is used to implement series and parallel features in those resistors.
auLvs	Netlisting view for DIVA and Assura
auCdl	Circuit Descriptive Language netlisting view typically used to generate a netlist for Dracula or third party simulators.
ivpcell	Device recognition symbol used in the extracted layout for netlisting purposes with DIVA and Assura
layout	Fixed cell or pcell used in Virtuoso Layout Editor.

12.1 Mosfets

- Four terminals (D, G, S, B)
- symbol, spectre, hspiceD, auLvs, auCdl, ivpcell, layout (Pcells)

12.2 Resistors

- Three terminals (PLUS, MINUS, B) for diffused and poly resistors
- Two terminals (PLUS, MINUS) for metal resistors
- symbol, schematic, auLvs, auCdl, ivpcell, layout (Pcells)
- Resistors called in schematic views include views for all simulators, symbol, spectre, hspiceD, auLvs, auCdl, ivpcell

12.3 Capacitor

- Two terminals (really four -- S/D/B overlapped) (G, D/S/B) for mos caps
- Three terminals (PLUS, MINUS, B) for metal mimcap capacitor
- symbol, spectre, hspiceD, auLvs, auCdl, ivpcell, layout (Pcells)

12.4 Diodes

- Two terminals (PLUS, MINUS)
- symbol, spectre, hspiceD, auLvs, auCdl, ivpcell, layout (Pcells)

12.5 Bipolars

- Three terminals (C, B, E)
- symbol, spectre, hspiceD, auLvs, auCdl, ivpcell, layout (Pcells)

12.6 Inductors

- Three terminals (PLUS, MINUS, B)
- symbol, spectre, auLvs, auCdl, ivpcell, layout (Pcells)

13 CDF parameters

13.1 Mosfets

Model Name - spectre model name (non-editable)

Multiplier - number of Parallel MOS devices

Length (M) - gate length in meters

Total Width (M) - gate width in meters (sum of all fingers)

Finger Width - width of each gate finger/stripe

Fingers - number of poly gate fingers/stripes used in layout

Threshold - finger width at which to apply device folding of the layout

Apply Threshold - button to apply threshold or not

Gate Connection - allow shorting of multi-fingered devices and addition of contact heads to gate ends

S/D Connection - allow shorting of sources and/or drains on multi-finger devices

S/D Metal Width - width of metal used to short sources/drains

Switch S/D - source is defined as left-most diffusion region and alternating regions to the right. Pins are not automatically permuted and can be switched using this parameter

Diff Cont - To switch on/off diffusion contact

Bodytie Type - None, Detached, or Integrated (butting source)

- For Detached, user may select Left, Right, Top, and/or Bottom to specify the located of bodyties. Selection of all four creates a guarding
- For Detached, the user may specify Tap Extension (in microns) which sets the distance from the bodytie to the device. Maximum distance is 100 microns
- For Integrated, the user may select Left or Right for a device with an odd number of fingers (1, 3, 5, ...). The user may select Left and Right for an even fingered device

Edit Area & Perim - allow Drain/Soure area and periphery be entered manually for simulation

Drain diffusion area, etc. - several simulation parameters are presented. The area and perimeter parameters are calculated and netlisted in accordance with the layouts or can be entered manually if “Edit Area & Perim” is checked

13.2 Resistors

Model Name - Spectre model name (non-editable)

Segments - number of series or parallel segments for a resistor

Segment Connection - cyclic field used for series or parallel segments

Calculated Parameter - radio button that determines whether resistance or Length is the calculated value when instantiating a new resistor device

Resistance - total resistance value equal to the sum of body resistance, contact resistance, end resistance, and grain resistance

Segment Width - resistor segment width in meters

Segment Length - resistor segment length in meters

Effective Width - effective resistor segment width in meters

Effective Length - effective resistor segment length in meters

Left Dummy - boolean value used to place a dummy resistor strip on the left side of the main resistor

Right Dummy - boolean value used to place a dummy resistor strip on the right side of the main resistor

Contact Rows - integer number of contact rows

Contact Columns - integer number of contact columns

Show Tap Params - boolean value allowing the user to set the visibility of the resistor tap properties

Left Tap - boolean value used to place a resistor tap on the left side of a device

Right Tap - boolean value used to place a resistor tap on the right side of a device

Top Tap - boolean value used to place a resistor tap on the top side of a device

Bottom Tap - boolean value used to place a resistor tap on the bottom side of a device

Tap Extension - float values to set where the left, right, top, and bottom taps would be to its original placements. This parameter is related to the stretch handle on the taps. The input format should be "left 1.3 right 1.0 top 0.0 bottom 2.0" without the quotes. If neither pair is not present, a zero is assumed

Sheet Resistivity - sheet rho value for body of resistor (non-editable)

End Resistance - resistance value for any salicided area near the contact heads in a non-salicided resistor (non-editable)

Contact Resistance - resistance value for the contact heads of a particular resistor (non-editable)

Grain Resistance - constant resistance value for any salicided area near the contact heads in a non-salicided resistor (non-editable)

Delta Width - resistor width process variation value in meters (non-editable)

Delta Length - resistor length process variation value in meters (non-editable)

Temperature Coefficient 1 - temperature coefficient #1 for resistor (non-editable)

Temperature Coefficient 2 - temperature coefficient #2 for resistor (non-editable)

13.3 MOScaps

Model Name - spectre model name (non-editable)

Multiplier - number of Parallel MOS devices

Calculated Parameter - Calculated parameter cyclic (capacitance, length, width)

Capacitance - total capacitance

Length (M) - gate length in meters

Total Width (M) - gate width in meters (sum of all fingers)

Finger Width - width of each gate finger/stripe

Fingers - number of poly gate fingers/stripes used in layout

Gate Connection - allow shorting of multi-fingered devices and addition of contact heads to gate ends

S/D Connection - allow shorting of sources and/or drains on multi-finger devices

S/D Metal Width - width of metal used to short sources/drains

Switch S/D - source is defined as left-most diffusion region and alternating regions to the right. Pins are not automatically permuted and can be switched using this parameter

Bodytie Type - None, Detached, or Integrated (butting source)

- For Detached, user may select Left, Right, Top, and/or Bottom to specify the located of bodyties. Selection of all four creates a guardring
- For Detached, the user may specify Tap Extension (in microns) which sets the distance from the bodytie to the device. Maximum distance is 100 microns
- For Integrated, the user may select Left or Right for a device with an odd number of fingers (1, 3, 5, ...). The user may select Left and Right for an even fingered device

Area capacitance - Capacitance per unit area used in parameter calculations (non-editable)

Fringe capacitance - Fringe Capacitance of perimeter used in parameter calculations (non-editable)

Temp rise from ambient, etc. - several simulation parameters are presented.

13.4 Bipolars

Model name	Model name used in simulation
Device Area	Emitter area in microns squared (non-editable)
Emitter width	Emitter width microns (non-editable)
Multiplier	Number of Parallel Bipolar devices
Estimated operating region	Simulation operating region

Diodes

Model name	Model used for simulation name
Calculate Parameter	Choices are 'area' , 'width' or 'length'
Device Area	Calculated junction area in meters squared (non-editable)
Length (M)	Diode length in meters
Width (M)	Diode width in meters
Multiplier	Number of Parallel Diode devices
Periphery of junction	Calculated junction periphery in meters (non-editable)

13.5 Inductors

Model	Model used for simulation name
Inductance (H)	Inductance value in Henry
Inner Radius	Inner Radius of Inductor in meters
Inductor Width	Width of each metal turns in meters.
Inductor Space	Space between each metal turns in meters (non-editable)
Multiplier	Number of Parallel Inductor devices
Number of Turns	Number of metal turns of Inductor in cyclic field.

14 Model Setup

This PRD supports the Cadence Spectre, Ultrasim, and AMS circuit simulators.

The following model sections are defined in the
<PRD_install_directory>/models/spectre/gpdk045.scs file.

Section

tt

ff

ss

fs

sf

mc

tt - Typical N and P model parameters

ff - Fast N and P model parameters

ss - Slow N and P model parameters

fs - Fast N and slow P model parameters

sf - Slow N and Fast P model parameters

mc - Monte Carlo model parameters

15 Techfile Layers

Cadence will provide a standard display setup, and will not support desired changes to the display. The customer is free to modify the display.drf file used on-site to achieve any desired display.

CDS #	GDS #	GDS type	CDS name	CDS purpose	Description
2	1	0	Oxide	drawing	Oxide
4	24	0	Oxide_thk	drawing	Thick Oxide
6	2	0	Nwell	drawing	Nwell
10	3	0	Poly	drawing	Poly
11	18	0	Nhvt	drawing	N+ high Vt implant
12	4	0	Nimp	drawing	N+ implant
13	23	0	Phvt	drawing	P+ high Vt implant
14	5	0	Pimp	drawing	P+ implant
15	52	0	Nzvt	drawing	Native Nmos
16	72	0	SiProt	drawing	Salicide Blocking
18	19	0	Nburied	drawing	N buried
20	6	0	Cont	drawing	Contact
26	26	0	Nlvt	drawing	N+ low Vt implant
27	27	0	Plvt	drawing	P+ low Vt implant
30	7	0	Metal1	drawing	Metal1
32	8	0	Via1	drawing	Via1
34	9	0	Metal2	drawing	Metal2
36	10	0	Via2	drawing	Via2
38	11	0	Metal3	drawing	Metal3
40	30	0	Via3	drawing	Via3
42	31	0	Metal4	drawing	Metal4
44	32	0	Via4	drawing	Via4
46	33	0	Metal5	drawing	Metal5
48	34	0	Via5	drawing	Via5
50	35	0	Metal6	drawing	Metal6
52	37	0	Via6	drawing	Via6
54	38	0	Metal7	drawing	Metal7
56	39	0	Via7	drawing	Via7
58	40	0	Metal8	drawing	Metal8
60	41	0	Via8	drawing	Via8
62	42	0	Metal9	drawing	Metal9
64	151	0	Via9	drawing	Via9
66	152	0	Metal10	drawing	Metal10
68	161	0	Via10	drawing	Via10
70	162	0	Metal11	drawing	Metal11
30	7	1	Metal1	pin	Pin purpose

34	9	1	Metal2	pin	Pin purpose
38	11	1	Metal3	pin	Pin purpose
42	31	1	Metal4	pin	Pin purpose
46	33	1	Metal5	pin	Pin purpose
50	35	1	Metal6	pin	Pin purpose
54	38	1	Metal7	pin	Pin purpose
58	40	1	Metal8	pin	Pin purpose
62	42	1	Metal9	pin	Pin purpose
66	152	1	Metal10	pin	Pin purpose
70	162	1	Metal11	pin	Pin purpose
30	7	3	Metal1	label	Label purpose
34	9	3	Metal2	label	Label purpose
38	11	3	Metal3	label	Label purpose
42	31	3	Metal4	label	Label purpose
46	33	3	Metal5	label	Label purpose
50	35	3	Metal6	label	Label purpose
54	38	3	Metal7	label	Label purpose
58	40	3	Metal8	label	Label purpose
62	42	3	Metal9	label	Label purpose
66	152	3	Metal10	label	Label purpose
70	162	3	Metal11	label	Label purpose
30	7	5	Metal1	fill	Fill purpose
34	9	5	Metal2	fill	Fill purpose
38	11	5	Metal3	fill	Fill purpose
42	31	5	Metal4	fill	Fill purpose
46	33	5	Metal5	fill	Fill purpose
50	35	5	Metal6	fill	Fill purpose
54	38	5	Metal7	fill	Fill purpose
58	40	5	Metal8	fill	Fill purpose
62	42	5	Metal9	fill	Fill purpose
66	152	5	Metal10	fill	Fill purpose
70	162	5	Metal11	fill	Fill purpose
71	7	2	Metal1	slot	Slot purpose
72	9	2	Metal2	slot	Slot purpose
73	11	2	Metal3	slot	Slot purpose
74	31	2	Metal4	slot	Slot purpose
75	33	2	Metal5	slot	Slot purpose
76	35	2	Metal6	slot	Slot purpose
77	38	2	Metal7	slot	Slot purpose
78	40	2	Metal8	slot	Slot purpose
79	42	2	Metal9	slot	Slot purpose
66	152	2	Metal10	slot	Slot purpose
70	162	2	Metal11	slot	Slot purpose
80	25	0	Psub	drawing	P substrate
82	22	0	DIOdummy	drawing	Recognition layer for diodes
84	21	0	PNPdummy	drawing	Recognition layer for pnp

85	85	0	PWdummy	drawing	Recognition layer for substrate
86	20	0	NPNdummy	drawing	Recognition layer for npn
87	60	0	VPNP2dum	drawing	Recognition layer for vnpnp2x2
88	17	0	IND2dummy	drawing	Recognition layer for inductor
89	61	0	VPNP5dum	drawing	Recognition layer for vnpnp5x5
90	16	0	INDdummy	drawing	Recognition layer for inductor
91	62	0	VPNP10dum	drawing	Recognition layer for vnpnp10x10
92	15	0	BJTdum	drawing	Recognition layer for vnpnp's
93	84	0	Cap3dum	drawing	Recognition layer for moscap
94	13	0	Resdum	drawing	Recognition layer for resistor
95	36	0	Bondpad	drawing	Recognition layer for bondpad
96	12	0	Capdum	drawing	Recognition layer for moscap
97	14	0	CapMetal	drawing	Recognition layer for moscap
98	71	0	ResWdum	drawing	Recognition layer for resistor
99	75	0	M1Resdum	drawing	Recognition layer for metal res1
100	76	0	M2Resdum	drawing	Recognition layer for metal res2
101	77	0	M3Resdum	drawing	Recognition layer for metal res3
102	78	0	M4Resdum	drawing	Recognition layer for metal res4
103	79	0	M5Resdum	drawing	Recognition layer for metal res5
104	80	0	M6Resdum	drawing	Recognition layer for metal res6
105	81	0	M7Resdum	drawing	Recognition layer for metal res7
106	82	0	M8Resdum	drawing	Recognition layer for metal res8
107	83	0	M9Resdum	drawing	Recognition layer for metal res9
108	93	0	M10Resdum	drawing	Recognition layer for metal res10
109	103	0	M11Resdum	drawing	Recognition layer for metal res11
114	70	0	IND3dummy	drawing	Recognition layer for inductor
115	74	0	ESDdummy	drawing	Recognition layer for esd

16 Virtuoso L,XL,GXL

The standard Cadence Virtuoso Layout design flow will be implemented. This includes basic connectivity of connection layers, wells, and substrate, and symbolic contacts. The M factor will be used for device instance multiplier - there will be no conflict with the parameter used in cell operation. Names will be displayed on the layout views to aid in schematic-layout instance correlation. Auto-abutment of MOSFET devices is supported. Pin permuting of MOSFET and Resistor device is also supported. The skill pcell layouts are compiled into the PRD.

The users should follow the guidelines listed below for layout design:

- The Virtuoso Layout tool requires a separate license for operation.

- Users obtain maximum leverage from the PRD by doing schematic driven layout in the Virtuoso Layout environment. This flow will produce a correct by design layout. The Virtuoso Custom Router (VCR) and Virtuoso Shape Based Router (VSR) can be used to finish the unconnected interconnect in the layout.

- The Router rules file for the target process is provided with the PRD.

- Abutment is currently supported only for MOS transistors.

- Note, abutment will work only on schematic driven layouts.

- Schematic Driven Layout is recommended over Netlist Driven Layout.

NOTE: Skill pcell source code is not included in the PRD kit.

17 Diva Decks

These decks can be found in the extracted PRD directory tree located under the 'gpdk045' directory.

17.1 Diva Extract

Diva Extract file is for extraction of all devices in PDK.

The file is:

- divaEXT.rul

17.2 Diva LVS

Diva LVS ruldeck is for compare and parameter check.

The file is:

- divaLVS.rul

18 Assura Decks

Cadence has developed the Assura DRC, EXT, and LVS rule files from the documentation provided.

These decks can be found in the extracted PRD directory tree in the directory:

- `assura`

18.1 Assura DRC

The Assura DRC file includes DRC check as well as antenna and density checks

- `assuraDRC.rul`

18.2 Assura LVS

The Assura LVS files provided are named

- `extract.rul`
- `compare.rul`

19 PVS Decks

Cadence has developed the PVS DRC, Antenna, and LVS rule files from the documentation provided.

These decks can be found in the extracted PRD directory tree in the directory:

- pvs

19.1 pvl DRC

The PVS DRC file includes DRC check as well as density checks

- pvlDRC.rul

19.2 pvlLVS

The PVS LVS files provided is named as

- pvlLVS.rul

19.3 pvlAnt

The PVS antenna files are also located inside pvs directory.

- pvlAnt.rul

19.4 pvl Extraction

The PVS extraction based files are also located inside pvs directory.

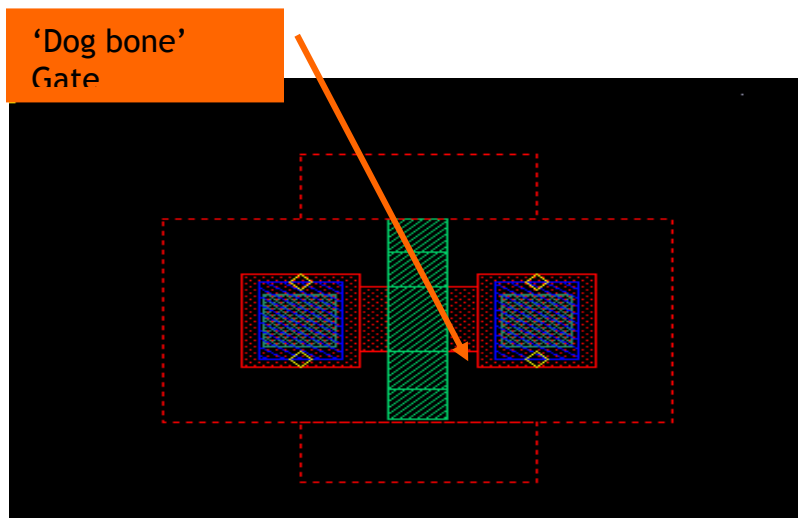
- extview.rul

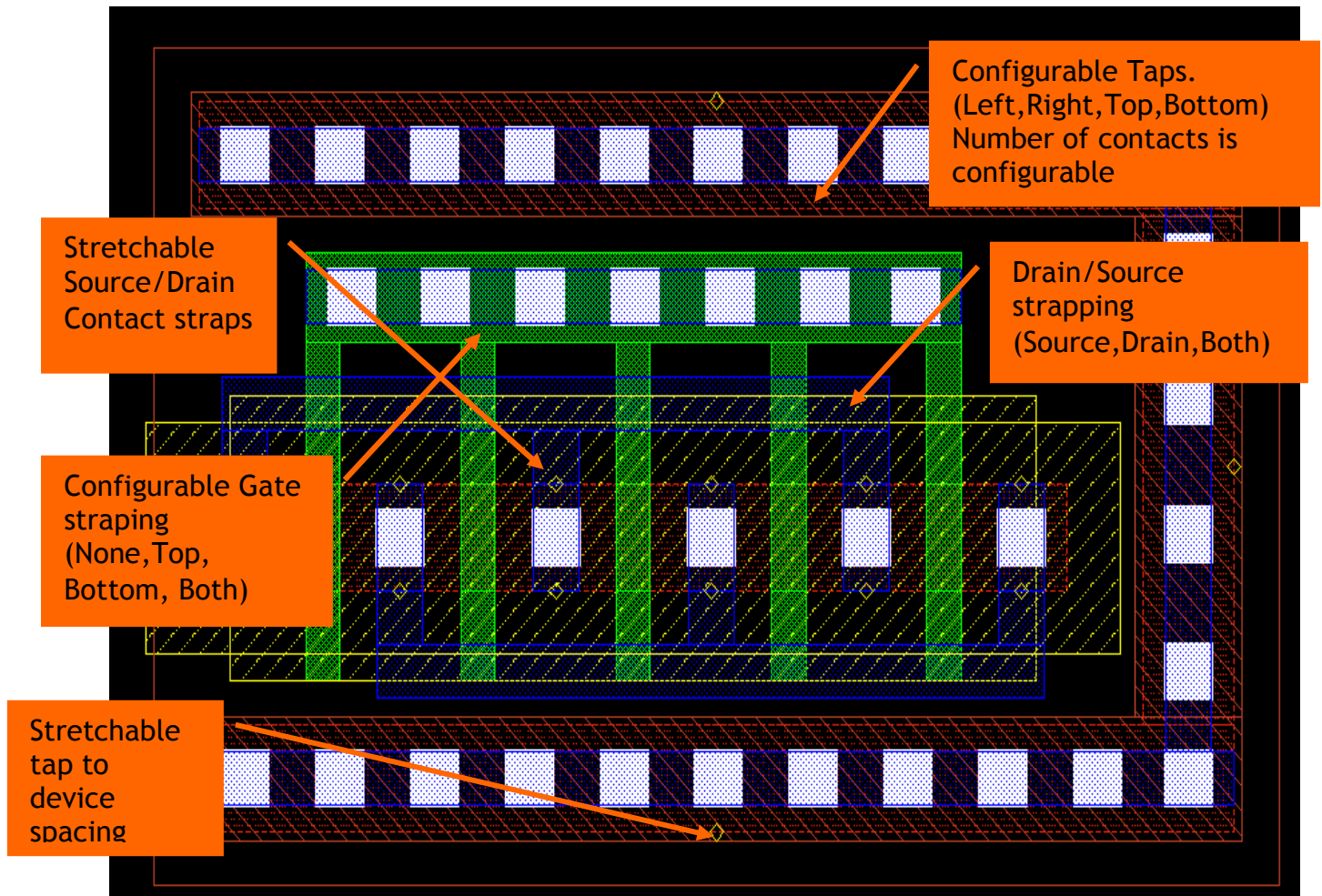
pvtech.lib is available in extracted PRD directory.

20 GPDK045 Pcell Layouts

20.1 Mos Pcell

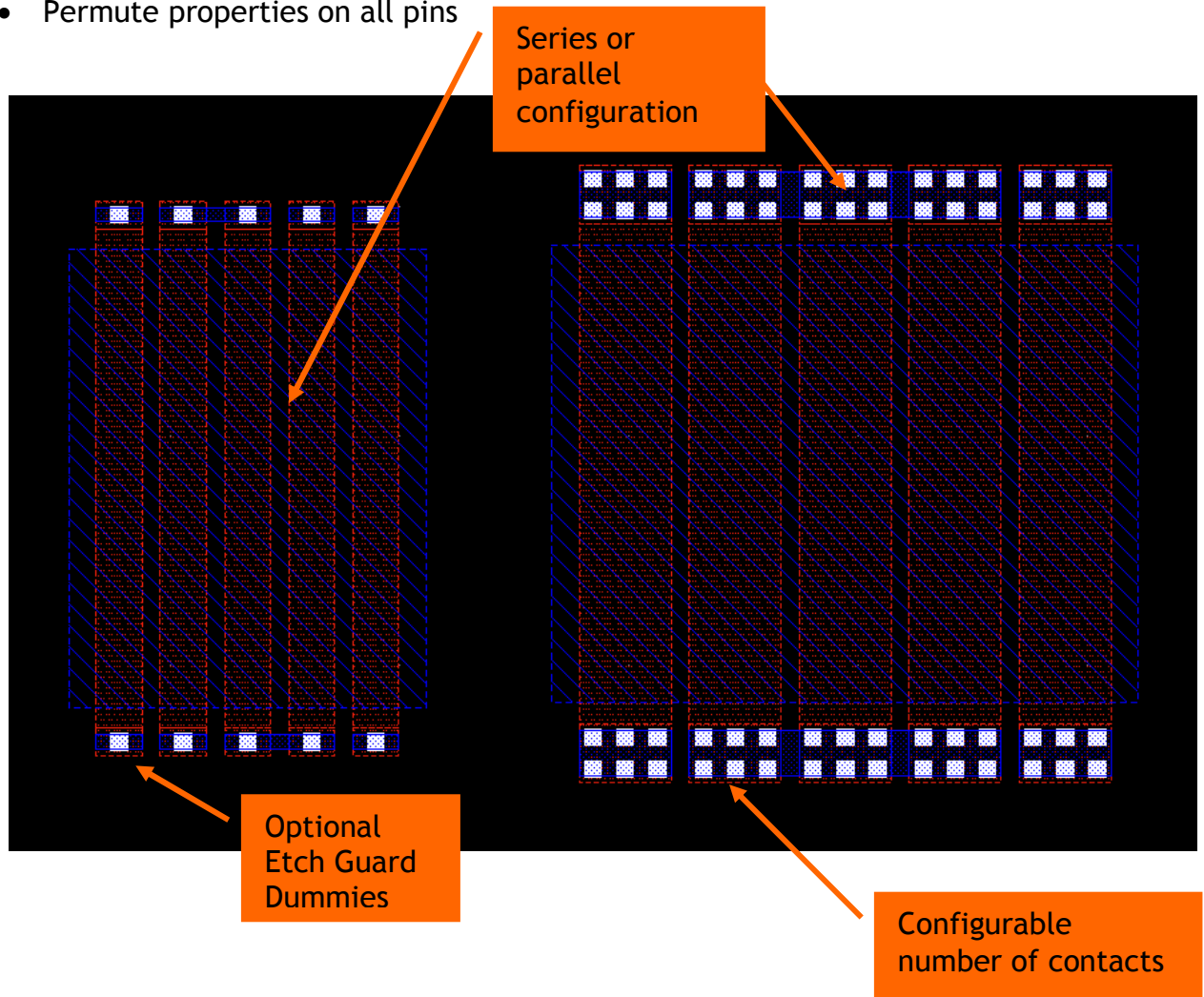
- Multi-fingered gates.
- Dog bone gate
- Configurable source-drain connection for multi fingered devices.(gate,source,both)
- Configurable gate straps for multi fingered devices (top, bottom, both, alternate)
- Variable source/drain contact coverage with stretch handles on each contact strap.
- Optional tap placement for four terminal device. Tab can be configured. (top,bottom,left,right) Spacing between device and tab can be controlled by stretch handle.
- Auto abutment enabled.
- VXL ready connectivity. Pins access direction needs to allow routing with CCAR.
- Permute properties on all pins.





20.2 Resistor Pcell

- Parallel/Series configuration with metal straps.
- Dog bone
- VXL ready connectivity
- Optional etch guard resistors
- Permute properties on all pins



20.3 Capacitor Pcell

- Configurable width, length or capacitance

Edit Instance Properties

OK Cancel Apply Next Previous

Attribute Connectivity Parameter Property ROD DFM

Model name:

Calculate Parameter:

Capacitance:

Width & Length:

Multiplier:

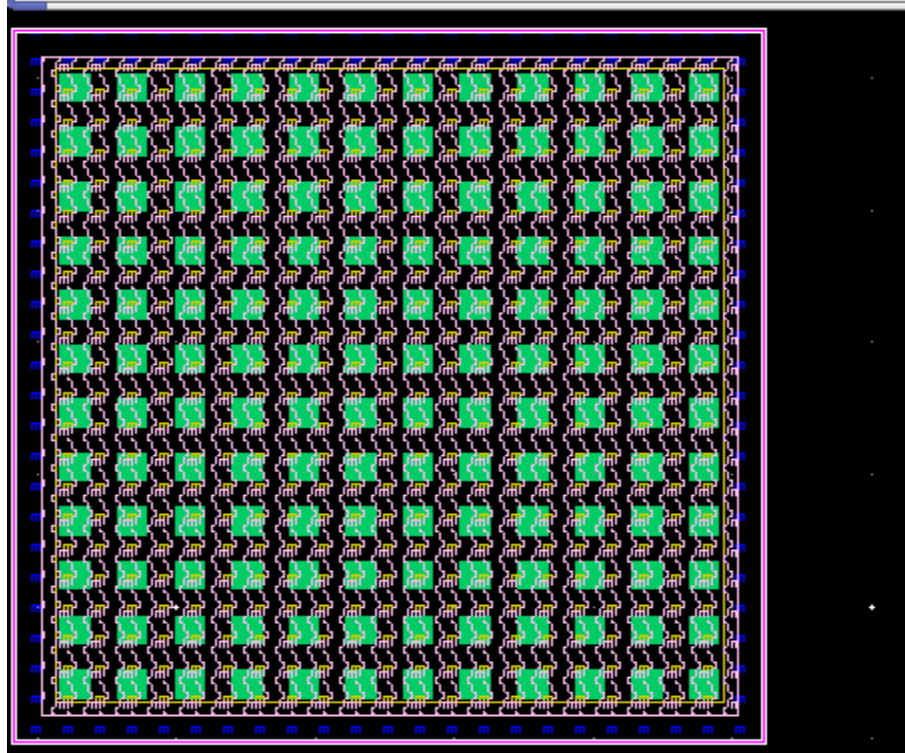
Show Sim Parameters: ☐

Area capacitance (F/M^2):

Fringe capacitance (F/M):

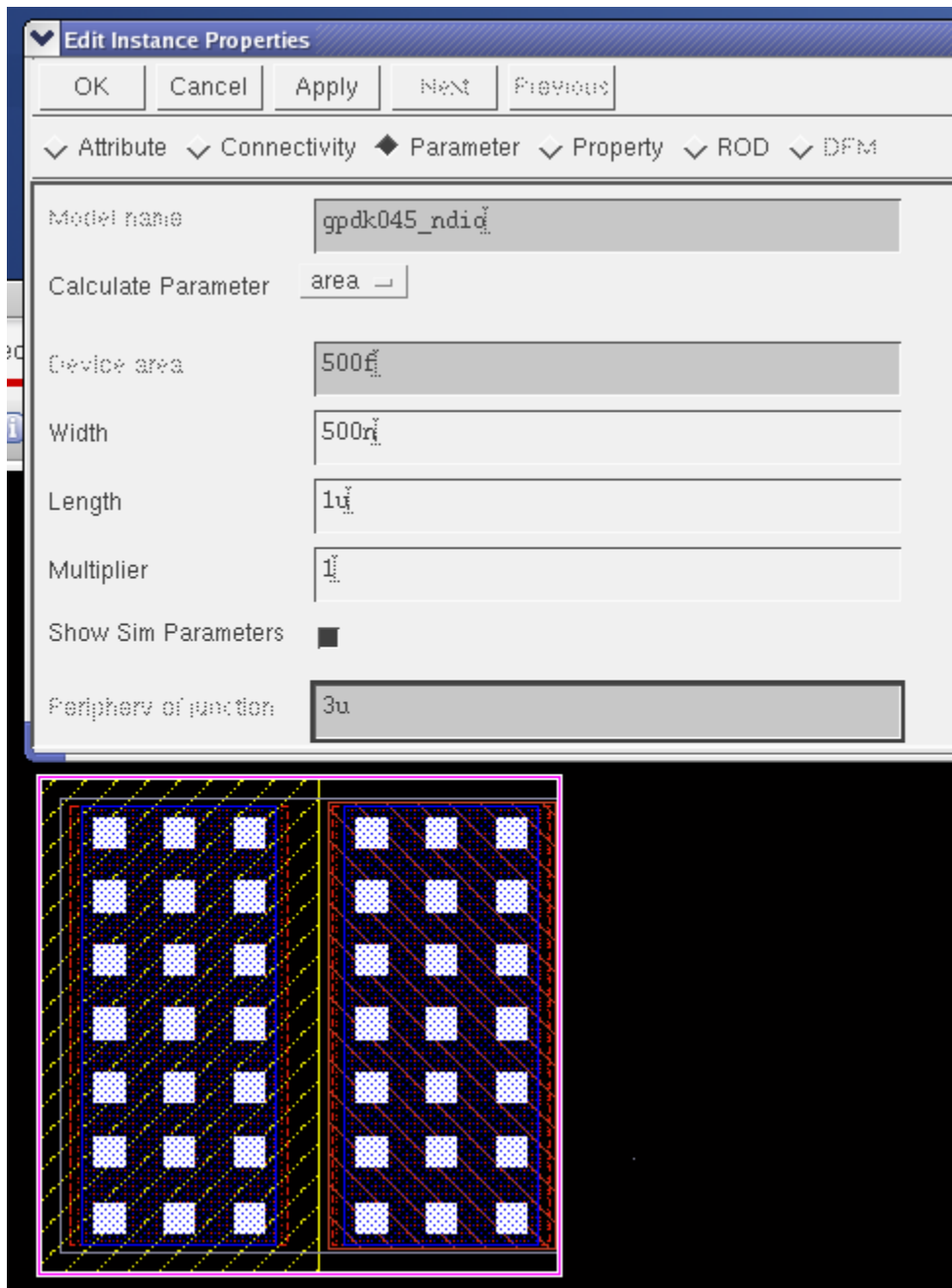
Area:

Perim:



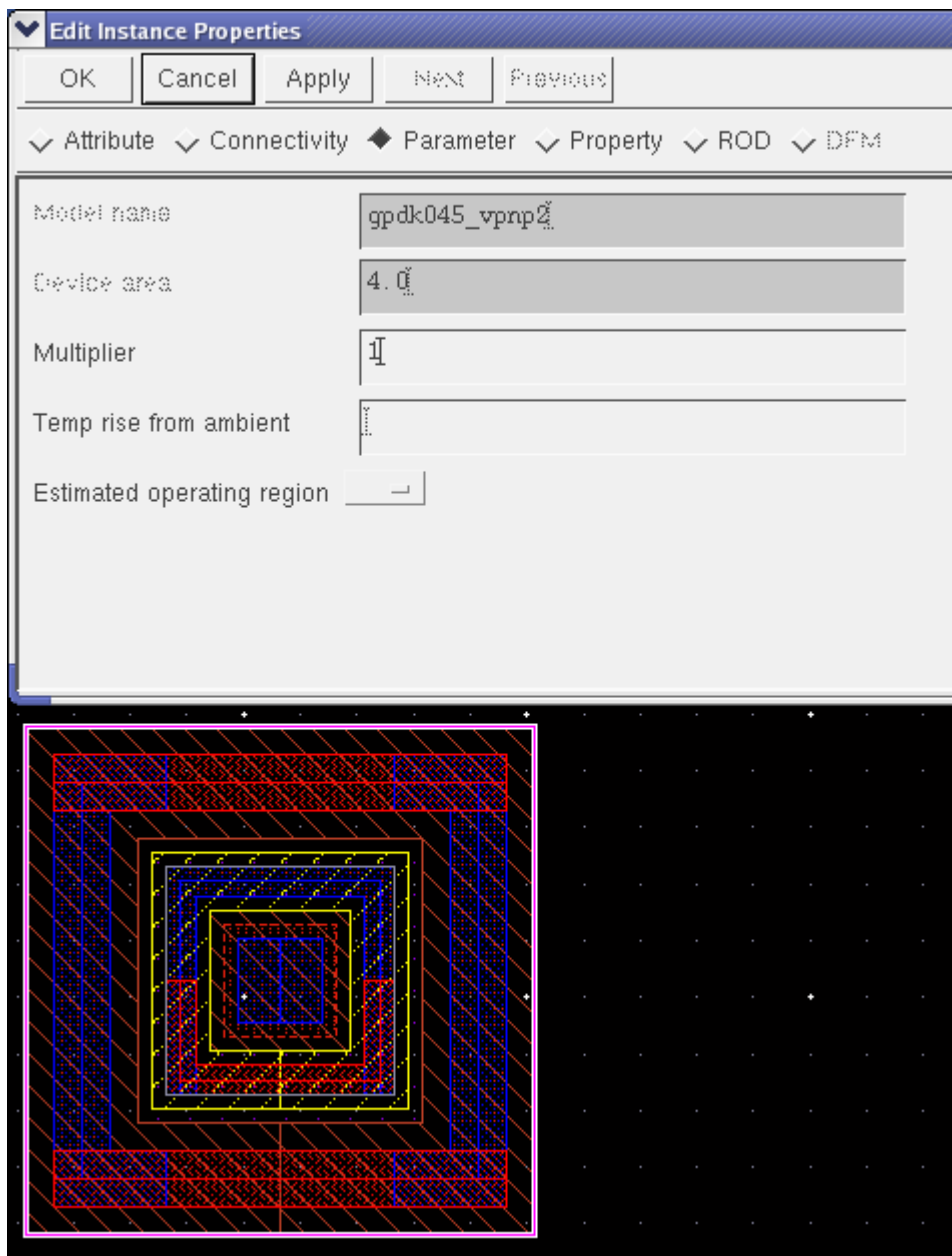
20.4 Diode Pcell

- Configurable width, length or area



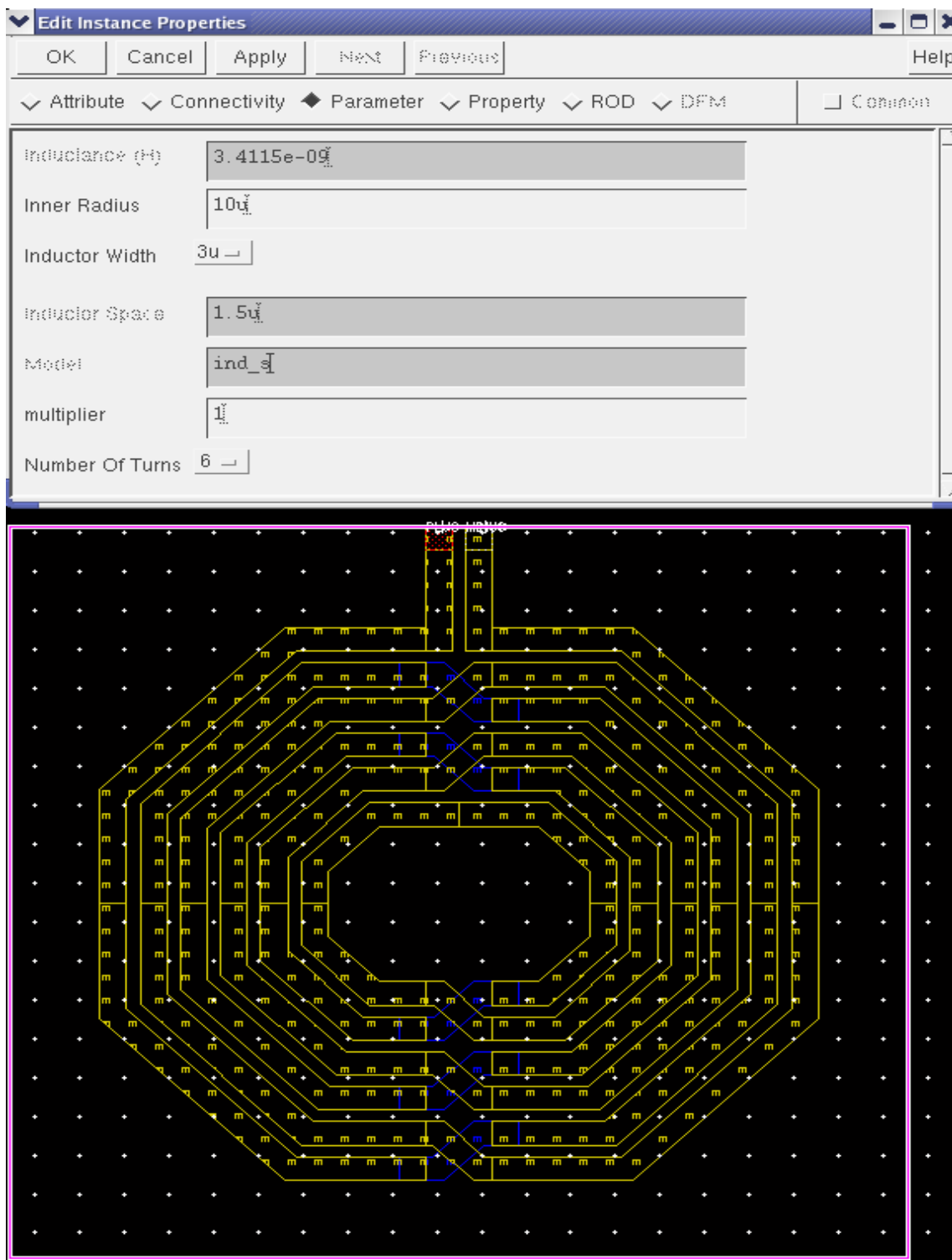
20.5 Bipolar Pcell

- BJT is a fixed layout



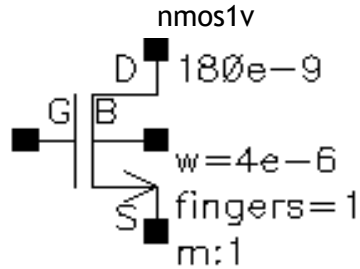
20.6 Inductor Pcell

- Configurable width, turns and radius



21 GPDK045 Device Datasheets

21.1 nmos1v datasheet



Spectre Netlist

Spectre Model Name = "g45n1svt"

```
NM1 (D G S B) g45n1svt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187
```

DIVA LVS Netlist

DIVA Device Name = "g45n1svt"

```
; g45n1svt Instance /NM1 = auLvs device M1
d g45n1svt D G S B (p D S)
i 1 g45n1svt D G S B " m 1.0 l 180e-9 w 4e-6 "
```

CDL Netlist

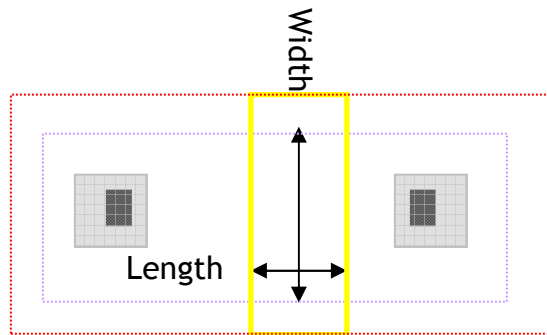
CDL Device Name = "g45n1svt"

```
MNM1 D G S B g45n1svt W=4u L=180.0n M=1.0
```

Assura Netlist

Assura auLvs Device Name = "g45n1svt"

```
c g45n1svt MOS D B      G B      S B      B B  ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n1svt D G S B ; m 1      l 1.8e-07 w 4e-06 ;
```



nmos1v - 1.2 volt nominal VT NMOS transistor

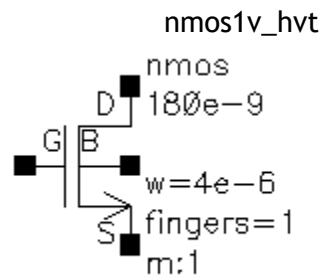
Device Layers	
Layer	Color and Fill
Oxide	
Nimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP CONTAINS POLY
G	Poly
D	Oxide AND Nimp NOT Poly
S	Oxide AND Nimp NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

- S and D are PERMUTABLE

21.2 nmos1v_hvt datasheet

**Spectre Netlist****Spectre Model Name = "g45n1hvt"**

```

NM1 (D G S B) g45n1hvt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187

```

DIVA LVS Netlist**DIVA Device Name = "g45n1hvt"**

```

; g45n1hvt Instance /NM1 = auLvs device M1
d g45n1hvt D G S B (p D S)
i 1 g45n1hvt D G S B " m 1.0 l 180e-9 w 4e-6 "

```

CDL Netlist**CDL Device Name = "g45n1hvt"**

```

MNM1 D G S B g45n1hvt W=4u L=180.0n M=1.0

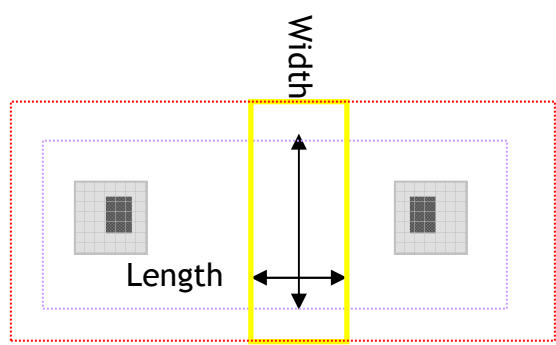
```

Assura Netlist**Assura auLvs Device Name = "g45n1hvt"**

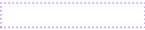
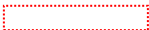
```

c g45n1hvt MOS D B      G B      S B      B B  ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n1hvt D G S B ; m 1      l 1.8e-07 w 4e-06 ;

```

nmos1v_hvt - 1.2 volt high VT NMOS transistor

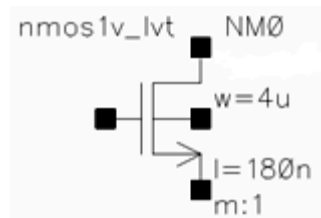
Device Layers	
Layer	Color and Fill
Oxide	
Nimp & Nhvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP AND NHVT CONTAINS POLY
G	Poly
D	Oxide AND Nimp AND Nhvt NOT Poly
S	Oxide AND Nimp AND Nhvt NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* S and D are PERMUTABLE

21.3 nmos1v_lvt datasheet

**Spectre Netlist****Spectre Model Name = "g45n1lvt"**

```

NM1 (D G S B) g45n1lvt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187

```

DIVA LVS Netlist**DIVA Device Name = "g45n1lvt"**

```

; g45n1lvt Instance /NM1 = auLvs device M1
d g45n1lvt D G S B (p D S)
i 1 g45n1lvt D G S B " m 1.0 l 180e-9 w 4e-6 "

```

CDL Netlist**CDL Device Name = "g45n1lvt"**

```

MNM1 D G S B g45n1lvt W=4u L=180.0n M=1.0

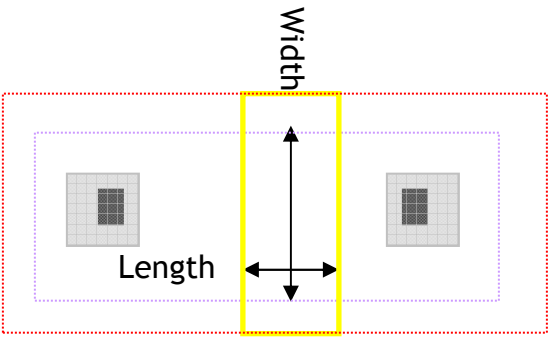
```

Assura Netlist**Assura auLvs Device Name = "g45n1lvt"**

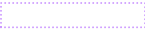
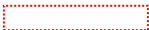
```

c g45n1lvt MOS D B      G B      S B      B B ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n1lvt D G S B ; m 1      l 1.8e-07 w 4e-06 ;

```



nmos1v_lvt - 1.2 volt low VT NMOS transistor

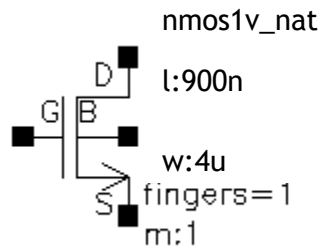
Device Layers	
Layer	Color and Fill
Oxide	
Nimp & Nlvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP AND NHVT CONTAINS POLY
G	Poly
D	Oxide AND Nimp AND Nlvt NOT Poly
S	Oxide AND Nimp AND Nlvt NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* S and D are PERMUTABLE

21.4 nmos1v_nat datasheet

**Spectre Netlist****Spectre Model Name = "g45n1nvt"**

```
NM1 (D G S B) g45n1nvt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187
```

DIVA LVS Netlist**DIVA Device Name = "g45n1nvt"**

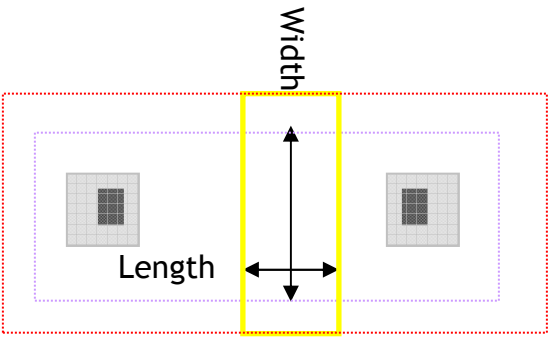
```
; g45n1nvt Instance /NM1 = auLvs device M1
d g45n1nvt D G S B (p D S)
i 1 g45n1nvt D G S B " m 1.0 l 900e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45n1nvt"**

```
MNM1 D G S B g45n1nvt W=4u L=900.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45n1nvt"**

```
c g45n1nvt MOS D B      G B      S B      B B ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n1nvt D G S B ; m 1      l 9.0e-07 w 4e-06 ;
```



nmos1v_nat - 1.2 volt native VT MOS transistor

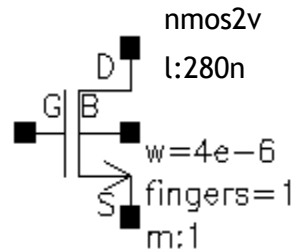
Device Layers	
Layer	Color and Fill
Oxide	
Nimp & Nzvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP AND NZVT CONTAINS POLY
G	Poly
D	Oxide AND Nimp AND Nzvt NOT Poly
S	Oxide AND Nimp AND NzvtT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* S and D are PERMUTABLE

21.5 nmos2v datasheet

**Spectre Netlist****Spectre Model Name = "g45n2svt"**

```
NM1 (D G S B) g45n2svt w=4u l=280.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187
```

DIVA LVS Netlist**DIVA Device Name = "g45n2svt"**

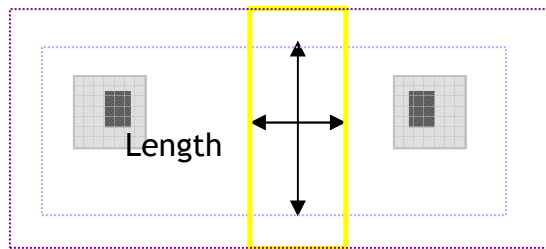
```
; g45n2svt Instance /NM1 = auLvs device M1
d g45n2svt D G S B (p D S)
i 1 g45n2svt D G S B " m 1.0 l 280e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45n2svt"**

```
MNM1 D G S B g45n2svt W=4u L=280.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45n2svt"**

```
c g45n2svt MOS D B G B S B B B ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n2svt D G S B ; m 1 l 2.8e-07 w 4e-06 ;
```



nmos2v - 1.8 volt nominal VT NMOS transistor

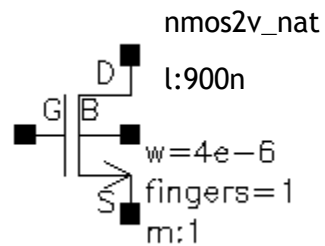
Device Layers	
Layer	Color and Fill
Oxide_thk	
Oxide	
Nimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE_THK AND OXIDE AND NIMP CONTAINS POLY
G	Poly
D	Oxide_thk AND Oxide AND Nimp NOT Poly
S	Oxide_thk AND Oxide AND Nimp NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

- S and D are PERMUTABLE

21.6 nmos2v_nat datasheet

**Spectre Netlist****Spectre Model Name = "g45n2nvt"**

```
NM1 (D G S B) g45n2nvt w=4u l=900.0n nf=1 as=6e-12 ad=6e-12 \
      ps=8.3u pd=8.3u m=1 nrs=37.5m nrd=37.5m sa=0.15e-9 sb=0.15e-9 sd=0.18e-9
      sca=76.0613 scb=0.0758094 scc=0.00958187
```

DIVA LVS Netlist**DIVA Device Name = "g45n2nvt"**

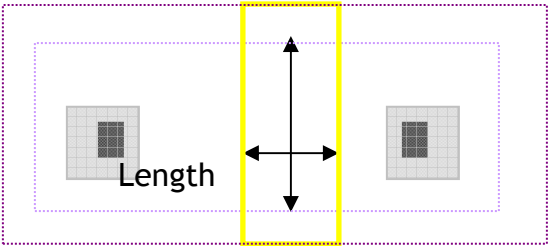
```
; g45n2nvt Instance /NM1 = auLvs device M1
d g45n2nvt D G S B (p D S)
i 1 g45n2nvt D G S B " m 1.0 l 900e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45n2nvt"**

```
MNM1 D G S B g45n2nvt W=4u L=900.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45n2nvt"**

```
c g45n2nvt MOS D B      G B      S B      B B ;;
* 4 pins
* 4 nets
S (p D S);
i NM1 g45n2nvt D G S B ; m 1      l 9.0e-07 w 4e-06 ;
```

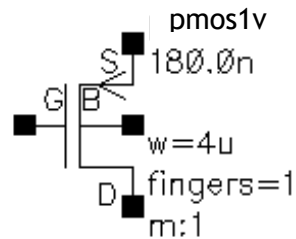
nmos2v_nat - 1.8 volt Native VT NMOS transistor

Device Layers	
Layer	Color and Fill
Oxide_thk	
Oxide	
Nimp & Nzvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE_THK AND OXIDE AND NIMP AND NZVT CONTAINS POLY
G	Poly
D	Oxide_thk AND Oxide AND Nimp AND Nzvt NOT Poly
S	Oxide_thk AND Oxide AND Nimp AND Nzvt NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)
S and D are PERMUTABLE	

21.7 pmos1v datasheet

**Spectre Netlist****Spectre Model Name = "g45p1svt"**

```
PM1 (D G S B) g45p1svt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=152.662 scb=0.0885908 scc=0.0183819
```

DIVA LVS Netlist**DIVA Device Name = "g45p1svt"**

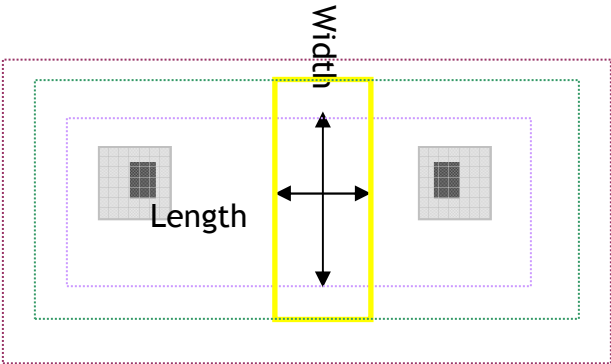
```
; g45p1svt Instance /PM1 = auLvs device M1
d g45p1svt D G S B (p D S)
i 1 g45p1svt D G S B " m 1.0 l 100e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45p1svt"**

```
MPM1 D G S B g45p1svt W=4u L=100.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45p1svt"**

```
c g45p1svt MOS D B      G B      S B      B B  ;;
* 4 pins
* 4 nets
S (p D S);
i MP1 g45p1svt D G S B ; m 1      l 1.0e-07 w 4e-06 ;
```



pmos1v - 1.2 volt nominal VT PMOS transistor

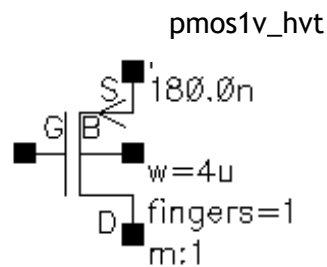
Device Layers	
Layer	Color and Fill
Nwell	
Oxide	
Pimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	NWELL AND OXIDE AND PIMP CONTAINS POLY
G	Poly
D	Nwell AND Oxide AND Pimp NOT Poly
S	Nwell AND Oxide AND Pimp NOT Poly
B	Nwell

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* S and D are PERMUTABLE

21.8 pmos1v_hvt datasheet

**Spectre Netlist****Spectre Model Name = "g45p1hvt"**

```
PM1 (D G S B) g45p1hvt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=152.662 scb=0.0885908 scc=0.0183819
```

DIVA LVS Netlist**DIVA Device Name = "g45p1hvt"**

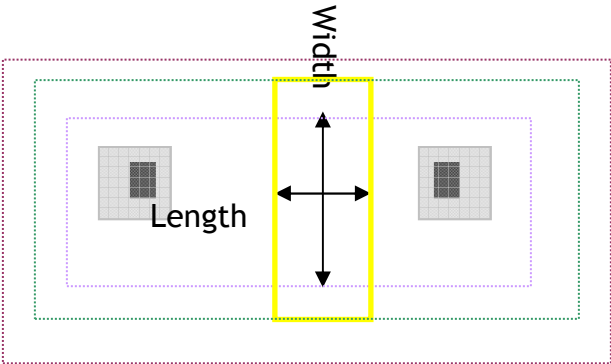
```
; g45p1hvt Instance /PM1 = auLvs device M1
d g45p1hvt D G S B (p D S)
i 1 g45p1hvt D G S B " m 1.0 l 100e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45p1hvt"**

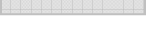
```
MPM1 D G S B g45p1hvt W=4u L=100.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45p1hvt"**

```
c g45p1hvt MOS D B      G B      S B      B B ;;
* 4 pins
* 4 nets
S (p D S);
i MP1 g45p1hvt D G S B ; m 1      l 1.0e-07 w 4e-06 ;
```



pmos1v_hvt - 1.2 volt high VT PMOS transistor

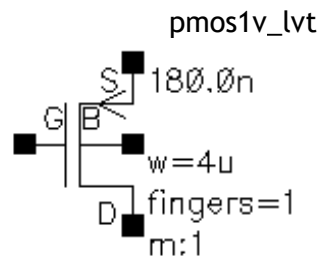
Device Layers	
Layer	Color and Fill
Nwell	
Oxide	
Pimp & Phvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	NWELL AND OXIDE AND PIMP AND PHVT CONTAINS POLY
G	Poly
D	Nwell AND Oxide AND Pimp AND Phvt NOT Poly
S	Nwell AND Oxide AND Pimp AND Phvt NOT Poly
B	Nwell

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

- S and D are PERMUTABLE

21.9 pmos1v_lvt datasheet

**Spectre Netlist****Spectre Model Name = "g45p1lvt"**

```
PM1 (D G S B) g45p1lvt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=152.662 scb=0.0885908 scc=0.0183819
```

DIVA LVS Netlist**DIVA Device Name = "g45p1lvt"**

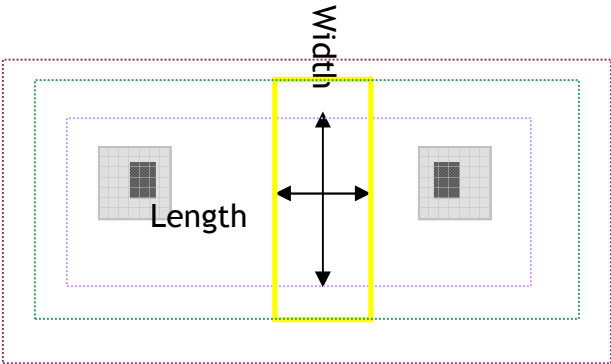
```
; g45p1lvt Instance /PM1 = auLvs device M1
d g45p1lvt D G S B (p D S)
i 1 g45p1lvt D G S B " m 1.0 l 100e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45p1lvt"**

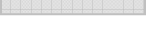
```
MPM1 D G S B g45p1lvt W=4u L=100.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45p1lvt"**

```
c g45p1lvt MOS D B      G B      S B      B B ;;
* 4 pins
* 4 nets
S (p D S);
i MP1 g45p1lvt D G S B ; m 1      l 1.0e-07 w 4e-06 ;
```



pmos1v_lvt - 1.2 volt low VT PMOS transistor

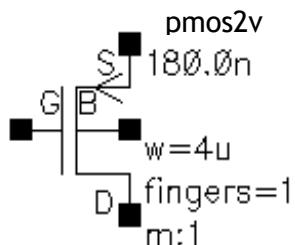
Device Layers	
Layer	Color and Fill
Nwell	
Oxide	
Pimp & Plvt	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	NWELL AND OXIDE AND PIMP AND PHVT CONTAINS POLY
G	Poly
D	Nwell AND Oxide AND Pimp AND Plvt NOT Poly
S	Nwell AND Oxide AND Pimp AND Plvt NOT Poly
B	Nwell

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

- S and D are PERMUTABLE

21.10 pmos2v datasheet

**Spectre Netlist****Spectre Model Name = "g45p2svt"**

```
PM1 (D G S B) g45p2svt w=4u l=180.0n nf=1 as=5.6e-12 ad=5.6e-12 \
      ps=8.28u pd=8.28u m=1 nrs=35m nrd=35m sa=0.14e-9 sb=0.14e-9 sd=0.16e-9
      sca=152.662 scb=0.0885908 scc=0.0183819
```

DIVA LVS Netlist**DIVA Device Name = "g45p2svt"**

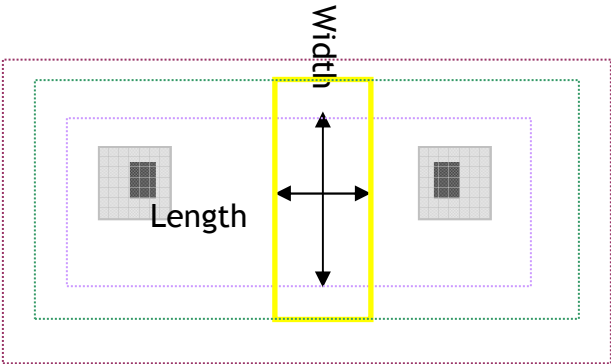
```
; g45p2svt Instance /PM1 = auLvs device M1
d g45p2svt D G S B (p D S)
i 1 g45p2svt D G S B " m 1.0 l 150e-9 w 4e-6 "
```

CDL Netlist**CDL Device Name = "g45p2svt"**

```
MPM1 D G S B g45p2svt W=4u L=150.0n M=1.0
```

Assura Netlist**Assura auLvs Device Name = "g45p2svt"**

```
c g45p2svt MOS D B      G B      S B      B B      ;;
* 4 pins
* 4 nets
S (p D S);
i MP1 g45p2svt D G S B ; m 1      l 1.5e-07 w 4e-06 ;
```

pmos2v - 1.8 volt nominal VT PMOS transistor

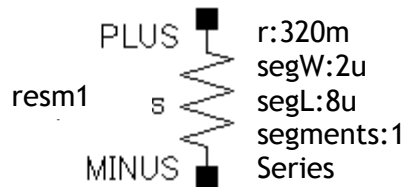
Device Layers	
Layer	Color and Fill
Nwell	
Oxide & Oxide_thk	
Pimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	NWELL AND OXIDE AND OXIDE_THK AND PIMP CONTAINS POLY
G	Poly
D	Nwell AND Oxide AND Oxide_thk AND Pimp NOT Poly
S	Nwell AND Oxide AND Oxide_thk AND Pimp NOT Poly
B	Nwell

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

- S and D are PERMUTABLE

21.11 resm1-resm11 datasheet

**Spectre Netlist****Spectre Model Name = "g45rm1"**

R1 (MINUS PLUS) resm1_pcell1 segL=8u segW=2u

Subckt resm1_pcell1 MINUS PLUS

Parameters segL=8u segW=2u

R0 (PLUS MINUS) g45rm1 l=segL w=segW

Ends resm1_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rm1"**

; g45rm1 Instance /R1 = auLvs device R1

d g45rm1 PLUS MINUS (p PLUS MINUS)

i 1 g45rm1 PLUS MINUS " r 320e-3 w 2e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rm1"**

RR1 PLUS MINUS \$[g45rm1] r=320m w=2u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rm1 "**

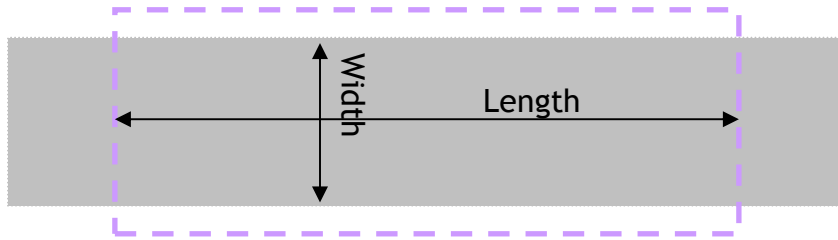
c g45rm1 RES PLUS B MINUS B ;;

* 2 pins

* 2 nets

S (p PLUS MINUS) ;

i R1 g45rm1 PLUS MINUS ; r 0.32 w 2e-06 l=8e-06;

**resm1 - Metal resistor**

Device Layers	
Layer	Color and Fill
M1Resdum (Marker Layer)	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	METAL1 AND M1RESDUM
PLUS	Metal1 TOUCHING M1Resdum
MINUS	Metal1 TOUCHING M1Resdum

LVS Comparison	
Parameter	Calculation
Length	METAL1 AND M1RESDUM LENGTH (ILLUSTRATED ABOVE)
Width	Metal1 AND M1Resdum Width (illustrated above)
Resistance	sheet resistance * Length / Width

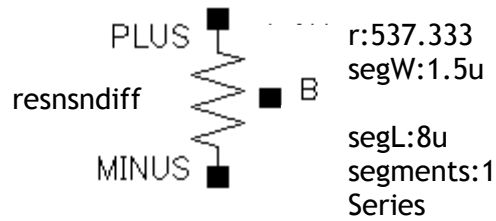
* PLUS and MINUS are PERMUTABLE

* The same format is used for resm2-resm11 as is used with resm1

* Only the metal sheet rho and device name changes

* Only the metal layer and recognition layer changes

21.12 resnsndiff datasheet

**Spectre Netlist****Spectre Model Name = "g45rnsnd"**

R1 (B MINUS PLUS) resnsndiff_pcell1 segL=8u segW=1.5u

Subckt resnsndiff_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnsnd l=segL w=segW

Ends resnsndiff_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rnsnd"**

; g45rnsnd Instance /R1 = auLvs device R1

d g45rnsnd PLUS MINUS B (p PLUS MINUS)

i 1 g45rnsnd PLUS MINUS B " r 537.333 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rnsnd"**

RR1 PLUS MINUS B \$[g45rnsnd] r=537.333 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rnsnd "**

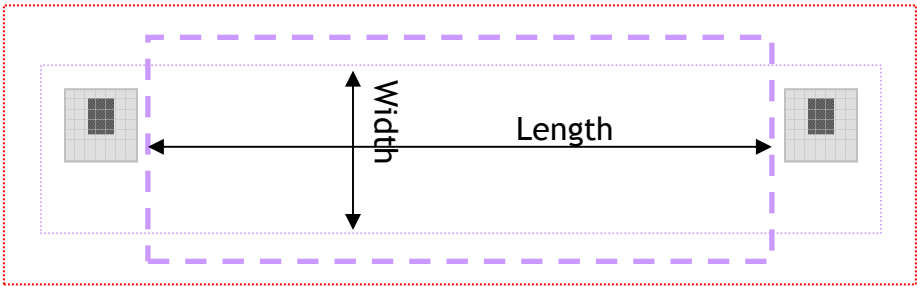
c g45rnsnd RES PLUS B MINUS B B I ;;

* 3 pins

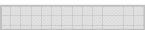
* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnsnd PLUS MINUS B ; r 537.333 w 1.5e-06 l=8e-06;



resnsndiff - N+ diffused resistor without salicide

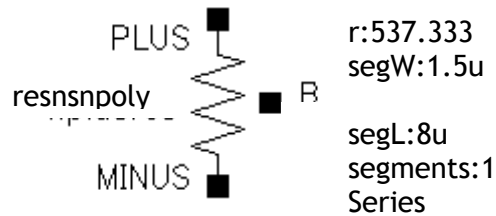
Device Layers	
Layer	Color and Fill
Oxide	
Nimp	
SiProt & Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP AND SIPROT AND RESDUM
PLUS	Oxide AND Nimp NOT Resdum
MINUS	Oxide AND Nimp NOT Resdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	SIPROT LENGTH (ILLUSTRATED ABOVE)
Width	Oxide Width (illustrated above)
Resistance	sheet resistance * Length / Width

* PLUS and MINUS are PERMUTABLE

21.13 resnsnpoly datasheet

**Spectre Netlist**

Spectre Model Name = "g45rnsnp"

R1 (B MINUS PLUS) resnsnpoly_pcell1 segL=8u segW=1.5u

Subckt resnsnpoly_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnsnp l=segL w=segW

Ends resnsnpoly_pcell1

DIVA LVS Netlist

DIVA Device Name = "g45rnsnp"

; g45rnsnp Instance /R1 = auLvs device R1

d g45rnsnp PLUS MINUS B (p PLUS MINUS)

i 1 g45rnsnp PLUS MINUS B " r 537.333 w 1.5e-6 l 8e-6"

CDL Netlist

CDL Device Name = "g45rnsnp"

RR1 PLUS MINUS B \$[g45rnsnp] r=537.333 w=1.5u l=8u

Assura Netlist

Assura auLvs Device Name = " g45rnsnp "

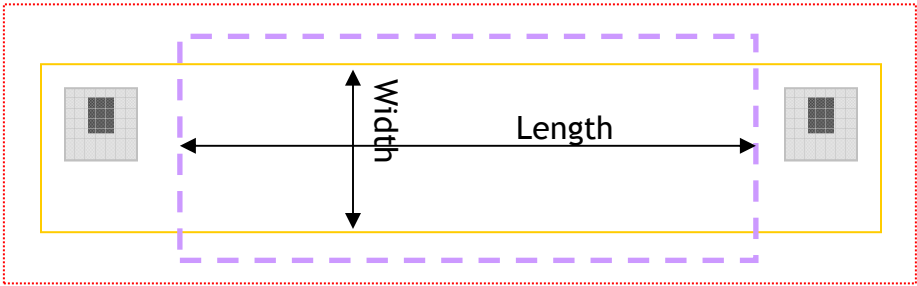
c g45rnsnp RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnsnp PLUS MINUS B ; r 537.333 w 1.5e-06 l=8e-06;



resnsnpoly - N+ poly resistor without salicide

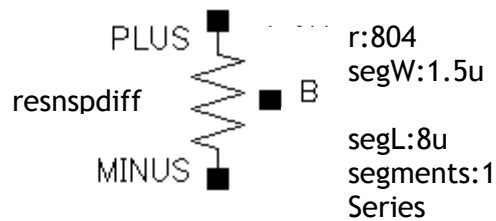
Device Layers	
Layer	Color and Fill
Poly	
Nimp	
SiProt & Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	POLY AND NIMP AND SIPROT AND RESDUM
PLUS	Poly AND Nimp NOT Resdum
MINUS	Poly AND Nimp NOT Resdum

LVS Comparison	
Parameter	Calculation
Length	SIPROT LENGTH (ILLUSTRATED ABOVE)
Width	Poly Width (illustrated above)
Resistance	sheet resistance * Length / Width

* PLUS and MINUS are PERMUTABLE

21.14 resnspdiff datasheet

**Spectre Netlist****Spectre Model Name = "g45rnspd"**

R1 (B MINUS PLUS) resnspdiff_pcell1 segL=8u segW=1.5u

Subckt resnspdiff_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnspd l=segL w=segW

Ends resnspdiff_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rnspd"**

; g45rnspd Instance /R1 = auLvs device R1

d g45rnspd PLUS MINUS B (p PLUS MINUS)

i 1 g45rnspd PLUS MINUS B " r 804 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rnspd"**

RR1 PLUS MINUS B \$[g45rnspd] r=804 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rnspd "**

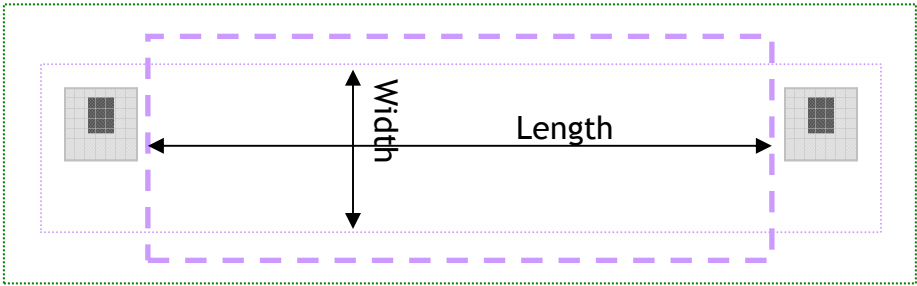
c g45rnspd RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnspd PLUS MINUS B ; r 804 w 1.5e-06 l=8e-06;



resnspdiff - P+ diffused resistor without salicide

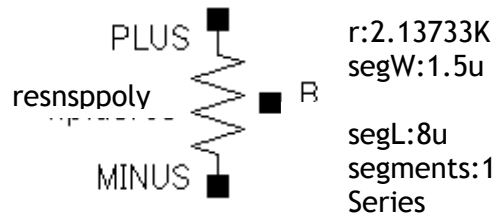
Device Layers	
Layer	Color and Fill
Oxide	
Pimp	
SiProt & Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND PIMP AND SIPROT AND RESDUM
PLUS	Oxide AND Pimp NOT Resdum
MINUS	Oxide AND Pimp NOT Resdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	SIPROT LENGTH (ILLUSTRATED ABOVE)
Width	Oxide Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.15 resnsppoly datasheet

**Spectre Netlist****Spectre Model Name = "g45rnspp"**

R1 (B MINUS PLUS) resnsppoly_pcell1 segL=8u segW=1.5u

Subckt resnsppoly_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnspp l=segL w=segW

Ends resnsppoly_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rnspp"**

; g45rnspp Instance /R1 = auLvs device R1

d g45rnspp PLUS MINUS B (p PLUS MINUS)

i 1 g45rnspp PLUS MINUS B " r 2137.33 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rnspp"**

RR1 PLUS MINUS B \$[g45rnspp] r=2137.33 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rnspp "**

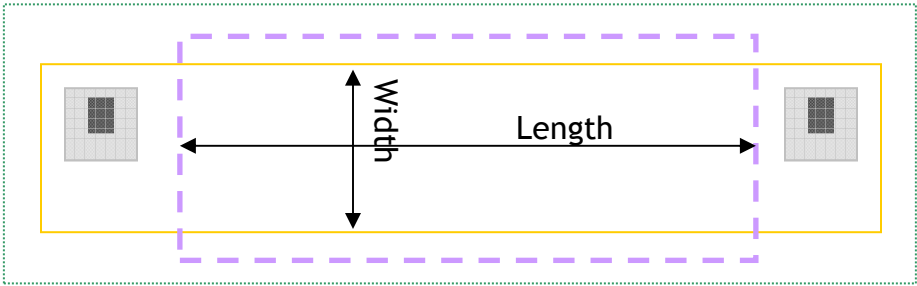
c g45rnspp RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnspp PLUS MINUS B ; r 2137.33 w 1.5e-06 l=8e-06;



resnsppoly - P+ poly resistor without salicide

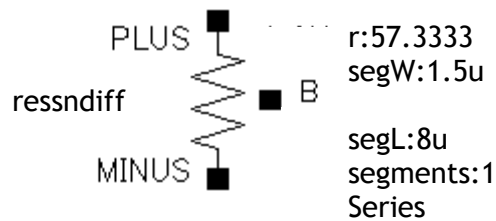
Device Layers	
Layer	Color and Fill
Poly	
Pimp	
SiProt & Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	POLY AND PIMP AND SIPROT AND RESDUM
PLUS	Poly AND Pimp NOT Resdum
MINUS	Poly AND Pimp NOT Resdum

LVS Comparison	
Parameter	Calculation
Length	SIPROT LENGTH (ILLUSTRATED ABOVE)
Width	Poly Width (illustrated above)
Resistance	sheet resistance * Length / Width

* PLUS and MINUS are PERMUTABLE

21.16 resndiff datasheet

**Spectre Netlist****Spectre Model Name = "g45rsnd"**

R1 (B MINUS PLUS) resndiff_pcell1 segL=8u segW=1.5u

Subckt resndiff_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rsnd l=segL w=segW

Ends resndiff_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rsnd"**

; g45rsnd Instance /R1 = auLvs device R1

d g45rsnd PLUS MINUS B (p PLUS MINUS)

i 1 g45rsnd PLUS MINUS B " r 57.3333 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rsnd"**

RR1 PLUS MINUS B \$[g45rsnd] r=57.3333 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rsnd "**

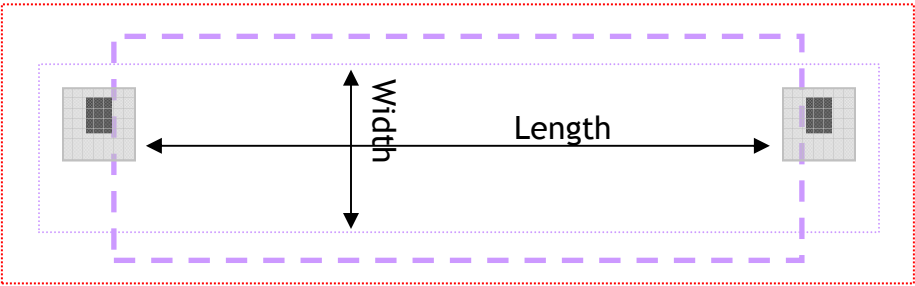
c g45rsnd RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rsnd PLUS MINUS B ; r 57.3333 w 1.5e-06 l=8e-06;



ressndiff - N+ diffused resistor with salicide

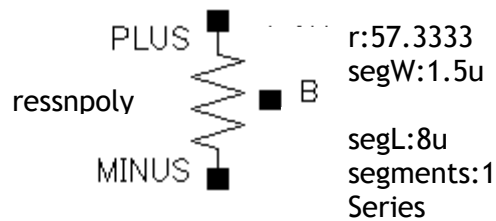
Device Layers	
Layer	Color and Fill
Oxide	
Nimp	
Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NIMP AND RESDUM NOT SIPROT
PLUS	Oxide AND Nimp NOT Resdum
MINUS	Oxide AND Nimp NOT Resdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	RESDUM LENGTH (ILLUSTRATED ABOVE)
Width	Oxide Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.17 ressnpoly datasheet

**Spectre Netlist****Spectre Model Name = "g45rsnp"**

R1 (B MINUS PLUS) ressnpoly_pcell1 segL=8u segW=1.5u

Subckt ressnpoly_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rsnp l=segL w=segW

Ends ressnpoly_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rsnp"**

; g45rsnp Instance /R1 = auLvs device R1

d g45rsnp PLUS MINUS B (p PLUS MINUS)

i 1 g45rsnp PLUS MINUS B " r 57.3333 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rsnp"**

RR1 PLUS MINUS B \$[g45rsnp] r=57.3333 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rsnp "**

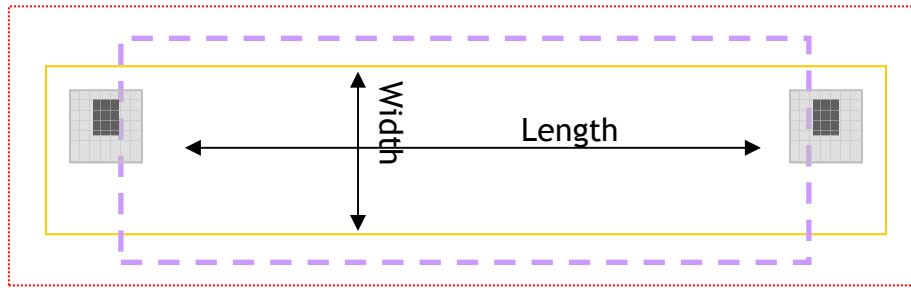
c g45rsnp RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rsnp PLUS MINUS B ; r 57.3333 w 1.5e-06 l=8e-06;



ressnpoly - N+ poly resistor with salicide

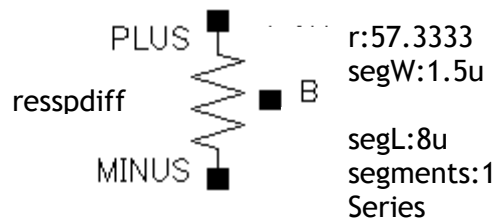
Device Layers	
Layer	Color and Fill
Poly	
Nimp	
Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	POLY AND NIMP AND RESDUM NOT SIPROT
PLUS	Poly AND Nimp NOT Resdum
MINUS	Poly AND Nimp NOT Resdum

LVS Comparison	
Parameter	Calculation
Length	RESDUM LENGTH (ILLUSTRATED ABOVE)
Width	Poly Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.18 resspdiff datasheet

**Spectre Netlist****Spectre Model Name = "g45rspd"**

R1 (B MINUS PLUS) resspdiff_pcell1 segL=8u segW=1.5u

Subckt resspdiff_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rspd l=segL w=segW

Ends resspdiff_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rspd"**

; g45rspd Instance /R1 = auLvs device R1

d g45rspd PLUS MINUS B (p PLUS MINUS)

i 1 g45rspd PLUS MINUS B " r 57.3333 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rspd"**

RR1 PLUS MINUS B \$[g45rspd] r=57.3333 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rspd "**

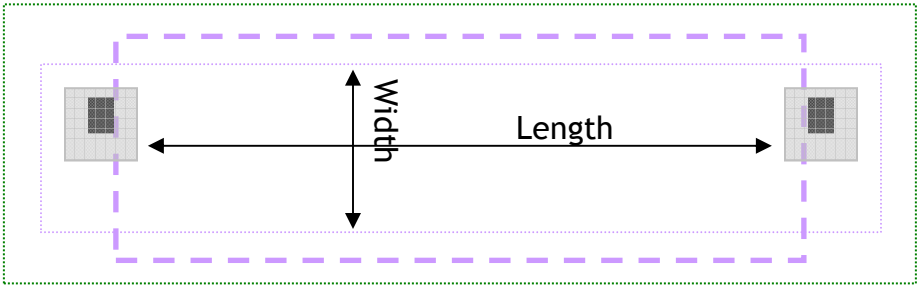
c g45rspd RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rspd PLUS MINUS B ; r 57.3333 w 1.5e-06 l=8e-06;



resspdiff - P+ diffused resistor with salicide

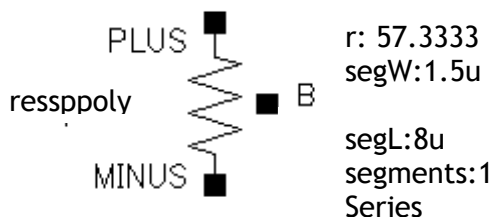
Device Layers	
Layer	Color and Fill
Oxide	
Pimp	
Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND PIMP AND RESDUM NOT SIPROT
PLUS	Oxide AND Pimp NOT Resdum
MINUS	Oxide AND Pimp NOT Resdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	RESDUM LENGTH (ILLUSTRATED ABOVE)
Width	Oxide Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.19 resppoly datasheet

**Spectre Netlist****Spectre Model Name = "g45rspp"**

R1 (B MINUS PLUS) resppoly_pcell1 segL=8u segW=1.5u

Subckt resppoly_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rspp l=segL w=segW

Ends resppoly_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rspp"**

; g45rspp Instance /R1 = auLvs device R1

d g45rspp PLUS MINUS B (p PLUS MINUS)

i 1 g45rspp PLUS MINUS B " r 57.3333 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rspp"**

RR1 PLUS MINUS B \$[g45rspp] r=57.3333 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rspp "**

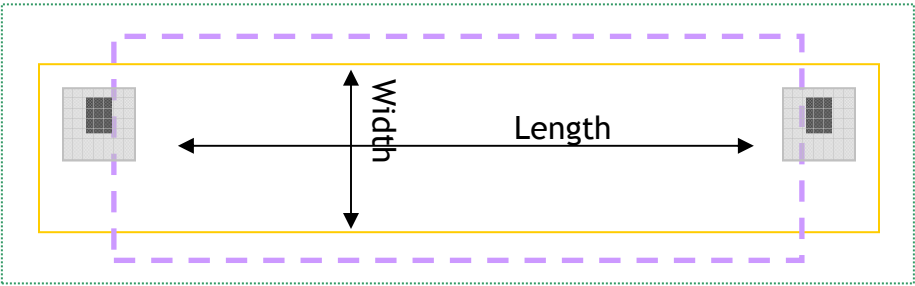
c g45rspp RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rspp PLUS MINUS B ; r 57.3333 w 1.5e-06 l=8e-06;



ressppoly - P+ poly resistor with salicide

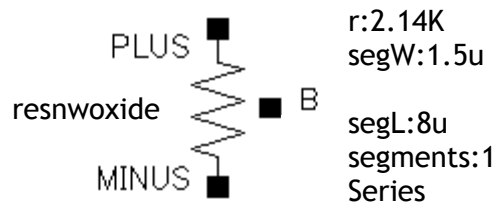
Device Layers	
Layer	Color and Fill
Poly	
Pimp	
Resdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	POLY AND PIMP AND RESDUM NOT SIPROT
PLUS	Poly AND Pimp NOT Resdum
MINUS	Poly AND Pimp NOT Resdum

LVS Comparison	
Parameter	Calculation
Length	RESDUM LENGTH (ILLUSTRATED ABOVE)
Width	Poly Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.20 resnwoxide datasheet

**Spectre Netlist****Spectre Model Name = "g45rnwo"**

R1 (B MINUS PLUS) resnwoxide_pcell1 segL=8u segW=1.5u

Subckt resnwoxide_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnwo l=segL w=segW

Ends resnwoxide_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rnwo"**

; g45rnwo Instance /R1 = auLvs device R1

d g45rnwo PLUS MINUS B (p PLUS MINUS)

i 1 g45rnwo PLUS MINUS B " r 2140 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rnwo"**

RR1 PLUS MINUS B \$[g45rnwo] r=2140 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rnwo "**

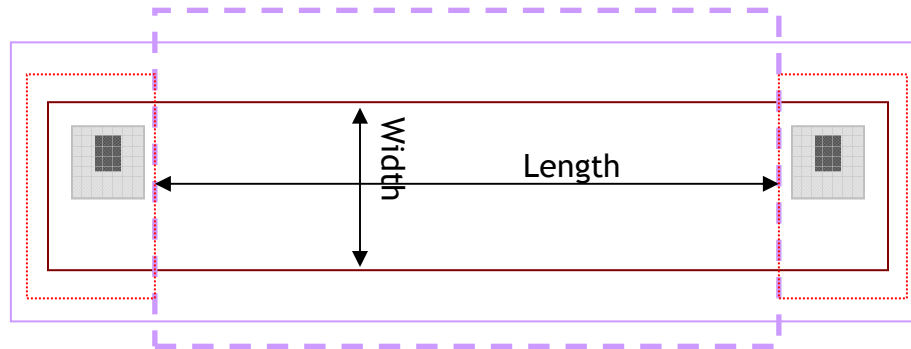
c g45rnwo RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnwo PLUS MINUS B ; r 2140 w 1.5e-06 l=8e-06;



resnwoxide - Nwell Resistor within diffusion

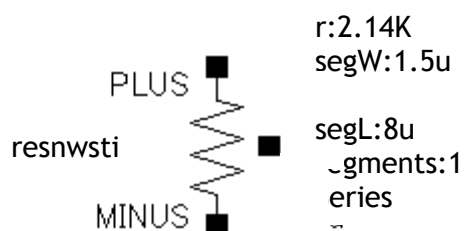
Device Layers	
Layer	Color and Fill
Oxide	
Nwell	
Nimp	
ResWdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	OXIDE AND NWELL AND SIPROT AND RESWDUM
PLUS	Oxide AND Nwell NOT ResWdum
MINUS	Oxide AND Nwell NOT ResWdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	RESWDUM LENGTH (ILLUSTRATED ABOVE)
Width	Nwell Width (illustrated above)
Resistance	sheet resistance * Length / Width

* PLUS and MINUS are PERMUTABLE

21.21 resnwsti datasheet

**Spectre Netlist****Spectre Model Name = "g45rnwi"**

R1 (B MINUS PLUS) resnwsti_pcell1 segL=8u segW=1.5u

Subckt resnwsti_pcell1 B MINUS PLUS

Parameters segL=8u segW=1.5u

R0 (PLUS MINUS B) g45rnwi l=segL w=segW

Ends resnwsti_pcell1

DIVA LVS Netlist**DIVA Device Name = "g45rnwi"**

; g45rnwi Instance /R1 = auLvs device R1

d g45rnwi PLUS MINUS B (p PLUS MINUS)

i 1 g45rnwi PLUS MINUS B " r 2140 w 1.5e-6 l 8e-6"

CDL Netlist**CDL Device Name = "g45rnwi"**

RR1 PLUS MINUS B \$[g45rnwi] r=2140 w=1.5u l=8u

Assura Netlist**Assura auLvs Device Name = " g45rnwi "**

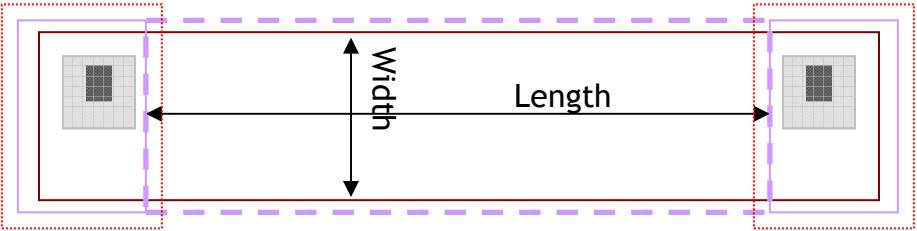
c g45rnwi RES PLUS B MINUS B B I ;;

* 3 pins

* 3 nets

S (p PLUS MINUS) ;

i R1 g45rnwi PLUS MINUS B ; r 2140 w 1.5e-06 l=8e-06;



resnwsti - Nwell Resistor Under STI

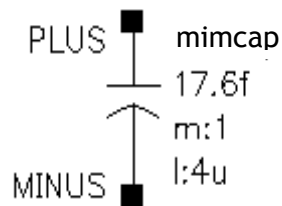
Device Layers	
Layer	Color and Fill
Oxide	
Nwell	
Nimp	
ResWdum (Marker Layer)	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	NWELL AND RESWDUM
PLUS	Nwell NOT ResWdum
MINUS	Nwell NOT ResWdum
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	RESWDUM LENGTH (ILLUSTRATED ABOVE)
Width	Nwell Width (illustrated above)
Resistance	sheet resistance * Length / Width

- PLUS and MINUS are PERMUTABLE

21.22 mimcap datasheet

**Spectre Netlist****Spectre Model Name = "g45cmim"**

C1 (PLUS MINUS) g45cmim area=16e-12 perim=16e-6 m=1

DIVA LVS Netlist**DIVA Device Name = "g45cmim"**

; g45cmim Instance /C1 = auLvs device C1

d g45cmim PLUS MINUS (p PLUS MINUS)

i 1 g45cmim PLUS MINUS " area 16e-12 m 1.0 "

CDL Netlist**CDL Device Name = "g45cmim"**

CC1 PLUS MINUS 16e-12 \$[g45cmim] m=1

Assura Netlist**Assura auLvs Device Name = "g45cmim"**

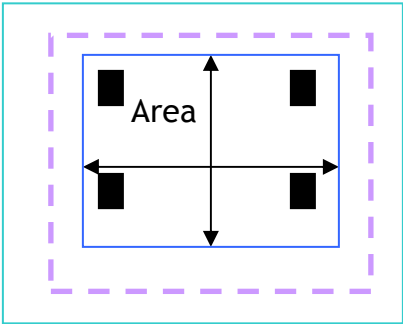
c g45cmim CAP PLUS B MINUS B ;;

* 2 pins

* 2 nets

S (p PLUS MINUS)

i C1 g45cmim PLUS MINUS ; area 16e-12 m 1 ;



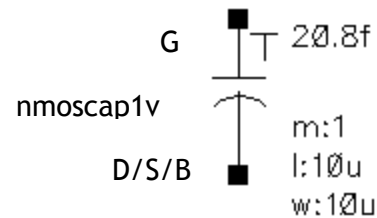
mimcap - Metal / Metal capacitor

Device Layers	
Layer	Color and Fill
Metal 2	
CapMetal	
Via2	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	CAPMETAL AND METAL2
PLUS	CapMetal
MINUS	Metal2 UNDER CapMetal

LVS Comparison	
Parameter	Calculation
Area	AREA OF CAPMETAL (ILLUSTRATED ABOVE)
Perimeter	Perimeter of CapMetal

21.23 nmoscap1v datasheet

**Spectre Netlist****Spectre Model Name = "g45ncap1"**

M1 (D G S B) g45ncap1 w=10u l=10u m=(1)*(1)

DIVA LVS Netlist**DIVA Device Name = "g45ncap1"**

; g45ncap1 Instance /M1 = auLvs device M1

d g45ncap1 D G S B (p D S)

i 1 g45ncap1 D G S B " m 1.0 l 10e-6 w 10e-6 "

CDL Netlist**CDL Device Name = "g45ncap1"**

MM1 D G S B g45ncap1 W=10u L=10u M=1

Assura Netlist**Assura auLvs Device Name = "g45ncap1"**

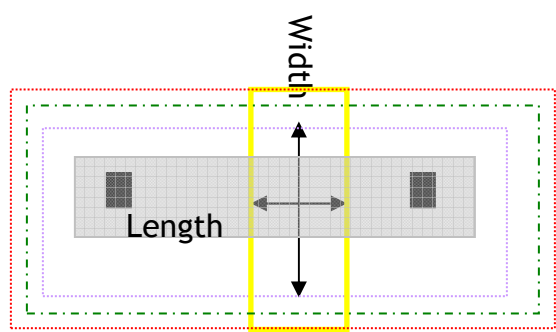
c g45ncap1 MOS D B G B S B B B ;;

* 4 pins

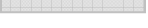
* 4 nets

S (p D S);

i M1 g45ncap1 D G S B ; m 1 l 1e-05 w 1e-05 ;



nmoscap1v - 1.2 volt NMOS capacitor

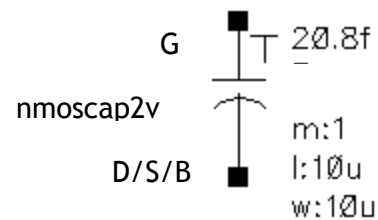
Device Layers	
Layer	Color and Fill
Capdum	
Oxide	
Nimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	CAPDUM AND OXIDE AND NIMP CONTAINS POLY
TOP	Poly
BOT	Capdum AND Oxide AND Nimp NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* Nmos Source, Drain, and Body are Shorted Together

21.24 nmoscap2v datasheet

**Spectre Netlist****Spectre Model Name = "g45ncap2"**

M1 (D G S B) g45ncap2 w=10u l=10u m=(1)*(1)

DIVA LVS Netlist**DIVA Device Name = "g45ncap2"**

; g45ncap2 Instance /M1 = auLvs device M1

d g45ncap2 D G S B (p D S)

i 1 g45ncap2 D G S B " m 1.0 l 10e-6 w 10e-6 "

CDL Netlist**CDL Device Name = "g45ncap2"**

MM1 D G S B g45ncap2 W=10u L=10u M=1

Assura Netlist**Assura auLvs Device Name = "g45ncap2"**

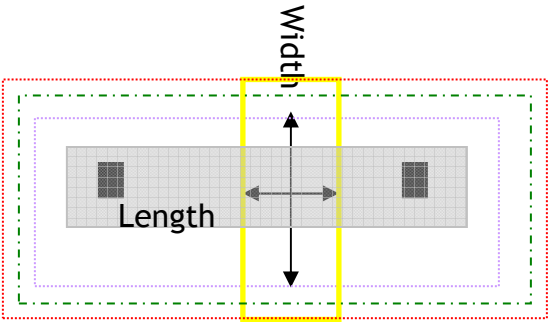
c g45ncap2 MOS D B G B S B B B ;;

* 4 pins

* 4 nets

S (p D S);

i M1 g45ncap2 D G S B ; m 1 l 1e-05 w 1e-05 ;



nmoscap2v - 1.8 volt NMOS capacitor

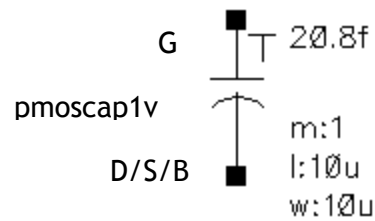
Device Layers	
Layer	Color and Fill
Capdum	
Oxide & Oxide_thk	
Nimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	CAPDUM AND OXIDE AND OXIDE_THK AND NIMP CONTAINS POLY
TOP	Poly
BOT	Capdum AND Oxide AND Oxide_thk AND Nimp NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* Nmos Source, Drain, and Body are Shorted Together

21.25 pmoscap1v datasheet

**Spectre Netlist****Spectre Model Name = "g45pcap1"**

M1 (D G S B) g45pcap1 w=10u l=10u m=(1)*(1)

DIVA LVS Netlist**DIVA Device Name = "g45pcap1"**

; g45pcap1 Instance /M1 = auLvs device M1

d g45pcap1 D G S B (p D S)

i 1 g45pcap1 D G S B " m 1.0 l 10e-6 w 10e-6 "

CDL Netlist**CDL Device Name = "g45pcap1"**

MM1 D G S B g45pcap1 W=10u L=10u M=1

Assura Netlist**Assura auLvs Device Name = "g45pcap1"**

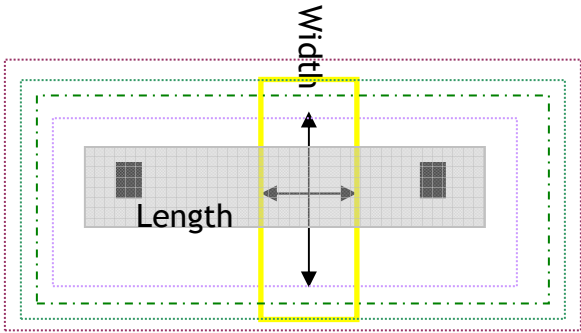
c g45pcap1 MOS D B G B S B B B ;;

* 4 pins

* 4 nets

S (p D S);

i M1 g45pcap1 D G S B ; m 1 l 1e-05 w 1e-05 ;



pmoscap1v - 1.2 volt PMOS capacitor

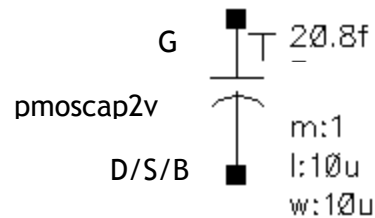
Device Layers	
Layer	Color and Fill
Nwell	
Capdum	
Oxide	
Pimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	CAPDUM AND OXIDE AND PIMP AND NWELL CONTAINS POLY
TOP	Poly
BOT	Capdum AND Oxide AND Pimp AND Nwell NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* Pmos Source, Drain, and Body are Shorted Together

21.26 pmoscap2v datasheet

**Spectre Netlist****Spectre Model Name = "g45pcap2"**

M1 (D G S B) g45pcap2 w=10u l=10u m=(1)*(1)

DIVA LVS Netlist**DIVA Device Name = "g45pcap2"**

; g45pcap2 Instance /M1 = auLvs device M1

d g45pcap2 D G S B (p D S)

i 1 g45pcap2 D G S B " m 1.0 l 10e-6 w 10e-6 "

CDL Netlist**CDL Device Name = "g45pcap2"**

MM1 D G S B g45pcap2 W=10u L=10u M=1

Assura Netlist**Assura auLvs Device Name = "g45pcap2"**

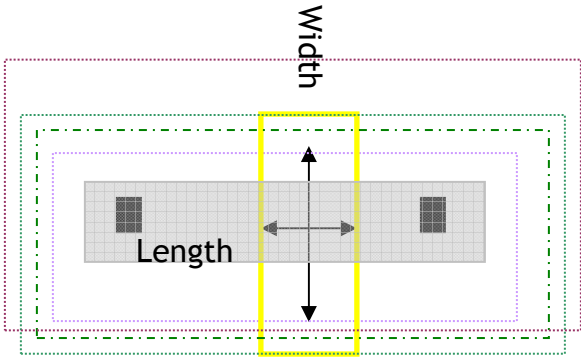
c g45pcap2 MOS D B G B S B B B ;;

* 4 pins

* 4 nets

S (p D S);

i M1 g45pcap2 D G S B ; m 1 l 1e-05 w 1e-05 ;



pmoscap2v - 1.8 volt PMOS capacitor

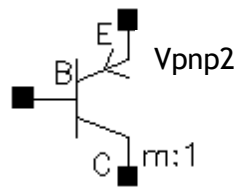
Device Layers	
Layer	Color and Fill
Nwell	
Capdum	
Oxide	
Pimp	
Poly	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	CAPDUM AND OXIDE AND PIMP AND NWELL CONTAINS POLY
TOP	Poly
BOT	Capdum AND Oxide AND Pimp AND Nwell NOT Poly
B	Substrate

LVS Comparison	
Parameter	Calculation
Length	POLY INTERSECTING OXIDE (ILLUSTRATED ABOVE)
Width	Poly inside Oxide (illustrated above)

* Pmos Source, Drain, and Body are Shorted Together

21.27 vnpnp2 datasheet

**Spectre Netlist****Spectre Model Name = "g45vnpnp2"**

Q0 (C B E) g45vnpnp2 area=4 m=1

DIVA LVS Netlist**DIVA Device Name = "g45vnpnp2"**

; g45vnpnp2 Instance /Q0 = auLvs device Q0

d vnpnp C B E

i 0 g45vnpnp2 C B E " area 4.0 m 1.0 "

CDL Netlist**CDL Device Name = " g45vnpnp2"**

QQ0 C B E g45vnpnp2 M=1 area=4.0

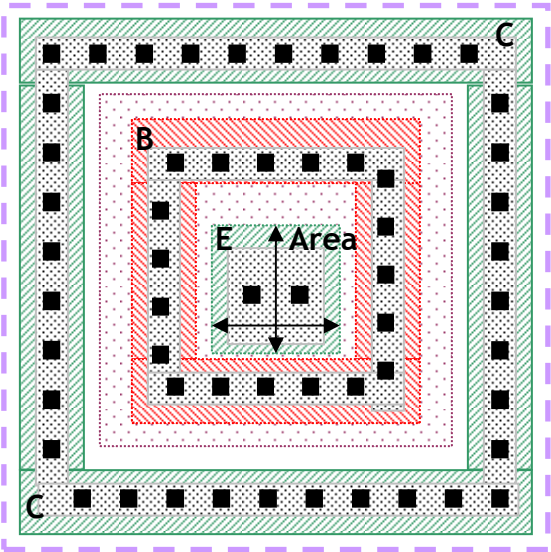
Assura Netlist**Assura auLvs Device Name = " g45vnpnp2"**

c g45vnpnp2 BJT C B B B E B ;;

* 3 pins

* 3 nets

i Q0 g45vnpnp2 C B E; area 4 m 1 ;



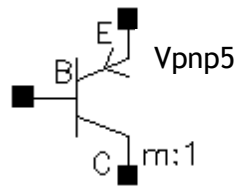
vpnp2 - 1.2 volt vertical substrate PNP with 2x2 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.28 vnp5 datasheet

**Spectre Netlist**

Spectre Model Name = “g45vnp5”

Q0 (C B E) g45vnp5 area=25 m=1

DIVA LVS Netlist

DIVA Device Name = “g45vnp5”

; g45vnp5 Instance /Q0 = auLvs device Q0

d vnp C B E

i 0 g45vnp5 C B E " area 25.0 m 1.0 "

CDL Netlist

CDL Device Name = “ g45vnp5”

QQ0 C B E g45vnp5 M=1 area=25.0

Assura Netlist

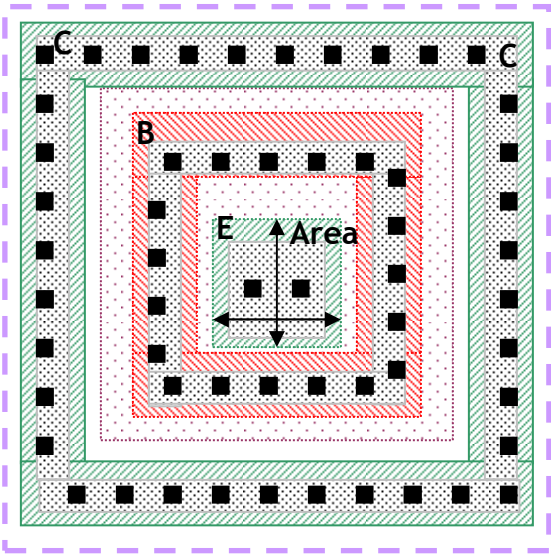
Assura auLvs Device Name = “ g45vnp5”

c g45vnp5 BJT C B B B E B ;;

* 3 pins

* 3 nets

i Q0 g45vnp5 C B E; area 25 m 1 ;



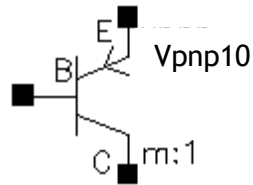
vpnp5 - 1.2 volt vertical substrate PNP with 5x5 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.29 vnpnp10 datasheet

**Spectre Netlist**

Spectre Model Name = “g45vnpnp10”

Q0 (C B E) g45vnpnp10 area=100 m=1

DIVA LVS Netlist

DIVA Device Name = “g45vnpnp10”

; g45vnpnp10 Instance /Q0 = auLvs device Q0

d vnpnp C B E

i 0 g45vnpnp10 C B E " area 100.0 m 1.0 "

CDL Netlist

CDL Device Name = “ g45vnpnp10”

QQ0 C B E g45vnpnp10 M=1 area=100.0

Assura Netlist

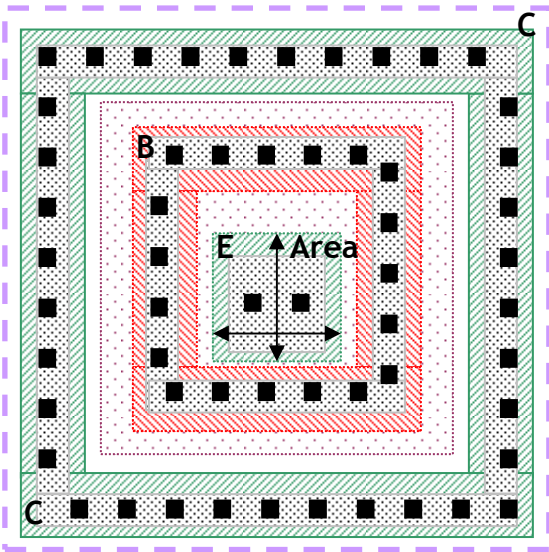
Assura auLvs Device Name = “ g45vnpnp10”

c g45vnpnp10 BJT C B B B E B ;;

* 3 pins

* 3 nets

i Q0 g45vnpnp10 C B E; area 100 m 1 ;



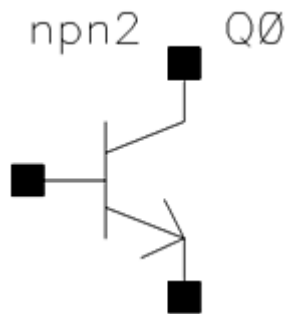
vpnp10 - 1.2 volt vertical substrate PNP with 10x10 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.30 npn2 datasheet

**Spectre Netlist**

Spectre Model Name = "g45vnnpn2"

Q1 (C B E) g45vnnpn2 area=4.0 m=1

DIVA LVS Netlist

DIVA Device Name = "g45vnnpn2"

; g45vnnpn2 Instance /Q1 = auLvs device Q1

d vnnp C B E

i 1 g45vnnpn2 C B E " area 4.0 m 1.0 "

CDL Netlist

CDL Device Name = "g45vnnpn2"

QQ1 C B E g45vnnpn2 area=4.0 m=1

Assura Netlist

Assura auLvs Device Name = "g45vnnpn2"

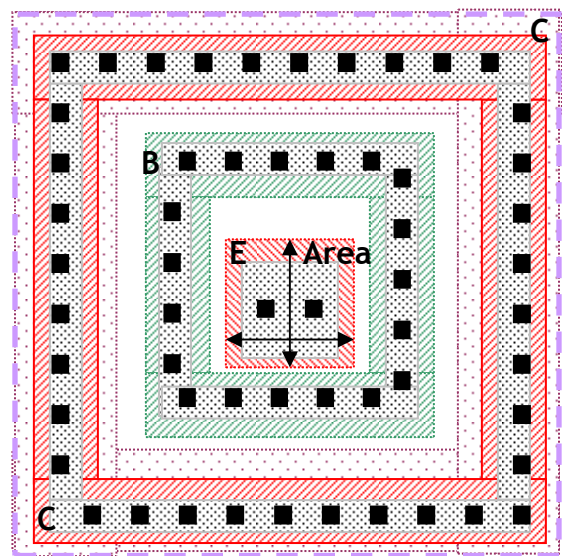
c g45vnnpn2 BJT C B B B E B ;;

* 3 pins

* 3 nets

* 0 instances

i Q1 g45vnnpn2 C B E ; area 4.0 m 1 ;



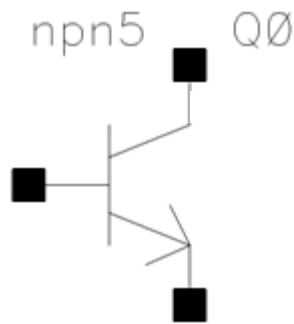
npn2 - 1.2 volt vertical substrate NPN with 2x2 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.31 npn5 datasheet

**Spectre Netlist****Spectre Model Name = "g45vnnpn5"**

Q1 (C B E) g45vnnpn5 area=25.0 m=1

DIVA LVS Netlist**DIVA Device Name = "g45vnnpn5"**

; g45vnnpn5 Instance /Q1 = auLvs device Q1

d vnnpn C B E

i 1 g45vnnpn5 C B E " area 25.0 m 1.0 "

CDL Netlist**CDL Device Name = "g45vnnpn5"**

QQ1 C B E g45vnnpn5 area=25.0 m=1

Assura Netlist**Assura auLvs Device Name = "g45vnnpn5"**

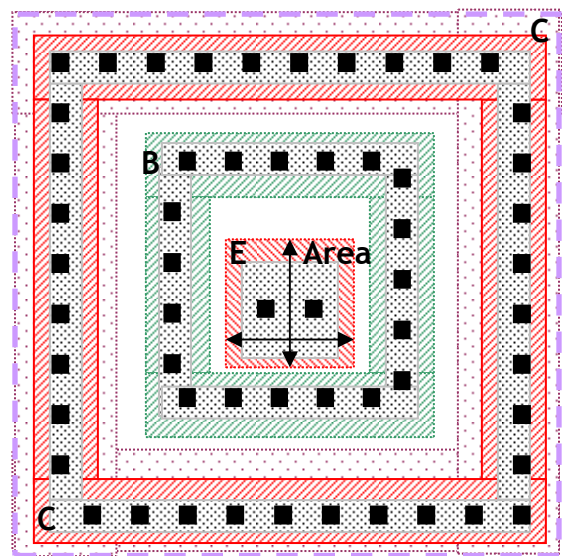
c g45vnnpn5 BJT C B B B E B ;;

* 3 pins

* 3 nets

* 0 instances

i Q1 g45vnnpn5 C B E ; area 25.0 m 1 ;



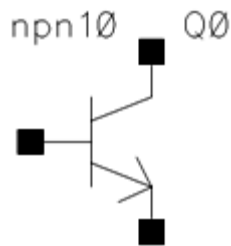
npn5 - 1.2 volt vertical substrate NPN with 5x5 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.32 npn10 datasheet

**Spectre Netlist****Spectre Model Name = "g45vnnpn10"**

Q1 (C B E) g45vnnpn10 area=100.0 m=1

DIVA LVS Netlist**DIVA Device Name = "g45vnnpn10"**

; g45vnnpn10 Instance /Q1 = auLvs device Q1

d vnnpn C B E

i 1 g45vnnpn10 C B E " area 100.0 m 1.0 "

CDL Netlist**CDL Device Name = "g45vnnpn10"**

QQ1 C B E g45vnnpn10 area=100.0 m=1

Assura Netlist**Assura auLvs Device Name = "g45vnnpn10"**

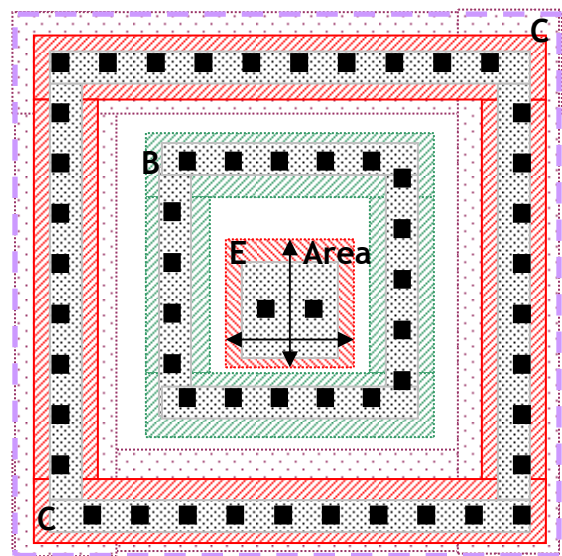
c g45vnnpn10 BJT C B B B E B ;;

* 3 pins

* 3 nets

* 0 instances

i Q1 g45vnnpn10 C B E ; area 100.0 m 1 ;



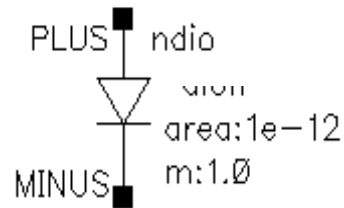
npn10 - 1.2 volt vertical substrate NPN with 10x10 fixed emitter

Device Layers	
Layer	Color and Fill
BJTdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal3	

Device Derivation	
Device	Layer Derivation
Recognition	BJTDUMMY CONTAINS NIMP AND PIMP
E	BJTDummy AND Pimp And Oxide AND Nwell
B	BJTDummy AND Nimp And Oxide AND Nwell
C	BJTDummy AND Pimp And Oxide ANDNOT Nwell

LVS Comparison	
Parameter	Calculation
Area	AREA OF EMITTER (ILLUSTRATED ABOVE)

21.33 ndio datasheet

**Spectre Netlist****Spectre Model Name = "g45nd1svt"**

D0 (PLUS MINUS) g45nd1svt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45nd1svt"**

; g45nd1svt Instance /D0 = auLvs device D0

d g45nd1svt PLUS MINUS

i 0 g45nd1svt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45nd1svt"**

DD0 PLUS MINUS g45nd1svt 160f 1.6u m=1

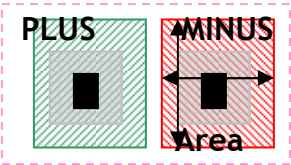
Assura Netlist**Assura auLvs Device Name = "g45nd1svt"**

c g45nd1svt DIO PLUS B MINUS B ;;

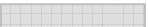
* 2 pins

* 2 nets

i D0 g45nd1svt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



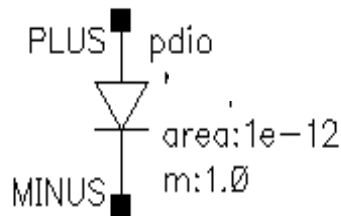
ndio - 1.2 volt N+/Psub diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of MINUS (illustrated above)

21.34 pdio datasheet

**Spectre Netlist****Spectre Model Name = "g45pd1svt"**

D0 (PLUS MINUS) g45pd1svt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45pd1svt"**

; g45pd1svt Instance /D0 = auLvs device D0

d g45pd1svt PLUS MINUS

i 0 g45pd1svt PLUS MINUS " area 160e-15 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45pd1svt"**

DD0 PLUS MINUS g45pd1svt 160f 1.6u m=1

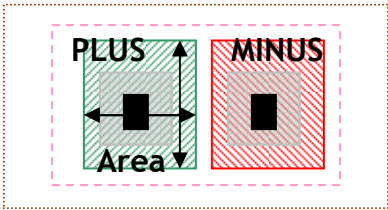
Assura Netlist**Assura auLvs Device Name = "g45pd1svt"**

c g45pd1svt DIO PLUS B MINUS B ;;

* 2 pins

* 2 nets

i D0 g45pd1svt PLUS MINUS ; area 1.6e-13 pj 1.6e-6 m 1 ;



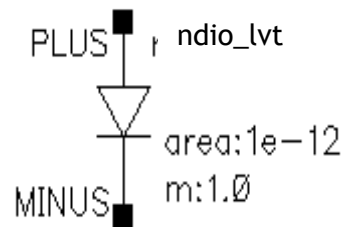
pdio - 1.2 volt P+/Nwell diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND PIMP AND NWELL
PLUS	DIOdummy AND Pimp AND Nwell
MINUS	DIOdummy AND Nimp AND Nwell

LVS Comparison	
Parameter	Calculation
area	Area of PLUS (illustrated above)

21.35 ndio_lvt datasheet

**Spectre Netlist****Spectre Model Name = "g45nd1lvt"**

D0 (PLUS MINUS) g45nd1lvt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45nd1lvt"**

; g45nd1lvt Instance /D0 = auLvs device D0

d g45nd1lvt PLUS MINUS

i 0 g45nd1lvt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45nd1lvt"**

DD0 PLUS MINUS g45nd1lvt 160f 1.6u m=1

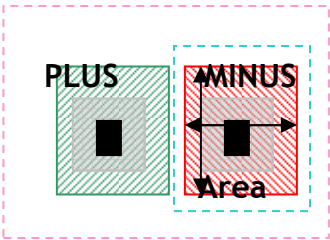
Assura Netlist**Assura auLvs Device Name = "g45nd1lvt"**

c g45nd1lvt DIO PLUS B MINUS B ;;

* 2 pins

* 2 nets

i D0 g45nd1lvt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



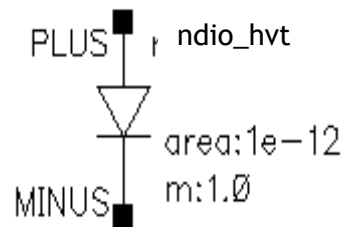
ndio_lvt - 1.2 volt low VT N+/Psub diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Nlvt	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of MINUS (illustrated above)

21.36 ndio_hvt datasheet

**Spectre Netlist****Spectre Model Name = "g45nd1hvt"**

D0 (PLUS MINUS) g45nd1hvt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45nd1hvt"**

; g45nd1hvt Instance /D0 = auLvs device D0

d g45nd1hvt PLUS MINUS

i 0 g45nd1hvt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45nd1hvt"**

DD0 PLUS MINUS g45nd1hvt 160f 1.6u m=1

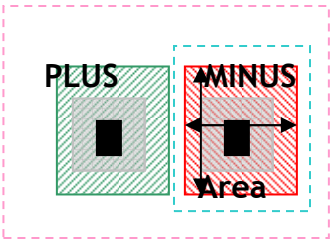
Assura Netlist**Assura auLvs Device Name = "g45nd1hvt"**

c g45nd1hvt DIO PLUS B MINUS B ;;

* 2 pins

* 2 nets

i D0 g45nd1hvt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



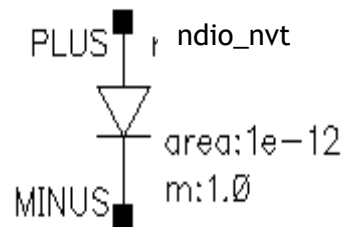
ndio_hvt - 1.2 volt high VT N+/Psub diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Nhvt	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of MINUS (illustrated above)

21.37 ndio_nvt datasheet

**Spectre Netlist****Spectre Model Name = "g45nd1nvt"**

D0 (PLUS MINUS) g45nd1nvt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45nd1nvt"**

; g45nd1nvt Instance /D0 = auLvs device D0

d g45nd1nvt PLUS MINUS

i 0 g45nd1nvt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45nd1nvt"**

DD0 PLUS MINUS g45nd1nvt 160f 1.6u m=1

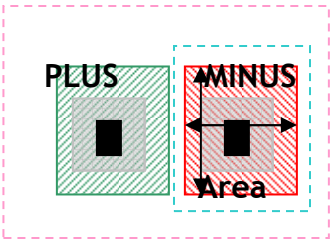
Assura Netlist**Assura auLvs Device Name = "g45nd1nvt"**

c g45nd1nvt DIO PLUS B MINUS B ;;

* 2 pins

* 2 nets

i D0 g45nd1nvt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



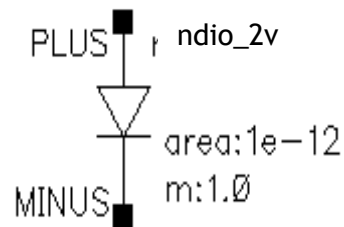
ndio_nvt - 1.2 volt native VT N+/Psub diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Nhvt	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of MINUS (illustrated above)

21.38 ndio_2v datasheet

**Spectre Netlist****Spectre Model Name = "g45nd2svt"**

D0 (PLUS MINUS) g45nd2svt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45nd2svt"**

; g45nd2svt Instance /D0 = auLvs device D0

d g45nd2svt PLUS MINUS

i 0 g45nd2svt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45nd2svt"**

DD0 PLUS MINUS g45nd2svt 160f 1.6u m=1

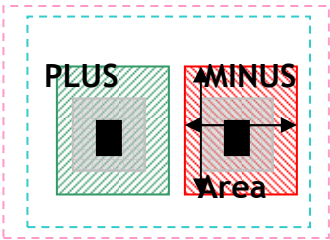
Assura Netlist**Assura auLvs Device Name = "g45nd2svt"**

c g45nd2svt DIO PLUS B MINUS B ;;

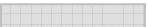
* 2 pins

* 2 nets

i D0 g45nd2svt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;

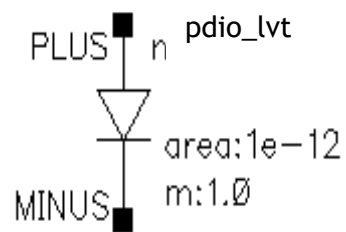


ndio_2v - 1.8 volt N+/Psub diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Oxide_thk	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of MINUS (illustrated above)

21.39 [pdio_lvt datasheet](#)**Spectre Netlist****Spectre Model Name = "g45pd1lvt"**

D0 (PLUS MINUS) g45pd1lvt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45pd1lvt"**

; g45pd1lvt Instance /D0 = auLvs device D0

d g45pd1lvt PLUS MINUS

i 0 g45pd1lvt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45pd1lvt"**

DD0 PLUS MINUS g45pd1lvt 160f 1.6u m=1

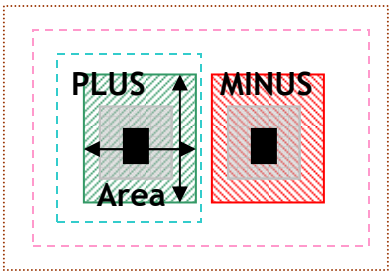
Assura Netlist**Assura auLvs Device Name = "g45pd1lvt"**

c g45pd1lvt DIO PLUS B MINUS B ;;

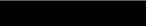
* 2 pins

* 2 nets

i D0 g45pd1lvt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;

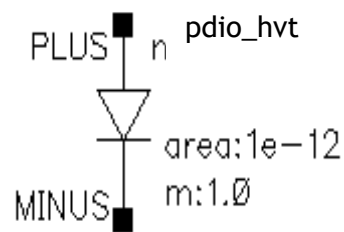


pdio_lvt - 1.2 volt low VT P+/Nwell diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Nlvt	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of PLUS (illustrated above)

21.40 [pdio_hvt datasheet](#)**Spectre Netlist****Spectre Model Name = "g45pd1hvt"**

D0 (PLUS MINUS) g45pd1hvt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45pd1hvt"**

; g45pd1hvt Instance /D0 = auLvs device D0

d g45pd1hvt PLUS MINUS

i 0 g45pd1hvt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45pd1hvt"**

DD0 PLUS MINUS g45pd1hvt 160f 1.6u m=1

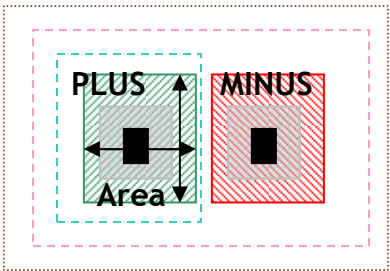
Assura Netlist**Assura auLvs Device Name = "g45pd1hvt"**

c g45pd1hvt DIO PLUS B MINUS B ;;

* 2 pins

* 2 nets

i D0 g45pd1hvt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



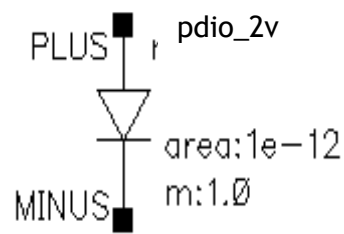
pdio_hvt - 1.2 volt high VT P+/Nwell diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Nhvt	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of PLUS (illustrated above)

21.41 pdio_2v datasheet

**Spectre Netlist****Spectre Model Name = "g45pd2svt"**

D0 (PLUS MINUS) g45pd2svt area=160f pj=1.6u m=1

DIVA LVS Netlist**DIVA Device Name = "g45pd2svt"**

; g45pd2svt Instance /D0 = auLvs device D0

d g45pd2svt PLUS MINUS

i 0 g45pd2svt PLUS MINUS " area 1e-12 pj 1.6e-6 m 1.0 "

CDL Netlist**CDL Device Name = "g45pd2svt"**

DD0 PLUS MINUS g45pd2svt 160f 1.6u m=1

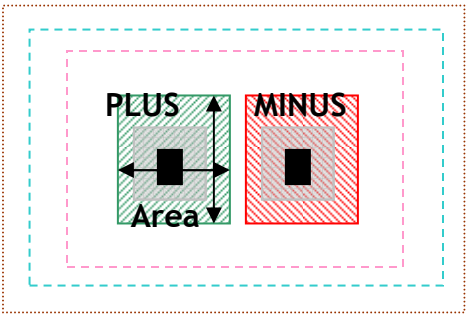
Assura Netlist**Assura auLvs Device Name = "g45pd2svt"**

c g45pd2svt DIO PLUS B MINUS B ;;

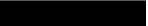
* 2 pins

* 2 nets

i D0 g45pd2svt PLUS MINUS ; area 1e-12 pj 1.6e-6 m 1 ;



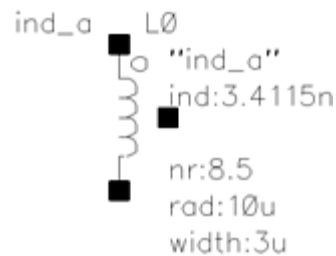
pdio_2v - 1.8 volt P+/Nwell diode

Device Layers	
Layer	Color and Fill
DIOdummy	
Nwell	
Nimp / Oxide	
Pimp / Oxide	
Cont	
Metal1	
Oxide_thk	

Device Derivation	
Device	Layer Derivation
Recognition	DIODUMMY AND NIMP
PLUS	DIOdummy AND Pimp
MINUS	DIOdummy AND Nimp

LVS Comparison	
Parameter	Calculation
area	Area of PLUS (illustrated above)

21.42 ind_a datasheet

**Spectre Netlist****Spectre Model Name = "g45inda"**

L0 (PLUS MINUS B) g45inda w=3u s=1.5u r=10u nr=8.5 m=1

DIVA LVS Netlist**DIVA Device Name = "g45inda"**

; g45inda Instance /L1 = auLvs device L0

d g45inda PLUS MINUS B

i 0 g45inda PLUS MINUS B "width 3u space 1.5e-6 rad 10e-6 nr 8.5 m 1.0
"**CDL Netlist****CDL Device Name = "g45inda"**

L0 PLUS MINUS B g45inda width 3u space 1.5e-6 rad 10e-6 nr 8.5 m 1

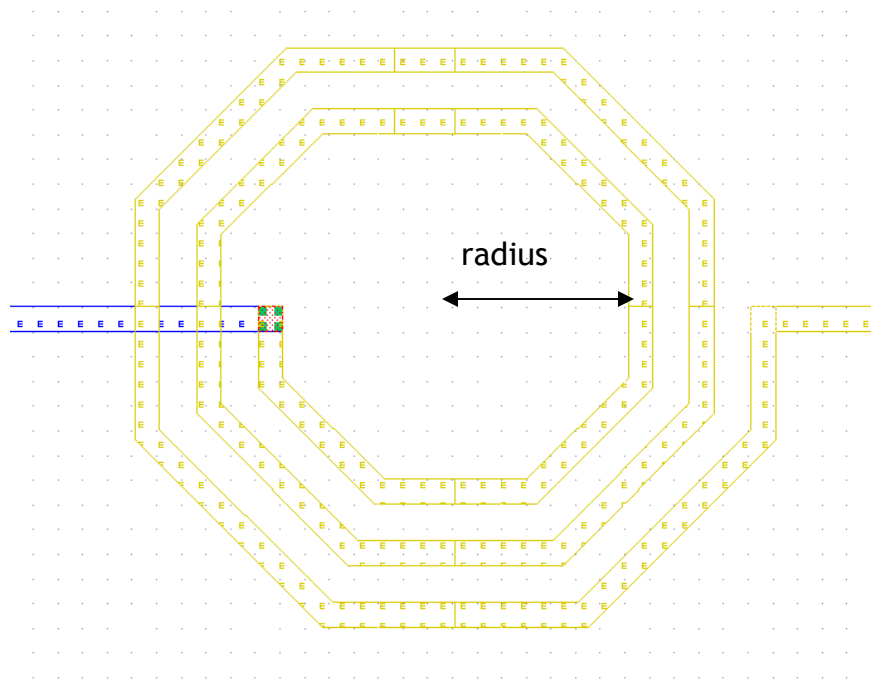
Assura Netlist**Assura auLvs Device Name = "g45inda"**

c g45inda IND PLUS B MINUS B B B ;;

* 3 pins

* 3 nets

i L0 g45inda PLUS MINUS B m 1 nr 8.5 width 3e-06 space 1.5e-06 rad 10e-06 ;



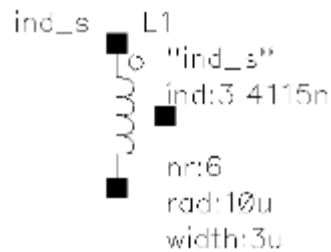
ind_a - Asymmetric Inductor

Device Layers	
Layer	Color and Fill
INDdummy	
Metal11	
Metal10	
IND2dummy	
Cont	

Device Derivation	
Device	Layer Derivation
Recognition	INDDUMMY AND METAL11
PLUS	INDdummy AND Metal11
MINUS	INDdummy AND Metal10

LVS Comparison	
Parameter	Calculation
radius	Inner radius of Metal11 turn (illustrated above)
width	Width of Metal11

21.43 ind_s datasheet

**Spectre Netlist****Spectre Model Name = "g45inds"**

L1 (PLUS MINUS B) g45inds w=3u s=1.5u r=10u nr=6 m=1

DIVA LVS Netlist**DIVA Device Name = "g45inds"**

; g45inds Instance /L2 = auLvs device L1

d g45inds PLUS MINUS B

i 1 g45inds PLUS MINUS B "width 3u space 1.5e-6 rad 10e-6 nr 6 m 1.0 "

CDL Netlist**CDL Device Name = "g45inds"**

L1 PLUS MINUS B g45inds width 3u space 1.5e-6 rad 10e-6 nr 6 m 1

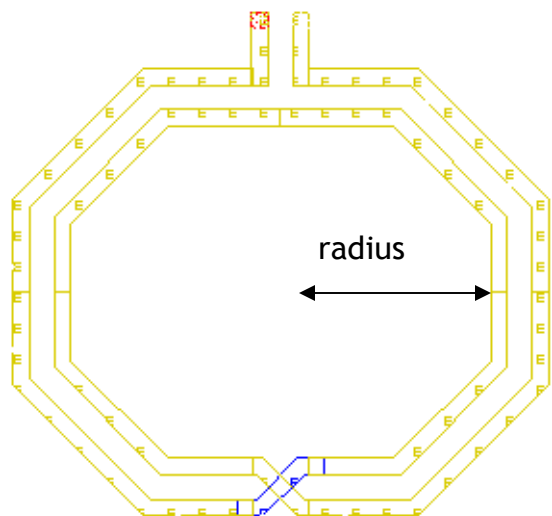
Assura Netlist**Assura auLvs Device Name = "g45inds"**

c g45inds IND PLUS B MINUS B B B ;;

* 3 pins

* 3 nets

i L1 g45inds PLUS MINUS B m 1 nr 6 width 3e-06 space 1.5e-06 rad 10e-06
;



ind_s - Symmetric Inductor

Device Layers	
Layer	Color and Fill
INDdummy	
Metal11	
Metal10	
IND2dummy	
Cont	

Device Derivation	
Device	Layer Derivation
Recognition	INDDUMMY AND METAL11
PLUS	IND2dummy AND Metal11
MINUS	IND3dummy AND Metal11

LVS Comparison	
Parameter	Calculation
radius	Inner radius of Metal11 turn (illustrated above)
width	Width of Metal11